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(54) **CD27-EXTRACELLULAR DOMAIN CAR TO
TARGET CD70-POSITIVE TUMORS**

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(71) Applicant: **BOARD OF REGENTS, THE
UNIVERSITY OF TEXAS SYSTEM,**
Austin, TX (US)

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(72) Inventors: **Katy REZVANI**, Houston, TX (US);
Elizabeth SHPALL, Houston, TX
(US); **Rafet BASAR**, Houston, TX
(US); **Sunil ACHARYA**, Houston, TX
(US); **Nadima UPRETY**, Houston, TX
(US); **David MARIN COSTA**,
Houston, TX (US); **Emily ENSLEY**,
Houston, TX (US)

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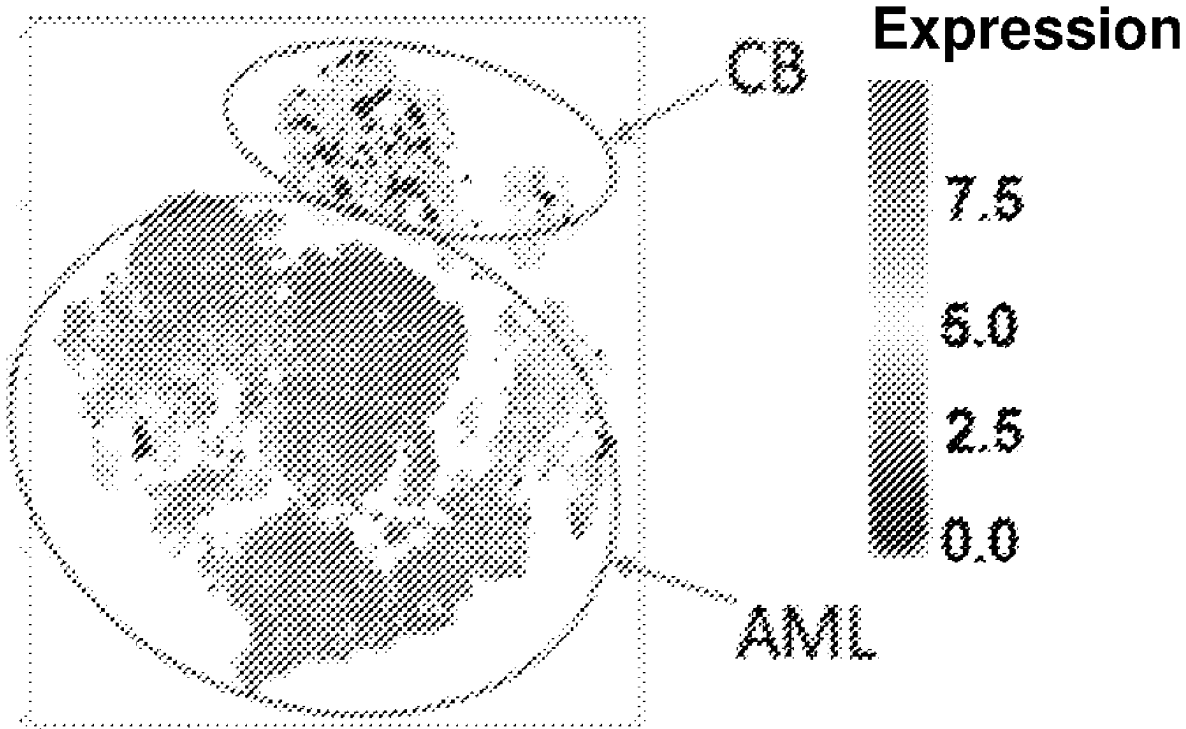
(60) Provisional application No. 63/141,016, filed on Jan.
25, 2021, provisional application No. 63/270,414,
filed on Oct. 21, 2021.

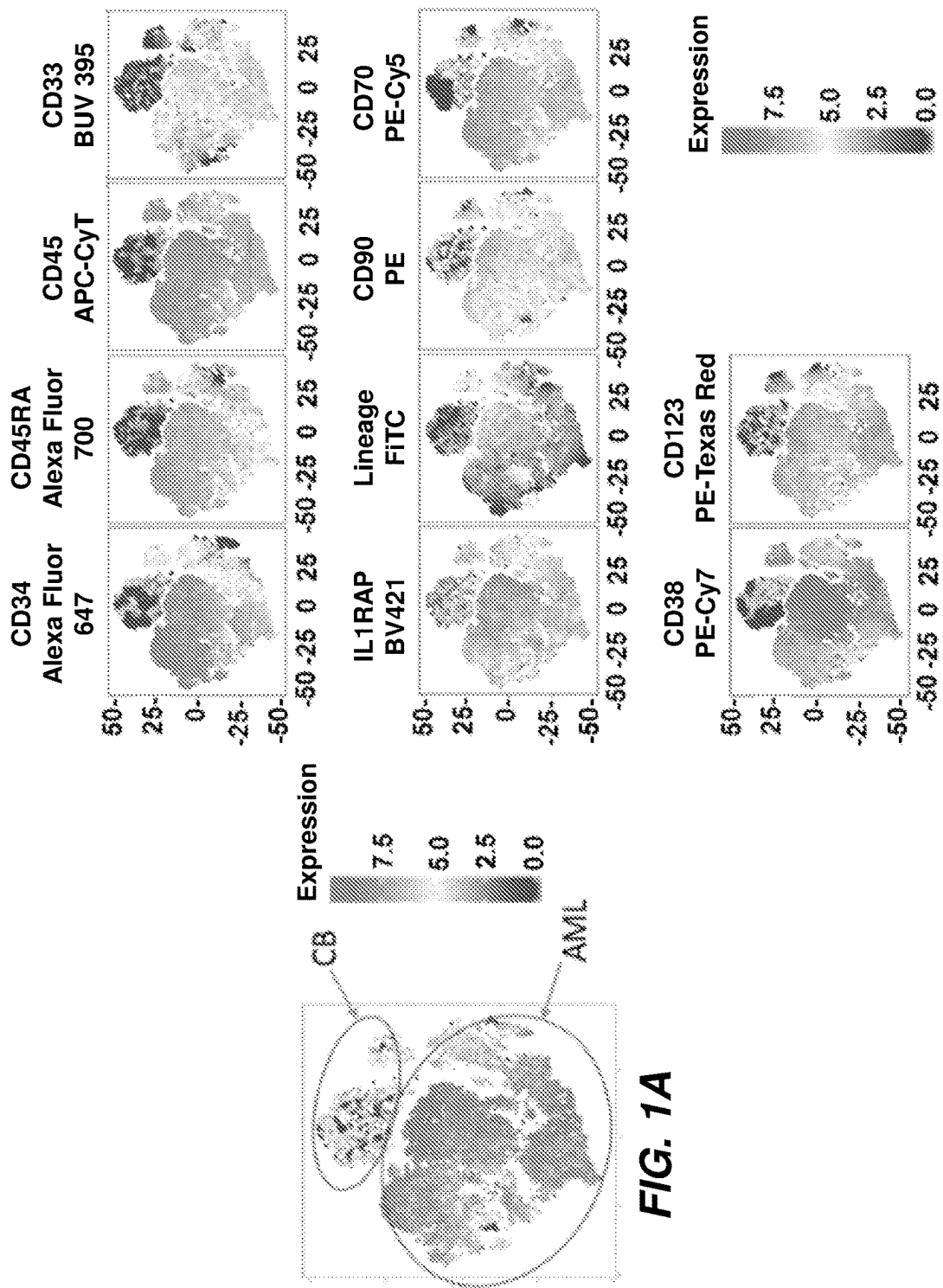
(57)

ABSTRACT

Embodiments of the disclosure encompass methods and compositions that utilize an anti-CD70 chimeric antigen receptor (CAR) that lacks an antibody or antibody fragment as part of the receptor, including as part of the antigen binding region of the CAR. In particular embodiments, instead of the CAR employing an antibody to bind CD70, the CAR molecule utilizes CD27, the receptor for the ligand CD70. In specific embodiments, the CAR comprises a truncated version of CD27 rather than full length CD27. In specific cases, the CAR lacks the CD27 transmembrane domain.

Specification includes a Sequence Listing.





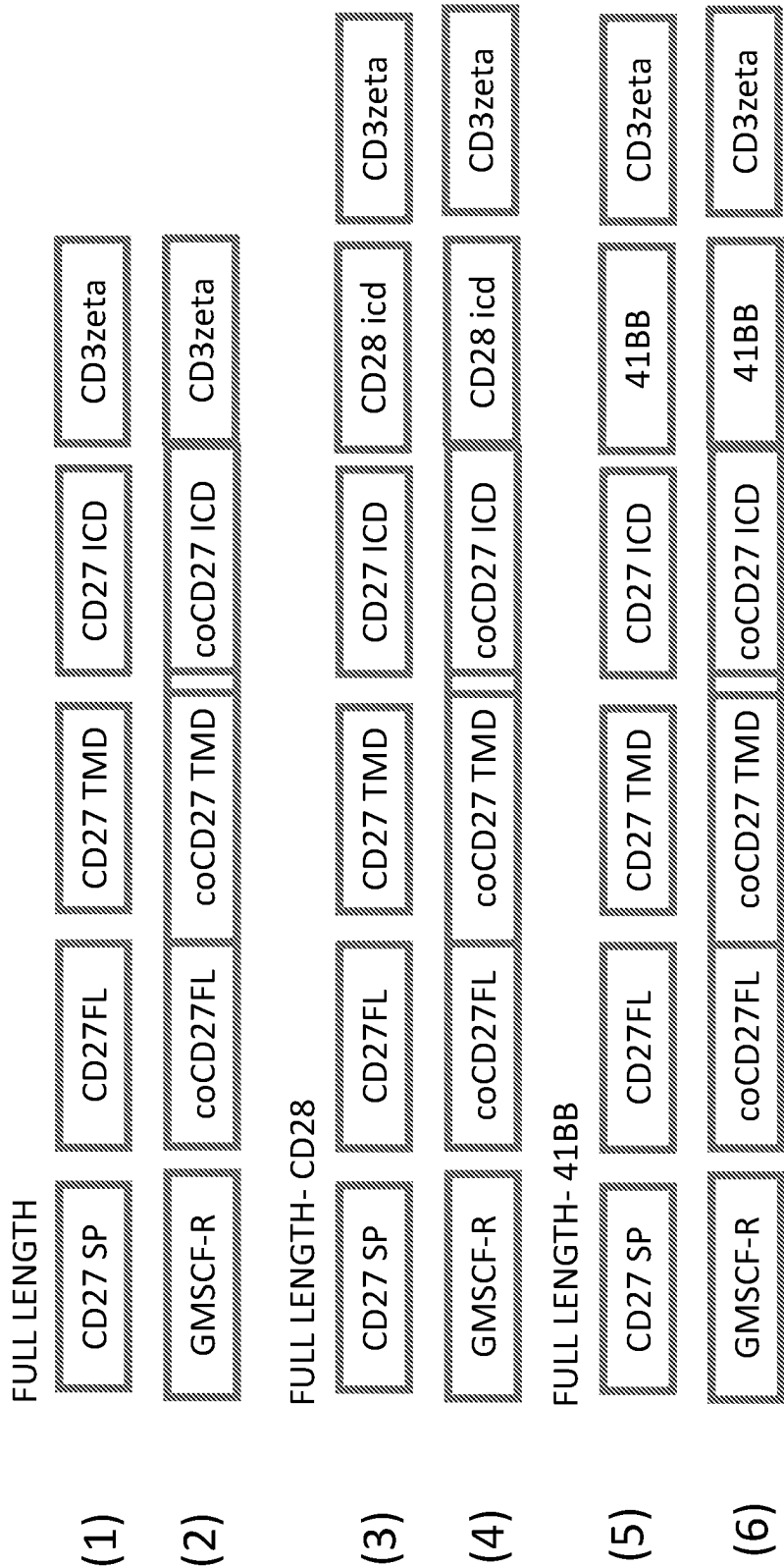


FIG. 2A

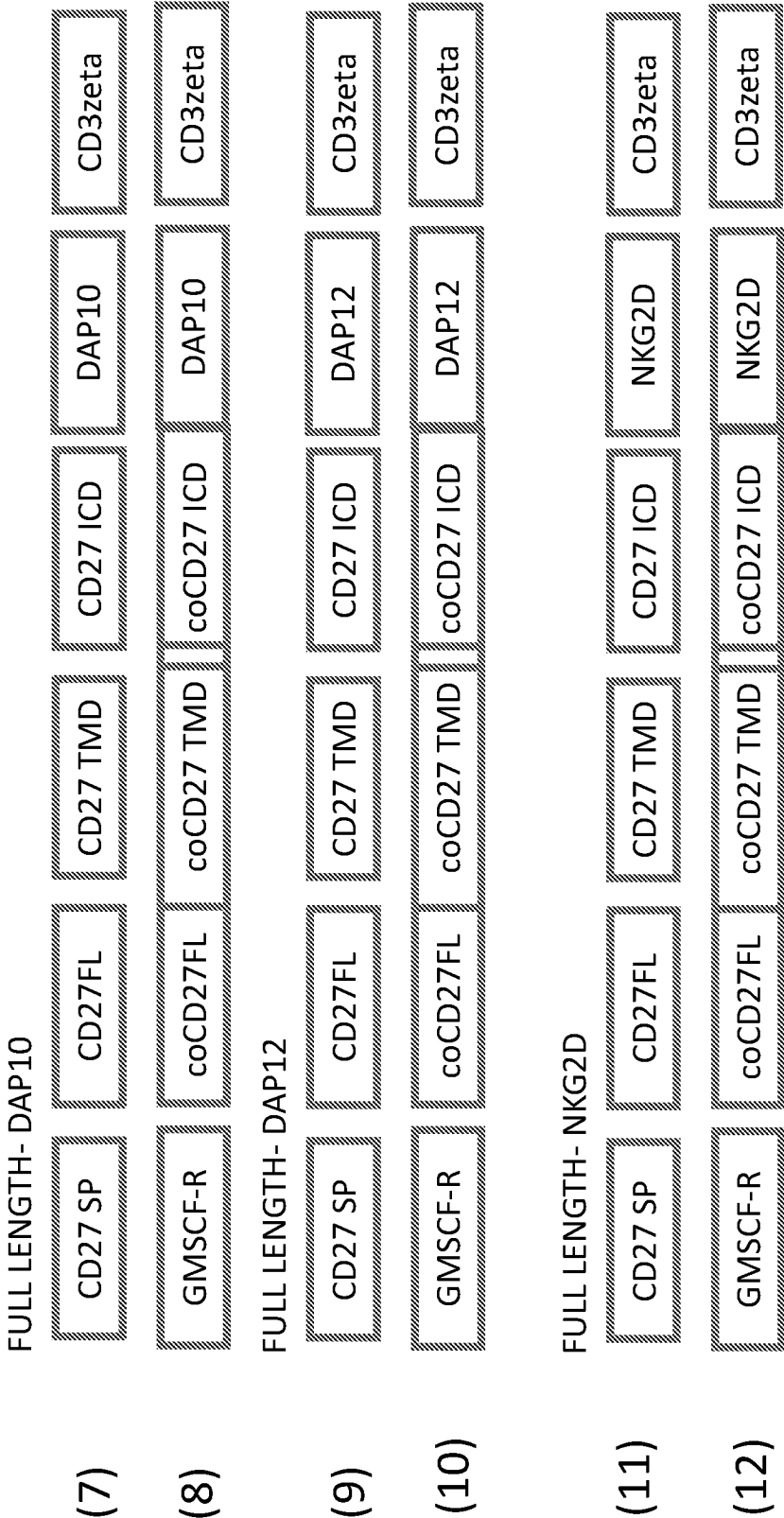


FIG. 2B

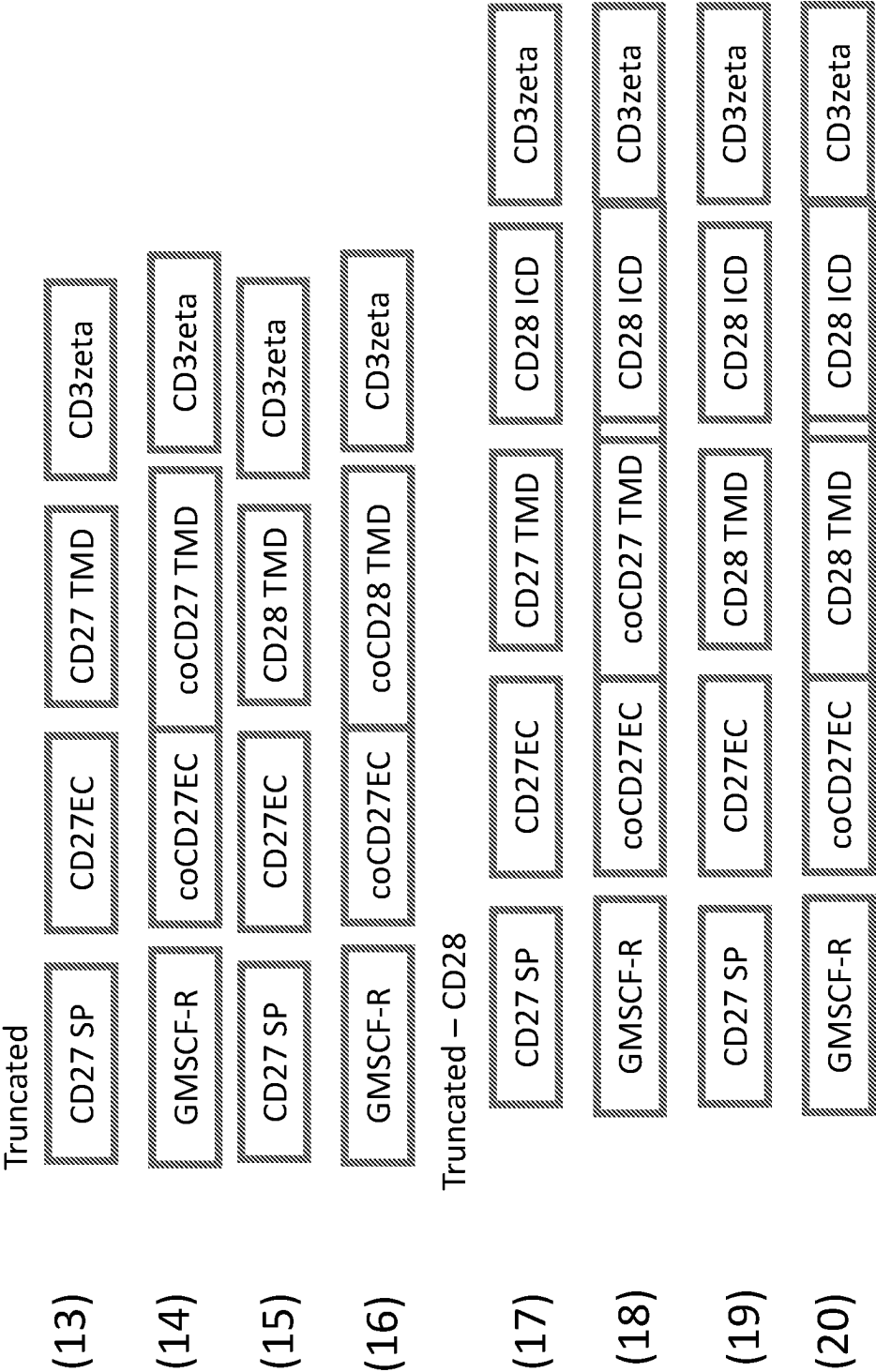


FIG. 2C

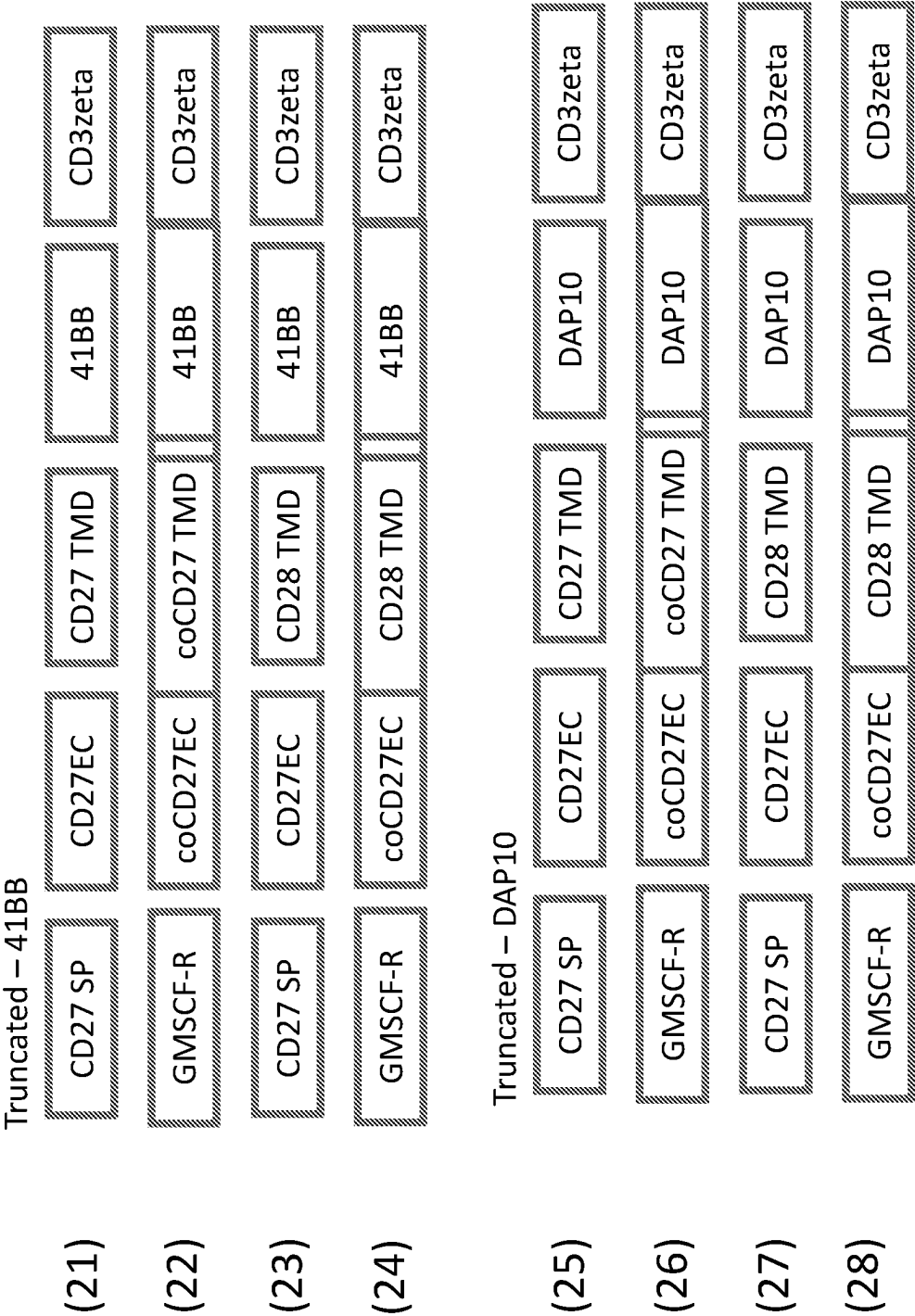


FIG. 2D

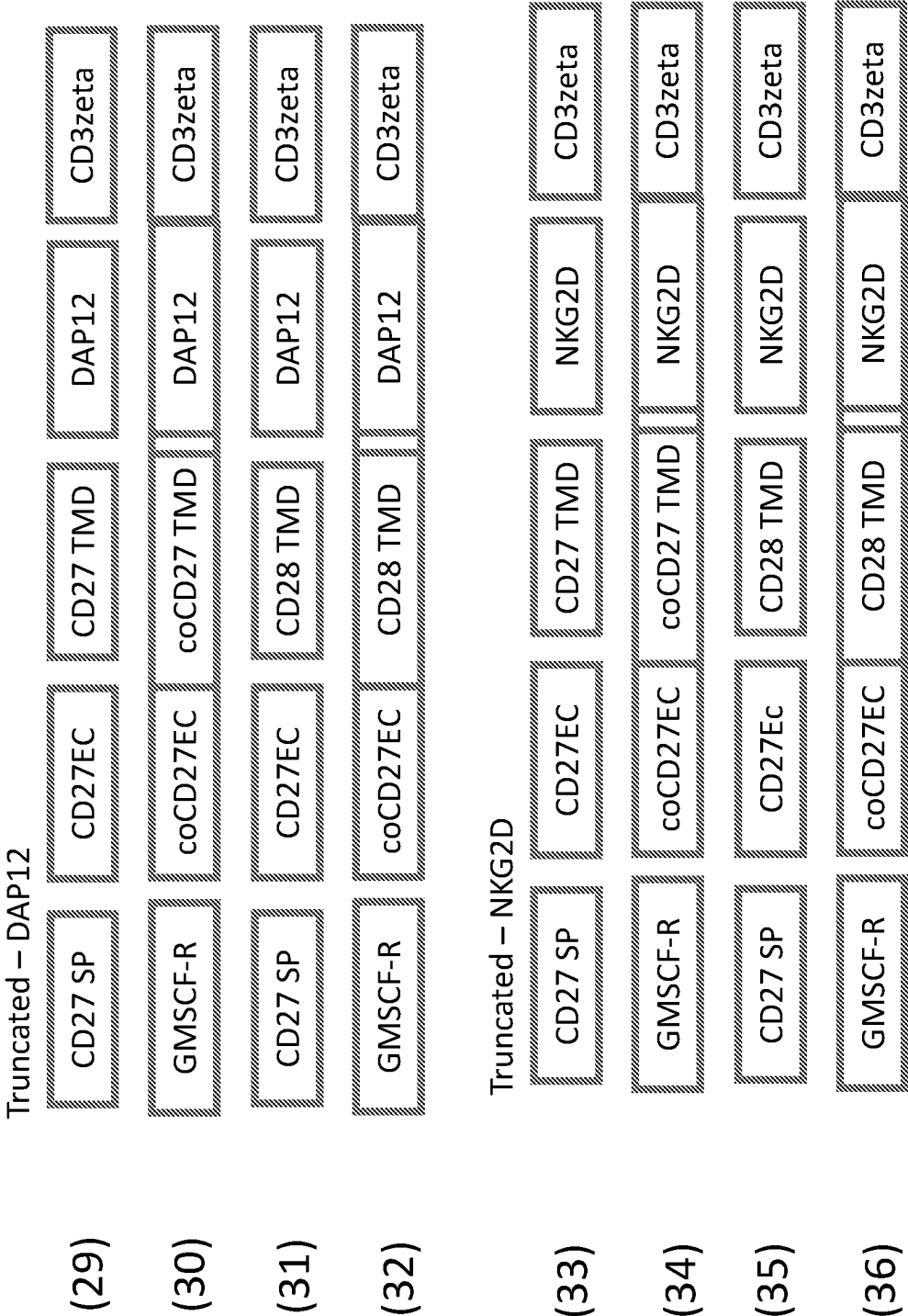


FIG. 2E

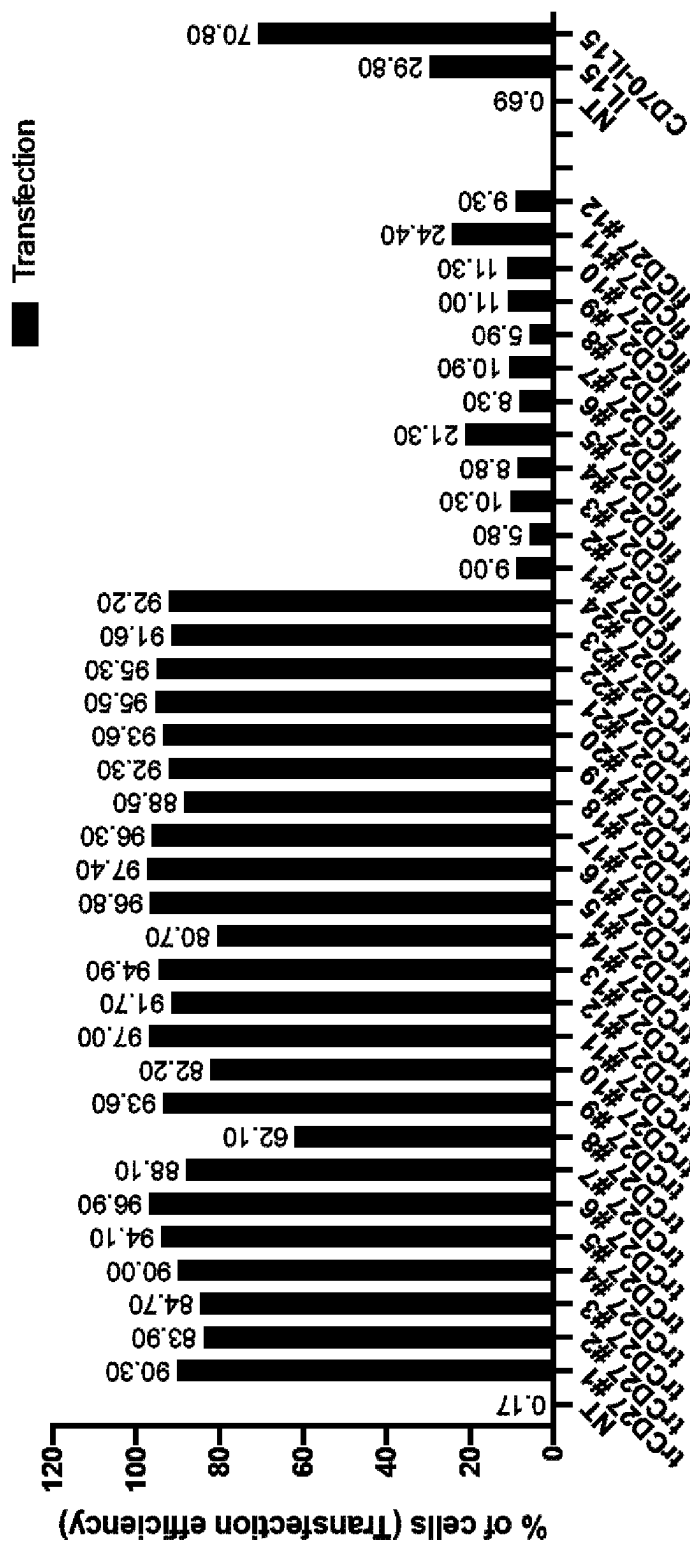


FIG. 3

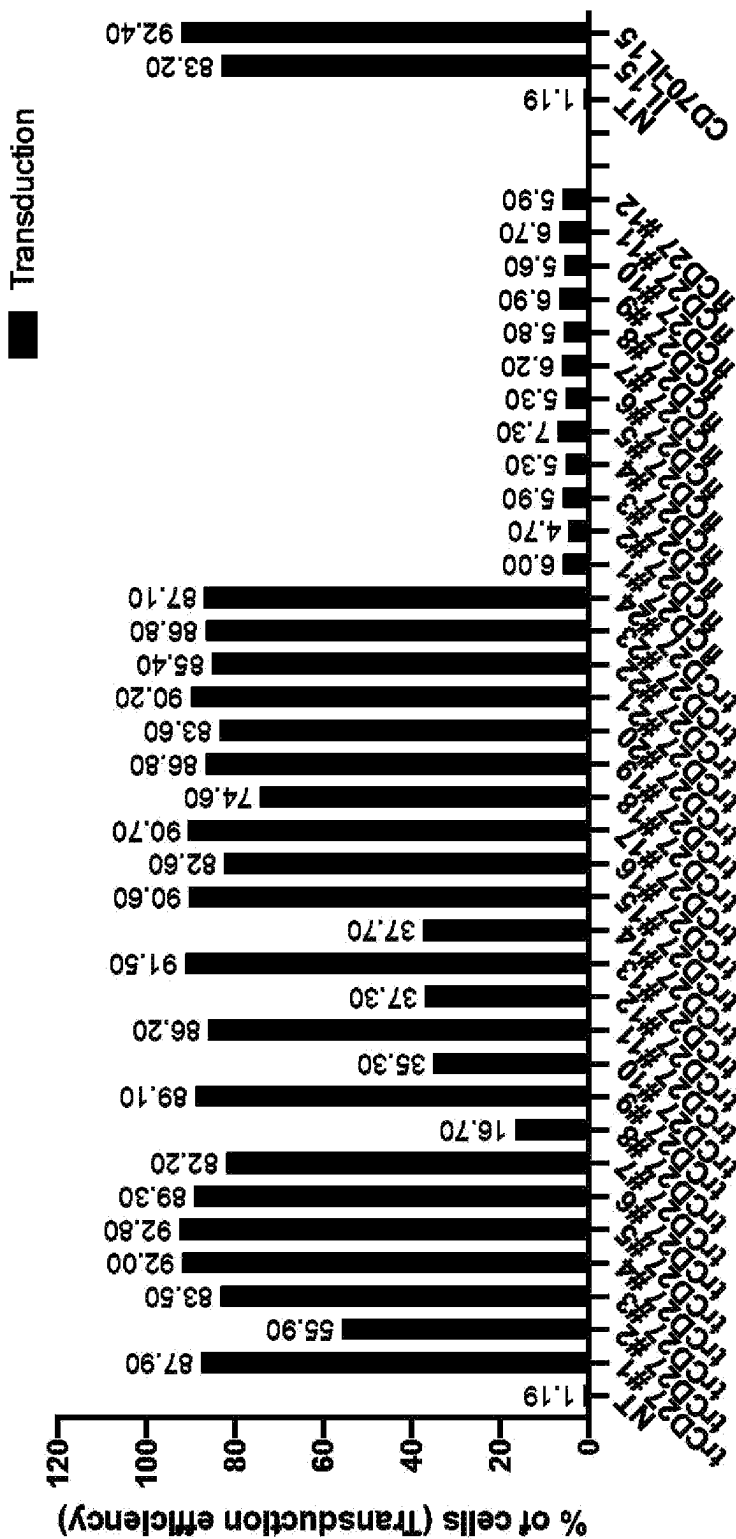


FIG. 4

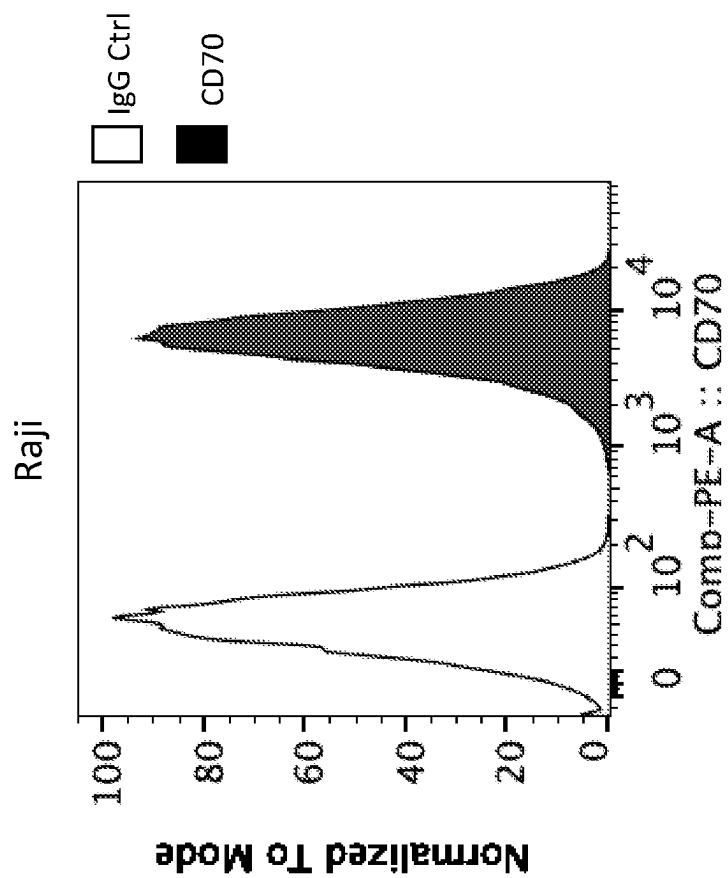


FIG. 5A

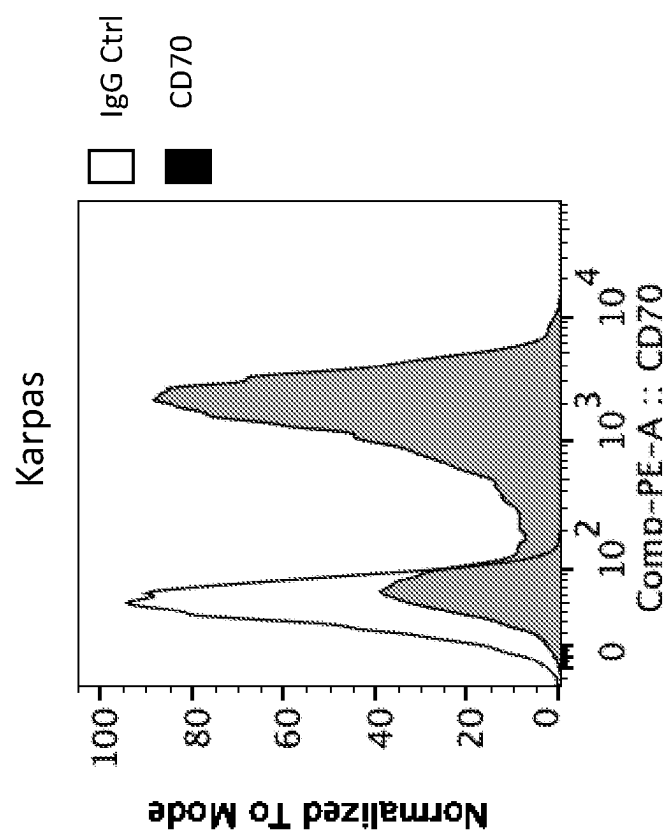


FIG. 5B

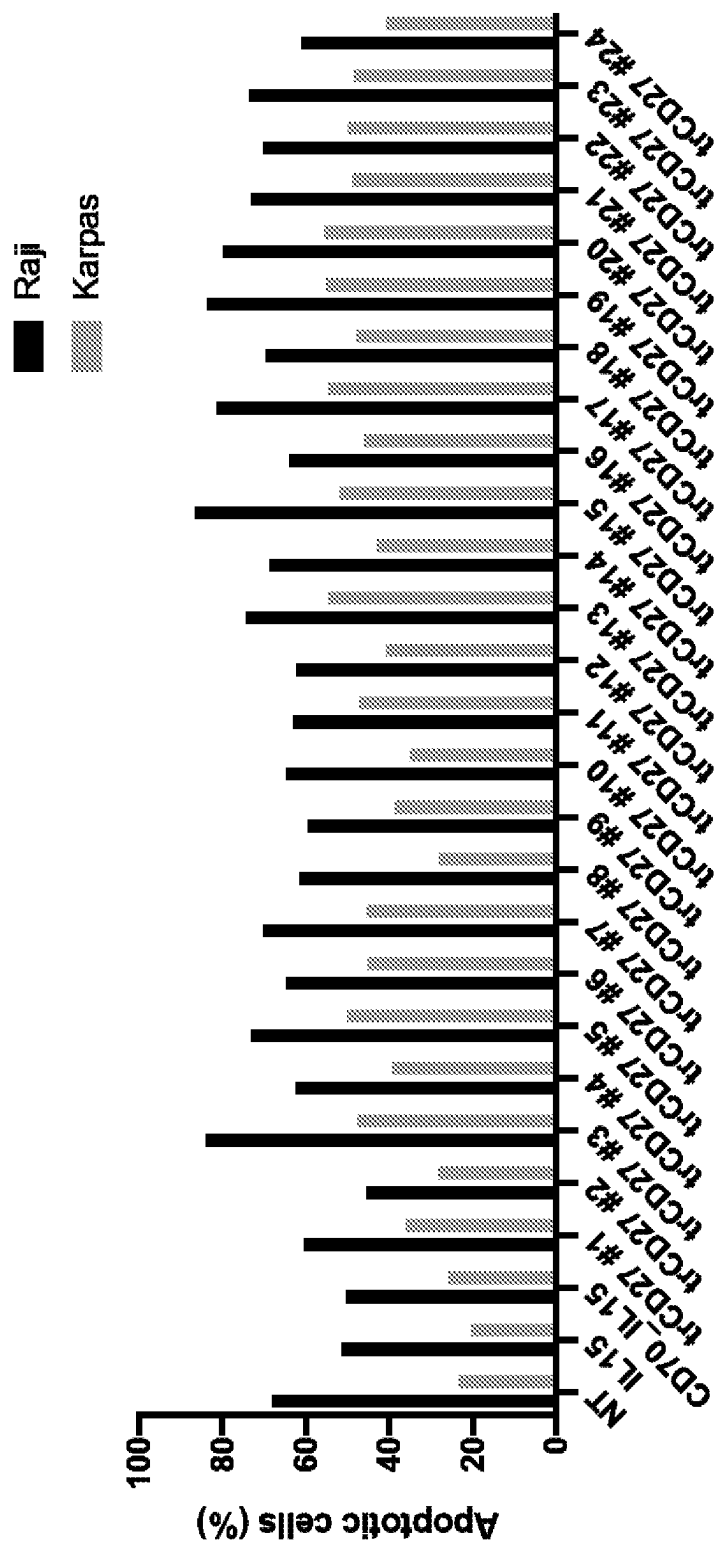


FIG. 5C

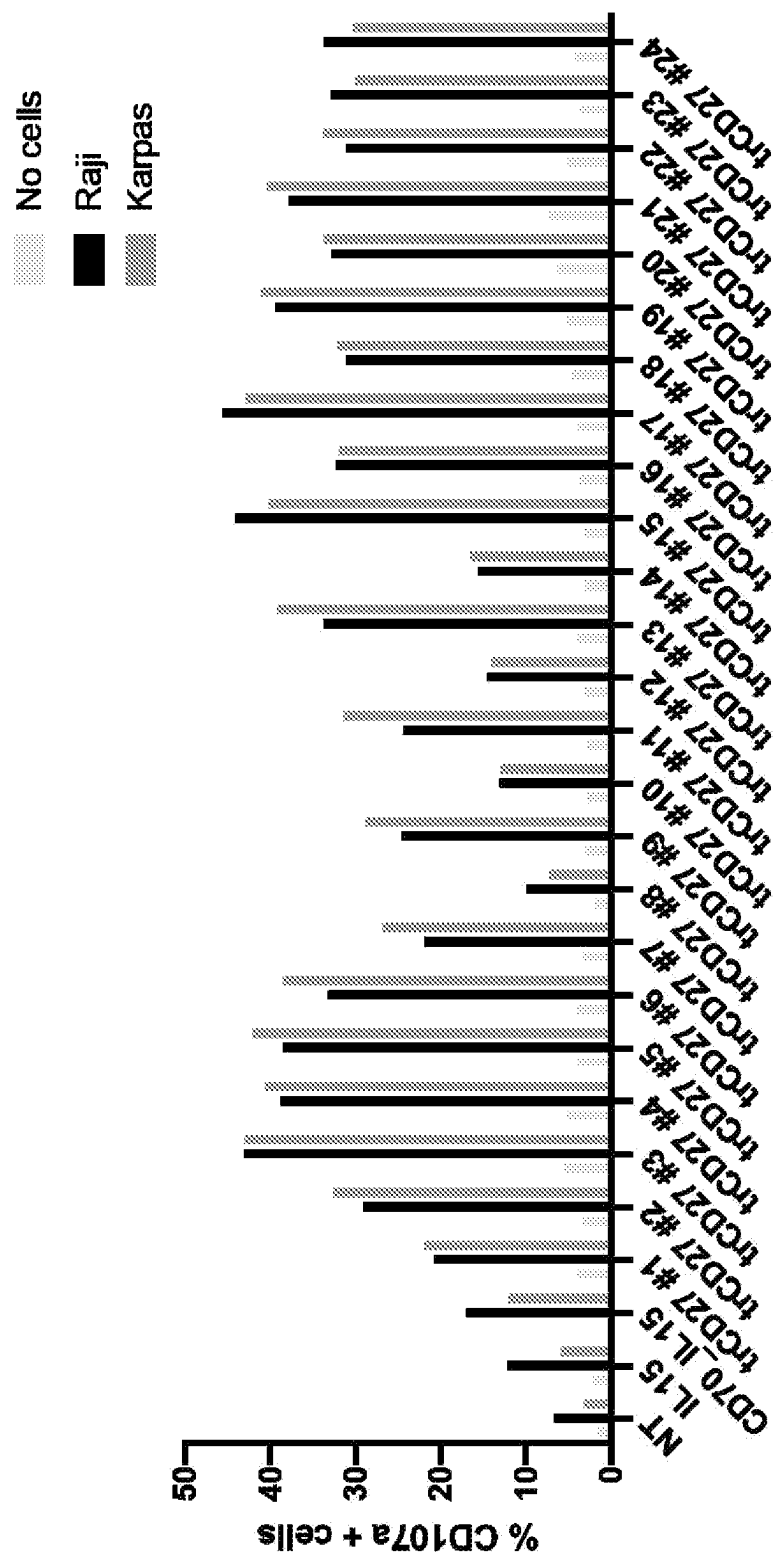


FIG. 6

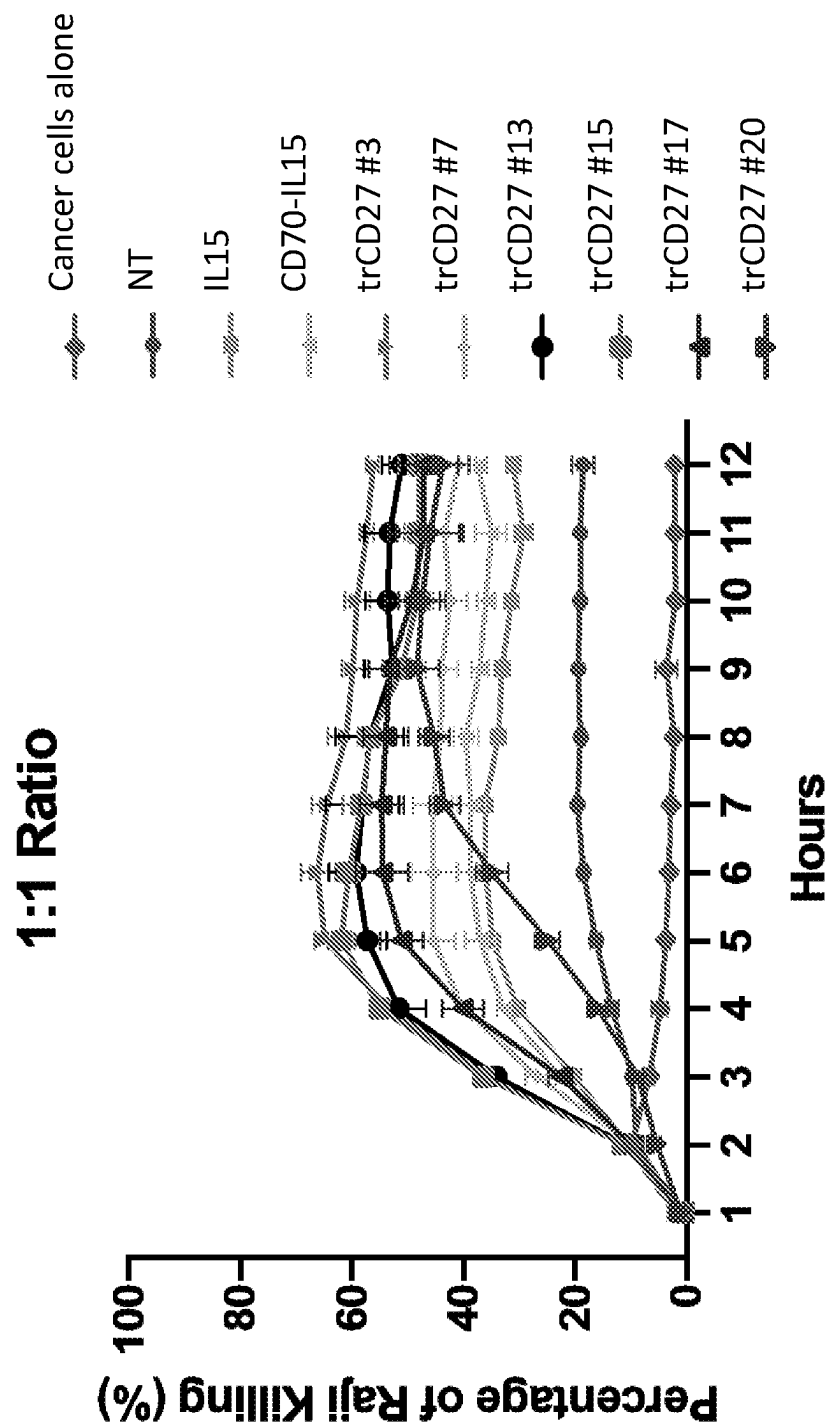
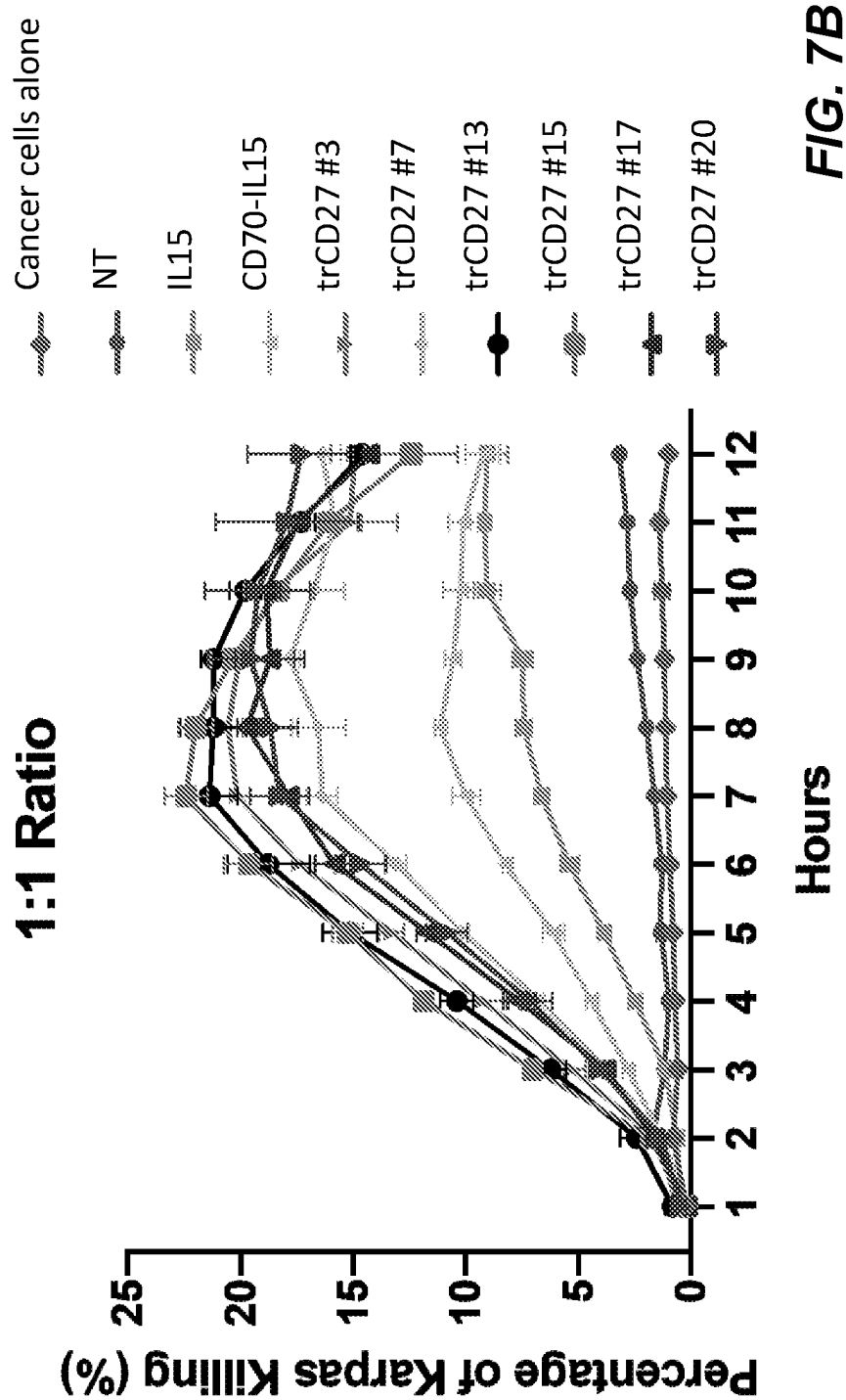
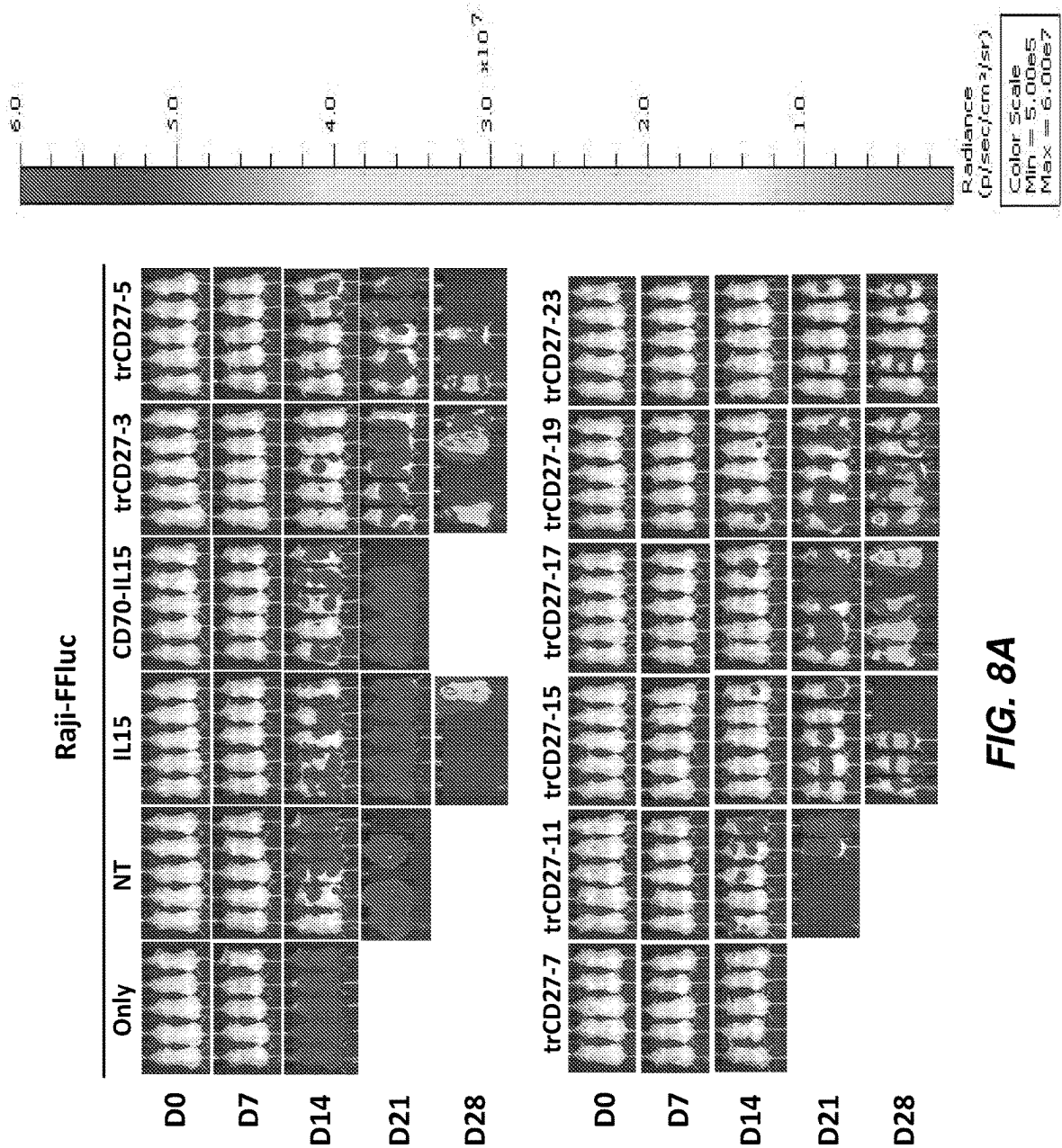


FIG. 7A





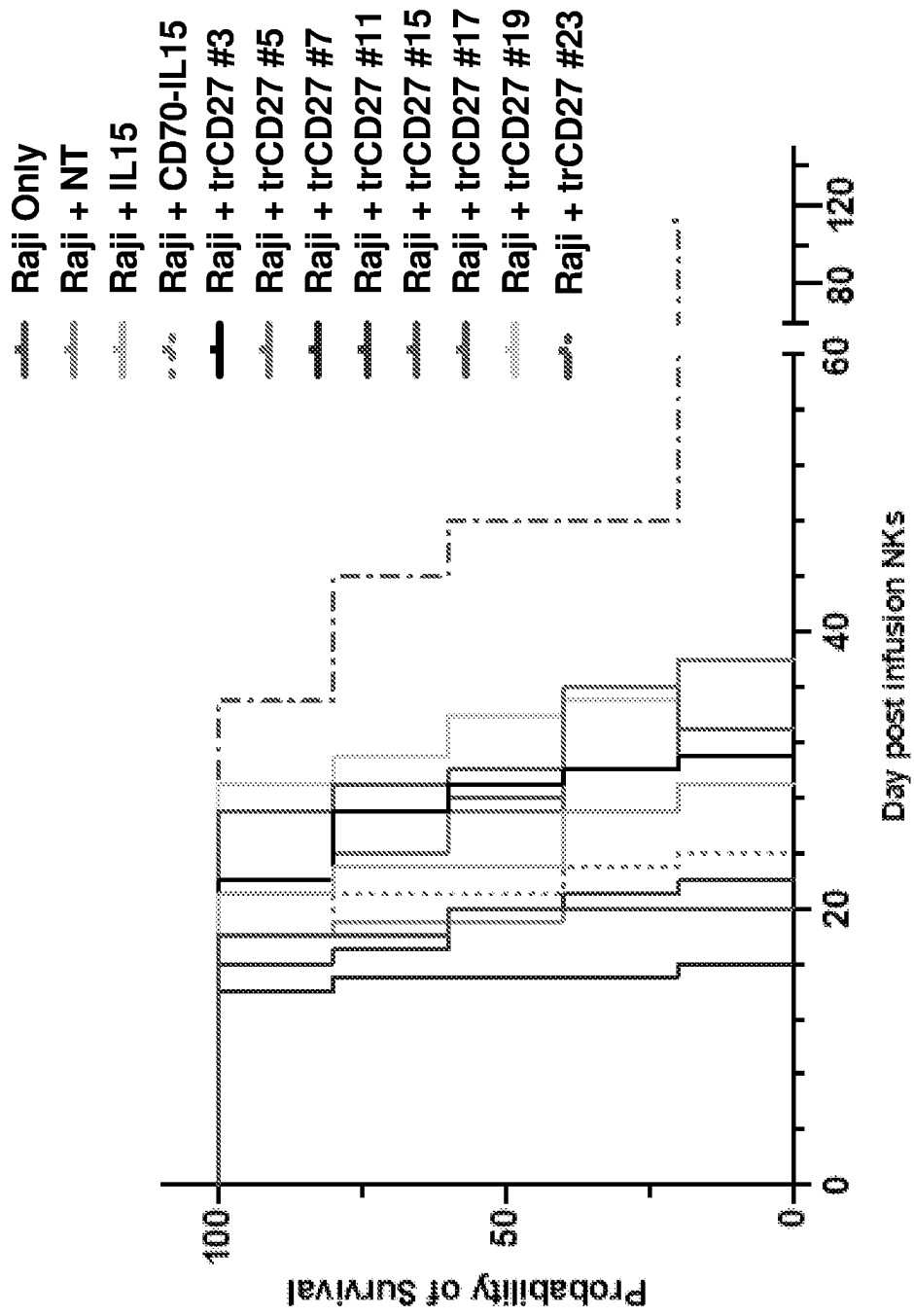


FIG. 8B

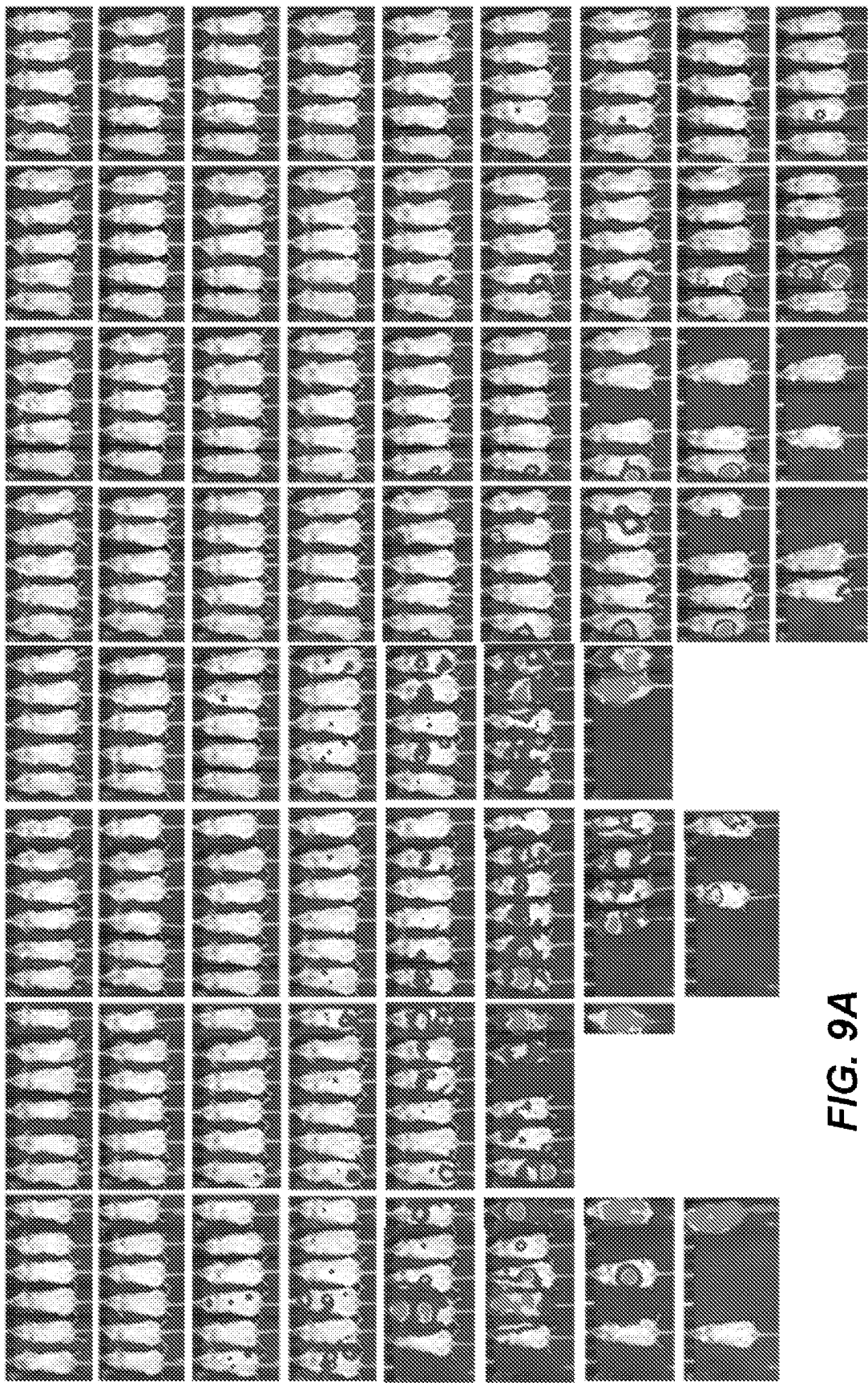


FIG. 9A

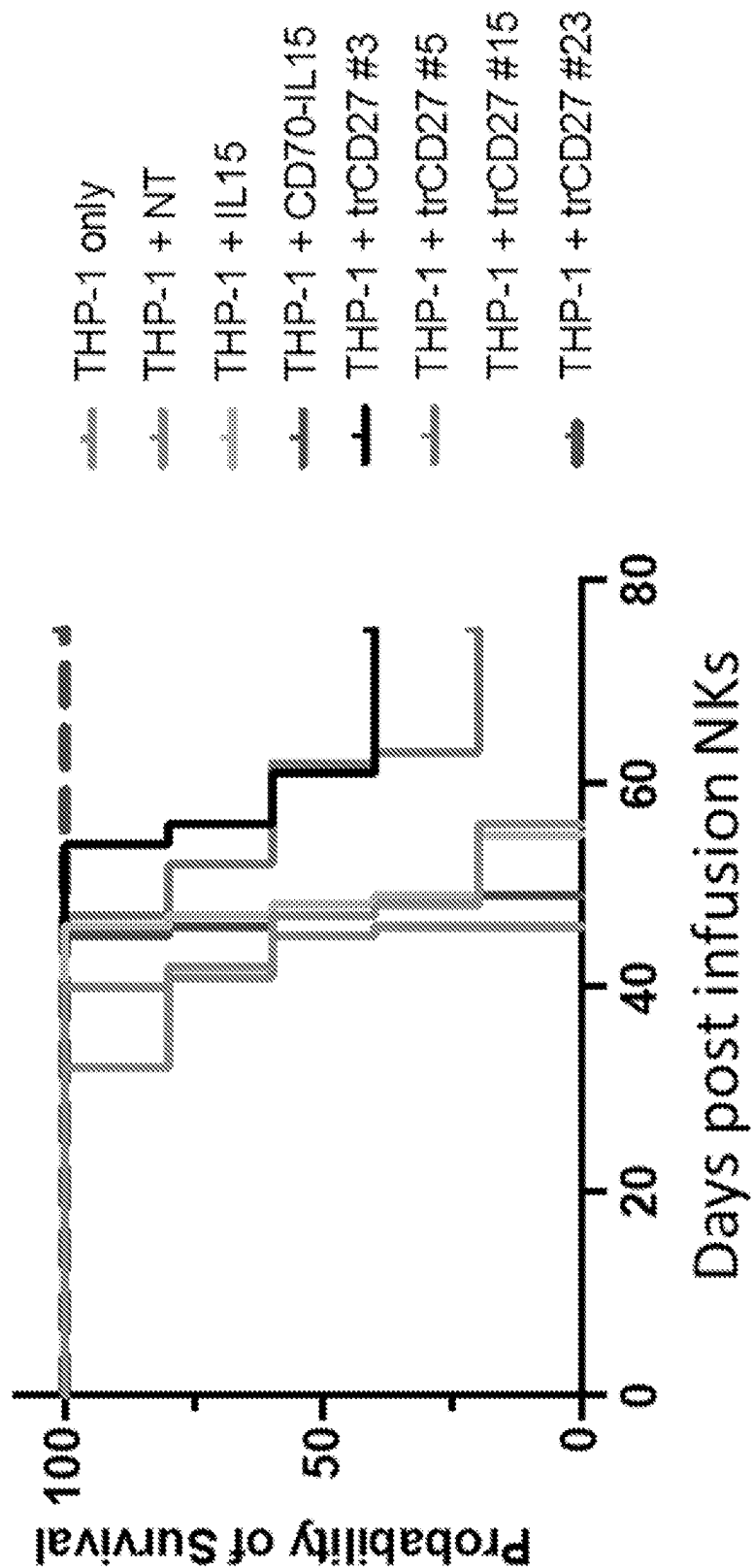


FIG. 9B

CD27-EXTRACELLULAR DOMAIN CAR TO TARGET CD70-POSITIVE TUMORS

[0001] This application is a national phase application under 35 U.S.C. § 371 of International Application No. PCT/US2022/013458, filed Jan. 24, 2022, which claims benefit of priority of U.S. Provisional Application No. 63/141,016, filed Jan. 25, 2021, and U.S. Provisional Application No. 63/270,414, filed Oct. 21, 2021, each of which are hereby incorporated by reference in their entirety.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jan. 18, 2022, is named "MDACP1261US_ST25.txt" and is 209,797 bytes in size.

TECHNICAL FIELD

[0003] Embodiments of the disclosure include at least the fields of cell biology, molecular biology, immunology, and medicine, including cancer medicine.

BACKGROUND

[0004] Genetic reprogramming of Natural Killer (NK) cells for adoptive cancer immunotherapy has clinically relevant applications and benefits such as 1) innate anti-tumor surveillance without prior need for sensitization 2) allogeneic efficacy without graft versus host reactivity and 3) direct cell-mediated cytotoxicity and cytolysis of target tumors. Human NK cell development and acquisition of self-tolerance, alloreactivity, and effector functions is an adaptive process of licensing, calibration, and arming. At the molecular level, specific activating and inhibitory receptors direct NK cellular functions by aggregating, balancing, and integrating extracellular signals into distinct effector functions. The functional activity of NK cells and responsiveness to extrinsic stimuli follow the 'rheostat' model of continuous education and thus are amenable to reprogramming. Genetic modification of NK cells to redirect their effector functions is an effective method to harness their cytotoxic capability to kill tumor cells.

[0005] CD70, the ligand for the cytokine receptor CD27, is an attractive 'pan-cancer antigen' since in addition to being expressed on hematologic malignancies such as acute myeloid leukemia (AML) and lymphoma, it also expressed on many solid tumors, including renal, bladder, lung, triple negative breast cancer, renal cell carcinoma, pancreatic cancer and melanoma, and it is only transiently found on activated T and B lymphocytes and dendritic cells. CD70 is especially advantageous as a target for the immunotherapy of AML as, unlike other AML targets, it is not expressed on normal hematopoietic stem cells and, therefore, as opposed to most of the CAR-T therapies for AML in clinic today, it is unlikely to result in prolonged cytopenias and the need for a hematopoietic stem cell transplant for the recipient after CAR therapy. The present disclosure provides a solution to long-felt needs in the art of cancer therapies for CD70-positive cancers.

BRIEF SUMMARY

[0006] The present disclosure is directed to method and compositions for the treatment or prevention of cancers

having CD70-expressing cells. The disclosure concerns particular anti-CD70 chimeric antigen receptors (CAR) and their use to target CD70-positive cancers of any kind. In particular embodiments, the antigen binding domain that targets CD70, including upon specific binding of CD70, is not an antibody. In particular embodiments, the anti-CD70 CAR lacks an antibody of any kind, including a scFv that binds CD70. In particular embodiments, the extracellular domain of the anti-CD70 CAR does not comprise an scFv or any other type of antibody.

[0007] In particular embodiments, the antigen binding domain that targets CD70 is a natural receptor of CD70, such as the receptor CD27. In specific cases, part or all of CD27 is employed in the CAR molecule. In aspects of the disclosure, the antigen binding domain present in the anti-CD70 CAR molecule comprises part or all of the extracellular domain of CD27, and in specific cases the CAR molecule does or does not utilize the transmembrane domain of CD27. In some cases, the CD28 transmembrane domain is employed instead of the CD27 transmembrane domain, despite the use of the CD27 extracellular domain in the CAR. Particular intracellular domains may be utilized for the CAR, such as CD3zeta alone or combinations of CD3zeta with either DAP10 or DAP12, and in some cases the intracellular domain does not comprise that of 4-1BB.

[0008] Embodiments of the disclosure include any polynucleotide that encodes part or all of the anti-CD70 CAR. Specific embodiments encompass polynucleotides that encode anti-CD70 CARs comprising an antigen binding domain, transmembrane domain, and at least one intracellular domain (including one or more costimulatory domains), wherein the CAR comprises an antigen binding domain that does not comprise an antibody. Vectors that harbor expression constructs that encode the anti-CD70 CARs, and optionally other genes, are contemplated. The vectors may or may not be viral. Cells harboring the vectors are also contemplated, including immune cells of any kind, such as at least NK cells or T cells. In some cases the cells are off-the-shelf cells, including NK cells, and these cells may or may not already be manipulated to express one or more heterologous genes, such as one or more exogenous cytokines.

[0009] Methods of treating or preventing cancer of any kind are encompassed herein, including by administering cells expressing particular anti-CD70 CARs at a therapeutically effective amount to ameliorate or prevent the cancer, or reduce the risk of the cancer, reduce the severity of the cancer, prevent metastasis or risk thereof, or delay the onset of the cancer.

[0010] Embodiments of the disclosure include methods of targeting CD70-expressing cells and, in specific embodiments, the cells are cancerous, whereas in other methods the CD70-expressing cells are immunoregulatory cells such as Tregs.

[0011] The present disclosure includes methods to genetically engineer human NK cells (including from cord blood (CB)) to target CD70+ve tumors using a CD27-ectodomain CAR. Encompassed herein are a variety of novel CAR constructs that fuse the extracellular binding portion of CD27 (CD27eCAR, or CD27 EC) with activating signaling endodomains that incorporate either CD3ζ alone or in combination with costimulatory or adaptor signaling domains, such as 4-1BB, CD28, DAP10 or DAP12, and a cytokine gene such as IL-15, IL-12, IL-18, or IL-21, to

support NK cell survival and proliferation. In a series of in vitro studies, the activity of CD27eCAR/IL-15 transduced CB-NK cells against AML and lymphoma targets is confirmed.

[0012] Embodiments of the disclosure include polynucleotides that encode an anti-CD70 chimeric antigen receptor (CAR) comprising a signal peptide, an anti-CD70 antigen binding domain, transmembrane domain, and at least one intracellular domain, wherein the anti-CD70 antigen binding domain does not comprise an antibody, and optionally wherein a transmembrane domain of the CAR is not a CD27 transmembrane domain. In specific embodiments, the anti-CD70 antigen binding domain is comprised in an extracellular domain of CD27 and/or may be codon optimized. In specific aspects, the antigen binding domain comprises, consists of, or consists essentially of SEQ ID NO:2. In some cases, the antigen binding domain and transmembrane domain comprise, consist of, or consist essentially of SEQ ID NO:1. The sequence that encodes the transmembrane domain may be codon optimized, and/or the transmembrane domain may be from CD28, the alpha chain of the T- cell receptor, beta chain of the T- cell receptor, zeta chain of the T- cell receptor, CD3 zeta, CD3 epsilon, CD3 gamma, CD3 delta, CD45, CD4, CD5, CD8, CD9, CD 16, CD22, CD33, CD37, CD64, CD80, CD86, CD 134, CD137, CD154, ICOS/CD278, GITR/CD357, NKG2D, DAP10, or DAP12. In a specific case, the CD28 transmembrane domain comprises, consists of, or consists essentially of SEQ ID NO:3. The CAR may comprise two or more, or three or more, intracellular domains. An intracellular domain may comprise any ITAM-containing signaling domain. An intracellular domain may comprise CD3zeta. The intracellular domain may be an intracellular costimulatory domain, such as one selected from the group consisting of CD27, CD28, 4-1BB, DAP12, NKG2D, OX-40 (CD134), DAP10, CD40L, 2B4, DNAM, CS1, CD48, NKp30, NKp44, NKp46, NKp80, and a combination thereof.

[0013] In some embodiments, the signal peptide is from CD27 or granulocyte-macrophage colony-stimulating factor receptor (GMSCF-R), and the sequence of the CD27 signal peptide may comprise, consist of, or consist essentially of SEQ ID NO:6. The signal peptide may be from CD27, the anti-CD70 antigen binding domain may be from CD27, the transmembrane domain may be from CD27, and an intracellular domain is from CD27. The signal peptide may comprise, consist of, or consist essentially of SEQ ID NO:6. The CAR may comprise, consist of, or consist essentially of SEQ ID NO:4. In some cases, the transmembrane domain comprises, consists of, or consists essentially of SEQ ID NO:3 or SEQ ID NO:7. In certain aspects, the intracellular domain comprises one or more of SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:11, or SEQ ID NO:12. The CAR may comprise, consist of, or consist essentially of SEQ ID NO:5. The CAR may comprise, consist of, or consist essentially of SEQ ID NO:1.

[0014] In specific embodiments, the polynucleotide encodes a CAR, and the CAR comprises one or more of the following:

- [0015]** (a) CD27 signal peptide (SP), CD27 extracellular domain (EC), CD27 transmembrane domain (TMD), DAP12 intracellular domain (ICD), and CD3zeta;
- [0016]** (b) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, DAP12 ICD;

- [0017]** (c) CD27 SP, CD27 EC, CD28 TMD, DAP12 ICD, and CD3zeta;
- [0018]** (d) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, DAP12 ICD, and CD3zeta;
- [0019]** (e) CD27 SP, CD27 EC, CD27 TMD, Natural killer group 2 member D (NKG2D) ICD, and CD3zeta;
- [0020]** (f) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, NKG2D ICD, and CD3zeta;
- [0021]** (g) CD27 SP, CD27 EC, CD28 TMD, Natural killer group 2 member D (NKG2D) ICD, and CD3zeta;
- [0022]** (h) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, NKG2D ICD, and CD3zeta;
- [0023]** (i) CD27 SP, CD27 EC, CD27 TMD, 4-1BB ICD, and CD3zeta;
- [0024]** (j) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, 4-1BB ICD, and CD3zeta;
- [0025]** (k) CD27 SP, CD27 EC, CD28 TMD, 4-1BB ICD, and CD3zeta;
- [0026]** (l) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, 4-1BB ICD, and CD3zeta;
- [0027]** (m) CD27 SP, CD27 EC, CD27 TMD, DAP10 ICD, and CD3zeta;
- [0028]** (n) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, DAP10 ICD, and CD3zeta;
- [0029]** (o) CD27 SP, CD27 EC, CD28 TMD, DAP10 ICD, and CD3zeta;
- [0030]** (p) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, DAP10 ICD, and CD3zeta;
- [0031]** (q) CD27 SP, CD27 full length (FL), CD27 TMD, CD27 ICD, and CD3zeta;
- [0032]** (r) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, and CD3zeta;
- [0033]** (s) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, CD28 ICD, and CD3zeta;
- [0034]** (t) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, CD28 ICD, and CD3zeta;
- [0035]** (u) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, 4-1BB ICD, and CD3zeta;
- [0036]** (v) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, 4-1BB ICD, and CD3zeta;
- [0037]** (w) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, DAP10 ICD, and CD3zeta;
- [0038]** (x) GMSCF-R SP, CD27 FL, CD27 TMD, CD27 ICD, DAP10 ICD, and CD3zeta;
- [0039]** (y) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, DAP12 ICD, and CD3zeta;
- [0040]** (z) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, DAP12 ICD, and CD3zeta;
- [0041]** (aa) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, NKG2D ICD, and CD3zeta;
- [0042]** (bb) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, NKG2D ICD, and CD3zeta;
- [0043]** (cc) CD27 SP, CD27 EC, CD27 TMD, and CD3zeta;
- [0044]** (dd) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, and CD3zeta;

- [0045] (ee) CD27 SP, CD27 EC, CD28 TMD, and CD3zeta;
 [0046] (ff) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD28 TMD, and CD3zeta;
 [0047] (gg) CD27 SP, CD27 EC, CD27 TMD, CD28 ICD, and CD3zeta;
 [0048] (hh) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, CD28 ICD, and CD3zeta;
 [0049] (ii) CD27 SP, CD27 EC, CD28 TMD, CD28 ICD, and CD3zeta; or
 [0050] (jj) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, and CD3zeta.

[0051] Embodiments of the disclosure include cells that comprise any polynucleotides encompassed herein. The cells may or may not be immune cells. Examples of immune cells include Natural Killer (NK) cells, T cell, gamma delta T cells, invariant NKT (iNKT) cells, B cells, macrophages, MSCs, or dendritic cells. When the cells are NK cells, they may be derived from cord blood, peripheral blood, induced pluripotent stem cells, bone marrow, hematopoietic stem cells (HSCs), from a cell line, or a mixture thereof. The NK cell line may be a NK-92 cell line or another NK cell line derived from a tumor or from a healthy NK cell, or is a progenitor cell. In specific embodiments, the NK cell is derived from a cord blood mononuclear cell. The NK cell may be a CD56+ NK cell. In some cases, the NK cells express one or more exogenously provided cytokines, such as IL-15, IL-2, IL-12, IL-18, IL-21, IL-7, or a combination thereof.

[0052] Embodiments of the disclosure include populations of immune cells of the disclosure, said cells present in a suitable medium. In specific cases, the immune cells are NK cells. The immune cells may or may not be obtained from cryopreserved storage.

[0053] In particular embodiments, there are methods of killing CD70-positive cells in an individual, comprising the step of administering to the individual a therapeutically effective amount of cells harboring any polynucleotide encompassed herein. The cells may be NK cells, T cells, gamma delta T cells, invariant NKT (iNKT) cells, B cells, macrophages, gamma delta T cells, or dendritic cells. The NK cells may be derived from cord blood, peripheral blood, induced pluripotent stem cells, bone marrow, HSCs, or from a cell line. The NK cells may be derived from cord blood mononuclear cells. In some cases, the CD70-positive cells are not cancer cells and may be T regulatory cells. The individual may have or be suspected of having any CD70-expressing cancer. The individual may have or be suspected of having, for example, acute myeloid leukemia, lymphoma, lung cancer, renal cancer, bladder cancer, melanoma, glioblastoma, breast cancer, head and neck cancer, mesothelioma, multiple myeloma, pancreatic cancer or a combination thereof. The cells may be allogeneic or autologous with respect to the individual. The individual may be a human. In some embodiments, the cells are administered to the individual once or more than once. The duration of time between administrations of the cells to the individual may be 1-24 hours, 1-7 days, 1-4 weeks, 1-12 months, or one or more years.

[0054] In certain embodiments of methods encompassed herein, the method further comprises the step of providing to the individual an effective amount of an additional therapy, such as surgery, radiation, gene therapy, immunotherapy,

and/or hormone therapy. In certain cases, the additional therapy comprises one or more antibodies.

[0055] In particular embodiments, the cells are administered to the individual by injection, intravenously, intraarterially, intraperitoneally, intratracheally, intratumorally, intramuscularly, endoscopically, intralesionally, intracranially, percutaneously, subcutaneously, regionally, by perfusion, in a tumor microenvironment, or a combination thereof. In specific embodiments, the method further comprises the step of identifying CD70-positive cells in the individual.

[0056] The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter which form the subject of the claims herein. It should be appreciated by those skilled in the art that the conception and specific embodiments disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present designs. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope as set forth in the appended claims. The novel features which are believed to be characteristic of the designs disclosed herein, both as to the organization and method of operation, together with further objects and advantages will be better understood from the following description when considered in connection with the accompanying figures. It is to be expressly understood, however, that each of the figures is provided for the purpose of illustration and description only and is not intended as a definition of the limits of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0057] For a more complete understanding of the present disclosure, reference is now made to the following descriptions taken in conjunction with the accompanying drawings.

[0058] FIGS. 1A-1B show CD70 expression on AML, patient samples using Tsne plots.

[0059] FIGS. 2A-2E illustrate a variety of anti-CD70 CAR constructs.

[0060] The following table provides information regarding the truncated (tr) and full length (fl) CD27 constructs and their respective reference numbers used in data encompassed herein. S. No refers to serial number.

S. No	Construct Name	ID
1	CD27TrCD28icdCD3ZIL1	trCD27 #1
2	GSPco27Tr28CD3z15	trCD27 #2
3	CD27tr28tdmCD3zIL15	trCD27 #3
4	GSPco27Tr28tmd3z15	trCD27 #4
5	CD27TrCD3ZIL15	trCD27 #5
6	GSPco27Tr3z15	trCD27 #6
7	CD27TrNKG2Dic3z15	trCD27 #7
8	GSPco27TrNKG2Dcd3z1	trCD27 #8
9	CD27Tr28tmd NKG2Dic 3z	trCD27 #9
10	GSPco27Tr28tmdNKG2Di	trCD27 #10
11	CD27Tr28tmdDAP12icd3	trCD27 #11
12	GSPco27Tr28tmdap12i	trCD27 #12
13	CD27TrDAP12icd3z15	trCD27 #12
14	GSPco27Trda p12icd3z1	trCD27 #14
15	CD27Tr28tmdDAP10icd3	trCD27 #15
16	GSPco27Tr28tmdDAP10i	trCD27 #16
17	CD27TrDAP10icd3z15	trCD27 #17
18	GSPco27FLdap10icd3z1	trCD27 #18

-continued

S. No	Construct Name	ID
19	CD27Tr28tmd41BBicd3z	trCD27 #19
20	GSPco27Tr28tmd41BBic	trCD27 #20
21	CD27Tr41BBicd3z15	trCD27 #21
22	GSPco27Tr41BBicd3z15	trCD27 #22
23	CD27TrCD28tmdicd3z15	trCD27 #24
24	GSPco27Tr28tmdicd3z	trCD27 #25
25	CD27FLDAP10icd3z15	flCD27 #1
26	SPco27FLdap10icd3z15	flCD27 #2
27	CD27FLDAP12icd3z15	flCD27 #3
28	SPco27FLdap12icd3z15	flCD27 #4
29	CD27FLNKG2Dic3z15	flCD27 #5
30	SPco27FLNKG2Dic3z15	flCD27 #6
31	CD27FLCD3zIL15	flCD27 #7
32	GSPco27FLcd3z15	flCD27 #8
33	CD27FLCD28icd3z15	flCD27 #9
34	GSPco27FL28CD3z15	flCD27 #10
35	CD27FL41BBicd3z15	flCD27 #11
36	GSPco27FL41BBicd3z15	flCD27 #12

[0061] With respect to this table, information regarding the truncated (tr) and full length (fl) CD27 constructs and their respective identification numbers used in the figures are as follows:

- [0062]** GSP: GMCSF receptor signal peptide
- [0063]** co: codon optimized
- [0064]** CD28 tmd: CD28 transmembrane domain
- [0065]** CD27 tmd: CD27 transmembrane domain
- [0066]** 3z: CD3 zeta signal
- [0067]** CD28 ic: CD28 intracellular domain
- [0068]** Natural killer group 2 member D (NKG2D) ic: NKG2D intracellular domain
- [0069]** Dap12 icd: DAP12 intracellular domain
- [0070]** Dap10 icd: DAP12 intracellular domain
- [0071]** 41bb icd: 41bb intracellular domain

[0072] FIG. 3 demonstrates transfection efficiency of truncated (tr) and full length (fl) CD27 constructs in 293T cells.

[0073] FIG. 4 demonstrates transduction efficiency of truncated (tr) and full length (fl) CD27 constructs in cord blood derived NK cells.

[0074] FIGS. 5A-5C demonstrate CD70 expression on tumor targets and killing activity of cord blood natural killer (CB-NK) cells transduced with various trCD27 constructs against CD70-positive Raji (FIG. 5A) and Karpas cells (FIG. 5B) using a 4 hr Annexin-V assay (FIG. 5C). FIG. 5C shows better killing for various trCD70 transduced CB-NK cells of both Raji (black bar) and Karpas (grey bar) cells, when compared to NT (non-transduced), IL15 (construct with only IL15 cytokine gene) or CD70-IL15 (construct with CD70 scFv together with IL15) CAR-transduced CBNK cells.

[0075] FIG. 6 demonstrates CD107a degranulation (a marker for the cytotoxic activity of NK cells) by CB-NK cells transduced with various trCD27 constructs against Raji and Karpas cells. TrCD27 CAR transduced CBNK cells and cancer cells (Raji and Karpas) were co-cultured at a 1:1 ratio in the presence of CD107a antibody for 6 hrs. After incubation, CD107a expression is assessed by flow cytometry. In the bars grouped in three, the left (green) bar is the control with CB-NK cells in the absence of co-culture with cancer cells, the middle (black) bar is trCD70-transduced CB-NK cells co-cultured with Raji cells, and the right (gray) bar is trCD70-transduced CB-NK cells co-cultured with Karpas cells. Most of the trCD27 transduced CB-NK cells showed

enhanced cytotoxic activity against Raji and Karpas cells, when compared to NT, IL15 or CD70-IL15 CAR-transduced CB-NK cells.

[0076] FIGS. 7A-7B show killing activity of CB-NK cells transduced with various trCD27 constructs against Raji cells (FIG. 7A) and Karpas cells (FIG. 7B) using Incucyte live imaging. TrCD27 CAR transduced CBNK cells and Raji or Karpas cells were co-cultured at 1:1 ratio, and real-time cytotoxicity of NK cells against Raji and Karpas cells was measured every hour over a 12 hour period. TrCD27 transduced CB-NK cells showed enhanced cytotoxic activity against Raji and Karpas cells, when compared to NT, IL15 or CD70-IL15 CAR-transduced CB-NK cells. Cancer cells cultured in the absence of CBNK cells (cancer cells alone) was used as control.

[0077] FIGS. 8A-8B show in vivo anti-tumor activity of CB-NK cells transduced with various trCD27 constructs against Raji cells having high CD70 expression. FIG. 8A shows bioluminescence imaging of Raji tumors after infusion with CB-NK cells transduced with various trCD27 CARs. FIG. 8B shows survival curves after infusion with CB-NK cells transduced with various trCD27 CARs.

[0078] FIGS. 9A-9B show in vivo anti-tumor activity of CB-NK cells transduced with various trCD27 constructs against THP-1 cells having high CD70 expression. FIG. 9A shows bioluminescence imaging of THP-1 tumors after infusion with CB-NK cells transduced with various trCD27 CARs. FIG. 9B shows survival curves after infusion with CB-NK cells transduced with various trCD27 CARs.

DETAILED DESCRIPTION

[0079] In keeping with long-standing patent law convention, the words “a” and “an” when used in the present specification in concert with the word comprising, including the claims, denote “one or more.” Some embodiments of the disclosure may consist of or consist essentially of one or more elements, method steps, and/or methods of the disclosure. It is contemplated that any method or composition described herein can be implemented with respect to any other method or composition described herein and that different embodiments may be combined.

[0080] Throughout this specification, unless the context requires otherwise, the words “comprise”, “comprises” and “comprising” will be understood to imply the inclusion of a stated step or element or group of steps or elements but not the exclusion of any other step or element or group of steps or elements. By “consisting of” is meant including, and limited to, whatever follows the phrase “consisting of” Thus, the phrase “consisting of” indicates that the listed elements are required or mandatory, and that no other elements may be present. By “consisting essentially of” is meant including any elements listed after the phrase, and limited to other elements that do not interfere with or contribute to the activity or action specified in the disclosure for the listed elements. Thus, the phrase “consisting essentially of” indicates that the listed elements are required or mandatory, but that no other elements are optional and may or may not be present depending upon whether or not they affect the activity or action of the listed elements.

[0081] Reference throughout this specification to “one embodiment,” “an embodiment,” “a particular embodiment,” “a related embodiment,” “a certain embodiment,” “an additional embodiment,” or “a further embodiment” or combinations thereof means that a particular feature, struc-

ture or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the foregoing phrases in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0082] As used herein, the terms “or” and “and/or” are utilized to describe multiple components in combination or exclusive of one another. For example, “x, y, and/or z” can refer to “x” alone, “y” alone, “z” alone, “x, y, and z,” “(x and y) or z,” “x or (y and z),” or “x or y or z.” It is specifically contemplated that x, y, or z may be specifically excluded from an embodiment.

[0083] Throughout this application, the term “about” is used according to its plain and ordinary meaning in the area of cell and molecular biology to indicate that a value includes the standard deviation of error for the device or method being employed to determine the value.

[0084] The term “engineered” as used herein refers to an entity that is generated by the hand of man, including a cell, nucleic acid, polypeptide, vector, and so forth. In at least some cases, an engineered entity is synthetic and comprises elements that are not naturally present or configured in the manner in which it is utilized in the disclosure. In specific embodiments, a vector is engineered through recombinant nucleic acid technologies, and a cell is engineered through transfection or transduction of an engineered vector.

[0085] As used herein, “prevent,” and similar words such as “prevented,” “preventing” etc., indicate an approach for preventing, inhibiting, or reducing the likelihood of the occurrence or recurrence of, a disease or condition, e.g., cancer. It also refers to delaying the onset or recurrence of a disease or condition or delaying the occurrence or recurrence of the symptoms of a disease or condition. As used herein, “prevention” and similar words also includes reducing the intensity, effect, symptoms and/or burden of a disease or condition prior to onset or recurrence of the disease or condition.

[0086] The term “sample,” as used herein, generally refers to a biological sample. The sample may be taken from tissue or cells from an individual. In some examples, the sample may comprise, or be derived from, a tissue biopsy, blood (e.g., whole blood), blood plasma, extracellular fluid, dried blood spots, cultured cells, discarded tissue. The sample may have been isolated from the source prior to collection. Non-limiting examples include blood (including but not limited to cord blood), serum, plasma, cerebral spinal fluid, pleural fluid, amniotic fluid, lymph fluid, saliva, urine, stool, tears, sweat, bone marrow, or mucosal excretions, and other bodily fluids isolated from the primary source prior to collection. In some examples, the sample is isolated from its primary source (cells, tissue, bodily fluids such as blood, environmental samples, etc.) during sample preparation. The sample may or may not be purified or otherwise enriched from its primary source. In some cases the primary source is homogenized prior to further processing. The sample may be filtered or centrifuged to remove buffy coat, lipids, or particulate matter. The sample may also be purified or enriched for nucleic acids, or may be treated with RNases. The sample may contain tissues or cells that are intact, fragmented, or partially degraded.

[0087] The term “subject,” as used herein, generally refers to an individual having a biological sample that is undergoing processing or analysis and, in specific cases, has or is suspected of having cancer. The subject can be any organism or animal subject that is an object of a method or material, including mammals, e.g., humans, laboratory animals (e.g., primates, rats, mice, rabbits), livestock (e.g., cows, sheep, goats, pigs, turkeys, and chickens), household pets (e.g., dogs, cats, and rodents), horses, and transgenic non-human animals. The subject can be a patient, e.g., have or be suspected of having a disease (that may be referred to as a medical condition), such as benign or malignant neoplasias, or cancer. The subject may be undergoing or having undergone treatment. The subject may be asymptomatic. The subject may be healthy individuals but that are desirous of prevention of cancer. The term “individual” may be used interchangeably, in at least some cases. The “subject” or “individual”, as used herein, may or may not be housed in a medical facility and may be treated as an outpatient of a medical facility. The individual may be receiving one or more medical compositions via the internet. An individual may comprise any age of a human or non-human animal and therefore includes both adult and juveniles (i.e., children) and infants and includes in utero individuals. It is not intended that the term connote a need for medical treatment, therefore, an individual may voluntarily or involuntarily be part of experimentation whether clinical or in support of basic science studies.

[0088] As used herein “treatment” or “treating,” includes any beneficial or desirable effect on the symptoms or pathology of a disease or pathological condition, and may include even minimal reductions in one or more measurable markers of the disease or condition being treated, e.g., cancer. Treatment can involve optionally either the reduction or amelioration of symptoms of the disease or condition, or the delaying of the progression of the disease or condition. “Treatment” does not necessarily indicate complete eradication or cure of the disease or condition, or associated symptoms thereof.

[0089] The disclosure includes the targeting of cancers expressing CD70 using an anti-CD70 CAR, and in some cases the CAR is expressed on NK cells derived from cord blood. A number of novel CAR molecules are described herein that were produced by fusing the extracellular binding portion of CD27 (CD27eCAR) with an activating signaling endodomain that incorporates either CD3 ζ alone or in combination with costimulatory or adaptor signaling domains such as 4-1BB, CD28, DAP10 or DAP12, and, at least in some cases, a cytokine gene such as IL-15, IL-12, IL-18 or IL-21. In some embodiments, the CAR is a fusion of the extracellular domain of CD27 (the natural ligand for CD70) with specificity for human CD70 antigen. In addition, in some cases codon optimization was utilized to improve the surface expression of CD27 on the surface of the transduced cells. In some case, the vector comprising the expression construct that encodes the anti-CD70 CAR also carries a cytokine gene, such as to produce human interleukin 15 (IL-15), IL-12, IL-18 or IL-21, to aid in the survival and maintenance of the cells harboring the vector, including NK cells as examples.

I. Chimeric Antigen Receptors (CARs)

[0090] Immune cells of the present disclosure can be genetically engineered to express one or more antigen-

binding receptors that target one or more antigens, e.g., CD70, such as engineered CARs. For example, the immune cells may be immune cells that are modified to express a CAR having antigenic specificity for CD70. Other CARs may be expressed by the same cells as the CD70 antigen receptor-expressing cells, and they may be directed to different antigens. In some aspects, the immune cells are engineered to express the CD70-specific CAR by knock-in of the CAR using, for example, CRISPR/Cas technology.

[0091] Suitable methods of modification of cells are known in the art. See, for instance, Sambrook and Ausubel, supra. For example, the cells may be transduced to express a CAR having antigenic specificity for a cancer antigen using transduction techniques described in Heemskerk et al., 2008 and Johnson et al., 2009.

[0092] In some embodiments, the cells comprise one or more nucleic acids introduced via genetic engineering that encode one or more antigen-targeting receptors (at least one of which may be directed against CD70), and genetically engineered products of such nucleic acids. In some embodiments, the nucleic acids are heterologous, i.e., normally not present in a cell or sample obtained from the cell, such as one obtained from another organism or cell, which for example, is not ordinarily found in the cell being engineered and/or an organism from which such cell is derived. In some embodiments, the nucleic acids are not naturally occurring, such as a nucleic acid not found in nature (e.g., chimeric).

[0093] The present disclosure encompasses CD70-specific CARs that comprise: a) an extracellular domain comprising an antigen binding region that targets, including specifically binds to, CD70; b) a transmembrane domain; and c) one or more intracellular domains, including intracellular signaling domains. In some aspects, the antigen-specific binding or recognition component is linked to a transmembrane domain and one or more intracellular signaling domains. In some embodiments, the CAR includes a transmembrane domain fused to the extracellular domain of the CAR that comprises the antigen binding region or domain. In one embodiment, the transmembrane domain that naturally is associated with one of the domains in the CAR is used, although in other cases it is not used. In some instances, the transmembrane domain is selected or modified by amino acid substitution to avoid binding of such domains to the transmembrane domains of the same or different surface membrane proteins to minimize interactions with other members of the receptor complex.

CAR, including using the extracellular domain of CD27 as the antigen binding domain of the CAR, in some cases.

[0095] As a matter of reference, the *Homo sapiens* CD27 molecule (CD27) on chromosome 12 is provided at National Center for Biotechnology Information (NCBI) GenBank® Accession No. NG_031995.1, which is incorporated by reference herein in its entirety. A nucleotide sequence that encodes the entire CD27 protein includes the following:

(SEQ ID NO: 9)

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ATGGCACGGCCACATCCCTGGTGGCTGTGCGTTCTGGGGACCCCTGG
TGGGGCTCTCAGCTACTCCAGCCCCAAGAGCTGCCAGAGAGGCACTA
CTGGGCTCAGGGAAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTC
CTCGTGAAGGACTGTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTT
GCATACCGGGGCTCTCTTCTCTCTGACCACCACACCCGGCCCCACTG
TGAGAGCTGTGCGCACTGTAAGTCTGGTCTTCTCGTTTCGCAACTGCACC
ATCACTGCCAATGCTGAGTGTGCCTGTGCGAATGGCTGGCAGTGCAGGG
ACAAGGAGTGCACCGAGTGTGATCCTCTTCCAAACCCCTTCGCTGACCGC
TCGGTCGTCTCAGGCCCTGAGCCACACCCCTCAGCCACCCCACTTACCT
TATGTCAGTGAGATGCTGGAGCCAGGACAGCTGGGCACATGCAGACTC
TGGGTGACTTCAGGCAGCTGCCTGCCCGGACTCTCTACCCACTGGCC
ACCCCAAAGATCCCTGTGCGAGCTCCGATTTTATTTCGATCCTTGTGATC
TTCTCTGGAATGTTCTTGTTTTACCCTGGCCGGGGCCCTGTTCTCTCC
ATCAACGAAGGAAATATAGATCAAAACAAGGAGAAAGTCTGTGGAGCC
TGCAGAGCCTTGTGTTACAGCTGCCCCAGGGAGGAGGAGGGCAGCACC
ATCCCCATCCAGGAGGATTACCGAAAACCGGAGCCTGCCTGCTCCCCCT
GA

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[0096] Any polynucleotide encompassed by the present disclosure may utilize SEQ ID NO:9 or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:9.

[0097] One example of a full wild-type CD27 protein sequence is at NCBI GenBank® Accession No. P26842 (and is also identical to the amino acid sequence in Accession No. NG_031995.1):

(SEQ ID NO: 4)

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1  MARPHPWLC VLGLVLGLSA TPAPKSCPER HYWAQGLCC QMCEPGTFLV KDCCQHRKAA
61  QCDPCIPGVG FSPDHHTPRH CESCRCNSG LLVRNCTITA NAEACACRNG QCRDKECTEC
121 DPLPNPSLTA RSSQALSHPH QPHTLPYVSE MLEARTAGHM QTLADFROLF ARTLSTHWPP
181 QRS LCSSDFI RILVIFSGMF LVFTLAGALF LHORRKYRSN KGESPVEPAE PCHYSCPREE
241 EGSTIPIQED YRKPEPACSP

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[0094] The present disclosure regards anti-CD70 CARs that comprise an antigen binding domain that does not employ an antibody or antibody fragment. In particular embodiments, instead of the anti-CD70 CARs utilizing an antibody or antibody fragment as an antigen binding domain, the CAR instead utilizes part or all of CD27 in the

[0098] In particular embodiments, part or all of SEQ ID NO:4 is utilized in the CAR. Although in certain embodiments the full length wild-type (or codon optimized) CD27 protein is utilized in the CAR, in other embodiments a truncated version of the wild-type CD27 protein is utilized

as the antigen binding domain in the CAR. In specific examples, part or all of the extracellular domain of CD27 is employed, such that the truncated CD27 that is used as the antigen binding domain is a truncated version of CD27 that is truncated at the C-terminus of the natural protein.

[0099] A nucleotide sequence that encodes a truncated version of the wild-type CD27 protein includes the following:

(SEQ ID NO: 63)

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GCTACTCCAGCCCCAAGAGCTGCCAGAGAGGCACTACTGGGCTC
AGGGAAGAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCTCGTGAA
GGACTGTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTTGACATACC
GGGGTCTCCTTCTCTCTGACCAACACCCGGCCCCACTGTGAGAGCT
GTCCGGCACTGTAACCTCTGGTCTTCTCGTTCGCAACTGCACCATCACTGC
CAATGCTGAGTGTGCCTGTGCAATGGCTGGCAGTGCAGGGACAAGGAG
TGACCCGAGTGTGATCCTCTTCCAAACCTTCGCTGACCGCTCGGTCGT
CTCAGGCCCTGAGCCACACCTCAGCCACCCACTTACCTTATGTACG
TGAGATGCTGGAGGCCAGGACAGCTGGGCACATGCAGACTCTGGCTGAC
TTCAGGCAGCTGCCTGCCCGGACTCTCTTACCCACTGGCCACCCCAA
GATCCCTGTGCAGCTCCGATTTTATTCGC
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[0100] Any polynucleotide encompassed by the present disclosure may utilized SEQ ID NO:63 or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:63.

[0101] A polypeptide sequence that encodes a truncated version of the wild-type CD27 protein includes the following:

(SEQ ID NO: 5)

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ATPAPKSCPERHYWAQGLCCQMCEPGTFLVKDCDQHRKAAQCDPCI
PGVSFSPDHHTRPHCESCRHCNSGLLVNCTITANAECACRNGWCQRDK
ECTCEDPLPNPSLTARSSQALSPHPQPTLPHYVSEMLEARAGHMOTLA
DPRQLPARTLSTHWPPQRSLSDFIR
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[0102] Any polypeptide encompassed by the present disclosure may utilized SEQ ID NO:5 or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO: 5.

[0103] In specific cases wherein the full length CD27 protein is used in the CAR molecule, an additional transmembrane domain is not utilized. In specific cases wherein the full length CD27 protein is used in the CAR molecule, which comprises its own intracellular signaling domain, an additional intracellular signaling domain may or may not be used in the CAR. In specific cases wherein the full length CD27 protein is used in the CAR molecule, although it comprises its own signal peptide, an additional signal peptide may or may not be utilized. In particular embodiments, CD3zeta (and/or DAP10, DAP12, 2B4, CD28, NKG2D, 41BB, or any ITAM-containing signaling domain) is utilized in the CAR.

[0104] In one example, the region of CD27 that is utilized as the antigen binding domain in the CAR molecule comprises, consists of, or consists essentially of amino acids

1-211 (includes CD27 TMD) of SEQ ID NO:4. In another example, the region of CD27 that is utilized in the CAR molecule as the antigen binding domain comprises, consists of, or consists essentially of amino acids 1-191 (lacks CD27 TMD) of SEQ ID NO:4. In specific examples, the region of CD27 that is utilized in the CAR molecule comprises, consists of, or consists essentially of amino acids 1-50, 1-51, 1-52, 1-53, 1-54, 1-55, 1-56, 1-57, 1-58, 1-59, 1-60, 1-61, 1-62, 1-63, 1-64, 1-65, 1-66, 1-67, 1-68, 1-69, 1-70, 1-71, 1-72, 1-73, 1-74, 1-75, 1-76, 1-77, 1-78, 1-79, 1-80, 1-81, 1-82, 1-83, 1-84, 1-85, 1-86, 1-87, 1-88, 1-89, 1-90, 1-91, 1-92, 1-93, 1-94, 1-95, 1-96, 1-97, 1-98, 1-99, 1-100, 1-101, 1-102, 1-103, 1-104, 1-105, 1-106, 1-107, 1-108, 1-109, 1-110, 1-111, 1-112, 1-113, 1-114, 1-115, 1-116, 1-117, 1-118, 1-119, 1-120, 1-121, 1-122, 1-123, 1-124, 1-125, 1-126, 1-127, 1-128, 1-129, 1-130, 1-131, 1-132, 1-133, 1-134, 1-135, 1-136, 1-137, 1-138, 1-139, 1-140, 1-141, 1-142, 1-143, 1-144, 1-145, 1-146, 1-147, 1-148, 1-149, 1-150, 1-151, 1-152, 1-153, 1-154, 1-155, 1-156, 1-157, 1-158, 1-159, 1-160, 1-161, 1-162, 1-163, 1-164, 1-165, 1-166, 1-167, 1-168, 1-169, 1-170, 1-171, 1-172, 1-173, 1-174, 1-175, 1-176, 1-177, 1-178, 1-179, 1-180, 1-181, 1-182, 1-183, 1-184, 1-185, 1-186, 1-187, 1-188, 1-189, 1-190, 1-191, 1-192, 1-193, 1-194, 1-195, 1-196, 1-197, 1-198, 1-199, 1-200, 1-201, 1-202, 1-203, 1-204, 1-205, 1-206, 1-207, 1-208, 1-209, 1-210, 1-211, 1-212, 1-213, 1-214, 1-215, 1-216, 1-217, 1-218, 1-219, 1-220, 1-221, 1-222, 1-223, 1-224, 1-225, 1-226, 1-227, 1-228, 1-229, 1-230, 1-231, 1-232, 1-233, 1-234, 1-235, 1-236, 1-237, 1-238, 1-239, 1-240, 1-241, 1-242, 1-243, 1-244, 1-245, 1-246, 1-247, 1-248, 1-249, 1-250, 1-251, 1-252, 1-253, 1-254, 1-255, 1-256, 1-57, 1-258, 1-259 or all (1-260) of SEQ ID NO:4; in specific embodiments, such amino acids in these ranges are contiguous. In some embodiments, a region of SEQ ID NO:4 is utilized that has truncation at the N-terminus, such as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more amino acids from the N-terminus. In certain cases, there is truncation at that N-terminus of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more amino acids and there is truncation at the C-terminus, such as the truncation endpoints as noted in the list above.

[0105] In certain embodiments, any region of the CAR molecule is codon optimized.

[0106] In some embodiments, the antigen binding domain comprises any of the examples of CD27 sequences encompassed herein but is codon optimized. In specific cases, the antigen binding domain comprises a C-terminal truncated CD27 (e.g., the extracellular domain) and is also codon optimized. In certain aspects, the antigen binding domain comprises, consists of, or consists essentially of amino acids 1-191 of CD27 and is also codon optimized. In other aspects, the antigen binding domain and transmembrane domain comprises, consists of, or consists essentially of amino acids 1-211 of CD27 and is also codon optimized.

[0107] In some cases, a specific component of the CAR molecule is utilized. In particular embodiments, the CAR comprises a particular transmembrane domain. Examples include the transmembrane domains from CD28 or from CD27. In some embodiments, the CD27 transmembrane domain or the CD28 transmembrane domain are not utilized in the CAR, whereas in alternative embodiments the CD27 transmembrane domain is utilized in the CAR. In particular embodiments, signal peptides are employed for the CAR,

and examples include the CD27 or GMCSF-R signal peptides, or both may be used. Examples of intracellular signaling domains for the CAR include at least CD27, CD28, DAP10, DAP12, NKG2D, 4-1BB, or a combination thereof.

[0108] In some cases, the CD27 signal peptide is utilized (MARPHPWWLCVLTGLVGLS (SEQ ID NO:6); ATGGCACGGCCA-CATCCCTGGTGGCTGTGCGTTCTGGGGACCCTGG-TGGGGCTCT CA (SEQ ID NO:61)) in the CAR, or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:6 or 61. In some cases, the GMCSF-R signal peptide (MLLLVTSLLLCEL-PHPAFLIP (SEQ ID NO:14); ATGCTTCTCCTGGTGACAAGCCTTCTGCTCTGT-GAGTTACCACACCCAGCATTCTCT CCTGATCCCA (SEQ ID NO:62)) is used in the CAR, or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:14 or 62.

[0109] The CAR may or may not comprise CD27 TM (ILVIFSGMFLVFTLAGALFLH (SEQ ID NO:7); ATCCTTGTGATCTTCTCTGGAATGTTCTTGTGTTT-CACCCTGGCCGGGGCCCTGTT CCTCCAT (SEQ ID NO:66)), or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:7 or 66.

[0110] The CAR may or may not comprise the ICD of CD27, which is QRRKYRSNKGESPVEPAEPCHY-SCPREEEGSTIPIQEDYRKPEPACSP (SEQ ID NO:8), or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:8.

[0111] In some embodiments, the CD3zeta ICD is RVKFSRSAD APAYQQGNQ LYNELNLGRR EEY-DVLDKRR GRDPEMGGKPRRKNPQEGLYNELQDKMA EAYSEIGMKG ERRRGKGHDGLYQGLSTATKDTYDALHMQA LPPRG (SEQ ID NO:10), CGCGTGAAGTTCAGCAGGAGCGCA-GACGCCCCCGCGTACCAGCAGGGCCAGAAC CAGCTCTATAACGAGCT-CAATCTAGGACGAAGAGAGGAGTAC-GATGTTTTGGAC AAAAGACGTGGCCGGGACCCT-GAGATGGGGGAAAGCCGAGAAGGAAGAACCCTCAGGAAGGCCTGTACAATGAACTGCAGAAAGATAAGATGGCGGAGGCCTACAG TGAGATTGGGAT-GAAAGGCGAGCGCCGGAGGGGCAAGGGGAC-GATGGCCTTTA CCAGGGTCTCAGTACAGCCACCAAGGACACC-TACGACGCCCTTCACATGCAGGC CCTGCCCCCTCGCGGAC (SEQ ID NO:67), and the sequence may be used in the CAR, or a sequence that is at

least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:10 or 67.

[0112] In some embodiments, the CD28 ICD (RSKRSRLHSD YMNMTPRRPGPTRKHYQPYA PPRDFAAYRS; SEQ ID NO:11) is utilized in the CAR molecule, or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:11.

[0113] In some embodiments, the 4-1BB ICD (KR-GRKKL LYIFKQPFMR PVQTTQEEDG CSCRFPEEEEGGCEL (SEQ ID NO:12); AAACGGGGCAGAAAGAACTCCTGTATATATT-CAAACAACCATTTATGAGACCA GTACAACTACT-CAAGAGGAAGATGGCTGTAGCTGCCGAT-TTCCAGAAGAAGAA GAAGGAGGATGTGAACTG (SEQ ID NO:68)) is used in the CAR molecule, or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:12 or 68.

[0114] In some embodiments, the DAP10 ICD (LCAR-PRRSPAQEDGKVIYNMPGRG (SEQ ID NO:72); CTTTGCGCACGCCACGCCGACGCCCGCC-CAAGAAGATGGCAAAGTCTACATC AACATGCCAGGCAGGGG (SEQ ID NO:69)) is used in the CAR molecule, or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:69 or 72.

[0115] In some embodiments, the DAP12 ICD (YFL-GRVPRGR-GAAEAATRKQRITETESPYQELQQRSDVYSDLN-TQRPYYK (SEQ ID NO:74); TACTTCTGGGCCGGCTGGTCCCTCGGGGGCGA-GGGGCTGCGGAGGCAGCGACC CGGAAACAGCGTATCACTGAGACCGAGTCGCCT-TATCAGGAGCTCCAGGGTCAG AGGTCCGATGTC-TACAGCGACCTCAACACACAGAGGCCGTATTA-CAAA (SEQ ID NO:71)) is used in the CAR molecule, or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:71 or 74.

[0116] In some embodiments, the NKG2D ICD (SANER-CKSKVVPQRQKQWRTSFDSSKLLDLNHNHFS-MEWSHRSRGRWGM (SEQ ID NO:73); AGCGCGAACGAACGCTGCAAAAGCAAAGTGTT-GCCGTGCCGCCAGAAACAGTG GCGCACCAGCTTT-GATAGCAAAAACTGGATCTGAACTATAACCAT-TTTGAAAGC ATGGAATGGAGC-CATCGCAGCCGCCGCGGCCGCGCATTTGGGGCATG (SEQ ID NO:70)) is used in the CAR molecule, or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:70 or 73.

[0117] In particular embodiments, codon optimized CD27 is utilized, and one example is as follows:

(SEQ ID NO: 13)

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ATG GCG AGA CCA CAC CCT TGG TGG CTG TGC GTA CTC GGC ACA
CTT GTA GGT CTG TCC GCT ACA CCG GCT CCG AAG TCC TGC CCG GAG CGG
CAT TAT TGG GCA CAG GGC AAG TTG TGT TGT CAA ATG TGT GAG CCG GGA
ACC TTT CTC GTG AAG GAT TGC GAT CAG CAT CGG AAG GCC GCG CAG TGC
GAC CCA TGT ATA CCA GGG GTC TCA TTT TCC CCA GAT CAC CAT ACG AGG
CCG CAC TGT GAG TCT TGC AGG CAT TGT AAT TCC GGC TTG TTG GTC CGC AAC
TGT ACT ATT ACT GCG AAT GCA GAG TGT GCT TGT AGA AAC GGA TGG CAG
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TGC AGG GAC AAA GAA TGT ACG GAG TGT GAT CCA CTG CCT AAC CCC AGT
 CTT ACA GCA AGA TCT TCA CAG GCC CTC AGC CCG CAT CCT CAA CCA ACA
 CAT CTT CCT TAC GTG TCA GAA ATG TTG GAG GCG CGA ACC GCA GGC CAT
 ATG CAG ACC CTG GCG GAC TTT CGG CAG CTG CCA GCA CGC ACA CTT AGT
 ACA CAC TGG CCA CCA CAA CGC AGC TTG TGC TCT TCC GAT TTC ATC CGC
 ATA CTG GTC ATC TTT TCT GGA ATG TTC CTT GTG TTC ACC CTG GCA GGA GCC
 CTG TTC CTT CAC CAG AGA CGC AAG TAC AGG TCA AAC AAG GGT GAG AGC
 CCC GTT GAA CCC GCA GAG CCG TGT AGA TAC TCA TGT CCT AGA GAA GAA
 GAG GGC TCT ACT ATC CCT ATT CAG GAA GAT TAT AGA AAA CCC GAA CCC
 GCG TGC AGC CCC.

[0118] Any polynucleotide encompassed by the present disclosure may utilized SEQ ID NO:13 or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:13.

[0119] In particular embodiments, the codon optimized extracellular domain of CD27 is utilized, and one example is as follows:

(SEQ ID NO: 64)
 GCTACACCGGCTCCGAAGTCTGCCCGGAGCGGCATTATTGGGCAC
 AGGGCAAGTTGTGTGTGTCAAATGTGTGAGCCGGGAACCTTTCTCGTGAA
 GGATTGCGATCAGCATCGGAAGCCGCGCAGTGCAGCCATGTATACCA
 GGGGTCTCATTTTCCCCAGATCACCATACGAGGCCGCACTGTGAGTCTT
 GCAGGCATTGTAATTCGGCTTGTGTGGTCCGCAACTGTACTATTACTGC
 GAATGCAGAGTGTGCTTGTAGAACGGATGGCAGTGCAGGGACAAAGAA
 TGTACGGAGTGTGATCCACTGCCTAACCCAGTCTTACAGCAAGATCTT
 CACAGGCCCTCAGCCCGCATCTCAACCAACACATCTTCCTTACGTGTC
 AGAAATGTTGGAGGCGCGAACCAGCAGGCCATATGCAGACCTGGCGGAC
 TTTCGGCAGCTGCCAGCAGCACACTTAGTACACACTGGCCACCAACA
 GCAGCTTGTGCTCTTCCGATTTCATCCGC

[0120] Any polynucleotide encompassed by the present disclosure may utilized SEQ ID NO:64 or a sequence that is at least 70, 75, 80, 85, 90, 95, 96, 97, 98, 99, or more % identical to SEQ ID NO:64.

[0121] FIGS. 2A-2E provide specific examples of CAR constructs comprising particular components. These are specific examples in which the CAR construct may comprise full length CD27 (FL; all 260 amino acids). In alternative cases, when the full length CD27 is not utilized, the extracellular domain of CD27 (CD27 EC) may be utilized, which is residues 20-191 of CD27. In particular embodiments, when a truncated CD27 is utilized in the CAR, the CD27 intracellular costimulatory domain (ICD) may not be utilized in the CAR and/or the CD27 transmembrane domain (TMD) may not be utilized in the CAR.

[0122] In FIG. 2A, example (1) of an anti-CD70 CAR comprises the CD27 signal peptide (SP), the full length CD27 (FL), the CD27 TMD, the CD27 ICD, and CD3zeta. Example (2) of FIG. 2A includes the granulocyte-macro-

phage colony-stimulating factor receptor (GMSCF-R) signal peptide, a codon optimized (co) full length CD27, a codon optimized CD27 transmembrane domain, a codon optimized CD27 ICD, and CD3zeta. Example (3) of FIG. 2A includes the CD27 SP, full length CD27, the CD27 TMD, the CD27 ICD, the CD28 ICD, and CD3 zeta. Example (4) of FIG. 2A includes the GMSCF-R SP, the codon optimized full length CD27, the codon optimized full length CD27 TMD, a codon optimized CD27 ICD, CD28 ICD, and CD3zeta. Example (5) of FIG. 2A includes the CD27 SP, full length CD27, CD27 TMD, CD27 ICD, 4-1BB ICD, and CD3zeta. Example (6) includes the GMSCF-R SP, codon optimized full length CD27, codon optimize CD27 TMD, codon optimized CD27 ICD, 4-1BB ICD, and CD3zeta.

[0123] Example (7) in FIG. 2B includes the CD27 SP, full length CD27, CD27 TMD, CD27 ICD, DAP10 ICD, and CD3zeta. Example (8) of FIG. 2B includes the GMSCF-R SP, codon optimized full length CD27, codon optimized CD27 TMD, codon optimized CD27 ICD, DAP10 ICD, and CD3zeta. Example (9) in FIG. 2B includes the CD27 SP, full length CD27, CD27 TMD, CD27 ICD, DAP12 ICD, and CD3zeta. Example (10) of FIG. 2B includes the GMSCF-R SP, codon optimized full length CD27, codon optimized CD27 TMD, codon optimized CD27 ICD, DAP12 ICD, and CD3zeta. Example (11) of FIG. 2B includes the CD27 SP, full length CD27, CD27 TMD, CD27 ICD, NKG2D ICD, and CD3zeta. Example (12) of FIG. 2B includes the GMSCF-R SP, codon optimized full length CD27, codon optimized CD27 TMD, codon optimized CD27 ICD, NKG2D ICD, and CD3zeta.

[0124] Example (13) in FIG. 2C includes the CD27 SP, CD27 extracellular domain (EC), the CD27 TMD, and CD3zeta. Example (14) in FIG. 2C includes the GMSCF-R SP, codon optimized CD27 EC, codon optimized TMD, and CD3zeta. Example (15) in FIG. 2C includes the CD27 SP, CD27 EC, CD28 TMD, and CD3zeta. Example (16) in FIG. 2C includes GMSCF-R SP, codon optimized CD27 EC, codon optimized CD28 TMD, and CD3zeta. Example (17) in FIG. 2C includes CD27 SP, CD27 EC, CD27 TMD, CD28 ICD, and CD3zeta. Example (18) of FIG. 2C includes GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, CD28 ICD, and CD3zeta. Example (19) of FIG. 2C includes CD27SP, CD27 EC, CD28 TMD, CD28 ICD, and CD3zeta. Example (20) of FIG. 2C includes GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, CD28 ICD, and CD3zeta.

[0125] Example (21) of FIG. 2D includes CD27 SP, CD27 EC, CD27 TMD, 4-1BB ICD, and CD3zeta. Example (22) of FIG. 2D includes GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, 4-1BB ICD, and CD3zeta. Example (23) of FIG. 2D includes CD27 SP, CD27 EC, CD28 TMD, 4-1BB ICD, and CD3zeta. Example (24) of FIG. 2D includes GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, 4-1BB ICD, and CD3zeta. Example (25) of FIG. 2D includes CD27 SP, CD27 EC, CD27 TMD, DAP10 ICD, and CD3zeta. Example (26) of FIG. 2D includes GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, DAP10 ICD, and CD3zeta. Example (27) of FIG. 2D includes CD27 SP, CD27 EC, CD28 costimulatory domain, DAP10 ICD, and CD3zeta. Example (28) of FIG. 2D includes GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, DAP10 ICD, and CD3zeta.

[0126] Example (29) of FIG. 2E includes CD27 SP, CD27 EC, CD27 TMD, DAP12 ICD, and CD3zeta. Example (30) of FIG. 2E includes GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, DAP12 ICD, and CD3zeta. Example (31) of FIG. 2E includes CD27 SP, CD27 EC, CD28 TMD, DAP12 ICD, and CD3zeta. Example (32) of FIG. 2E includes GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, DAP12 ICD, and CD3zeta. Example (33) of FIG. 2E includes CD27 SP, CD27 EC, CD27 TMD, NKG2D ICD, and CD3zeta. Example (34) of FIG. 2E includes GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, NKG2D ICD, and CD3zeta. Example (35) of FIG. 2E includes CD27 SP, CD27 EC, CD28 TMD, NKG2D ICD, and CD3zeta. Example (36) of FIG. 2E includes GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, NKG2D ICD, and CD3zeta.

[0127] One example of an anti-CD70 CAR of the disclosure comprises SEQ ID NO:1 that comprises a CD27 extracellular domain (not underlined in SEQ ID NO:1 below) operably linked to a CD28 transmembrane domain (underlined in SEQ ID NO:1 below) and a CD27 signal peptide (bolded in SEQ ID NO:1 below):

(SEQ ID NO: 1)

MARHPHWLCLVLGLSATPAPKSCSPERHYWAQGLCCQMCEP

GTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTPHCESCRHCNSGLLVR

NCTITANAECACRNGWQCRDKETCDPLPNPSLTARSSQALSHPHQPT

HLPYVSEMLEARTAGHMQTLADFRQLPARTLSTHWPQRLSCSSDFIRF

WVLVVVGGVLACYSLLVTVAFIIFWV

[0128] The CD27 extracellular domain of SEQ ID NO:1 comprises the following as SEQ ID NO:2: MARPHPWWLCVLGTLVGLSATPAPKSCPERHY-WAQQGLCCQMCEPGTFLVKDCDQ HRKAAQCDCIPGVSFSPDHHTRPHC-ESCRHCNSGLLRNCTTTANAECACRNGWQC RDKE-CTECDPLPNPSLTARSSQALSPHPQPTHLPYVSEM-LEARTAGHMQTLADFRQL PARTI STHWPPORSLCSSDFIR.

[0129] The CD28 transmembrane domain of SEQ ID NO:1 comprises the following as SEQ ID NO:3: FWVLVVVGGVLACYSLLVTVAFIIFWV (TTTTGGGTGCTGGTGGTGGTGGTG-GAGTCCTGGCTTGCTATAGCTTGCTAGTAA CAGTGGCCCTTATTATTTCTGGGTG (SEQ ID NO:65)).

[0130] The CD27 signal peptide of SEQ ID NO:1 comprises the following as SEQ ID NO:6: MARPHPWWLCVLTGLVGLS (ATGGCACGGCCA-CATCCCTGGTGGCTGTGCGTTCTGGGGACCCTGG-TGGGGCTC TCA (SEQ ID NO:61)).

[0131] The present disclosure provides for cells (including immune cells, such as NK cells) that harbor a vector that encodes at least one CAR, and the CAR may be first generation, second generation, or third or a subsequent generation, for example. The CAR may or may not be bispecific for two or more different antigens, one of which is CD70. In particular embodiments, the CAR is not a bivalent tandem CAR. The CAR may comprise one or more costimulatory domains. Each costimulatory domain may comprise the costimulatory domain of any one or more of, for example, members of the TNFR superfamily, CD28, CD137 (4-1BB), CD134 (OX40), DAP10, DAP12, CD27, CD2, CD5, ICAM-1, LFA-1 (CD11a/CD18), Lck, TNFR-I, TNFR-II, Fas, CD30, CD40, any ITAM-containing signaling domain, or combinations thereof, for example. In specific embodiments, the CAR comprises CD3zeta. In certain embodiments, the CAR lacks one or more specific costimulatory domains; for example, the CAR may lack 4-1BB and/or lack CD28.

[0132] In particular embodiments, the CAR polypeptide in the cells comprises an extracellular spacer domain that links the antigen binding domain and the transmembrane domain. Extracellular spacer domains may include, but are not limited to, Fc fragments of antibodies or fragments or derivatives thereof, hinge regions of antibodies or fragments or derivatives thereof, CH2 regions of antibodies, CH3 regions antibodies, artificial spacer sequences or combinations thereof. Examples of extracellular spacer domains include but are not limited to CD8-alpha hinge, CD28, artificial spacers made of polypeptides such as Gly3, or CH1, CH3 domains of IgGs (such as human IgG1 or IgG4). In specific cases, the extracellular spacer domain may comprise (i) a hinge, CH2 and CH3 regions of IgG4, (ii) a hinge region of IgG4, (iii) a hinge and CH2 of IgG4, (iv) a hinge region of CD8-alpha, (v) a hinge, CH2 and CH3 regions of IgG1, (vi) a hinge region of IgG1 or (vii) a hinge and CH2 of IgG1, (viii) a hinge region of CD28, or a combination thereof.

[0133] In specific embodiments, the hinge is from IgG1 and in certain aspects the CAR polypeptide comprises a particular IgG1 hinge amino acid sequence or is encoded by a particular IgG1 hinge nucleic acid sequence.

[0134] In specific embodiments, the hinge is from CD28 and in certain aspects the CAR polypeptide comprises a particular CD28 hinge amino acid sequence or is encoded by a particular CD28 hinge nucleic acid sequence.

[0135] The transmembrane domain in some embodiments is derived either from a natural or from a synthetic source. Where the source is natural, the domain in some aspects is derived from any membrane-bound or transmembrane protein. Transmembrane regions include those derived from (i.e. comprise at least the transmembrane region(s) of) the alpha, beta or zeta chain of the T-cell receptor, CD28, CD3 zeta, CD3 epsilon, CD3 gamma, CD3 delta, CD45, CD4, CD5, CD8, CD9, CD 16, CD22, CD33, CD37, CD64, CD80, CD86, CD 134, CD137, CD154, ICOS/CD278, GITR/CD357, NKG2D, and DAP molecules, such as DAP10 or DAP12. Alternatively the transmembrane domain in some embodiments is synthetic. In some aspects, the synthetic transmembrane domain comprises predominantly

hydrophobic residues such as leucine and valine. In some aspects, a triplet of phenylalanine, tryptophan and valine may be found at each end of a synthetic transmembrane domain.

[0136] In certain embodiments, the CD70-specific CAR may be co-expressed with one or more cytokines to improve persistence when there is a low amount of tumor-associated antigen. For example, the CAR may be co-expressed with one or more cytokines, such as IL-7, IL-2, IL-15, IL-12, IL-18, IL-21, IL-7, GM-CSF, or a combination thereof.

[0137] In specific examples, an anti-CD70 CAR may have the following nucleotide sequence:

CD27tr28tdmCD3zIL15:

(SEQ ID NO: 15)
 ATGGCACGGCCACATCCCTGGTGGCTGTGCGTTCTGGGGACCCCTGG
 TGGGGCTCTCAGCTACTCCAGCCCCAAGAGCTGCCCAGAGAGGCACTA
 CTGGGCTCAGGGAAAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTC
 CTCGTGAAGGACTGTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTT
 GCATACCGGGGTCTCCTTCTCTCTGACCACCACACCGGCCCCACTG
 TGAGAGCTGTGCGCACTGTAACCTCTGGTCTTCTCGTTCGCAACTGCACC
 ATCACTGCCAATGCTGAGTGTGCTGTGCGCAATGGCTGGCAGTGCAGGG
 ACAAGGAGTGCACCGAGTGTGATCCTCTTCCAAACCCCTTCGCTGACCGC
 TCGGTCTCTCAGGCCCTGAGCCACACCCCTCAGCCACCCACTTACCT
 TATGTGAGTGTGAGTGTGAGGCGCAGGACAGCTGGGCACATGCAGACTC
 TGGCTGACTTCAGGCAGCTGCTGCGCGGACTCTCTTACCCACTGGCC
 ACCCCAAGATCCCTGTGAGCTCCGATTTTATTCGCTTTTGGGTGCTG
 GTGGTGGTGGTGGAGTCTGGCTTGTATAGCTTGTAGTAACAGTGG
 CCTTTATTATTTTCTGGGTG

[0138] A corresponding amino acid sequence for CD27tr28tdmCD3zIL15 is as follows:

(SEQ ID NO: 38)
 MARPHPWLLCVLGLTLVGLSATPAPKSCPERHYWAQGLCCQMCEPG
 TFLVKDCDQHRKAAQDPCIPGVSFSPDHHRPHCESCRHCNSGLLVNR
 CTITANAECACRNGWQCRDKECTECPLPNPSLTARSSQALSPHPQPTH
 LPYVSEMLEARTAGHMQLADFRQLPARTLSTHWPPQRSLSDFIRFW
 VLVVVGVLACYSLLVTVAFIIFWV

CD27Tr28tmd41BBid3zIL15:

(SEQ ID NO: 16)
 ATGACAAGAGTTACTAACGCCCTCTCTCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACCTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGC
 GACACAGTGTGGGTCCGCCGACACCAGACTAAGAACCTAGAACCTCGCT

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GGAAAGGACCTTACACAGTCCCTGCTGACCACCCCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACGCCGCCACGTGAAGGCTGCCGAC
 CCCGGGGGTGGACCATCCTCTAGACTGCCATGCTCGAGGATGGCACGGC
 CACATCCCTGGTGGCTGTGCGTTCTGGGGACCCCTGGTGGGGCTCTCAGC
 TACTCCAGCCCCAAGAGCTGCCCAGAGAGCACTACTGGGCTCAGGGA
 AAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCTCTGTAAGGACT
 GTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTTGATACCGGGGGT
 CTCCTTCTCTCTGACCACCACACCGGCCCCACTGTGAGAGCTGTCTGG
 CACTGTAACCTCTGGTCTTCTCGTTCGCAACTGCACCATCACTGCCAATG
 CTGAGTGTGCTGTGCGCAATGGCTGGCAGTGCAGGGACAAGGAGTGCAC
 CGAGTGTGATCCTCTTCCAAACCCCTTCGCTGACCGCTCGGTCTGCTCAG
 GCCCTGAGCCACACCCCTCAGCCACCCACTTACCTTATGTGAGTGTGAGA
 TGCTGGAGGCCAGGACAGCTGGGCACATGCAGACTCTGGCTGACTTCAG
 GCAGCTGCTGCGCGGACTCTCTTACCCACTGGCCACCCCAAGATCC
 CTGTGCAGCTCCGATTTTATTCGCTTTTGGGTGCTGGTGGTGGTGGTG
 GAGTCTGGCTTGCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTTT
 CTGGGTGAAACGGGGCAGAAAGAACTCCTGTATATATTCAAACAACCA
 TTTATGAGACCAGTACAACTACTCAAGAGGAAGATGGCTGTAGCTGCC
 GATTTCAGAGAAGAAGAAGAGGAGTGTGAACAGAAACGCGTGAAGTT
 CAGCAGGAGCGCAGACGCCCCCGGTACAGCAGGGCCAGAACAGCTC
 TATAACGAGCTCAATCTAGGACGAAGAGAGTACGATGTTTTGGACA
 AAAGACGTGGCCGGACCTGAGATGGGGGAAAGCCGAGAAGGAAGAA
 CCCCTCAGGAAGGCCCTGTACAATGAACAGCAAGATAAGATGGCGGAG
 GCCTACAGTGTGATTGGGATGAAAGCGAGCGCGGAGGGGCAAGGGGC
 ACGATGGCCTTTACCAGGGTCTCAGTACAGCCACCAAGGACACCTACGA
 CGCCCTTCACATGCAGGCCCTGCCCCCTCGCGGACCGCAGTGTACTAAT
 TATGCTCTCTTGAAATTGGCTGGAGATGTTGAGAGCAATCCCGGCCCA
 TGCGCATTAGCAAGCCCCACCTGCGGAGCATCAGATCCAGTGTACCT
 GTGCTGTGCTGTAACAGCCACTTCTGACCGAGGCCGGCATCCACGTG
 TTCATCCTGGGTGCTTTCAGCGCGGACTGCCCCAAGACCGAGGCCAACT
 GGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGGACCTGATCCAGAG
 CATGCACATCGACGCCACCTGTACACCGAGAGCGAGTGCACCCAGC
 TGCAAGGTGACCGCCATGAAGTCTTCTGCTGGAACAGCAGGTGATCA
 GCCTGGAAAGCGGCGACGCCAGCATCCAGACACCGTGGAGAACCTGAT
 CATCTGGCCAAACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGC
 GGCTGCAAGAGTGCAGGAACTGGAAGAGAAGAACATCAAGAGTTTC
 TGCAGAGCTTCGTGCACATCGTGCAGATGTTTCATCAACACCGAGCTGA

[0139] A corresponding amino acid sequence for CD27Tr28tmd41BBic3zIL15 is as follows:

(SEQ ID NO: 39)
MTRVTNSPSLQAHLQALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
VGDTVVWRRHQTKNLEPRWKGPYTVLLTTPALKVDGIAAWIHAAHVKA
ADPGGGPSRLPCSRMARPHPWWLCVLGTLVGLSATPAPKSCPERHYWA
QGKLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPHCES
CRHCNSGLLVNRNCTITANAECACRNGWQCRDKECTECDPLPNPSLTARS
SQALSHPHPQTHLPYVSEMLEARTAGHMQLADFRQLPARTLSTHWPPQ
RSLCSSDFIRFWLVVVGGVLACYSLLVTVAFIIFWVKRGRKKLLYIFK
QPFMRPVQTTQEEDGCSRFPEEEEGGCELKRVKFSRSADAPAYQQQN
QLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKM
AEAYSEIGMKGERRRGKGGHGLYQGLSTATKDTYDALHMQUALPPRGQC
TNYALLKLAGDVESNPGPMRISKPHLRSISIQCYLCLLLNSHFLTEAGI
HVFILGCFISAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDVH
PSCVKVTAMKCFLELQVISLES GDASIHDVTENLIILANNSLSSNGNVT
ESGCKECELEEKNKEFLQSFVHIVQMFINTS

GSPco27Tr28tmd41BBicCD3zIL15:

(SEQ ID NO: 17)
ATGACAAGAGTTACTAACAGCCCTCTCTCCAAGCTCACTTACAGG
CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
CCAAGAACAACCTGGACCGACCGGTGGTACCTACCCCTTACCGAGTCGGC
GACACAGTGTGGGTCCGCCGACACACAGACTAAGAACCTAGAACCTCGCT
GGAAAGGACCTTACACAGTCTGTGACCAACCCACCGCCCTCAAAGT
AGACGGCATCGCAGCTTGATACACGCCGCCACCGTGAAGGTGCGGAC
CCCGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGCTTCTCC
TGGTGACAAGCCTTCTGCTCTGTGAGTTACCAACCCAGCATTCTCTCT
GATCCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCGGCATTATTGG
GCACAGGGCAAGTTGTGTGTCAAATGTGTGAGCCGGGAACCTTTCTCG
TGAAGGATTGCGATCAGCATCGGAAGGCCGCGCAGTGCAGCCATGTAT
ACCAGGGTCTCATTTTCCCAGATACCATACGAGGCCGCACTGTGAG
TCTTGCAGGCATTGTAAATTCGGCTTGTGTGTCGCAACTGTACTATTA
CTGCGAATGCAGAGTGTGCTTGTAGAAACGGATGGCAGTGCAGGGACAA
AGAATGTACGGAGTGTGATCCACTGCCTAACCCAGTCTTACAGCAAGA
TCTTCACAGGCCCTCAGCCGCATCTCAACCAACACATCTTCCTTACG
TGTCAGAAATGTTGGAGCGCGAACCGCAGGCATATGCAGACCTTGGC
GGACTTTCGGCAGCTGCCAGCAGCACACTTAGTACACACTGGCCACCA
CAACGCAGCTTGTGCTCTTCCGATTTCATCCGCTTTTGGGTGCTGGTGG
TGGTTGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAACAGTGGCCTT

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TATTATTTTCTGGGTGAAACGGGGCAGAAAGAACTCCTGTATATATTC
AAACAACCATTATGAGACCAGTACAACTACTCAAGAGGAAGATGGCT
GTAGCTGCCGATTTCAGAAGAAGAAGAGGAGATGTGAAGTAAACG
CGTGAAGTTCAGCAGGAGCGCAGACGCCCCCGCTACCAGCAGGGCCAG
AACCAGCTCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACGATG
TTTTGGACAAAAGACGTGGCCGGGACCTGAGATGGGGGAAAGCCGAG
AAGGAAGAACCCTCAGGAAGGCTGTACAATGAATGCAGAAAGATAAG
ATGGCGGAGGCTACAGTGAGATTGGGATGAAAGGCGAGCGCCGGAGGG
GCAAGGGGCACGATGGCCTTTACCAGGCTCTCAGTACAGCCACCAAGGA
CACCTACGACGCCCTTCACATGCAGGCCCTGCCCTCGCGGACCGCAG
TGTACTAATTATGCTCTCTTGAATTGGCTGGAGATGTTGAGAGCAATC
CCGGGCCCATGCGCATTAGCAAGCCCCACCTGCGGAGCATCAGCATCCA
GTGCTACCTGTGCTGCTGTGAACAGCCACTTCTGACCGAGGCCGGC
ATCCACGTGTTTCATCTGGGTGCTTTCAGCGCCGAGTGCCTCAAGACCG
AGGCCAACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCAGGACCT
GATCCAGAGCATGCACATCGACGCCACCCTGTACACCGAGAGCGACGTG
CACCCAGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGCTGGAAGTGC
AGGTGATCAGCCTGGAAAGCGGCGACGCCAGCATCCACGACACCGTGGA
GAACCTGATCATCTGGCCAACAACAGCCTGAGCAGCAACGGCAACGTG
ACCGAGAGCGGCTGCAAAGAGTGCAGGAACTGGAAGAAGAAGAACATCA
AAGAGTTTCTGCAGAGCTTCGTGCACATCGTGCAGATGTTTCATCAACAC
CAGCTGA

[0140] A corresponding amino acid sequence for GSPco27Tr28tmd41BBicCD3zIL15 is as follows:

(SEQ ID NO: 40)
MTRVTNSPSLQAHLQALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
VGDTVVWRRHQTKNLEPRWKGPYTVLLTTPALKVDGIAAWIHAAHVKA
ADPGGGPSRLPCSRMLLVTSLLLCELPHPAFLLIIPATPAPKSCPERH
YWAQGLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPH
CESCRHCNSGLLVNRNCTITANAECACRNGWQCRDKECTECDPLPNPSLT
ARSSQALSHPHPQTHLPYVSEMLEARTAGHMQLADFRQLPARTLSTHW
PPQRSLSDFIRFWLVVVGGVLACYSLLVTVAFIIFWVKRGRKKLLY
IFKQPFMRPVQTTQEEDGCSRFPEEEEGGCELKRVKFSRSADAPAYQQ
QQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK
DKMAEAYSEIGMKGERRRGKGGHGLYQGLSTATKDTYDALHMQUALPPRG
PQCTNYALLKLAGDVESNPGPMRISKPHLRSISIQCYLCLLLNSHFLTE
AGIHVFILGCFISAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTES
DVHPSCVKVTAMKCFLELQVISLES GDASIHDVTENLIILANNSLSSNG
NVTESGCKECELEEKNKEFLQSFVHIVQMFINTS

CD27Tr28tmdDAP10icd3z15:

(SEQ ID NO: 18)

ATGACAAGAGTTACTAACAGCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACCTGGACCGACCGGTGGTACCTCACCTTACCAGAGTCGGC
 GACACAGTGTGGGTCCGCCGACACCAGACTAAGAACCCTAGAACCCTCGCT
 GGAAAGGACCTTACACAGTCTGTGTGACACCCCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACGCCGCCACGTGAAGGCTGCCGAC
 CCCGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGGCAGGC
 CACATCCCTGGTGGCTGTGCGTTCTGGGGACCTGGTGGGGCTCTCAGC
 TACTCCAGCCCCAAGAGCTGCCCAGAGAGGCACTACTGGGCTCAGGGA
 AAGCTGTGTGCTGCCAGATGTGTGAGCCAGGAACATTCCTCGTGAAGGACT
 GTGACCAGCATAGAAAGGCTGTCTAGTGTGATCCTTGATACACGGGGGT
 CTCTTCTCTCTGACCACCACACCCGGCCCCACTGTGAGAGCTGTCCG
 CACTGTAACCTCTGGTCTTCTCGTTTCGCAACTGCACCATCACTGCCAATG
 CTGAGTGTGCTGTGCAATGGCTGGCAGTGCAGGACAAGGAGTGCAC
 CGAGTGTGATCCTCTTCCAAACCTTTCGCTGACCGCTCGGTGCTCTCAG
 GCCCTGAGCCACACCTCAGCCACCCACTTACCTTATGTCTAGTGAGA
 TGCTGGAGGCCAGGACAGTGGGCACATGCAGACTCTGGCTGACTTCAG
 GCAGTGCCTGCCCGGACTCTCTTACCCACTGGCCACCCCAAGATCC
 CTGTGACAGCTCCGATTTTATTCTGCTTTTGGGTGCTGGTGGTGGTGGTG
 GAGTCTGCTTGTCTATAGCTTGTCTAGTAACAGTGGCCTTTATTTATTT
 CTGGGTGCTTTGCGCAGCCCCACGCCGAGCCCCGCCCAAGAAGATGGC
 AAAGTCTACATCAACATGCCAGGCAGGGGCAAACGCGTGAAGTTCAGCA
 GGAGCGCAGACGCCCCCGCTACCAGCAGGGCCAGAACCAGTCTATAA
 CGAGCTCAATCTAGGACGAAGAGAGGAGTACGATGTTTTGGACAAAAGA
 CGTGGCCGGGACCTGAGATGGGGGGAAGCCGAGAAGGAAGAACCTC
 AGGAAGGCCTGTACAATGAATGCAGAAAGATAAGATGGCGGAGGCCA
 CAGTGAGATTGGGATGAAAGGCGAGCGCCGAGGGGCAAGGGGCACGAT
 GGCCCTTACCAGGGTCTCAGTACAGCCACCAAGGACACCTACGACGCC
 TTCACATGACGCCCCTGCCCTCGCGGACCGCAGTGTACTAATTATGC
 TCTCTTGAATTTGGCTGGAGATGTTGAGAGCAATCCCGGGCCATGCGC
 ATTAGCAAGCCCCACTGCGGAGCATCAGCATCCAGTGCTACCTGTGCC
 TGCTGCTGAACAGCCACTTCTGACCGAGGCCGATCCAGTGTTCAT
 CCTGGGCTGCTTTCAGCGCCGACTGCCCAAGACCGAGGCCAAGTGGTG
 AACGTGATCAGCGACCTGAAGAAGATCGAGGACCTGATCCAGAGCATGC
 ACATCGACGCCACCTGTACACCGAGAGCGACGTGCACCCAGCTGCAA
 GGTGACCGCCATGAAGTGCTTTCTGCTGGAAGTGCAGGTGATCAGCCTG
 GAAAGCGCGACGCCAGCATCCAGACACCGTGGAGAACCTGATCATCC

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TGGCCAACAACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTG
 CAAAGAGTGCAGGAACTGGAAGAGAAGAACATCAAAGAGTTTCTGCAG
 AGCTTCGTGCACATCGTGCAGATGTTTCATCAACACCAGCTGA

[0141] A corresponding amino acid sequence for CD27Tr28tmdDAP10icd3z15 is as follows:

(SEQ ID NO: 41)

MTRVTNSPSLQAHLQALYLQHEVVRPLAAAYQEQLDRPVVPHPYR
 VGDTVVVRRHQTKNLEPRWKGPTVLLTTPALKVDGIAAWIHAAHVKA
 ADPGGGPSRLPCSRMARPHPWWLCVLGTLVGLSATPAPKSCPERHYWA
 QGKLCCQMCEPGTFLVKDCDQHRKAAQCDPCI PGVSFSPDHHTRPHCES
 CRHCNSGLLVNRCTITANAECACRNGWQCRDKECTEDPLPNPSLTARS
 SQALSPHPQPTHLPYVSEMLEARTAGHMOTLADFRQLPARTLSHWPPQ
 RSLCSDFI RFWVLVVVGGVLACYSLLVTVAFIIFWVLCARPRRSPAQE
 DGKVIYNMPGRGKRVKFSRSADAPAYQQGQNQLYNELNLRREEYDVL
 KRRGRDPEMGGKPRKPNQEGLYNELQKDKMAEAYSEIGMKGERRRGKG
 HDGLYQGLSTATKDTYDALHMQALPPRPQCTNYALLKLADVESNPGP
 MRISKPHLRSISIQCYLCLLLNSHFLTEAGIHVFI LGCFSAGLPKTEAN
 WVNISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVI
 SLESGDASIHDTVENLILANNLSNNGNVTESGCKECEELEEKNIKEF
 LQSFVHIVQMFINTS

GSPco27Tr28tmdDAP10IL15:

(SEQ ID NO: 19)

ATGACAAGAGTTACTAACAGCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACCTGGACCGACCGGTGGTACCTCACCTTACCAGAGTCGGC
 GACACAGTGTGGGTCCGCCGACACCAGACTAAGAACCCTAGAACCCTCGCT
 GGAAAGGACCTTACACAGTCTGTGTGACACCCCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACGCCGCCACGTGAAGGCTGCCGAC
 CCCGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGCTTCTCC
 TGGTGACAAGCCTTCTGCTCTGTGAGTTACCACACCCAGCATTCTCTCT
 GATCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCGGCATTATTGG
 GCACAGGGCAAGTTGTGTGTCAAATGTGTGAGCGGGAACCTTCTCG
 TGAAGGATTGCGATCAGCATCGGAAGGCCGCGCAGTGCGACCCATGTAT
 ACCAGGGGTCTCATTTTCCCCAGATCACCATACGAGGCCGCACTGTGAG
 TCTTGACGGCATTGTAATTCCGGCTTGTGGTCCGCAACTGTACTATTA
 CTGCGAATGCAGAGTGTGCTTGTAGAAACGGATGGCAGTGCAGGGACAA
 AGAATGTACGGAGTGTGATCCACTGCCTAACCCAGCTTACAGCAAGA
 TCTTCACAGGCCCTCAGCCGCATCCTCAACCAACACATCTTCTTACG

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TGTCAGAAATGTTGGAGGCGCGAACCGCAGGCCATATGCAGACCTGGC
GGACTTTCGGCAGCTGCCAGCAGCACACTTAGTACACACTGGCCACCA
CAACGCAGCTTGTGCTCTCCGATTTCATCCGCTTTTGGGTGCTGGTGG
TGGTTGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAACAGTGGCCTT
TATTATTTTTCGGGTGCTTTGCGCACGCCACGCCGAGCCCCGCCCAA
GAAGATGGCAAAGTCTACATCAACATGCCAGGCAGGGGCAAACGCGTGA
AGTTCAGCAGGAGCGCAGACGCCCCCGGTACCAGCAGGGCCAGAACCA
GCTCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACGATGTTTTG
GACAAAAGACGTGGCCGGGACCTTGAGATGGGGGAAAGCCGAGAAGGA
AGAACCCTCAGGAAGGCTGTACAATGAACGTCAGAAAGATAAGATGGC
GGAGGCTTACAGTGAGATTGGGATGAAAGGCGAGCGCCGAGGGGCAAG
GGGCACGATGGCCTTTACCAGGGTCTCAGTACAGCCACCAAGGACACCT
ACGACGCCCTTACATGCAGGCCCTGCCCCCTCGCGGACCGCAGTGATC
TAATTATGCTCTCTTGAAATTGGCTGGAGATGTTGAGAGCAATCCCGGG
CCCATGCGCATTAGCAAGCCCCACCTGCGGAGCATCAGCATCCAGTGCT
ACCTGTGCCTGCTGCTGAACAGCCACTTCTGACCGAGGCCGGCATCCA
CGTGTTATCCTGGGTGCTTACGCGCCGACTGCCCAAGACCGAGGCC
AACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGGACCTGATCC
AGAGCATGCACATCGACGCCACCTGTACACCGAGAGCGAGTGCACCC
CAGCTGCAAGGTGACGCCCATGAAGTGCTTTCTGCTGGAACGTCAGGTG
ATCAGCCTGAAAAGCGGCGACGCCAGCATCCACGACACCGTGAGAACC
TGATCATCCTGGCCAACAACAGCCTGAGCAGCAACGGCAACGTGACCGA
GAGCGGCTGCAAGAGTGCAGGAACTGGAAGAGAAGAACATCAAAGAG
TTTCTGCAGAGCTTCGTGCACATCGTGAGATGTTTCATCAACACAGCT
GA

[0142] A corresponding amino acid sequence for GSPco27Tr28tmdDAP10IL15 is as follows:

(SEQ ID NO: 42)

MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
VGDTVWVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKA
ADPGGPPSSRLPCSRMLLLVTSLLLCELPHPAFLLI PATPAPKSCPERH
YWAQGLKLCQMCEPFTFLVKDQDHRKAAQCDPCIPGVSFSPDHHTPH
CESCRHCNSGLLRNCTITANAECACRNGWQCRDKECTCDPLNPNSLT
ARSSQALSHPHPQTHLPYVSEMLEARAGHMQTLADFRQLPARTLSTHW
PPQRSLSCSSDFIRFWVLVVVGVLACYSLLVTVAFIIFWVLCARPRRSP
AQEDGKVYINMPGRKRVKFSRSADAPAYQQQNQLYNELNLGRREEYD
VLDKRRGRDPEMGKPRRKNPQEGLYNELQDKMAEAYSEIGMKGERRR
GKGHDGLYQGLSTATKDYDALHMQALPPRGPCQTNALLKLAGDVESN
PGPMRISKPHLRSISIQCYLCLLLNSHFLTEAGIHVFILGCFSAGLPKT

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EANWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCVKVTAMKCFLLEL
QVISLESGDASIHDVTENLIILANNSLSSNGNVTESGCKECELEEKNI
KEFLQSFVHIVQMFINTS

CD27Tr28tmdDAP12icd3z15:

(SEQ ID NO: 20)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
CCAAGAACAACCTGGACCGACCGGTGTACCTCACCTTACCAGTTCGGC
GACACAGTGTGGGTCCGCCGACACAGACTAAGAACC TAGAACCTCGCT
GGAAAGGACCTTACACAGTCTCTGCTGACACCCCCACCGCCCTCAAAGT
AGACGGCATCGCAGCTTGATACACGCGCCACGTGAAGGCTGCCGAC
CCCCGGGGTGAGCATCCTCTAGACTGCCATGTCTGAGGATGGCAGGCG
CACATCCCTGGTGGCTGTGCGTTCTGGGGACCTGGTGGGGCTCTCAGC
TACTCCAGCCCCCAAGAGCTGCCCAGAGAGCACTACTGGGCTCAGGGA
AAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCTCGTGAAGGACT
GTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTTGATACCGGGGGT
CTCCTTCTCTCTGACCACACACCCGGCCCCACTGTGAGAGCTGTGCGG
CACTGTAACCTGGTCTTCTCGTTCGCAACTGCACCATCACTGCCAATG
CTGAGTGTGCTGTGCGCAATGGCTGGCAGTGCAGGACAAAGAGTGCAC
CGAGTGTGATCCTCTTCAAACCCCTTCTGCTGACCGCTCGGTCTGCTCAG
GCCCTGAGCCCAACCCCTCAGCCCAACCCACTTACCTTATGTGAGTGA
TGCTGGAGGCCAGGACAGCTGGGCACATGCAGACTCTGGCTGACTTCAG
GCAGCTGCCTGCCCGGACTCTCTTACCCTGAGCCACCCCAAAGATCC
CTGTGACGCTCCGATTTTATTCGCTTTTGGGTGCTGGTGGTGGTGGTG
GAGTCTGGCTTGCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTTT
CTGGGTGTACTTCTGGGCGGCTGGTCCCTCGGGGGCAGGGGCTGCG
GAGGCAGCGACCCGAAACAGCGTATCACTGAGACCGAGTGCCTTATC
AGGAGCTCCAGGGTCAGAGGTCCGATGTCTACAGCGACCTCAACACACA
GAGGCCGTATTACAAAAACGCGTGAAGTTCAGCAGGAGCGCAGACGCC
CCCGCGTACCAGCAGGGCCAGAACCAGCTCTATAACGAGCTCAATCTAG
GACGAAGAGAGGAGTACGATGTTTGGACAAAAGACGTGGCCGGGACCC
TGAGATGGGGGAAAGCCGAGAAGGAAGAACCTCAGGAAGGCTGTAC
AATGAACTGCAGAAAGATAAGATGGCGGAGGCCTACAGTGAGATTGGGA
TGAAAGGCGAGCGCCGAGGGGCAAGGGGCACGATGGCCTTTACCAGGG
TCTCAGTACAGCCACCAAGGACACCTACGACGCCCTTACATGCAGGCC
CTGCCCCCTCGCGGACCGCAGTGTACTAATTATGCTCTCTTGAAATTGG
CTGGAGATGTTGAGAGCAATCCCGGGCCCATGCGCATTAGCAAGCCCCA
CCTGCGGAGCATCAGCATCCAGTGCTACCTGTGCTGCTGCTGAACAGC

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CACTTCCTGACCGAGGCCGGCATCCACGTGTTTCATCCTGGGTGCTTCA
 GCGCCGGACTGCCAAGACCGAGGCCAACTGGGTGAACGTGATCAGCGA
 CCTGAAGAAGATCGAGGACCTGATCCAGAGCATGCACATCGACGCCACC
 CTGTACACCGAGAGCGACGTGACCCCACTGCAAGGTGACCGCCATGA
 AGTGCTTTCTGCTGGAACGTGACGGTATCAGCCTGGAAGCGGCGACGC
 CAGCATCCACGACACCGTGAGAACCTGATCATCCTGGCCAACAACAGC
 CTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTGCAAGAGTGCGAGG
 AACTGGAAGAAGAACAATCAAGAGTTTCTGAGAGCTTCGTGCACAT
 CGTGAGATGTTTCATCAACACAGCTGA

[0143] A corresponding amino acid sequence for CD27Tr28tmdDAP12icd3z15 is as follows:

(SEQ ID NO: 43)

MTRVTNPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
 VGDVWVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKA
 ADPGGGPSRLPCSRMARPHPWWLCLVGLTSLVGLSATPAPKSCPERHYWA
 QGKLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPHCES
 CRHCNSGLLVNRNCTITANAECACRNGWQCRDKECTCDPLPNPSLTARS
 SQALSPHPQPTHLPYVSEMLEARTAGHMQLADFRQLPARTLSHWPPQ
 RSLCSSDFIRFVWLVVVGVLACYSLLVTAFIIFWVYFLGRLVPRGRG
 AAEEATRKRQITETESPYQELQGRSDVYDLNTQRPYKRVKPSRSA
 DAPAYQQGQNQLYNELNLRREEYDVLDKRRGRDPEMGGKPRRKNPQEG
 LYNELQKDKMAEAYSEIGMKGERRRGKGDGLYQGLSTATKDTYDALHM
 QALPPRGPQCTNYALLKLKAGDVESNPGPMRISKPHLRSISIQCYLCLLL
 NSHFLTEAGIHVFILGCFISAGLPKTEANWVNVISDLKKIEDLIQSMHID
 ATLYTESDVHPSCKVTAMKCFLELQVISLESGDASIHDTVENLIILAN
 NSLSSNGNVTESGCKECEELEEKNIKEFLQSPFHIVQMFINTS

GSPco27Tr28tmdDAP12icd15:

(SEQ ID NO: 21)

ATGACAAGAGTTACTAACAGCCCTCTCTCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACCTGGACCGACCGGTGGTACCTACCCCTACCGAGTCGGC
 GACACAGTGTGGGTCGCGGACACAGACTAAGAACCTAGAACCTCGCT
 GGAAAGGACCTTACACAGTCTGTGACCAACCCCAACCGCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACCGCGCCACGTGAAGGCTGCCGAC
 CCCGGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGCTTCTCC
 TGGTGACAAGCCTTCTGCTCTGTGAGTTACACACCCAGCATTCCTCCT
 GATCCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCGGCATTATTGG
 GCACAGGGCAAGTTGTGTGTCAAATGTGTGAGCCGGGAACCTTTCTCG

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TGAAGGATTGCGATCAGCATCGGAAGGCCGCGCAGTGCGACCCATGTAT
 ACCAGGGGTCTCATTTTCCCCAGATCACCATACGAGGCCGCGACTGTGAG
 TCTTGCAGGCATTGTAATTCGGCTTGTGGTCCGCAACTGTACTATTA
 CTGCGAATGCAGAGTGTGCTTGTAGAAACGGATGGCAGTGCAGGGACAA
 AGAATGTACGGAGTGTGATCCACTGCCTAACCCAGCTTACAGCAAGA
 TCTTACAGGCCCTCAGCCCGCATCCTCAACCAACACATCTTCCTTACG
 TGTGAGAAATGTTGGAGGCGGAACCGCAGGCCATATGAGACCCCTGGC
 GGACTTTCGGCAGCTGCCAGCACGCACTTAGTACACACTGGCCACCA
 CAACGCAGCTTGTGCTCTTCCGATTCATCCGCTTTTGGGTGCTGGTGG
 TGGTTGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAACAGTGGCCTT
 TATTATTTTCTGGGTGTACTTCTGGGCCGGCTGGTCCCTCGGGGCGCA
 GGGGCTGCGGAGGCAGCGACCCGGAACAGCGTATCACTGAGACCGAGT
 CGCCTTATCAGGAGCTCCAGGGTCAGAGGTGGATGTCTACAGCGACCT
 CAACACACAGAGGCCGTATTACAAAAACCGTGAAGTTTACGAGGAGC
 GCAGACGCCCCCGCTACCAGCAGGGCCAGAACAGCTCTATAACGAGC
 TCAATCTAGGACGAAGAGAGGAGTACGATGTTTGGACAAAAGACGTGG
 CCGGGACCTTGAGATGGGGGGAAGCCGAGAAGGAAGAACCTCAGGAA
 GGCTGTACAATGAAGTGCAGAAAGATAAGATGGCGGAGGCTACAGTG
 AGATTGGGATGAAAGGCGAGCGCCGAGGGGCAAGGGGCACGATGGCCT
 TTACCAGGCTCTCAGTACAGCCACCAAGGACACCTACGACGCCCTTAC
 ATGACAGGCCCTGCCCCCTCGCGGACCGCAGTGTACTAATTATGCTCTCT
 TGAAATTGGCTGGAGATGTTGAGAGCAATCCCGGGCCCATGCGCATTAG
 CAAGCCCCACCTGCGGAGCATCAGCATCCAGTGTACCTGTGCCTGCTG
 CTGAACAGCCACTTCTTGACCGAGGCCGCGCATCCACGTGTTTCATCCTGG
 GCTGCTTCAGCGCCGACTGCCCAAGACCGAGGCCAACTGGGTGAACGT
 GATCAGCGACCTGAAGAAGATCGAGGACCTGATCCAGAGCATGCACATC
 GACGCCACCTGTACACCGAGAGCGACGTGACCCAGCTGCAAGGTGA
 CCGCCATGAAGTGCTTTCTGCTGGAACTGCAGGTGATCAGCCTGGAAAG
 CGGCGACCGCAGCATCCACGACACCGTGGAGAACCTGATCATCTGGCC
 AACACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTGCAAG
 AGTGCGAGGAACTGGAAGAGAAGAACATCAAGAGTTTCTGAGAGCTT
 CGTGACATCGTGAGATGTTTCATCAACACAGCTGA

[0144] A corresponding amino acid sequence for GSPco27Tr28tmdDAP12icd15 is as follows:

(SEQ ID NO: 44)

MTRVTNPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
 VGDVWVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKA
 ADPGGGPSRLPCSRMLLVTSLLLCELPHPFLILPATPAPKSCPERH
 YWAQGLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPH

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CESCRHCNSGLLVRNCTITANAECACRNGWQCRDKECTCDPLPNPSLT
 ARSSQALSPHPQPTHLPYVSEMLEARTAGHMOTLADFRQLPARTLSTHW
 PPQRLCSSDFIRFWLVVVGGVLACYSLLVTVAFIIFWVYFLGRLVPR
 GRGAAEAATRKQRI TETESPYQELQGQSRDVS DLNTQRPYKRVKFS
 RSADAPAYQQGNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNP
 QEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDA
 LHMQUALPPRGPQCTNYALLKLAGDVESNPGPMRISKPHLSISIQCYLC
 LLLNSHFLTEAGIHVFILGCF SAGLPKTEANWVNVISDLKKIEDLIQSM
 HIDATLYTESDVHPSCKVTAMKCFLELQVISLES GDASIHDTVENLI I
 LANNSLSSNGNVTESGCKECELEEKNIKEFLQSFVHIVQMFINTS

CD27Tr28tmdNKG2Dic3z15:

(SEQ ID NO: 22)

ATGACAAGAGTTACTAACAGCCCCCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGC
 GACACAGTGTGGGTCGCCGACACACAGACTAAGAACCTAGAACCTCGCT
 GGAAAGGACCTTACACAGTCTGTGTCACACCCCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACCGCCGCCACGTGAAGGTGCCGAC
 CCGGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGGCAGGC
 CACATCCCTGGTGGCTGTGCGTTCTGGGACCTGGTGGGGCTCTCAGC
 TACTCCAGCCCCAAGAGCTGCCCAGAGAGGCAC TACTGGGCTCAGGGA
 AAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCCTCGTGAAGGACT
 GTGACCAGCATAGAAAGCTGCTCAGTGTGATCCTTGATACCGGGGT
 CTCTCTCTCTCTGACCACACACCGGCCCTACTGTGAGAGCTGTCCG
 CACTGTAACCTCTGTCTTCTCGTTCGCAACTGCACCATCACTGCCAATG
 CTGAGTGTGCTGTGCAATGGCTGGCAGTGACGGACAAGGAGTGAC
 CGAGTGTGATCCTCTTCCAAACCTTCGCTGACCGCTCGGTCTCTCAG
 GCCCTGAGCCACACCTCAGCCACCCACTTACCTTATGTCTAGTGA
 TGCTGGAGGCCAGGACAGCTGGGCACATGCAGACTCTGGCTGACTTCA
 GCAGTGCCTGCCCGACTCTCTCTACCCACTGGCCACCCCAAAGATCC
 CTGTGACGCTCCGATTTTATTCTGCTTTTGGGTGCTGGTGGTGGTGGT
 GAGTCTGGCTTGCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTTT
 CTGGGTGAGCGCGAACGAACGCTGCAAAAGCAAAGTGGTCCGTCGCCG
 CAGAAACAGTGGCGCACCACTTTGATAGCAAAAACTGGATCTGAAC
 ATAACCATTTTGAAAGCATGGAATGGAGCCATCGCAGCGCGCGGCCG
 CATTTGGGGCATGAAACGCGTGAAGTTCAGCAGGAGCGCAGACGCCCC
 GCGTACCAGCAGGCGCAGAACAGCTCTATAACGAGCTCAATCTAGGAC
 GAAGAGAGGAGTACGATGTTTTGGACAAAAGAGCTGGCCGGGACCTGA

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GATGGGGGGAAGCCGAGAAGGAAGAACCTCAGGAAGGCCTGTACAAT
 GAAGTGCAGAAAGATAAGATGGCGGAGGCCTACAGTGAGATTGGGATGA
 AAGGCGAGCGCCGGAGGGGCAAGGGGCACGATGGCCTTTACCAGGTCT
 CAGTACAGCCACCAAGGACACCTACGACGCCCTTACATGCAGGCCCTG
 CCCCCTCGCGGACCGCAGTGTACTAATTATGCTCTCTTGAAATTGGCTG
 GAGATGTTGAGAGCAATCCCGGGCCCATGCGCATTAGCAAGCCCCACCT
 GCGGAGCATCAGCATCCAGTGTCTACCTGTGCTGCTGCTGAACAGCCAC
 TTCCTGACCGAGGCCGGCATCCAGTGTTCATCTGGGCTGCTTCAGCG
 CCGGACTGCCCAAGACCGAGGCCAACTGGGTGAACGTGATCAGCGACCT
 GAAGAAGATCGAGGACCTGATCCAGAGCATGCACATCGACGCCACCCTG
 TACACCGAGAGCGACGTGCACCCAGCTGCAAGGTGACCGCCATGAAGT
 GCTTCTGCTGGAAGTGCAGGTGATCAGCCTGGAAAGCGGCAGCCAG
 CATCCACGACACCGTGGAGAACCTGATCATCTGGCCAAACAGCCTG
 AGCAGCAACGGCAACGTGACCGAGAGCGGCTGCAAGAGTGCAGGGAAC
 TGGAAGAGAAGAACATCAAAGAGTTTCTGCAGAGCTTCGTGCACATCGT
 GCAGATGTTTCATCAACACCAGCTGA

[0145] A corresponding amino acid sequence for CD27Tr28tmdNKG2Dic3z15 is as follows:

(SEQ ID NO: 45)

MTRVTNSPSLQAHLQALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
 VGDTVWVRRHQTKNLEPRWKGPTVLLTTPALKVDGIAAWIHAAHVKA
 ADPGGGPSRLPCSRMARPHPWWLCVLGTLVGLSATPAPKSCPERHYWA
 QGKLCCQMCEPGTFLVKDCDQHRKAAQCDPCI PGVSFSPDHHTRPHCES
 CRHCNSGLLVRNCTITANAECACRNGWQCRDKECTCDPLPNPSLTARS
 SQALSPHPQPTHLPYVSEMLEARTAGHMOTLADFRQLPARTLSTHWPPQ
 RSLCSSDFIRFWLVVVGGVLACYSLLVTVAFIIFWVSANERCKSKVVP
 CRQKQWRTSFD SKKLDLNYNHFESMEWSHRSRRRIWGMKRVKFSRSAD
 APAYQQGNQLYNELNLGRREEYDVL DKRRGRDPEMGGKPRRKNPQEGL
 YNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQ
 ALPPRGPQCTNYALLKLAGDVESNPGPMRISKPHLSISIQCYLCLLL
 SHFLTEAGIHVFILGCF SAGLPKTEANWVNVISDLKKIEDLIQSMHIDA
 TLYTESDVHPSCKVTAMKCFLELQVISLES GDASIHDTVENLI I LAN
 SLSSNGNVTESGCKECELEEKNIKEFLQSFVHIVQMFINTS

GSPco27Tr28tmdNKG2Dicd3z15:

(SEQ ID NO: 23)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTACCAAGA
ACAACTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGCGACACAGTGTG
GGTCCGCCGACACCAGACTAAGAACC TAGAACCTCGCTGGAAGGACCTTACAC
AGTCCTGTGACCACCCCCACGCCCTCAAAGTAGACGGCATCGCAGCTTGGATA
CACGCCGCCACGTGAAGGCTGCCGACCCCGGGGTGGACCATCTCTAGACTG
CCATGCTCGAGGATGCTTCTCCTGGTGACAAGCCTTCTGCTCTGTGAGTTACCAC
ACCCAGCATTCTCTGTATCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCG
GCATTATTGGGCACAGGGCAAGTTGTGTTGTCAAATGTGTGAGCCGGAACCTTT
CTCGTGAAGGATTGCGATCAGCATCGGAAGGCCGCGCAGTGCACCCATGTATA
CCAGGGTCTCATTITCCCCAGATCACCATACGAGGCCGCACTGTGAGTCTTGCA
GGCATTGTAAATCCGGCTTGTGGTCCGCACTGTACTATTACTGCGAATGCAGA
GTGTGCTTGTAGAAACGGATGGCAGTGCAGGGACAAAGAATGTACGGAGTGTGA
TCCACTGCCTAACCCAGTCTTACAGCAAGATCTTCACAGGCCCTCAGCCCGCAT
CCTCAACCAACACATCTTCCTTACGTGTCAGAAATGTTGGAGGCGCAACCGCAG
GCCATATGCAGACCCTGGCGGACTTTCGGCAGCTGCCAGCACGCACACTTAGTAC
ACACTGGCCACCACAACGCAGCTTGTGCTCTTCCGATTTCATCCGCTTTTGGGTGC
TGGTGGTGGTTGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAACAGTGGCCTT
TATTATTTTCTGGGTGAGCGCAACGAACGCTGCAAAAGCAAAGTGGTGCCGTG
CCGCCAGAAACAGTGGCGCACAGCTTTGATAGCAAAAACCTGGATCTGAAC TA
TAACCATTTTGAAGCATGGAATGGAGCCATCGCAGCCGCCGCGCCGCAATTG
GGGCATGAAACGCGTGAAGTTCAGCAGGAGCGCAGACGCCCCCGCTACCAGCA
GGGCCAGAACAGCTCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACGA
TGTTTTGGACAAAAGAGCTGGCCGGGACCTGAGATGGGGGAAAGCCGAGAAG
GAAGAACCTCAGGAAGGCCTGTACAATGAACTGCAGAAAGATAAGATGGCGG
AGGCCTACAGTGAGATTGGGATGAAAGGCGAGCGCCGGAGGGGCAAGGGGCAC
GATGGCCTTTACCAGGGTCTCAGTACAGCCACCAAGGACACCTACGACGCCCTTC
ACATGCAGGCCCTGCCCCCTCGCGACCGCAGTGTAATTATGCTCTCTTGAA
ATTGGCTGGAGATGTTGAGAGCAATCCCGGGCCCATGCGCATTAGCAAGCCCCA
CCTCGGAGCATCAGCATCCAGTGCTACCTGTGCTGCTGCTGAACAGCCACTTC
CTGACCGAGGCCGGCATCCACGTGTTTCATCCTGGGCTGCTTCAGCGCCGACTGC
CCAAGACCGAGGCCAACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGG
ACCTGATCCAGAGCATGCACATCGACGCCACCCTGTACACCGAGAGCGACGTGC
ACCCAGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGCTGGAACGCAGGTGAT
CAGCCTGGAAGCGGCGACGCCAGCATCCACGACACCGTGGAGAACCTGATCAT
CCTGGCCAACAACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTGCAA
AGAGTGCGAGGAACTGGAAGAGAAGAATCAAAGAGTTTCTGCAGAGCTTCGT
GCACATCGTCAGATGTTTCATCAACACCGCTGA

[0146] A corresponding amino acid sequence for GSPco27Tr28tmdNKG2Dcd3z15 is as follows:

(SEQ ID NO: 46)

MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
 VGDTVVVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKAADPGGG
 PSSRLPCSRMLLLVTSLLLCELPHPAFLLIPATPAPKSCPERHYWAQGLCCQMCEPG
 TFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPHCESCRHCNSGLLVNRNCTITANAE
 CACRNGWQCRDKECTECDPLPNPSLTARSSQALSHPHPQTHLPYVSEMLEARAGH
 MQTLADFRQLPARTLSTHWPPQRSLSQSDFIKFWVLVVGGVLACYSLLVTVAFIIF
 WVSANERCKSKVVPQRQKQWRTSFDKSKLDLNYNHFESMEWSHRSGRIWGMK
 RVKFSRSADAPAYQQQNQLYNELNLGRREEYDVLDRGRDPEMGGKPRKPNQ
 EGLYNELQDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALP
 PRGPQCTNYALLKLAGDVESNPGPMRISKPHLSISIQCYLCLLNSHFLTEAGIHVFI
 LGCFSAAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFI
 LELQVISLESGLASIHDTVENLIILANNSLSSNGNVTESGCKECEELEKNIKEFLQSFV
 HIVQMFINTS

CD27Tr41BBcd3z15:

(SEQ ID NO: 24)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTACCAAGA
 ACAACTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGCGACACAGTGTG
 GGTCCGCCGACACCAGACTAAGAACCTAGAACCTCGCTGAAAGGACCTTACAC
 AGTCCTGTGACCACCCCCACCGCCCTCAAAGTAGACGGCATCGCAGCTTGGATA
 CACGCCGCCACGTGAAGGCTGCCGACCCCGGGGTGGACCATCTCTAGACTG
 CCATGCTCGAGGATGGCACGGCCACATCCCTGGTGGCTGTGCGTTCTGGGGACCC
 TGGTGGGGCTCTCAGCTACTCCAGCCCCAAGAGCTGCCAGAGAGGCACTACT
 GGGCTCAGGGAAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCCTCGTGA
 AGGACTGTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTTGACATACCGGGG
 TCTCCTTCTCTCTGACCACCACACCGGCCCACTGTGAGAGCTGTGGGCACTG
 TAACTCTGGTCTTCTCGTTTCGCAACTGCACCATCACTGCCAATGCTGAGTGTGCCT
 GTCGCAATGGCTGGCAGTGACGGACAAGGAGTGACCGAGTGTGATCCTCTTC
 CAAACCTTCGCTGACCGCTCGGTCTGTCAGGCCCTGAGCCACACCTCAGCC
 CACCCACTTACCTTATGTCAGTGAGATGCTGGAGGCCAGGACAGCTGGGCACAT
 GCAGACTCTGGCTGACTTCAGGCAGCTGCCTGCCCGGACTCTCTTACCCACTGG
 CCACCCCAAAGATCCCTGTGCAGCTCCGATTTTATTTCGATCCTTGTGATCTTCTC
 TGGATGTTCTTGTGTTTACCCCTGGCCGGGGCCCTGTTCTCCATAAACGGGGC
 AGAAAGAACTCCTGTATATATTCAAACAACCATTTATGAGACCAGTACAACT
 ACTCAAGAGGAAGATGGCTGTAGCTGCCGATTTCCAGAAGAAGAAGAGAGG
 ATGTGAAC TGAAACGCGTGAAGTTCAGCAGGAGCGCAGACGCCCCCGGTACCA

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GCAGGGCCAGAACCAGCTCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTA
CGATGTTTTGGACAAAAGACGTGGCCGGGACCCCTGAGATGGGGGAAAGCCGAG
AAGGAAGAACCCTCAGGAAGGCCTGTACAATGAACTGCAGAAAGATAAGATGG
CGGAGGCCTACAGTGAGATTGGGATGAAAGGCGAGCGCCGGAGGGGCAAGGGG
CACGATGGCCTTTACCAGGGTCTCAGTACAGCCACCAAGGACACCTACGACGCC
CTTCACATGCAGGCCCTGCCCCCTCGCGGACCGCAGTGCTACTAATTATGCTCTCTT
GAAATTGGCTGGAGATGTTGAGAGCAATCCCGGGCCATGCGCATTAGCAAGCC
CCACCTGCGGAGCATCAGCATCCAGTGCTACCTGTGCCTGCTGCTGAACAGCCAC
TTCCTGACCGAGGCCCGGCATCCACGTGTTTCATCCTGGGCTGCTTCAGCGCCGGAC
TGCCCAAGACCGAGGCCAACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCG
AGGACCTGATCCAGAGCATGCACATCGACGCCACCCGTACACCGAGAGCGACG
TGCACCCAGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGCTGGAAGTGCAGGT
GATCAGCCTGGAAGCGGCGACGCCAGCATCCACGACACCGTGGAGAACCTGAT
CATCTGGCCAACAACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTG
CAAAGAGTGCAGGAACTGGAAGAGAAGAACATCAAAGAGTTTCTGCAGAGCTT
CGTGACATCGTGCAGATGTTTCATCAACACCAGCTGA

[0147] A corresponding amino acid sequence for CD27Tr41BBicd3z15 is as follows:

(SEQ ID NO: 47)

MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
VGDTVWVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKAADPGGG
PSSRLPCSRMARPHPWLCVLGTLVGLSATPAPKSCPERHYWAQGLCCQMCEPGT
FLVKDCDQHRKAAQCDPCIPGVNSFDPDHHTRPHCESRHCNSGLLVNRNCTITANAEC
ACRNGWQCRDKECTECDPLPNPSLTARSSQALSPHPQPTHLPYVSEMLEARAGHM
QTLADFRQLPARTLSTHWPPQRSCLSSDFIRILVIFSGMFLVFTLAGALFLHKGGRKKL
LYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELKRVKFSRSADAPAYQQGQNQLY
NELNLGRREEYDVLDKRRGRDPEMGKPRRKNPQEGLYNELQKDKMAEAYSEIGM
KGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPRGPQCTNYALLKLAGDVESN
PGPMRISKPHLRISISIQCYLCLLLNSHFLTEAGIHVFILGCFSAGLPKTEANWVNVISDL
KKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLES GDASIHDTVENLIIL
ANNSLSNNGNVTESGCKECEELEEKNIKEFLQSFVHIVQMFINTS

GSPco27Tr41BBicd3z15:

(SEQ ID NO: 25)

ATGACAAGAGTTACTAACAGCCCTCTCTCCAAGCTCACTTACAGG
CTCTCTACTTAGTCCAGCACGAAGCTGGGAGACCTCTGGCGGAGCCTACCAAGA
ACAACTGGACCGACCGGTGGTACCTCACCCTTACCGAGTCGGCGACACAGTGTG
GGTCCGCCGACACCAGACTAAGAACCTAGAACCTCGCTGGAAGGACCTTACAC
AGTCCTGCTGACCACCCACCGCCCTCAAAGTAGACGGCATCGCAGCTTGGATA

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CACGCCGCCACGTGAAGGCTGCCGACCCCGGGGTGGACCATCCTCTAGACTG
CCATGCTCGAGGATGCTTCTCTGGTGACAAGCCTTCTGCTCTGTGAGTTACCAC
ACCCAGCATTCTCTGATCCCAGCTACACGGCTCCGAAGTCTGCCCGGAGCG
GCATTATTGGGCACAGGGCAAGTTGTGTTGTCAAATGTGTGAGCCGGGAACCTTT
CTCGTGAAGGATTGCGATCAGCATCGGAAGGCCGCGCAGTGCAGCCCATGTATA
CCAGGGGTCTCATTTTCCCAGATCACCATACGAGGCCGCACTGTGAGTCTTGCA
GGCATTGTAATCCGGCTTGTGTCGCAACTGTACTATTACTGCGAATGCAGA
GTGTGCTTGTAGAAACGGATGGCAGTGCAGGGACAAAGAATGTACGGAGTGTGA
TCCACTGCCTAACCCAGTCTTACAGCAAGATCTTCACAGGCCCTCAGCCCGCAT
CCTCAACCAACACATCTTCCTTACGTGTCAGAAATGTTGGAGGCGCAACCGCAG
GCCATATGCAGACCCCTGGCGGACTTTCGGCAGCTGCCAGCACGCACACTTAGTAC
ACACTGGCCACCACAACGCAGCTTGTGCTCTCCGATTTCATCCGCATACTGGTC
ATCTTTTCTGGAATGTTCTTGTGTTACCCCTGGCAGGAGCCCTGTTCTTCACAA
ACGGGGCAGAAAGAACTCCTGTATATATTCAAACAACCATTTATGAGACCAGT
ACAAACTACTCAAGAGGAAGATGGCTGTAGCTGCGGATTTCAGAAGAAGAAGA
AGGAGGATGTGAACTGAAACGCGTGAAGTTCAGCAGGAGCGCAGACGCCCCGC
GTACCAGCAGGGCCAGAACCAGCTCTATAACGAGCTCAATCTAGGACGAAGAGA
GGAGTACGATGTTTTGGACAAAAGACGTGGCCGGGACCTGAGATGGGGGAAA
GCCGAGAAGGAAGAACCTCAGGAAGGCCTGTACAATGAACTGCAGAAAGATA
AGATGGCGGAGGCCCTACAGTGAGATTGGGATGAAAGGCGAGCGCCGGAGGGGC
AAGGGGCACGATGGCCTTACCAGGGTCTCAGTACAGCCACCAAGGACACCTAC
GACGCCCTTCACATGCAGGCCCTGCCCCCTCGCGGACCGCAGTGTACTAATTATG
CTCTCTTGAAATTGGCTGGAGATGTTGAGAGCAATCCCGGGCCCATGCGCATTAG
CAAGCCCCACCTGCGGAGCATCAGCATCCAGTGCTACCTGTGCTGCTGCTGAAC
AGCCACTTCTTGACCGAGGCCGCATCCACGTGTTTCATCCTGGGCTGCTTCAGCG
CCGACTGCCCAAGACCGAGGCCAACTGGGTGAACGTGATCAGCGACCTGAAGA
AGATCGAGGACCTGATCCAGAGCATGCACATCGACGCCACCTGTACACCGAGA
GCGACGTGCACCCAGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGCTGGAAC
GCAGGTGATCAGCCTGGAAGCGGCGACGCCAGCATCCACGACACCGTGAGAGAA
CCTGATCATCCTGGCCAACAACAGCCTGAGCAGCAACGGCAACGTGACCGAGAG
CGGCTGCAAGAGTGCAGGAACGGGAAGAGAAGAACATCAAAGAGTTTCTGCA
GAGCTTCGTGCACATCGTGCAGATGTTTCATCAACACCAGCTGA

[0148] A corresponding amino acid sequence for GSPco27Tr41BBid3z15 is as follows:

(SEQ ID NO: 48)

MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
VGDTVWVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKAADPGGG
PSSRLPCSRMLLLVTSLLLCELPHPAFLLIPATPAPKSCPERHYWAQGLCCQMCEPG

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TFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPHCESCRHCNSGLLVRNCTITANAE
CACRNGWQCRDKECTECDPLPNPSLTARSSQALSHPHQPTHLPYVSEMLEARAGH
MQTLADFRQLPARTLSTHWPPQRSLSDFIRILVIFSGMFLVFTLAGALFLHKRGRK
KLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCELKRVKFSRSADAPAYQQGQNO
LYNELNLGRREYDVLDKRRGRDPENGGKPRRKNPQEGLYNELQKDKMAEAYSEI
GMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPRGPQCTNYALLKLAGDV
ESNPGPMRISKPHLRSISIQCYLCLLNHSHFLTEAGIHVFILGCF SAGLPKTEANWVNVI
SDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLESGDASIHDVTEN
LIILANNSLSSNGNVTESGCKECEELEEKNIKEFLQSFVHIVQMFINTS

CD27TrCD3ZIL15:

(SEQ ID NO: 26)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTACCAAGA
ACAACTGGACCGACCGGTGGTACCTCACCCTTACCGAGTCGGCGACACAGTGTG
GGTCCGCCGACACCAGACTAAGAACCTAGAACCTCGCTGGAAGGACCTTACAC
AGTCCTGTGACCACCCCCACGCCCTCAAAGTAGACGGCATCGCAGCTTGGATA
CACGCCGCCACGTGAAGGCTGCCGACCCCGGGGTGGACCATCCTCTAGACTG
CCATGCTCGAGGATGGCACGGCCACATCCCTGGTGGCTGTGCGTTCTGGGGACCC
TGGTGGGGCTCTCAGTACTCCAGCCCCAAGAGCTGCCAGAGAGGCACTACT
GGGCTCAGGGAAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCCTCGTGA
AGGACTGTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTTGACATACCGGGG
TCTCCTTCTCTCTGACCACACACCCGGCCCCACTGTGAGAGCTGTGCGCACTG
TAACTCTGGTCTTCTCGTTCGCAACTGCACCATCACTGCCAATGCTGAGTGTGCCT
GTCGCAATGGCTGGCAGTGCAGGGACAAGGAGTGACCGAGTGTGATCCTCTTC
CAAACCCCTCGCTGACCGCTCGGTCTGCTCAGGCCCTGAGCCACACCCCTCAGCC
CACCCACTTACCTTATGTCAGTGAGATGCTGGAGGCCAGGACAGCTGGGCACAT
GCAGACTCTGGCTGACTTCAGGCAGCTGCCTGCCGGACTCTCTCTACCCACTGG
CCACCCCAAAGATCCCTGTGCAGCTCCGATTTTATTGCGATCCTTGTGATCTTCTC
TGGAATGTTCTTGTTTTACCCCTGGCCGGGGCCCTGTTCTCCATAAACGCGTGA
AGTTCAGCAGGAGCGCAGACGCCCCGCGTACCAGCAGGGCCAGAACCAGCTCT
ATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACGATGTTTGGACAAAAGAC
GTGGCCGGGACCCCTGAGATGGGGGAAAGCCGAGAAGGAAGAACCTCAGGAA
GGCCTGTACAATGAACTGCAGAAAGATAAGATGGCGGAGGCCTACAGTGAGATT
GGGATGAAAGGCGAGCGCCGGAGGGGCAAGGGGCACGATGGCCTTTACCAGGG
TCTCAGTACAGCCACCAAGGACACCTACGACGCCCTTCACATGCAGGCCCTGCC
CCTCGCGGACCGCAGTGTAATAATTATGCTCTCTTGAAATTGGCTGGAGATGTTG
AGAGCAATCCCGGGCCCATGCGCATTAGCAAGCCCCACCTGCGGAGCATCAGCA
TCCAGTGCTACCTGTGCCTGCTGCTGAACAGCCACTTCCTGACCGAGGCCGGCAT

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CCACGTGTTTCATCCTGGGCTGCTTCAGCGCCGACTGCCCCAAGACCGAGGCCAAC
 TGGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGGACCTGATCCAGAGCATG
 CACATCGACGCCACCCTGTACACCGAGAGCGACGTGCACCCAGCTGCAAGGTG
 ACCGCCATGAAGTGCTTTCTGCTGGAATGCAGGTGATCAGCCTGGAAAGCGGC
 GACGCCAGCATCCACGACACCGTGGAGAACCTGATCATCCTGGCCACAACAGC
 CTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTGCAAAGAGTGCAGGAACT
 GGAAGAGAAGAACATCAAAGAGTTTCTGCAGAGCTTCGTGCACATCGTGCAGAT
 GTTCATCAACACCGAGTGA

[0149] A corresponding amino acid sequence for CD27TrCD3ZIL15 is as follows:

(SEQ ID NO: 49)

MTRVTNSPSLQAHLQALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
 VGDTVVVRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAHVKAADPGGG
 PSSRLPCSRMARPHPWMLCVLGLVGLSATPAPKSCPERHYWAQKLCQMCPEGT
 FLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPHCESCRHCNSGLLVRNCTITANAEC
 ACRNGWQCRDKECTECDPLPNPSLTARSSQALSPHPQPTHLPYVSEMLEARTAGHM
 QTLADFRQLPARTLSTHWPPQRSCLSSDFIRILVIFSGMFLVFTLAGALFLHKRVKFSR
 SADAPAYQQGQNQLYNELNLGRREEYDVLDRKRRDPEMGGKPRKPNQEGLYNE
 LQKDMAEAYSEIGMKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPRGPQC
 TNYALLKLAGDVESNPGPMRISKPHLSISIQCYLCLLNSHFLTEAGIHVFILGCFSA
 GLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVIS
 LESGDASIHDVTVENLIILANNSLSSNGNVTESGCKECELEEKNIKEFLQSFVHIVQMFI
 NTS

GSPco27Tr3z15:

(SEQ ID NO: 27)

ATGACAAGAGTTACTAACAGCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGACGCTACCAAGA
 ACAACTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGCGACACAGTGTG
 GGTCCGCCGACACCAGACTAAGAACCTAGAACCTCGCTGGAAGGACCTTACAC
 AGTCCTGCTGACACCCCCACGCCCTCAAAGTAGACGGCATCGCAGCTTGGATA
 CACGCCGCCACGTGAAGGCTGCCGACCCCGGGGTGGACCATCCTCTAGACTG
 CCATGCTCGAGGATGCTTCTCCTGGTGACAAGCCTTCTGCTCTGTGAGTTACCAC
 ACCAGCATTCTCTGATCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCG
 GCATTATTGGGCACAGGGCAAGTTGTGTTGTCAAATGTGTGAGCCGGGAACCTTT
 CTCGTGAAGATTGCGATCAGCATCGGAAGGCCGCGCAGTGCACCCATGTATA
 CCAGGGGTCTCATTTTCCCCAGATCACCATACGAGGCCGCACTGTGAGTCTTGCA
 GGCATTGTAATTCCGGCTTGTGGTCCGCAACTGTACTATTACTGCGAATGCAGA
 GTGTGCTTGTAGAAACGGATGGCAGTGCAGGACAAAGAATGTACGGAGTGTGA

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TCCACTGCCTAACCCAGTCTTACAGCAAGATCTTCACAGGCCCTCAGCCCGCAT
 CCTCAACCAACACATCTTCCTTACGTGTCAGAAATGTTGGAGGCGGAACCGCAG
 GCCATATGCAGACCCTGGCGGACTTTCGGCAGCTGCCAGCACGCACACTTAGTAC
 ACACTGGCCACCACAACGCAGCTTGTGCTCTTCCGATTTTCATCCGCATACTGGTC
 ATCTTTTCTGGAATGTTCTTGTGTTACCCCTGGCAGGAGCCCTGTTCTTCACCG
 CGTGAAGTTCAGCAGGAGCGCAGACGCCCCCGGTACCAGCAGGGCCAGAACCA
 GCTCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACGATGTTTTGGACAA
 AAGACGTGGCGGGACCCCTGAGATGGGGGAAAGCCGAGAAGGAAGAACCTC
 AGGAAGGCCTGTACAATGAAGTGCAGAAAGATAAGATGGCGGAGGCCTACAGTG
 AGATTGGGATGAAAGGCGAGCGCCGGAGGGCAAGGGGCACGATGGCCTTTACC
 AGGGTCTCAGTACAGCCACCAAGGACACCTACGACGCCCTTCACATGCAGGCCC
 TGCCCCCTCGCGGACCGCAGTGTACTAATTATGCTCTCTTGAAATGGCTGGAGA
 GTTGAGAGCAATCCCGGGCCCATGCGCATTAGCAAGCCCCACCTGCGGAGCAT
 CAGCATCCAGTGCTACCTGTGCCTGCTGCTGAACAGCCACTTCTTGACCGAGGCC
 GGCATCCACGTGTTTCATCCTGGGCTGCTTCAGCGCCGACTGCCAAGACCGAGG
 CCAACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGGACCTGATCCAGA
 GCATGCACATCGACGCCACCTGTACACCGAGAGCGACGTGCACCCCAGCTGCA
 AGGTGACCGCCATGAAGTGCTTTCTGCTGGAAGTGCAGGTGATCAGCCTGGAAA
 GCGGCGACGCCAGCATCCACGACACCGTGAGAACCTGATCATCCTGGCCAAACA
 ACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTGCAAGAGTGCGAG
 GAACTGGAAGAGAAGAACATCAAGAGTTTCTGCAGAGCTTCGTGCACATCGTG
 CAGATGTTTCATCAACACCAGCTGA

[0150] A corresponding amino acid sequence for GSPco27Tr3z15 is as follows:

(SEQ ID NO: 50)

MTRVTNPSLSLQHLQALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
 VGDTVWVRRHQTKNLEPRWGPYTVLLTPTALKVDGIAAWIHAAHVKAADPGGG
 PSSRLPCSRMLLLVTSLLLCELPHPAFLIPATPAPKSCPERHYWAQKLCQMCPEG
 TFLVKDCDQHRKAAQCDPCIPGVSFSPDHTRPHCESCRHCNSGLLVNRNCTITANAE
 CACRNGWQCRDKECTECDPLPNPSLTARSSQALSHPQPTHLPYVSEMLEARAGH
 MQTLADFRQLPARTLSTHWPPQRSLSDFIRILVIFSGMFLVFTLAGALFLHKRVKFS
 RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYN
 ELQKDKMAEAYSEIGMKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPRGPQ
 CTNYALLKLAGDVESNPGPMRISKPHLRSISIQCYLCLLLNSHFLTEAGIHVFILGCFPS
 AGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVAMKCFLELQVI
 SLESGDASIHDTVENLILANNSLSNGNVTESGCKECELEEKNIKEFLQSFVHIVQM
 FINTS

CD27TrCD28icd3z15:

(SEQ ID NO: 28)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTACCAAGA
ACAACTGGACCGACCGGTGGTACCTCACCCCTACCGAGTCGGCGACACAGTGTG
GGTCCGCCGACACCAGACTAAGAACC TAGAACCTCGCTGGAAGGACCTTACAC
AGTCCTGTGACCACCCCCACGCCCTCAAAGTAGACGCGCATCGCAGCTTGGATA
CACGCCGCCACGTGAAGGCTGCCGACCCCGGGGTGGACCATCCTCTAGACTG
CCATGCTCGAGGATGGCACGGCCACATCCCTGGTGGCTGTGCGTTCTGGGGACCC
TGGTGGGGCTCTCAGCTACTCCAGCCCCAAGAGCTGCCAGAGAGGCACTACT
GGGCTCAGGGAAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCTCTGTGA
AGGACTGTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTTGCATACCGGGGG
TCTCCTTCTCTCTGACCACCACACCGGCCCACTGTGAGAGCTGTGCGCACTG
TAACTCTGGTCTTCTCGTTTCGCAACTGCACCATCACTGCCAATGCTGAGTGTGCCT
GTCGCAATGGCTGGCAGTGCAGGGACAAGGAGTGACCGAGTGTGATCCTCTTC
CAAACCCCTCGCTGACCGCTCGGTCTGCTCAGGCCCTGAGCCACACCCCTCAGCC
CACCCACTTACCTTATGTCTAGTGAGATGCTGGAGGCCAGGACAGCTGGGCACAT
GCAGACTCTGGCTGACTTCAGGCAGCTGCCTGCCCGGACTCTCTTACCCACTGG
CCACCCCAAAGATCCCTGTGCAGCTCCGATTTTATTTCGCATCCTTGTGATCTTCTC
TGGAATGTTCTTGTTTTACCCCTGGCCGGGGCCCTGTTCTCCATAGGAGTAAG
AGGAGCAGGCTCCTGCACAGTGACTACATGAACATGACTCCCCGCCGCCCGGG
CCCCACCGCAAGCATTACCAGCCCTATGCCCCACCACGCGACTTCGCAGCCTATC
GCTCACGCGTGAAGTTCAGCAGGAGCGCAGACGCCCCGCGTACCAGCAGGGCC
AGAACCAGCTCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACGATGTTT
TGGACAAAAGACGTGGCCGGGACCCTGAGATGGGGGAAAGCCGAGAAGGAAG
AACCCTCAGGAAGGCCTGTACAATGAACTGCAGAAAGATAAGATGGCGGAGGCC
TACAGTGAGATTGGGATGAAAGGCGAGCGCCGGAGGGGCAAGGGGCACGATGG
CCTTTACCAGGGTCTCAGTACAGCCACCAAGGACACCTACGACGCCCTTCACATG
CAGGCCCTGCCCCCTCGCGGACCGCAGTGTAATAATTATGCTCTCTTGAAATTGG
CTGGAGATGTTGAGAGCAATCCCGGGCCCATGCGCATTAGCAAGCCCCACCTGC
GGAGCATCAGCATCCAGTGCTACCTGTGCCTGCTGCTGAACAGCCACTTCCTGAC
CGAGGCCGGCATCCACGTGTTTCATCCTGGGCTGCTTACGCGCCGACTGCCCAAG
ACCGAGGCCAACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGGACCTG
ATCCAGAGCATGCACATCGACGCCACCCTGTACACCGAGAGCGACGTGCACCCC
AGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGCTGGAACCTGAGGTGATCAGCC
TGGAAGCGGCGACGCCAGCATCCACGACACCGTGGAGAACCTGATCATCCTGG
CCAACAACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTGCAAAGAG
TGCGAGGAAC TGAAGAGAAGAACATCAAAGAGTTTCTGCAGAGCTTCGTGCAC
ATCGTGCAGATGTTTCATCAACACCAGCTGA

[0151] A corresponding amino acid sequence for CD27TrCD28icd3z15 is as follows:

(SEQ ID NO: 51)
 MTRVTNSPSLQAHLQALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
 VGDTVVWRRHQTKNLEPRWKGYPYTVLLTPTALKVDGIAAWIHAAHVKA
 ADPGGGPSSRLPCSRMARPHPWWLCVLGTLVGLSATPAPKSCPERHYWA
 QGKLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPCHES
 CRHCNSGLLVNRCTITANAECACRNGWQCRDKECTECDPLNPSTLARS
 SQALSPHPQPTHLPYVSEMLEAR TAGHMQTLADFRQLPARTLSTHWPPQ

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RSLCSSDFIRILVIFSGMFLVFTLAGALFLHRSKRSLHSDYMMTPR
 RGPTRKHYQPYAPPRDFAAYRSRVKFSRSADAPAYQQGQNQLYNELNL
 GRREEYDVLDRKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIG
 MKGERRRGKGGHDGLYQGLSTATKDTYDALHMQALPPRGPQCTNYALLKL
 AGDVESNPGPMRISKPHLRISISIQCYLCLLNSHFLTEAGIHVFILGCF
 SAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAM
 KCFLELQVISLESGDASIHDVTENLIILANNSLSSNGNVTESGCKECE
 ELEEKNIKEFLQSFVHIVQMFINTS

GSPco27Tr28CD3z15:

(SEQ ID NO: 29)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGCTTGGAGACCTCTGGCGGCAGCCTACCAAGA
 ACAACTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGCGACACAGTGTG
 GGTCCGCCGACACAGACTAAGAACCTAGAACCTCGCTGAAAGGACCTTACAC
 AGTCTGTGTGACCACCCCCACCGCCCTCAAAGTAGACGGCATCGCAGCTTGGATA
 CACGCCGCCACGTGAAGGCTGCCGACCCCGGGGTGGACCATCTCTAGACTG
 CCATGCTCGAGGATGCTTCTCTGGTGACAAGCTTCTGCTCTGTGAGTTACCAC
 ACCCAGCATCTCTCTGATCCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCG
 GCATTATTGGGCACAGGGCAAGTTGTGTGTTCAAATGTGTGAGCCGGGAACCTTT
 CTCGTGAAGGATTGCGATCAGCATCGGAAGCCGCGCAGTGCAGCCCATGTATA
 CCAGGGGTCTCATTTTCCCAGATCACCATACGAGGCCGCACTGTGAGTCTTGCA
 GGCAATTGTAATTCCGGCTTGTGGTCCGCACTGTACTATTACTGCGAATGCGAGA
 GTGTGCTTGTAGAAACGGATGGCAGTGCAGGGACAAAGAATGTACGGAGTGTGA
 TCCACTGCCTAACCCAGTCTTACAGCAAGATCTTACAGGCCCTCAGCCCGCAT
 CCTCAACCAACACATCTTCCTTACGTGTGAGAAATGTTGGAGGCGCAACCGCAG
 GCCATATGCAGACCTGGCGGACTTTCGGCAGCTGCCAGCACGCACCTTAGTAC
 ACACTGGCCACCACAACGCAGCTTGTGCTCTTCGATTTATCCGCATACTGGTC
 ATCTTTTCTGGAATGTTCTTGTGTTACCCCTGGCAGGAGCCCTGTTCTTACAG
 GAGTAAGAGGAGCAGGCTCCTGCACAGTGACTACATGAACATGACTCCCCGCCG
 CCCCCGGCCCCACCGCAAGCATTACCAGCCCTATGCCCCACACGCGACTTCGCA
 GCCTATCGCTCACGCGTGAAGTTCAGCAGGAGCGCAGACGCCCCCGGTACCAG
 CAGGGCCAGAACCCAGCTCTATAACGAGCTCAATCTAGGACGAAGAGAGAGTAC
 GATGTTTTGGACAAAAGACGTGGCCGGGACCCTGAGATGGGGGGAAGCCGAGA
 AGGAAGAACCCTCAGGAAGGCCTGTACAATGAACTGCAGAAAGATAAGATGGC
 GGAGGCCCTACAGTGAGATTGGGATGAAAGGCGAGCGCCGAGGGGCAAGGGGC
 ACGATGGCCTTTACCAGGGTCTCAGTACAGCCACCAAGGACACCTACGACGCCCT
 TCACATGCAGGCCCTGCCCTCGCGGACCGCAGTGACTAATTATGCTCTCTTG
 AAATTGGCTGGAGATGTTGAGAGCAATCCCGGGCCCATGCGCATTAGCAAGCCC

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CACCTGCGGAGCATCAGCATCCAGTGCTACCTGTGCTGCTGCTGAACAGCCACT
 TCCTGACCGAGCCGGCATCCACGTGTTTCATCCTGGGCTGCTTCAGCGCCGACT
 GCCCAAGACCGAGGCCAACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCGA
 GGACCTGATCCAGAGCATGCACATCGACGCCACCCTGTACACCGAGAGCGACGT
 GCACCCAGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGCTGGAACGACAGGTG
 ATCAGCCTGGAAGCGGCGACGCCAGCATCCAGACACCGTGAGAACCTGATC
 ATCCTGGCCAAACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTGC
 AAAGAGTGCGAGGAACGGAAGAGAAGAATCAAGAGTTTCTGCAGAGCTTC
 GTGCACATCGTGAGATGTTTCATCAACACCGAGCTGA

[0152] A corresponding amino acid sequence for
 GSPco27Tr28CD3z15 is as follows:

(SEQ ID NO: 52)
 MTRVTNSPSLQAHLQALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
 VGDTVVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKA
 ADPGGGPSSRLPCSRMLLLVTSLLLCELPHPAFLLIPATPAPKSCPERH
 YWAQGLKCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRP
 CESCRHCNSGLLVRNCTITANAECACRNGWQCRDKECTECPDLPNPSLT
 ARSSQALSPPHQPHTLPYVSEMLEARTAGHMQLADFRQLPARTLSTHW

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PPQRSLSDFIRILVIFSGMFLVFTLAGALFLHRSKRSRLHSDYMM
 TPRRPGPTRKHYPYAPPRDFAAYRSRVKFSRSADAPAYQQGQNQLYNE
 LNLGRREEYDVLDKRRGRDPEMGGKPRKPNQEGLYNELQDKMAEAYS
 EIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPRGPQCTNYAL
 LKLAGDVESNPGPMRISKPHLRSISIQCYLCLLLNSHFLTEAGIHVFIL
 GCFSAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDVHPCKV
 TAMKCFLELQVISLESQDASIHDTVENLIILANNSLSSNGNVTESGCK
 ECEELEEKNIKEFLQSFVHIVQMFINTS

CD27TrCD28tmdied3z15:

(SEQ ID NO: 30)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTACCAAGA
 ACAACTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGCGACACAGTGTG
 GGTCCGCCGACACAGACTAAGAACCTAGAACCTCGCTGGAAGGACCTTACAC
 AGTCCTGCTGACCACCCCCACCGCCCTCAAAGTAGACGGCATCGCAGCTTGGATA
 CACGCGCCACGTGAAGGCTGCCGACCCCGGGGTGGACCATCTCTAGACTG
 CCATGCTCGAGGATGGCACGGCCACATCCCTGGTGGCTGTGCGTTCTGGGGACCC
 TGGTGGGGCTCTCAGCTACTCCAGCCCCAAGAGCTGCCAGAGAGGCACCTACT
 GGGCTCAGGGAAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCTCTCGTGA
 AGGACTGTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTTGACATACCGGGG
 TCTCCTTCTCTCTGACCACCACACCGGCCCACTGTGAGAGCTGTGCGCACTG
 TAACTCTGGTCTTCTCGTTGCAACTGCACCATCACTGCCAATGTGAGTGTGCTC
 GTCGCAATGGCTGGCAGTGCAGGACAAGGAGTGCACCGAGTGTGATCCTCTTC
 CAAACCCCTCGCTGACCGCTCGGTGCTCTCAGGCCCTGAGCCACACCCCTCAGCC
 CACCCACTTACCTTATGTCAGTGAGATGCTGGAGGCCAGGACAGCTGGGCACAT
 GCAGACTCTGGCTGACTTCAGGCAGCTGCCTGCCCGACTCTCTTACCCACTGG
 CCACCCCAAAGATCCCTGTGCAGCTCCGATTTTATTCGCTTTTGGGTGCTGGTGGT
 GGTGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT

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TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTACATGAACATGA
 CTCCCCCGCGCCCCGGGCCACCCGCAAGCATTACCAGCCCTATGCCCCACCACG
 CGACTTCGCAGCCTATCGCTCACGCGTGAAGTTCAGCAGGAGCGCAGACGCCCC
 CGCGTACCAGCAGGGCCAGAACCAGCTCTATAACGAGCTCAATCTAGGACGAAG
 AGAGGAGTACGATGTTTTGGACAAAAGACGTGGCCGGACCCCTGAGATGGGGGG
 AAAGCCGAGAAGGAAGAACCCCTCAGGAAGGCCTGTACAATGAACCTGCAGAAAG
 ATAAGATGGCGGAGGCCTACAGTGAGATTGGGATGAAAGGCGAGCGCCGGAGG
 GGCAAGGGGCACGATGGCCTTTACCAGGGTCTCAGTACAGCCACCAAGGACACC
 TACGACGCCCTTCACATGCAGGCCCTGCCCCCTCGCGGACCGCAGTGTACTAATT
 ATGCTCTCTTGAAATTGGCTGGAGATGTTGAGAGCAATCCCGGGCCCATGCGCAT
 TAGCAAGCCCCACCTGCGGAGCATCAGCATCCAGTGCTACCTGTGCTGCTGCTG
 AACAGCCACTTCTTGACCGAGGCCGGCATCCACGTGTTTATCCTGGGCTGCTTCA
 GCGCCGACTGCCCCAAGACCGAGGCCAACTGGGTGAACGTGATCAGCGACCTGA
 AGAAGATCGAGGACCTGATCCAGAGCATGCACATCGACGCCACCCCTGTACACCG
 AGAGCGACGTGCACCCAGCTGCAAGGTGACCGCATGAAGTGCTTTCTGCTGG
 AACTGCAGGTGATCAGCCTGGAAAGCGGCGACGCCAGCATCCACGACACCGTGG
 AGAACCTGATCATCTTGCCAAACAACAGCCTGAGCAGCAACGGCAACGTGACCG
 AGAGCGGTGCAAAGAGTGCAGGAACCTGGAAGAGAAGAACATCAAAGAGTTT
 CTGCAGAGCTTCGTGCACATCGTGCAGATGTTTATCAACACCAGCTGA

[0153] A corresponding amino acid sequence for
 CD27TrCD28tmdicd3z15 is as follows:

GSPco27Tr28tmdicCD3z15:

	(SEQ ID NO: 53)	(SEQ ID NO: 31)
MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR	ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG	
VGDTVWVRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKA	CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA	
ADPGGGPSRLPCSRMARPHPWLCVLGTLVGLSATPAPKSCPERHYWA	CCAAGAACAACCTGGACCGACCGGTGGTACCTCACCCCTACCAGATCGGC	
QGLKCCQMCEPGTFLVKDCQHRKAAQCDPCIPGVSFSPDHHTRPHCES	GACACAGTGTGGGTCCGCGGACACCAGACTAAGAACCTAGAACCTCGCT	
CRHCNSGLLVNRCTITANAECACRNGWQCRDKECTECDPLPNPSLTARS	GGAAAGGACCTTACACAGTCTGTGTGACCACCCCCACCGCCCTCAAAGT	
SQALSPHPQPTHLPYVSEMLEARHTAGHMQLADFRQLPARTLSTHWPPQ	AGACGGCATCGCAGCTTGGATACACGCGCCACGTGAAGGCTGCCGAC	
RLCSDSDFIRFWVLVVVGVLACYLLVTVAFIIFWVRSKRSLLLHSDY	CCCGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGCTTCTCC	
MNMTPRRPGPTRKHYPYAPPRDFAAYRSRVKFSRSADAPAYQQQNQL	TGGTGACAAGCCTTCTGCTCTGTGAGTTACCACACCCAGCATTCTCTCT	
YNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAE	GATCCCAGCTACACCGGCTCCGAAGTCTTGCCGGAGCGGCATTATTGG	
AYSEIGMKGERRRGKHGDLGYGLSTATKDTYDALHMQALPRGPQCTN	GCACAGGGCAAGTTGTGTGTCAAATGTGTGAGCCGGGAACCTTTCTCG	
YALLKLADVESNPGPMRISKPHLRSISIQCYLCLLNSHFLTEAGIHV	TGAAGGATTGCGATCAGCATCGGAAGCGCGCAGTGCGACCCATGTAT	
FILGCFsAGLPKTEANVWNVISDLKKIEDLIQSMHIDATLYTESDVHPS	ACCAGGGGTCTCATTTTCCCAGATCACCATACGAGGCGGCACTGTGAG	
CKVTAMKCFLELQVISLESGLDASIHDTVENLIILANNSLSNNGVTES	TCTTGACAGGCATTGTAATCCGGCTTGTGGTCCGCAACTGTACTATTA	
GCKECEEELEEKNIKEFLQSFVHIVQMFINTS	CTGCGAATGCAGAGTGTGCTTGTAGAAACGGATGGCAGTGACGGACAA	
	AGAATGTACGGAGTGTGATCCACTGCCTAACCCAGTCTTACAGCAAGA	
	TCTTCACAGGCCCTCAGCCCGCATCTCAACCAACACATCTTCTTACG	
	TGTCAGAAATGTTGGAGGCGGAACCGCAGGCCATATGCAGACCCCTGGC	

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GGACTTTCGGCAGCTGCCAGCAGCACACTTAGTACACACTGGCCACCA
 CAACGCAGCTTGTGCTCTTCCGATTTCATCCGCTTTTGGGTGCTGGTGG
 TGGTTGGTGGAGTCTGGCTTGTCTAGCTTGTAGTAACAGTGGCCTT
 TATTATTTTCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCACAGTGAC
 TACATGAACATGACTCCCCCGCCCGGGGCCACCCGCAAGCATTACC
 AGCCCTATGCCCCACCACGCGACTTCGAGCCTATCGCTCACGCGTGAA
 GTTCAGCAGGAGCGCAGACGCCCGCGTACCAGCAGGGCCAGAACCAG
 CTCTATAACGAGCTCAATCTAGGACGAAGAGGAGTACGATGTTTTGG
 ACAAAGAGCTGGCCGGGACCTGAGATGGGGGAAAGCCGAGAAGGAA
 GAACCCCTCAGGAAGCCTGTACAATGAAGTGCAGAAAGATAAGATGGCG
 GAGGCCTACAGTGAGATTGGGTGAAGGCGAGCGCCGAGGGGCAAGG
 GGCACGATGGCCTTTACCAGGCTCTCAGTACAGCCACCAAGGACACCTA
 CGACGCCCTTACATGCAGGCCCTGCCCTCGCGGACCGCAGTGTACT
 AATTATGCTCTCTTGAATTTGGTGTGAGATGTTGAGAGCAATCCCGGC
 CCATGCGCATTAGCAAGCCCCACCTGCGGAGCATCAGCATCCAGTGCTA
 CCTGTGCCTGTGCTGAACAGCCACTTCTGACCGAGGCCGGCATCCAC
 GTGTTATCCTGGGCTGCTTCAGCGCCGACTGCCAAGACCGAGGCCA
 ACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGGACCTGATCCA
 GAGCATGCACATCGACGCCACCTGTACACCGAGAGCGACGTGCACCCC
 AGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGCTGGAAGTGCAGGTGA
 TCAGCCTGGAAAGCGGCGACGCCAGCATCCACGACACCGTGAGAACCT
 GATCATCTGGCCAACAACAGCCTGAGCAGCAACGGCAACGTGACCGAG
 AGCGGCTGCAAGAGTGCAGGAACTGGAAGAGAAGAATCAAGAGT
 TTCTGCAGAGCTTCGTGCACATCGTGCAGATGTTATCAACACCAGCTG
 A

[0154] A corresponding amino acid sequence for GSPco27Tr28umdcd3z15 is as follows:

(SEQ ID NO: 54)

MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
 VGDTVWVRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAHVKA
 ADPGGPPSRLPCSRMLLLVTSLLLCELPHPAFLLI PATPAPKSCPERH
 YWAQGLKLCQMCEPFTFLVKDQHRKAAQCDPCIPGVSFSDPHHTRPH
 CESCRHCNSGLLRNCTITANAECACRNGWQCRDKECTEDPLNPNSLT
 ARSSQALSHPHPQTHLPYVSEMLEARAGHMQLADFRQLPARTLSTHW
 PPQRLSCSSDFIRFWVLVVGGVLACYSLLVTVAFIIFVRSKRSLRH
 SDYMNMTPRRPGPTRKHYPYAPPRDFAAYRSRVKFSRSADAPAYQQGQ
 NQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDK
 MAEAYSEIGMKGERRRGKHDGLYQGLSTATKDTYDALHMQALPPRGPQ
 CTNYALLKLAGDVESNPGPMRISKPHLRSISIQCYLCLLLNSHFLTEAG

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IHVFI LGCF SAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDV
 HPSCKVTAMKCFLELQVISLES GDASIHDTVENLIILANNSLSSNGNV
 TESGCKECEBELEEKNIKEFLQSFVHI VQMFINTS

CD27TrDAP10icd3z15:

(SEQ ID NO: 32)

ATGACAAGAGTTACTAACAGCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACCTGGACCGACCGGTGTACCTCACCTTACCAGTTCGGC
 GACACAGTGTGGTCCGCGCACCCAGACTAAGAACC TAGAACCTCGCT
 GGAAAGGACCTTACACAGTCTGTGACCAACCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGATACACGCCGCCACGTGAAGGCTGCCGAC
 CCGGGGGTGGACCATCCTCTAGACTGCCATGTCTGAGGATGGCACGGC
 CACATCCCTGGTGGCTGTGCGTTCTGGGGACCTGGTGGGCTCTCAGC
 TACTCCAGCCCCAAGAGCTGCCAGAGAGCACTACTGGGCTCAGGGA
 AAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATTCCTCGTGAAGGACT
 GTGACCAGCATAGAAAGGCTGCTCAGTGTGATCCTTGATACCGGGGT
 CTCCTTCTCTCTGACCACACACCCGCCCACTGTGAGAGCTGTGCGG
 CACTGTAACTCTGGTCTTCTCGTTCGCAACTGCACCATCACTGCCAATG
 CTGAGTGTGCTGTGCAATGGCTGGCAGTGCAGGACAAAGAGTGCAC
 CGAGTGTGATCCTCTTCAAACCTTCGCTGACCGCTCGGTCTGTCTCAG
 GCCCTGAGCCCCACCCCTCAGCCCACTTACCTTATGTGAGTGA
 TGCTGGAGGCCAGGACAGCTGGGCACATGCAGACTCTGGCTGACTTCAG
 GCAGCTGCCTGCCCGGACTCTCTTACCCTGAGCCCAAGATCC
 CTGTGACGCTCCGATTTTATTCGCATACTGGTCATCTTTCTGGAATGT
 TCCTTGTGTTACCCCTGGCAGGAGCCCTGTTCTTACCTTTGCGCACG
 CCCACGCGCAGCCCCGCCAAGAAGATGGCAAAGTCTACATCAACATG
 CCAGGCAGGGGCAACGCGTGAAGTTCAGCAGGAGCGCAGACGCCCCCG
 CGTACCAGCAGGGCCAGAACCAGCTCTATAACGAGCTCAATCTAGGACG
 AAGAGAGGAGTACGATGTTTGGACAAAAGACGTGGCCGGGACCTGAG
 ATGGGGGAAAGCCGAGAAGGAAGAACCTCAGGAAGCCTGTACAATG
 AACTGCAGAAAGATAAGATGGCGGAGGCTACAGTGAGATTGGGATGAA
 AGGCAGCGCGGAGGGGCAAGGGGCACGATGGCCTTTACCAGGCTCTC
 AGTACAGCCACCAAGGACACCTACGACGCCCTTACATGCAGGCCCTGC
 CCCCTCGCGGACCGCAGTGTACTAATTATGCTCTCTTGAATTGGCTGG
 AGATGTTGAGAGCAATCCCGGCCCATGCGCATTAGCAAGCCCCACCTG
 CGGAGCATCAGCATCCAGTGTACCTGTGCTGTGCTGAACAGCCACT
 TCCTGACCGAGCGCGCATCCAGTGTTCATCCTGGGCTGCTTACGCGC
 CGGACTGCCCAAGACCGAGGCCAACTGGGTGAACGTGATCAGCGACCTG

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AAGAAGATCGAGGACCTGATCCAGAGCATGCACATCGACGCCACCCCTGT
 ACACCGAGAGCGACGTGCAACCCAGCTGCAAGGTGACCGCCATGAAGTG
 CTTTCTGCTGGAACATGCAGGTGATCAGCCTGGAAAGCGGCGACGCCAGC
 ATCCACGACACCGTGGAGAACCTGATCATCCTGGCCAACAACAGCCTGA
 GCAGCAACGGCAACGTGACCGAGAGCGGCTGCAAGAGTGCGAGGAACT
 GGAAGAGAAGAATCAAAAGAGTTCTGCAGAGCTTCGTGCACATCGTG
 CAGATGTTTCATCAACACCAGCTGA

[0155] A corresponding amino acid sequence for CD27TrDAP10icd3z15 is as follows:

(SEQ ID NO: 55)

MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
 VGDTVWVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAHVKA
 ADPGGPPSRLPCSRMARPHWVLVGLSATPAPKSCPERHYWA
 QGKLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPHCES
 CRHCNSGLLVNRCTITANAECACRNGWQCRDKECTECLPLNPSTARS
 SQALSPHPQPTHLPIVSEMLEARTAGHMQLADFRQLPARTLSTHWPPQ
 RSLCSSDFIRILVIFSGMFLVFTLAGALFLHLCARPRRSPAQEDGKVYI
 NMPGRGKRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRD
 PEMGGKPRRKNPQEGLYNELQDKMAEAYS EIGMKGERRRGKHGDGLYQ
 GLSTATKDTYDALHMQALPPRGQCTNYALLKLAGDVESNPGPMRISKP
 HLRISIQYCLLLNSHFLTEAGIHVFILGCF SAGLPKTEANWVNVIS
 DLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLLLEQVISLES GD
 ASIHDVTENLII LANNLSL SNGVNTESGCKECEELEEKNIKEFLQSFVH
 IVQMFINTS

GSPco27FLdap10icd3z15:

(SEQ ID NO: 33)

TGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCAGCAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACATGGACCGACCGGTGGTACCTCACCTTACCAGATCGGC
 GACACAGTGTGGGTCGCGGACACCGAGCTAAGAACCTAGAACCTCGCT
 GGAAAGGACCTTACACAGTCTGCTGACCAACCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACGCCGCCACGTGAAGGTGCGGAC
 CCCGGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGCTTCTCC
 TGGTGACAAGCCTTCTGCTCTGTGAGTTACCAACCCAGCATTCTCTCT
 GATCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCGGCATTATTGG
 GCACAGGGCAAGTTGTGTTGTCAAATGTGTGAGCCGGGAACCTTTCTCG
 TGAAGGATTGCGATCAGCATCGGAAGCGCGCAGTGCGACCCATGTAT
 ACCAGGGGTCTCATTTTCCCAGATACCATACGAGGCCGCACTGTGAG

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TCTTGCAAGCATTGTAATTCGGCTTGTGGTCCGCAACTGTACTATTA
 CTGCGAATGCAGAGTGTGCTTGTAGAAACGGATGGCAGTGCAGGGACAA
 AGAATGTACGGAGTGTGATCCACTGCCTAACCCAGTCTTACAGCAAGA
 TCTTCACAGGCCCTCAGCCGCATCCTCAACCAACACATCTTCCTTACG
 TGTGAGAAATGTTGGAGGCGCAACCGCAGGCCATATGCAGACCTTGGC
 GGACTTTCGGCAGCTGCCAGCACGCACACTTAGTACACACTGGCCACCA
 CAACGCAGCTTGTGCTCTTCCGATTTTCATCCGCATACTGGTCATCTTTT
 CTGGAATGTTCTTGTGTTCACTTGGCAGGAGCCCTGTTCTTTCACCT
 TTGGCGACGCCCCACGCCGAGCCCCGCCAAGAAGATGGCAAAGTCTAC
 ATCAACATGCCAGGCAGGGGCAACCGCGTGAAGTTCAGCAGGAGCGCAG
 ACGCCCCCGCGTACCAGCAGGGCCAGAACCAGCTCTATAACGAGCTCAA
 TCTAGGACGAAGAGAGGAGTACGATGTTTTGGACAAAAGACGTGGCCGG
 GACCTTGAGATGGGGGAAAGCCGAGAAGGAAGAACCCTCAGGAAGGCC
 TGTACAATGAACTGCAGAAAGATAAGATGGCGGAGGCCATCAGTGAGAT
 TGGGATGAAAGCGAGCGCGCGAGGGGCAAGGGGCACGATGGCCTTTAC
 CAGGGTCTCAGTACAGCCACCAAGGACACCTACGACGCCCTTCACATGC
 AGGCCCTGCCCTCGCGGACCGCAGTGTACTAATTATGCTCTCTTGAA
 ATTGCTGGAGATGTTGAGAGCAATCCCGGCCCATCGCATAGCAAG
 CCCACCTGCGGAGCATCAGCATCCAGTGCTACCTGTGCTGCTGCTGA
 ACAGCCACTTCTTGACCGAGCGCGCATCCAGTGTTCATCCTGGGCTG
 CTTACAGCGCGGACTGCCAAGACCGAGGCCAAGTGGTGAACGTGATC
 AGCGACCTGAAGAAGATCGAGGACCTGATCCAGAGCATGCACATCGAGC
 CCACCTGTACACCGAGAGCGACGTGCACCCAGCTGCAAGGTGACCGC
 CATGAAGTGCTTTCTGCTGGAACGTGAGGTGATCAGCTGGAAAGCGGC
 GACGCCAGCATCCACGACACCGTGGAGAACCTGATCATCCTGGCCAACA
 ACAGCCTGAGCAGCAACGGCAACGTGACCGAGAGCGGCTGCAAAGAGTG
 CGAGGAAGTGAAGAGAAGAATCAAAAGAGTTCTGCAGAGCTTCGTG
 CACATCGTGAGATGTTTCATCAACACCAGCTGA

[0156] A corresponding amino acid sequence for GSPco27FLdap10icd3z15 is as follows:

(SEQ ID NO: 56)

MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHYPYR
 VGDTVWVRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAHVKA
 ADPGGPPSRLPCSRMLLVTSLLLCELPHPAFLLIPATPAPKSCPERH
 YWAQGLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPH
 CESRHCNSGLLVNRCTITANAECACRNGWQCRDKECTECLPLNPSTLT
 MQTLADFRQLPARTLSTHWPPQRSLS SDFIRILVIFSGMFLVFTLAGA
 LARSSQALSPHPQPTHLPIVSEMLEARTAGHFLHLCARPRRSPAQEDGK
 VYINMPGRGKRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRR

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GRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHDG
 LYQGLSTATKDTYDALHMQALPPRGPOCTNYALLKLAGDVESNPGPMRI
 SKPHLRISISIQCYLCLLLNSHFLTEAGIHVFILGCFISAGLPKTEANWVN
 VISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLELQVISLE
 SGDASIHDTVENLIILANNSLSSNGNVTESGCKECELEEKNIKEFLQS
 FVHIVQMFINTS

CD27TrDAP12icd3z15:

(SEQ ID NO: 34)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACCTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGC
 GACACAGTGTGGGTCCGCCGACACCAGACTAAGAACCTAGAACCTCGCT
 GGAAAGGACCTTACACAGTCTGTGACACCCCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACGCCGCCACGTGAAGGCTGCCGAC
 CCCGGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGGCACGGC
 CACATCCCTGGTGGCTGTGCGTTCTGGGACCTCGTGGGGCTCTCAGC
 TACTCCAGCCCCAAGAGCTGCCCAGAGAGGCACTACTGGGCTCAGGGA
 AAGCTGTGTGCCAGATGTGTGAGCCAGGAACATCTCTCGTGAAGGACT
 GTGACCAGCATAGAAAGGCTGTCTAGTGTGATCTTGATACCGGGGT
 CTCTTCTCTCTGACCACCACACCGGCCCTGTGAGAGCTGTGCG
 CACTGTAACCTCTGGTCTTCTCGTTTCGCAACTGCACCATCACTGCCAATG
 CTGAGTGTGCTGTGCAATGGCTGGCAGTGCAGGACAAGGAGTGCAC
 CGAGTGTGATCTCTTCAAACCTTCGCTGACCGCTCGGTCTGTCTCAG
 GCCCTGAGCCACACCTCAGCCACCCACTTACCTTATGTCTAGTGAGA
 TGCTGGAGGCCAGGACAGCTGGGCACATGCAGACTCTGGCTGACTTCTCAG
 GCAGTGCCTGCCCGACTCTCTCTACCCACTGGCCACCCCAAGATCC
 CTGTGCAGCTCCGATTTTATTCGCATCCTTGTGATCTTCTCTGGAATGT
 TCCTTGTTTTTACCCTGGCCGGGGCCTGTTCTCCATTACTTCTTGGG
 CCGGCTGTTCCTCGGGGCGAGGGGCTGCGGAGGCAGCGACCCGGA
 CAGCGTATCACTGAGACCGAGTCGCCTTATCAGGAGCTCCAGGGTCAGA
 GGTCGGATGTCTACAGCGACCTCAACACACAGAGCCGTATTACAAAA
 ACGCGTGAAGTTGACGAGGAGCGCAGACGCCCCCGTACCAGCAGGGC
 CAGAACCAGCTCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACG
 ATGTTTTGGACAAAAGAGCTGGCCGGGACCTGAGTGGGGGAAAGCC
 GAGAAGGAAGAACCTCAGGAAGGCTGTACAATGAAGTGCAGAAAGAT
 AAGATGGCGGAGGCTACAGTGTGATTTGGGATGAAAGGCGAGCGCCGGA
 GGGGCAAGGGGCACGATGGCTTTTACCAGGGTCTCAGTACAGCCACCA
 GGACACCTACGACGCCCTTCACATGCAGGCCCTGCCCTCGCGGACCG

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CAGTGCTACTAATTATGCTCTCTTGAAATTGGCTGGAGATGTTGAGAGCA
 ATCCCGGGCCCATGCGCATTAGCAAGCCCCACCTGCGGAGCATCAGCAT
 CCAGTGCTACCTGTGCCTGCTGCTGAACAGCCACTTCTGACCGAGGCC
 GGCATCCACGTGTTTCATCCTGGGCTGCTTACGCGCCGACTGCCCAAGA
 CCGAGGCCAACTGGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGGA
 CCTGATCCAGAGCATGCACATCGACGCCACCTGTACACCGAGAGCGAC
 GTGCACCCAGCTGCAAGGTGACCGCCATGAAGTGTCTTCTGCTGGAAC
 TGCAGGTGATCAGCTTGAAAGCGGCGACGCCAGCATCCACGACACCGT
 GGAGAACCTGATCATCTTGGCCAACAACAGCCTGAGCAGCAACGGCAAC
 GTGACCGAGAGCGGCTGCAAAGAGTGCAGGAACTGGAAGAGAAGAACA
 TCAAAGAGTTTCTGCAGAGCTTCGTGCACATCGTGCAGATGTTTCATCAA
 CACCAGCTGA

[0157] A corresponding amino acid sequence for CD27TrDAP12icd3z15 is as follows:

(SEQ ID NO: 57)

MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
 VGDTVWVRRHQTKNLEPRWKGPYTVLLTTPALKVDGIAAWIHAAHVKA
 ADPGGGPSRLPCSRMARPHPWLCVLGTLVGLSATPAPKSCPERHYWA
 QGKLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRPHCES
 CRHCNSGLLVNRCTITANAECACRNGWQCRDKECTEDPLPNPSLTARS
 SQALSPHPQPTHLPYVSEMLEARTAGHMOTLADFRQLPARTLSTHWPPQ
 RSLCSSDFIRILVIFSGMFLVFTLAGALFLHYFLGRLVPRGRGAAEAAT
 RKQRITETESPYQELQGQRSDVYSDLNTQRPPYKRVKFSRSADAPAYQ
 QGQNQLYNELNLGRREEYDVLDRKRRDPPEMGGKPRRKNPQEGLYNELQ
 KDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
 GPQCTNYALLKLAGDVESNPGPMRISKPHLRISISIQCYLCLLLNSHFLT
 EAGIHVFILGCFISAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTE
 SDVHPSCKVTAMKCFLELQVISLESAGDASIHDTVENLIILANNSLSSN
 GNVTESGCKECELEEKNIKEFLQSFVHIVQMFINTS

GSPco27FLdap12icd3z15:

(SEQ ID NO: 35)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACCTGGACCGACCGGTGGTACCTCACCTTACCGAGTCGGC
 GACACAGTGTGGGTCCGCCGACACCAGACTAAGAACCAGAACCTCGCT
 GGAAAGGACCTTACACAGTCTGTGACACCCCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACGCCGCCACGTGAAGGCTGCCGAC
 CCCGGGGGTGGACCATCTCTAGACTGCCATGCTCGAGGATGCTTCTCC

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TGGTGACAAGCCTTCTGCTCTGTGAGTTACACACCCAGCATTCTCTCT
 GATCCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCGGCATTATTGG
 GCACAGGGCAAGTTGTGTGTCAAATGTGTGAGCCGGAACTTTCTCG
 TGAAGGATTGCATCAGCATCGGAAGGCCGCGCAGTGCACCCATGTAT
 ACCAGGGGTCTCATTTTCCCCAGATCACCATACGAGGCCGCACTGTGAG
 TCTTGACAGCATTGTAATTCCGGCTTGTGTGTCGCACTGTACTATTA
 CTGCGAATGCAGAGTGTGCTTGTAGAAACGGATGCGAGTGCAGGGACAA
 AGAATGTACGGAGTGTGATCCACTGCCTAACCCAGTCTTACAGCAAGA
 TCTTCACAGGCCCTCAGCCCGCATCTCAACCAACACATCTTCTCTACG
 GTGCAGAAATGTTGGAGGCGCAACCGCAGGCCATATGCAGACCTTGGC
 GGACTTTCGGCAGCTGCCAGCAGCGACACTTAGTACACACTGGCCACCA
 CAACGCAGCTTGTGCTCTTCCGATTTCATCCGCATACTGGTCATCTTTT
 CTGGAATGTTCTTGTGTTACCCCTGGCAGGAGCCCTGTTCTTCACTA
 CTTCTTGGCCCGGCTGGTCCCTCGGGGCGAGGGGCTGCGGAGGCAGCG
 ACCCGGAAACAGCGTATCACTGAGACCGAGTGCCTTATCAGGAGCTCC
 AGGGTCAGAGGTCGGATGTCTACAGCGACCTCAACACACAGAGGCCGTA
 TTACAAAAAACCGTGAAGTTTACGAGGAGCGCAGACGCCCCCGGTAC
 CAGCAGGGCCAGAACCAGCTCTATAACGAGCTCAATCTAGGACGAAGAG
 AGGAGTACGATGTTTTGGACAAAAGACGTGGCCGGGACCCCTGAGATGGG
 GGGAAAGCCGAGAAGGAAGAACCTCAGGAAGGCCTGTACAATGAAGT
 CAGAAAGATAAGATGGCGAGGCCCTACAGTGAAGTTGGGATGAAAGGCG
 AGCGCCGGAGGGGCAAGGGGCACGATGGCCTTACCAGGGTCTCAGTAC
 AGCCACCAAGGACACCTACGACGCCCTTACATGCAGGCCCTGCCCCCT
 CGCGGACCGCAGTGTACTAATTATGCTCTCTTGAAATGGGTGGAGATG
 TTGAGAGCAATCCCGGCCCATGCGCATTAGCAAGCCCCACCTGCGGAG
 CATCAGCATCCAGTGTCTACCTGTGCTGCTGCTGAACAGCCATTCTCTG
 ACCGAGGCCGGCATCCAGTGTTCATCTGGGCTGCTTCAGCGCCGGAC
 TGCCCAAGACCGAGGCCAAGTGGGTGAACGTGATCAGCGACCTGAAGAA
 GATCAGGAGACCTGATCCAGAGCATGCACATCGACGCCACCTGTACACC
 GAGAGCGACGTGACCCCCAGCTGCAAGGTGACCGCCATGAAGTGTCTTC
 TGCTGGAACTGCAGGTGATCAGCCTGGAAAGCGGCGACGCCAGCATCCA
 CGACACCGTGGAGAACCTGATCATCTGCGCAACACAGCCTGAGCAGC
 AACGGCAACGTGACCGAGAGCGGCTGCAAGAGTGCAGGAACTGGAAG
 AGAAGAACATCAAAGAGTTTCTGCAGAGCTTCTGTCACATCGTGCAGAT
 GTTCATCAACACCGAGTGA

[0158] A corresponding amino acid sequence for GSPco27FLdap12icd3z15 is as follows:

(SEQ ID NO: 58)

MTRVTNSPSLQAHLQALYLQHEVVRPLAAAYQEQLDRPVVPHPYR
 VGDTVWVRRHQTKNLEPRWGPYTVLLTTPALKVDGIAAWIHAAHVKA
 ADPGGGPSSRLPCSRMLLVTSLLLCELPHPAFLLIPATPAPKSCPERH
 YWAQKLCQMCPEPTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTRP
 CESCRHCNSGLLVNRNCTITANAECACRNGWCQRCDECTEDPLPNPSLT
 ARSSQALSPHPQPTHLPYVSEMLEARTAGHMOTLADFRQLPARTLSTHW
 PPQSLCSDFIRILVIFSGMFLVFTLAGALFLHYFLGRVLPGRGAEE
 AATRQIRITETESPYQELQGQRSDVYSDLNTQRPYKRVKFSRSADAP
 AYQQQONQLYNELNLGRREEYDVLDKRGRDPEMGGKPRRKNPQEGLYN
 ELQDKMAEAYSEIGMKGERRRGKHGDLGYQLSTATKTDYDALHMQAL
 PPRGPQCTNYALLKLAGDVESNPGPMRISKPHLRSISIQCYLCLLNSH
 FLTEAGIHVFILGCFSAGLPKTEANWVNVISDLKKIEDLIQSMHIDATL
 YTESDVHPSCKVTAMKCFLELQVISLESGLASIHDTVENLII LANNSL
 SSNGNVTESGCKECEEELEKNIKEFLQSFVHI VQMFINTS

CD27TrNKG2Dic3z15:

(SEQ ID NO: 36)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACTGGACCGACCGGTGGTACCTCACCCCTTACCGAGTCGGC
 GACACAGTGTGGGTCCGCCGACACAGACTAAGAACCTAGAACCTCGCT
 GGAAGGACCTTACACAGTCTGTGTGACCACCCCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGGATACACGCCGCCACGTGAAGGCTGCCGAC
 CCGGGGGTGGACCATCTCTAGACTGCCATGTCTGAGGATGGCAGGC
 CACATCCCTGGTGGCTGTGCGTTCTGGGGACCCCTGGTGGGCTCTCAGC
 TACTCCAGCCCCCAAGAGCTGCCAGAGGCACTACTGGGCTCAGGGA
 AAGCTGTGCTGCCAGATGTGTGAGCCAGGAACATCTCTCGTGAAGGACT
 GTGACCAGCATAGAAAGGCTGCTCAGTGTGATCTTGCATACCGGGGGT
 CTCCTTCTCTCTGACCACACACCGGCCCCACTGTGAGAGCTGTCTCG
 CACTGTAACTCTGGTCTTCTCGTTCGCAACTGCACCATCACTGCCAATG
 CTGAGTGTGCCTGTGCAATGGCTGGCAGTGCAGGGACAAGGAGTGCAC
 CGAGTGTGATCCTCTTCAAACCCTTCGCTGACCGCTCGGTCTGTCTCAG
 GCCCTGAGCCACACCCCTCAGCCACCCACTTACCTTATGTGAGTGA
 TGCTGGAGGCCAGGACAGCTGGGCACATGCAGACTCTGGTGACTTCAG
 GCAGCTGCCTGCCCGACTCTCTTACCCACTGGCCACCCCAAAGATCC
 CTGTGCAGCTCCGATTTTATTCGCATCCTTGTGATCTTCTCTGGAATGT
 TCCTTGTTTTACCCCTGGCCGGGGCCCTGTTCTCCATAGCGCGAACGA

- continued

ACGCTGCAAAAGCAAAGTGGTGCCTGCCGCCAGAAACAGTGGCGCACC
 AGCTTTGATAGCAAAAACCTGGATCTGAACTATAACCATTTTGAAAGCA
 TGGAAATGGAGCCATCGCAGCCGCCGCGCCGCATTTGGGGCATGAAACG
 CGTGAAGTTCAGCAGGAGCGCAGACGCCCCGCGTACCAGCAGGGCCAG
 AACCAGCTCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACGATG
 TTTTGGACAAAAGACGTGGCCGGGACCTGAGATGGGGGAAAGCCGAG
 AAGGAAGAACCCTCAGGAAGGCCTGTACAATGAACTGCAGAAAGATAAG
 ATGGCGGAGGCCACAGTGAATTTGGGATGAAAGCGAGCGCCGAGGG
 GCAAGGGGCACGATGGCCTTTACCAGGGTCTCAGTACAGCCACCAAGGA
 CACCTACGACGCCCTTCACATGCAGGCCCTGCCCCCTCGCGGACCGCAG
 TGTACTAATTATGCTCTCTTGAATTTGGCTGGAGATGTTGAGAGCAATC
 CCGGGCCCATGCGCATTAGCAAGCCCCACCTGCGGAGCATCAGCATCCA
 GTGCTACCTGTGCCTGCTGCTGAACAGCCACTTCCTGACCGAGGCCGCGC
 ATCCACGTGTTTACCTCGGGCTGCTTCAGCGCCGACTGCCCAAGACCG
 AGGCCAATGGGTGAACGTGATCAGCGACCTGAAGAAGATCGAGGACCT
 GATCCAGAGCATGCACATCGACGCCACCTGTACACCGAGAGCGACGTG
 CACCCAGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGCTGGAAGTGC
 AGGTGATCAGCCTGGAAAGCGCGCAGCGCAGCATCCACGACACCGTGGA
 GAACCTGATCATCCTGGCCAACAACAGCCTGAGCAGCAACGGCAACGTG
 ACCGAGAGCGGCTGCAAGAGTGCAGGGAAGTGAAGAGAAGAATCA
 AAGAGTTTCTGCAGAGCTTCGTGCACATCGTGCAGATGTTTATCAACAC
 CAGCTGA

[0159] A corresponding amino acid sequence for CD27TrNKG2Dic3z15 is as follows:

(SEQ ID NO: 59)

MTRVTNPSLQAHLQALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
 VGDVTVWRRHQTKNLEPRWKGPYTVLLTPTALKVDGIAAWIHAAHVKA
 ADPGGPPSRLPCSRMARPHPWWLCLVLTGLVLSATPAPKSCPERHYWA
 QGKLCCQMCEPGTFLVKDCDQHRKAAQCDPCIPGVSFSPDHHTPHCES
 CRHCNSGLLVNRNCTITANAECACRNGWQCRDKECTECDPLPNPSLTARS
 SQALSHPHPQTHLPVYSEMLEARHTAGHMQLADFRQLPARTLSHWPQ
 RSLCSDDFIRILVIFSGMFLVFTLAGALFLHSANERCKSKVVPQRQKQW
 RTSFDSKLLDLNHNHFPESMEWSHRSRGRRIWGMKRVKFSRSADAPAYQQ
 GQNQLYNELNLGRREEYDVLDRRGRDPEMGGKPRRKNPQEGLYNELQK
 DKMAEAYSEIGMKGERRRKGHDGLYQGLSTATKDTYDALHMQALPPRG
 PQCTNYALLKLADGVEDSNPGPMRISKPHLRSISIQCYLCLLLNSHFLTE
 AGIHVFILGCFAGLPKTEANWNVNISDLKKIEDLIQSMHIDATLYTES
 DVHPSCKVTAMKCFLELQVISLESGLDASIHDTVENLILANNLSLSSNG
 NVTESGCKECELEEKNKEFLQSPFHIVQMFINTS

GSPco27TrNKG2Dic3z15:

(SEQ ID NO: 37)

ATGACAAGAGTTACTAACAGCCCCCTCTCTCCAAGCTCACTTACAGG
 CTCTCTACTTAGTCCAGCACGAAGTCTGGAGACCTCTGGCGGCAGCCTA
 CCAAGAACAACCTGGACCGACCGGTGGTACCTCACCTTACCAGTCCGGC
 GACACAGTGTGGGTCCGCCGACACAGACTAAGAACCCTAGAACCTCGCT
 GGAAAGGACCTTACACAGTCTGTGTGACCACCCCCACCGCCCTCAAAGT
 AGACGGCATCGCAGCTTGATACACGCCGCCACGTGAAGGCTGCCGAC
 CCGGGGGTGGACCATCCTCTAGACTGCCATGCTCGAGGATGCTTCTCC
 TGGTGACAAGCCTTCTGTCTGTGAGTTACCACACCCAGCATTCCTCCT
 GATCCCAGCTACACCGGCTCCGAAGTCTGCCCGGAGCGGCATTATTGG
 GCACAGGGCAAGTTGTGTTGTCAAATGTGTGAGCCGGGAACCTTCTCG
 TGAAGGATTGCGATCAGCATCGGAAGGCCGCGCAGTGCAGCCCATGTAT
 ACCAGGGGTCTCATTTTCCCAGATCACCATACGAGGCCGCACTGTGAG
 TCTTGCAGGCATTGTAATTCGGCTTGTGGTCCGCAACTGTACTATTA
 CTGCGAATGCAGAGTGTGCTTGTAGAAACGGATGGCAGTGCAGGGACAA
 AGAATGTACGGAGTGTGATCCACTGCCTAACCCAGTCTTACAGCAAGA
 TCTTTCACAGGCCCTCAGCCCGCATCCTCAACCAACACATCTTCCTTACG
 TGTGAGAAATGTTGGAGCGCGAACCAGGCCATATGCAGACCTGGC
 GGACTTTCGGCAGCTGCCAGCACGCACACTTAGTACACACTGGCCACCA
 CAACGCAGCTTGTGCTCTTCCGATTTATCCGCATACTGGTCATCTTTT
 CTGGAATGTTCTTGTGTTTACCCTGGCAGGAGCCCTGTTCTCTACAG
 CGCGAACGAACGCTGCAAAAGCAAAGTGGTGCCGTGCCGCCAGAAACAG
 TGGCGCACACGCTTGTATAGCAAAAACCTGGATCTGAACTATAACCATT
 TTGAAAGCATGGAATGGAGCCATCGCAGCCGCCGCGCGCCGATTGGGG
 CATGAAACGCGTGAAGTTTACGAGGAGCGCAGACGCCCCCGGTACCAG
 CAGGGCCAGAACACGCTCTATAACGAGCTCAATCTAGGACGAAGAGAGG
 AGTACAGATGTTTGGACAAAAGACGTGGCCGGGACCTGAGATGGGGGG
 AAAGCCGAGAAGGAAGAACCTCAGGAAGGCCTGTACAATGAACTGCAG
 AAAGATAAGATGGCGGAGGCCCTACAGTGAATTTGGATGAAAGCGCAGC
 GCCGAGGGGCAAGGGGACGATGGCCTTTACCAGGGTCTCAGTACAGC
 CACCAAGGACACCTACGACGCCCTTACATGCAGGCCCTGCCCCCTCGC
 GGACCGCAGTGTACTAATTATGCTCTCTTGAAATTGGCTGGAGATGTTG
 AGAGCAATCCCGGGCCCATGCGCATTAGCAAGCCCCACCTGCGGAGCAT
 CAGCATCCAGTGTACCTGTGCCTGCTGCTGAACAGCCACTTCCTGACC
 GAGGCCGCGCATCCACGTGTTTATCTGGGTGCTTACGCGCCGACTGC
 CCAAGACCGAGGCCAAGTGGGTGAACGTGATCAGCGACCTGAAGAAGAT
 CGAGGACCTGATCCAGAGCATGCACATCGACGCCACCTGTACACCGAG
 AGCGACGTGCACCCAGCTGCAAGGTGACCGCCATGAAGTGCTTTCTGC

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TGGAAGTGCAGGTGATCAGCCTGGAAAGCGGCGACGCCAGCATCCACGA
CACCGTGGAGAACCTGATCATCTGGCCAAACAGCCTGAGCAGCAAC
GGCAACGTGACCGAGAGCGGCTGCAAAGAGTGCAGGAAGTGGAAAGAGA
AGAACATCAAAGAGTTTCTGCAGAGCTTCGTGCACATCGTGCAGATGTT
CATCAACACCAGCTGA

[0160] A corresponding amino acid sequence for GSPco27TrNKG2Dcd3z15 is as follows:

(SEQ ID NO: 60)
MTRVTNSPSLQAHQLALYLQHEVWRPLAAAYQEQLDRPVVPHPYR
VGDTVWVRHQTKNLEPRWKGYPYTVLLTPTALKVDGIAAWIHAHVKA
ADPGGGPSSRLPCSRMLLLVTSLLLCCLPHPAFLLIIPATPAPKSCPERH
YWAQGLKCCQMCPEGTFLVKDCDQHRKAAQCDPCIPGVSPDHHTRPH
CESCRHCNSGLLVNRCTITANAECACRNGWQCRDKECTECLPLNPSTLT
ARSSQALSPHPQPTHLPIVSEMLEARTAGHMQLADFRQLPARTLSTHW
PPQRLSCSSDFIRILVIFSGMFLVFTLAGALFLHSANERCKSKVVPQRQ
KQWRTSFDSKKLDLNNHFESMEWSHRSRGRGIWGMKRVKFSRSADAPA
YQQGQNQLYNELNLGRREEYDVLDRGRDPEMGGKPRRKNPQEGLYNE
LQKDKMAEAYSEIGMKGERRRKGHDGLYQGLSTATKDTYDALHMQLP
PRGPQCTNYALLKLKAGDVESNPGPMRISKPHLRSISIQCYLCLLLNSHF
LTEAGIHVFIILGCFSGAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLY
TESDVHPSCKVTAMKCFLELQVISLESGLDASIHDTVENLIILANNSLS
SNGNVTESGCKECEELEEKNIKEFLQSFVHVQMFINTS

[0161] The sequence of the open reading frame encoding the chimeric receptor can be obtained from a genomic DNA source, a cDNA source, or can be synthesized (e.g., via PCR), or combinations thereof. Depending upon the size of the genomic DNA and the number of introns, it may be desirable to use cDNA or a combination thereof as it is found that introns stabilize the mRNA. Also, it may be further advantageous to use endogenous or exogenous non-coding regions to stabilize the mRNA.

[0162] It is contemplated that the chimeric construct can be introduced into immune cells of any kind as naked DNA or in a suitable vector. Methods of stably transfecting cells by electroporation using naked DNA are known in the art. See, e.g., U.S. Pat. No. 6,410,319. Naked DNA generally refers to the DNA encoding a chimeric receptor contained in a plasmid expression vector in proper orientation for expression.

[0163] Alternatively, a viral vector (e.g., a retroviral vector, adenoviral vector, adeno-associated viral vector, or lentiviral vector) can be used to introduce the chimeric construct into immune cells. Suitable vectors for use in accordance with the method of the present disclosure are non-replicating in the immune cells. A large number of vectors are known that are based on viruses, where the copy number of the virus maintained in the cell is low enough to maintain the viability of the cell, such as, for example, vectors based on HIV, SV40, EBV, HSV, or BPV. Non-viral

vectors include plasmids, transposons, nanoparticles, liposome, lipids, metals, or a combination thereof.

II. Cytokines

[0164] In some embodiments, the cells expressing the anti-CD70 CAR are engineered to express one or more heterologous cytokines and/or are engineered to upregulate normal expression of one or more heterologous cytokines. The cells may or may not be transduced or transfected for one or more cytokines on the same vector as other genes.

[0165] One or more cytokines may be co-expressed from the vector as a separate polypeptide from the antigen receptor. Interleukin-15 (IL-15), for example, is tissue restricted and only under pathologic conditions is it observed at any level in the serum, or systemically. IL-15 possesses several attributes that are desirable for adoptive therapy. IL-15 is a homeostatic cytokine that induces development and cell proliferation of natural killer cells, promotes the eradication of established tumors via alleviating functional suppression of tumor-resident cells, and inhibits activation-induced cell death (AICD). In addition to IL-15, other cytokines are envisioned. These include, but are not limited to, cytokines, chemokines, and other molecules that contribute to the activation and proliferation of cells used for human application. NK cells expressing IL-15 are capable of continued supportive cytokine signaling, which is useful for their survival post-infusion.

[0166] In specific embodiments, the cells expresses one or more exogenously provided cytokines. As one example, the cytokine is IL-15, IL-12, IL-2, IL-18, IL-21, GM-CSF, or a combination thereof. As another example, the cytokine is IL-21. As another example, the cytokine is IL-12. The cytokine may be exogenously provided to the NK cells because it is expressed from an expression vector within the cell. In an alternative case, an endogenous cytokine in the cell is upregulated upon manipulation of regulation of expression of the endogenous cytokine, such as genetic recombination at the promoter site(s) of the cytokine. In cases wherein the cytokine is provided on an expression construct to the cell, the cytokine may be encoded from the same vector as a suicide gene and/or as the anti-CD70 CAR. In some embodiments, the present disclosure concerns co-utilization of the CAR with IL-15. In some embodiments, the present disclosure concerns co-utilization of the CAR with IL-21. In some embodiments, the present disclosure concerns co-utilization of the CAR with IL-12.

III. Suicide Genes

[0167] In particular embodiments, a suicide gene is utilized in conjunction with the anti-CD70 cell therapy to control its use and allow for termination of the cell therapy at a desired event and/or time. The suicide gene is employed in transduced cells for the purpose of eliciting death for the transduced cells when needed. The cells of the present disclosure that have been modified to harbor a vector encompassed by the disclosure may comprise one or more suicide genes. In some embodiments, the term "suicide gene" as used herein is defined as a gene which, upon administration of a prodrug or other agent, effects transition of a gene product to a compound which kills its host cell. In other embodiments, a suicide gene encodes a gene product that is, when desired, targeted by an agent (such as an antibody) that targets the suicide gene product.

[0168] In some cases, the cell therapy may be subject to utilization of one or more suicide genes of any kind when an individual receiving the cell therapy and/or having received the cell therapy shows one or more symptoms of one or more adverse events, such as cytokine release syndrome, neurotoxicity, anaphylaxis/allergy, and/or on-target/off tumor toxicities (as examples) or is considered at risk for having the one or more symptoms, including imminently. The use of the suicide gene may be part of a planned protocol for a therapy or may be used only upon a recognized need for its use. In some cases the cell therapy is terminated by use of agent(s) that targets the suicide gene or a gene product therefrom because the therapy is no longer required.

[0169] Utilization of the suicide gene may be instigated upon onset of at least one adverse event for the individual, and that adverse event may be recognized by any means, including upon routine monitoring that may or may not be continuous from the beginning of the cell therapy. The adverse event(s) may be detected upon examination and/or testing. In cases wherein the individual has cytokine release syndrome (which may also be referred to as cytokine storm), the individual may have elevated inflammatory cytokine(s) (merely as examples: interferon-gamma, granulocyte macrophage colony-stimulating factor, IL-10, IL-6 and TNF-alpha); fever; fatigue; hypotension; hypoxia, tachycardia; nausea; capillary leak; cardiac/renal/hepatic dysfunction; or a combination thereof, for example. In cases wherein the individual has neurotoxicity, the individual may have confusion, delirium, aplasia, and/or seizures. In some cases, the individual is tested for a marker associated with onset and/or severity of cytokine release syndrome, such as C-reactive protein, IL-6, TNF-alpha, and/or ferritin

[0170] Examples of suicide genes include engineered non-secretable (including membrane bound) tumor necrosis factor (TNF)-alpha mutant polypeptides (see PCT/US19/62009, which is incorporated by reference herein in its entirety), and they may be affected by delivery of an antibody that binds the TNF-alpha mutant. Examples of suicide gene/prodrug combinations that may be used are Herpes Simplex Virus-thymidine kinase (HSV-tk) and ganciclovir, acyclovir, or FIAU; oxidoreductase and cycloheximide; cytosine deaminase and 5-fluorocytosine; thymidine kinase thymidilate kinase (Tdk::Tmk) and AZT; and deoxycytidine kinase and cytosine arabinoside. The *E. coli* purine nucleoside phosphorylase, a so-called suicide gene that converts the prodrug 6-methylpurine deoxyriboside to toxic purine 6-methylpurine, may be utilized. Other suicide genes include CD20, CD52, inducible caspase 9, purine nucleoside phosphorylase (PNP), Cytochrome p450 enzymes (CYP), Carboxypeptidases (CP), Carboxylesterase (CE), Nitroreductase (NTR), Guanine Ribosyltransferase (XGRTP), Glycosidase enzymes, Methionine- α , γ -lyase (MET), and Thymidine phosphorylase (TP), as examples.

[0171] In particular embodiments, vectors that encode the CD70-targeting CAR, or any vector in a NK cell encompassed herein, include one or more suicide genes. The suicide gene may or may not be on the same vector as a CD70-targeting CAR. In cases wherein the suicide gene is present on the same vector as the CD70-targeting CAR, the suicide gene and the CAR may be separated by an IRES or 2A element, for example.

IV. Pharmaceutical Compositions

[0172] Pharmaceutical compositions of the present disclosure comprise an effective amount of cells expressing anti-CD70 CARs dissolved or dispersed in a pharmaceutically acceptable carrier. The phrases “pharmaceutical or pharmacologically acceptable” refers to molecular entities and compositions that do not produce an adverse, allergic or other untoward reaction when administered to an animal, such as, for example, a human, as appropriate. The preparation of a pharmaceutical composition that comprises cells expressing anti-CD70 CARs will be known to those of skill in the art in light of the present disclosure, as exemplified by Remington: The Science and Practice of Pharmacy, 21st Ed. Lippincott Williams and Wilkins, 2005, incorporated herein by reference. Moreover, for animal (e.g., human) administration, it will be understood that preparations should meet sterility, pyrogenicity, general safety and purity standards as required by FDA Office of Biological Standards.

[0173] As used herein, “pharmaceutically acceptable carrier” includes any and all solvents, dispersion media, coatings, surfactants, antioxidants, preservatives (e.g., antibacterial agents, antifungal agents), isotonic agents, absorption delaying agents, salts, preservatives, drugs, drug stabilizers, gels, binders, excipients, disintegration agents, lubricants, sweetening agents, flavoring agents, dyes, such like materials and combinations thereof, as would be known to one of ordinary skill in the art (see, for example, Remington’s Pharmaceutical Sciences, 18th Ed. Mack Printing Company, 1990, pp. 1289-1329, incorporated herein by reference). Except insofar as any conventional carrier is incompatible with the active ingredient, its use in the pharmaceutical compositions is contemplated.

[0174] The pharmaceutical compositions may comprise different types of carriers depending on whether it is to be administered in solid, liquid or aerosol form, and whether it need to be sterile for such routes of administration as injection. The presently disclosed compositions can be administered intravenously, intradermally, transdermally, intrathecally, intraarterially, intraperitoneally, intranasally, intravaginally, intrarectally, topically, intramuscularly, subcutaneously, mucosally, orally, topically, locally, inhalation (e.g., aerosol inhalation), injection, infusion, continuous infusion, localized perfusion bathing target cells directly, via a catheter, via a lavage, in cremes, in lipid compositions (e.g., liposomes), or by other method or any combination of the forgoing as would be known to one of ordinary skill in the art (see, for example, Remington’s Pharmaceutical Sciences, 18th Ed. Mack Printing Company, 1990, incorporated herein by reference).

[0175] The cells expressing anti-CD70 CARs may be formulated into a composition in a free base, neutral or salt form. Pharmaceutically acceptable salts, include the acid addition salts, e.g., those formed with the free amino groups of a proteinaceous composition, or which are formed with inorganic acids such as for example, hydrochloric or phosphoric acids, or such organic acids as acetic, oxalic, tartaric or mandelic acid. Salts formed with the free carboxyl groups can also be derived from inorganic bases such as for example, sodium, potassium, ammonium, calcium or ferric hydroxides; or such organic bases as isopropylamine, trimethylamine, histidine or procaine. Upon formulation, solutions will be administered in a manner compatible with the dosage formulation and in such amount as is therapeutically effective. The formulations are easily administered in a

variety of dosage forms such as formulated for parenteral administrations such as injectable solutions, or aerosols for delivery to the lungs, or formulated for alimentary administrations such as drug release capsules and the like.

[0176] Further in accordance with the present disclosure, the compositions of the present disclosure suitable for administration is provided in a pharmaceutically acceptable carrier with or without an inert diluent. The carrier should be assimilable and includes liquid, semi-solid, i.e., pastes, or solid carriers. Except insofar as any conventional media, agent, diluent or carrier is detrimental to the recipient or to the therapeutic effectiveness of a the composition contained therein, its use in administrable composition for use in practicing the methods of the present invention is appropriate. Examples of carriers or diluents include fats, oils, water, saline solutions, lipids, liposomes, resins, binders, fillers and the like, or combinations thereof. The composition may also comprise various antioxidants to retard oxidation of one or more component. Additionally, the prevention of the action of microorganisms can be brought about by preservatives such as various antibacterial and antifungal agents, including but not limited to parabens (e.g., methylparabens, propylparabens), chlorobutanol, phenol, sorbic acid, thimerosal or combinations thereof.

[0177] In accordance with the present disclosure, the composition is combined with the carrier in any convenient and practical manner, i.e., by solution, suspension, emulsification, admixture, encapsulation, absorption and the like. Such procedures are routine for those skilled in the art.

[0178] In a specific embodiment of the present disclosure, the composition is combined or mixed thoroughly with a semi-solid or solid carrier. The mixing can be carried out in any convenient manner such as grinding. Stabilizing agents can be also added in the mixing process in order to protect the composition from loss of therapeutic activity, i.e., denaturation in the stomach. Examples of stabilizers for use in an the composition include buffers, amino acids such as glycine and lysine, carbohydrates such as dextrose, mannose, galactose, fructose, lactose, sucrose, maltose, sorbitol, mannitol, etc.

[0179] In further embodiments, the present disclosure may concern the use of a pharmaceutical lipid vehicle compositions that include cells expressing anti-CD70 CARs and optionally an aqueous solvent. As used herein, the term "lipid" will be defined to include any of a broad range of substances that is characteristically insoluble in water and extractable with an organic solvent. This broad class of compounds are well known to those of skill in the art, and as the term "lipid" is used herein, it is not limited to any particular structure. Examples include compounds that contain long-chain aliphatic hydrocarbons and their derivatives. A lipid may be naturally occurring or synthetic (i.e., designed or produced by man). However, a lipid is usually a biological substance. Biological lipids are well known in the art, and include for example, neutral fats, phospholipids, phosphoglycerides, steroids, terpenes, lysolipids, glycosphingolipids, glycolipids, sulphatides, lipids with ether and ester-linked fatty acids and polymerizable lipids, and combinations thereof. Of course, compounds other than those specifically described herein that are understood by one of skill in the art as lipids are also encompassed by the compositions and methods of the present invention.

[0180] One of ordinary skill in the art would be familiar with the range of techniques that can be employed for

dispersing a composition in a lipid vehicle. For example, the cells expressing anti-CD70 CARs may be dispersed in a solution containing a lipid, dissolved with a lipid, emulsified with a lipid, mixed with a lipid, combined with a lipid, covalently bonded to a lipid, contained as a suspension in a lipid, contained or complexed with a micelle or liposome, or otherwise associated with a lipid or lipid structure by any means known to those of ordinary skill in the art. The dispersion may or may not result in the formation of liposomes.

[0181] The actual dosage amount of a composition of the present disclosure administered to an animal patient can be determined by physical and physiological factors such as body weight, severity of condition, the type of disease being treated, previous or concurrent therapeutic interventions, idiopathy of the patient and on the route of administration. Depending upon the dosage and the route of administration, the number of administrations of a preferred dosage and/or an effective amount may vary according to the response of the subject. The practitioner responsible for administration will, in any event, determine the concentration of active ingredient(s) in a composition and appropriate dose(s) for the individual subject.

[0182] In certain embodiments, pharmaceutical compositions may comprise, for example, at least about 0.1% of an active compound. In other embodiments, the an active compound may comprise between about 2% to about 75% of the weight of the unit, or between about 25% to about 60%, for example, and any range derivable therein. Naturally, the amount of active compound(s) in each therapeutically useful composition may be prepared in such a way that a suitable dosage will be obtained in any given unit dose of the compound. Factors such as solubility, bioavailability, biological half-life, route of administration, product shelf life, as well as other pharmacological considerations will be contemplated by one skilled in the art of preparing such pharmaceutical formulations, and as such, a variety of dosages and treatment regimens may be desirable.

[0183] In other non-limiting examples, a dose may also comprise from about 1 microgram/kg/body weight, about 5 microgram/kg/body weight, about 10 microgram/kg/body weight, about 50 microgram/kg/body weight, about 100 microgram/kg/body weight, about 200 microgram/kg/body weight, about 350 microgram/kg/body weight, about 500 microgram/kg/body weight, about 1 milligram/kg/body weight, about 5 milligram/kg/body weight, about 10 milligram/kg/body weight, about 50 milligram/kg/body weight, about 100 milligram/kg/body weight, about 200 milligram/kg/body weight, about 350 milligram/kg/body weight, about 500 milligram/kg/body weight, to about 1000 mg/kg/body weight or more per administration, and any range derivable therein. In non-limiting examples of a derivable range from the numbers listed herein, a range of about 5 mg/kg/body weight to about 100 mg/kg/body weight, about 5 microgram/kg/body weight to about 500 milligram/kg/body weight, etc., can be administered, based on the numbers described above.

A. Alimentary Compositions and Formulations

[0184] In particular embodiments of the present disclosure, the cells expressing anti-CD70 CARs are formulated to be administered via an alimentary route. Alimentary routes include all possible routes of administration in which the composition is in direct contact with the alimentary tract.

Specifically, the pharmaceutical compositions disclosed herein may be administered orally, buccally, rectally, or sublingually. As such, these compositions may be formulated with an inert diluent or with an assimilable edible carrier, or they may be enclosed in hard- or soft-shell gelatin capsule, or they may be compressed into tablets, or they may be incorporated directly with the food of the diet.

[0185] In certain embodiments, the active compounds may be incorporated with excipients and used in the form of ingestible tablets, buccal tables, troches, capsules, elixirs, suspensions, syrups, wafers, and the like (Mathiowitz et al., 1997; Hwang et al., 1998; U.S. Pat. Nos. 5,641,515; 5,580,579 and 5,792,451, each specifically incorporated herein by reference in its entirety). The tablets, troches, pills, capsules and the like may also contain the following: a binder, such as, for example, gum tragacanth, acacia, cornstarch, gelatin or combinations thereof; an excipient, such as, for example, dicalcium phosphate, mannitol, lactose, starch, magnesium stearate, sodium saccharine, cellulose, magnesium carbonate or combinations thereof; a disintegrating agent, such as, for example, corn starch, potato starch, alginic acid or combinations thereof; a lubricant, such as, for example, magnesium stearate; a sweetening agent, such as, for example, sucrose, lactose, saccharin or combinations thereof; a flavoring agent, such as, for example peppermint, oil of wintergreen, cherry flavoring, orange flavoring, etc. When the dosage unit form is a capsule, it may contain, in addition to materials of the above type, a liquid carrier. Various other materials may be present as coatings or to otherwise modify the physical form of the dosage unit. For instance, tablets, pills, or capsules may be coated with shellac, sugar, or both. When the dosage form is a capsule, it may contain, in addition to materials of the above type, carriers such as a liquid carrier. Gelatin capsules, tablets, or pills may be enterically coated. Enteric coatings prevent denaturation of the composition in the stomach or upper bowel where the pH is acidic. See, e.g., U.S. Pat. No. 5,629,001. Upon reaching the small intestines, the basic pH therein dissolves the coating and permits the composition to be released and absorbed by specialized cells, e.g., epithelial enterocytes and Peyer's patch M cells. A syrup of elixir may contain the active compound sucrose as a sweetening agent methyl and propylparabens as preservatives, a dye and flavoring, such as cherry or orange flavor. Of course, any material used in preparing any dosage unit form should be pharmaceutically pure and substantially non-toxic in the amounts employed. In addition, the active compounds may be incorporated into sustained-release preparation and formulations.

[0186] For oral administration the compositions of the present disclosure may alternatively be incorporated with one or more excipients in the form of a mouthwash, dentifrice, buccal tablet, oral spray, or sublingual orally-administered formulation. For example, a mouthwash may be prepared incorporating the active ingredient in the required amount in an appropriate solvent, such as a sodium borate solution (Dobell's Solution). Alternatively, the active ingredient may be incorporated into an oral solution such as one containing sodium borate, glycerin and potassium bicarbonate, or dispersed in a dentifrice, or added in a therapeutically-effective amount to a composition that may include water, binders, abrasives, flavoring agents, foaming agents, and humectants. Alternatively the compositions may be

fashioned into a tablet or solution form that may be placed under the tongue or otherwise dissolved in the mouth.

[0187] Additional formulations that are suitable for other modes of alimentary administration include suppositories. Suppositories are solid dosage forms of various weights and shapes, usually medicated, for insertion into the rectum. After insertion, suppositories soften, melt or dissolve in the cavity fluids. In general, for suppositories, traditional carriers may include, for example, polyalkylene glycols, triglycerides or combinations thereof. In certain embodiments, suppositories may be formed from mixtures containing, for example, the active ingredient in the range of about 0.5% to about 10%, and preferably about 1% to about 2%.

B. Parenteral Compositions and Formulations

[0188] In further embodiments, compositions may be administered via a parenteral route. As used herein, the term "parenteral" includes routes that bypass the alimentary tract. Specifically, the pharmaceutical compositions disclosed herein may be administered for example, but not limited to intravenously, intradermally, intramuscularly, intraarterially, intrathecally, subcutaneous, or intraperitoneally U.S. Pat. Nos. 6,613,308; 5,466,468; 5,543,158; 5,641,515; and 5,399,363 (each specifically incorporated herein by reference in its entirety).

[0189] Solutions of the active compounds as free base or pharmacologically acceptable salts may be prepared in water suitably mixed with a surfactant, such as hydroxypropylcellulose. Dispersions may also be prepared in glycerol, liquid polyethylene glycols, and mixtures thereof and in oils. Under ordinary conditions of storage and use, these preparations contain a preservative to prevent the growth of microorganisms. The pharmaceutical forms suitable for injectable use include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions (U.S. Pat. No. 5,466,468, specifically incorporated herein by reference in its entirety). In all cases the form must be sterile and must be fluid to the extent that easy injectability exists. It must be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms, such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (i.e., glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and/or vegetable oils. Proper fluidity may be maintained, for example, by the use of a coating, such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, sorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars or sodium chloride. Prolonged absorption of the injectable compositions can be brought about by the use in the compositions of agents delaying absorption, for example, aluminum monostearate and gelatin.

[0190] For parenteral administration in an aqueous solution, for example, the solution should be suitably buffered if necessary and the liquid diluent first rendered isotonic with sufficient saline or glucose. These particular aqueous solutions are especially suitable for intravenous, intramuscular, subcutaneous, and intraperitoneal administration. In this

connection, sterile aqueous media that can be employed will be known to those of skill in the art in light of the present disclosure. For example, one dosage may be dissolved in isotonic NaCl solution and either added hypodermoclysis fluid or injected at the proposed site of infusion, (see for example, "Remington's Pharmaceutical Sciences" 15th Edition, pages 1035-1038 and 1570-1580). Some variation in dosage will necessarily occur depending on the condition of the subject being treated. The person responsible for administration will, in any event, determine the appropriate dose for the individual subject. Moreover, for human administration, preparations should meet sterility, pyrogenicity, general safety and purity standards as required by FDA Office of Biologics standards.

[0191] Sterile injectable solutions are prepared by incorporating the active compounds in the required amount in the appropriate solvent with various of the other ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the various sterilized active ingredients into a sterile vehicle which contains the basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum-drying and freeze-drying techniques which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof. A powdered composition is combined with a liquid carrier such as, e.g., water or a saline solution, with or without a stabilizing agent.

C. Miscellaneous Pharmaceutical Compositions and Formulations

[0192] In other particular embodiments of the disclosure, the active compound cells expressing anti-CD70 CARs may be formulated for administration via various miscellaneous routes, for example, topical (i.e., transdermal) administration, mucosal administration (intranasal, vaginal, etc.) and/or inhalation.

[0193] Pharmaceutical compositions for topical administration may include the active compound formulated for a medicated application such as an ointment, paste, cream or powder. Ointments include all oleaginous, adsorption, emulsion and water-solubly based compositions for topical application, while creams and lotions are those compositions that include an emulsion base only. Topically administered medications may contain a penetration enhancer to facilitate adsorption of the active ingredients through the skin. Suitable penetration enhancers include glycerin, alcohols, alkyl methyl sulfoxides, pyrrolidones and luarocapram. Possible bases for compositions for topical application include polyethylene glycol, lanolin, cold cream and petrolatum as well as any other suitable absorption, emulsion or water-soluble ointment base. Topical preparations may also include emulsifiers, gelling agents, and antimicrobial preservatives as necessary to preserve the active ingredient and provide for a homogenous mixture. Transdermal administration of the present invention may also comprise the use of a "patch". For example, the patch may supply one or more active substances at a predetermined rate and in a continuous manner over a fixed period of time.

[0194] In certain embodiments, the pharmaceutical compositions may be delivered by eye drops, intranasal sprays, inhalation, and/or other aerosol delivery vehicles. Methods

for delivering compositions directly to the lungs via nasal aerosol sprays has been described e.g., in U.S. Pat. Nos. 5,756,353 and 5,804,212 (each specifically incorporated herein by reference in its entirety). Likewise, the delivery of drugs using intranasal microparticle resins (Takenaga et al., 1998) and lysophosphatidyl-glycerol compounds (U.S. Pat. No. 5,725,871, specifically incorporated herein by reference in its entirety) are also well-known in the pharmaceutical arts. Likewise, transmucosal drug delivery in the form of a polytetrafluoroethylene support matrix is described in U.S. Pat. No. 5,780,045 (specifically incorporated herein by reference in its entirety).

[0195] The term aerosol refers to a colloidal system of finely divided solid or liquid particles dispersed in a liquefied or pressurized gas propellant. The typical aerosol of the present invention for inhalation will consist of a suspension of active ingredients in liquid propellant or a mixture of liquid propellant and a suitable solvent. Suitable propellants include hydrocarbons and hydrocarbon ethers. Suitable containers will vary according to the pressure requirements of the propellant. Administration of the aerosol will vary according to subject's age, weight and the severity and response of the symptoms.

V. Combination Therapies

[0196] In certain embodiments, the compositions and methods of the present embodiments involve a cancer therapy that is additional to the compositions comprising cells expressing anti-CD70 CARs. The additional therapy may be radiation therapy, surgery (e.g., lumpectomy and a mastectomy), chemotherapy, gene therapy, DNA therapy, viral therapy, RNA therapy, immunotherapy, bone marrow transplantation, nanotherapy, monoclonal antibody therapy, hormone therapy, or a combination of the foregoing. The additional therapy may be in the form of adjuvant or neoadjuvant therapy.

[0197] In some embodiments, the additional therapy is the administration of small molecule enzymatic inhibitor(s) or anti-metastatic agent(s). In some embodiments, the additional therapy is the administration of side-effect limiting agents (e.g., agents intended to lessen the occurrence and/or severity of side effects of treatment, such as anti-nausea agents, etc.). In some embodiments, the additional therapy is radiation therapy. In some embodiments, the additional therapy is surgery. In some embodiments, the additional therapy is a combination of radiation therapy and surgery. In some embodiments, the additional therapy is gamma irradiation. In some embodiments, the additional therapy is therapy targeting PBK/AKT/mTOR pathway, HSP90 inhibitor, tubulin inhibitor, apoptosis inhibitor, and/or chemopreventative agent(s). The additional therapy may be one or more of the chemotherapeutic agents known in the art.

[0198] An immune cell therapy (in addition to the cell therapy of the disclosure) may be administered before, during, after, or in various combinations relative to an additional cancer therapy, such as immune checkpoint therapy. The administrations may be in intervals ranging from concurrently to minutes to days to weeks. In embodiments where the immune cell therapy is provided to a patient separately from the composition(s) of the disclosure, one would generally ensure that a significant period of time did not expire between the time of each delivery, such that the two compounds would still be able to exert an advantageously combined effect on the patient. In such instances, it

is contemplated that one may provide a patient with the immunotherapy therapy and the disclosed compositions within about 12 to 24 or 72 h of each other and, more particularly, within about 6-12 h of each other. In some situations it may be desirable to extend the time period for treatment significantly where several days (2, 3, 4, 5, 6, or 7) to several weeks (1, 2, 3, 4, 5, 6, 7, or 8) lapse between respective administrations.

[0199] Administration of any compound or cell therapy of the present embodiments to a patient will follow general protocols for the administration of such compounds, taking into account the toxicity, if any, of the agents. Therefore, in some embodiments there is a step of monitoring toxicity that is attributable to combination therapy.

[0200] The term “cancer,” as used herein, may be used to describe a solid tumor, metastatic cancer, or non-metastatic cancer. In certain embodiments, the cancer may originate in the bladder, blood, bone, bone marrow, brain, breast, colon, esophagus, duodenum, small intestine, large intestine, colon, rectum, anus, gum, head, kidney, liver, lung, nasopharynx, neck, ovary, pancreas, prostate, skin, stomach, testis, tongue, or uterus. In some embodiments, the cancer is recurrent cancer. In some embodiments, the cancer is Stage I cancer. In some embodiments, the cancer is Stage II cancer. In some embodiments, the cancer is Stage III cancer. In some embodiments, the cancer is Stage IV cancer.

[0201] The cancer may specifically be of the following histological type, though it is not limited to these: neoplasm, malignant; carcinoma; carcinoma, undifferentiated; giant and spindle cell carcinoma; small cell carcinoma; papillary carcinoma; squamous cell carcinoma; lymphoepithelial carcinoma; basal cell carcinoma; pilomatrix carcinoma; transitional cell carcinoma; papillary transitional cell carcinoma; adenocarcinoma; gastrinoma, malignant; cholangiocarcinoma; hepatocellular carcinoma; combined hepatocellular carcinoma and cholangiocarcinoma; trabecular adenocarcinoma; adenoid cystic carcinoma; adenocarcinoma in adenomatous polyp; adenocarcinoma, familial polyposis *coli*; solid carcinoma; carcinoid tumor, malignant; bronchioloalveolar adenocarcinoma; papillary adenocarcinoma; chromophobe carcinoma; acidophil carcinoma; oxyphilic adenocarcinoma; basophil carcinoma; clear cell adenocarcinoma; granular cell carcinoma; follicular adenocarcinoma; papillary and follicular adenocarcinoma; nonencapsulating sclerosing carcinoma; adrenal cortical carcinoma; endometroid carcinoma; skin appendage carcinoma; apocrine adenocarcinoma; sebaceous adenocarcinoma; ceruminous adenocarcinoma; mucoepidermoid carcinoma; cystadenocarcinoma; papillary cystadenocarcinoma; papillary serous cystadenocarcinoma; mucinous cystadenocarcinoma; mucinous adenocarcinoma; signet ring cell carcinoma; infiltrating duct carcinoma; medullary carcinoma; lobular carcinoma; inflammatory carcinoma; paget’s disease, mammary; acinar cell carcinoma; adenosquamous carcinoma; adenocarcinoma w/squamous metaplasia; thymoma, malignant; ovarian stromal tumor, malignant; thecoma, malignant; granulosa cell tumor, malignant; androblastoma, malignant; sertoli cell carcinoma; leydig cell tumor, malignant; lipid cell tumor, malignant; paraganglioma, malignant; extramammary paraganglioma, malignant; pheochromocytoma; glomangiosarcoma; malignant melanoma; amelanotic melanoma; superficial spreading melanoma; malignant melanoma in giant pigmented nevus; epithelioid cell melanoma; blue nevus, malignant; sarcoma; fibrosarcoma; fibrous his-

tiocytoma, malignant; myxosarcoma; liposarcoma; leiomyosarcoma; rhabdomyosarcoma; embryonal rhabdomyosarcoma; alveolar rhabdomyosarcoma; stromal sarcoma; mixed tumor, malignant; mullerian mixed tumor; nephroblastoma; hepatoblastoma; carcinosarcoma; mesenchymoma, malignant; brenner tumor, malignant; phyllodes tumor, malignant; synovial sarcoma; mesothelioma, malignant; dysgerminoma; embryonal carcinoma; teratoma, malignant; struma ovarii, malignant; choriocarcinoma; mesonephroma, malignant; hemangiosarcoma; hemangiopericytoma, malignant; kaposi’s sarcoma; hemangiopericytoma, malignant; lymphangiosarcoma; osteosarcoma; juxtacortical osteosarcoma; chondrosarcoma; chondroblastoma, malignant; mesenchymal chondrosarcoma; giant cell tumor of bone; ewing’s sarcoma; odontogenic tumor, malignant; ameloblastic odontosarcoma; ameloblastoma, malignant; ameloblastic fibrosarcoma; pinealoma, malignant; chordoma; glioma, malignant; ependymoma; astrocytoma; protoplasmic astrocytoma; fibrillary astrocytoma; astroblastoma; glioblastoma; oligodendroglioma; oligodendroblastoma; primitive neuroectodermal; cerebellar sarcoma; ganglioneuroblastoma; neuroblastoma; retinoblastoma; olfactory neurogenic tumor; meningioma, malignant; neurofibrosarcoma; neurilemmoma, malignant; granular cell tumor, malignant; malignant lymphoma; hodgkin’s disease; hodgkin’s; paragranuloma; malignant lymphoma, small lymphocytic; malignant lymphoma, large cell, diffuse; malignant lymphoma, follicular; mycosis fungoides; other specified non-hodgkin’s lymphomas; malignant histiocytosis; multiple myeloma; mast cell sarcoma; immunoproliferative small intestinal disease; leukemia; lymphoid leukemia; plasma cell leukemia; erythroleukemia; lymphosarcoma cell leukemia; myeloid leukemia; basophilic leukemia; eosinophilic leukemia; monocytic leukemia; mast cell leukemia; megakaryoblastic leukemia; myeloid sarcoma; and hairy cell leukemia.

A. Chemotherapy

[0202] A wide variety of chemotherapeutic agents may be used in accordance with the present embodiments. The term “chemotherapy” refers to the use of drugs to treat cancer. A “chemotherapeutic agent” is used to connote a compound or composition that is administered in the treatment of cancer. These agents or drugs are categorized by their mode of activity within a cell, for example, whether and at what stage they affect the cell cycle. Alternatively, an agent may be characterized based on its ability to directly cross-link DNA, to intercalate into DNA, or to induce chromosomal and mitotic aberrations by affecting nucleic acid synthesis.

[0203] Examples of chemotherapeutic agents include alkylating agents, such as thiopeta and cyclophosphamide; alkyl sulfonates, such as busulfan, improsulfan, and piposulfan; aziridines, such as benzodopa, carboquone, meturedopa, and uredopa; ethylenimines and methylamelamines, including altretamine, triethylenemelamine, triethylenephosphoramide, triethylenethiophosphoramide, and trimethylolomelamine; acetogenins (especially bullatacin and bullatacinone); a camptothecin (including the synthetic analogue topotecan); bryostatin; callystatin; CC-1065 (including its adozelesin, carzelesin and bizelesin synthetic analogues); cryptophycins (particularly cryptophycin 1 and cryptophycin 8); dolastatin; duocarmycin (including the synthetic analogues, KW-2189 and CB1-TM1); eleutherobin; pancratistatin; a sarcodictyin; spongistatin; nitrogen mustards, such as chlorambucil, chlornaphazine, cholophosphamide, estra-

mustine, ifosfamide, mechlorethamine, mechlorethamine oxide hydrochloride, melphalan, novembichin, phenesterine, prednimustine, trofosfamide, and uracil mustard; nitrosureas, such as carmustine, chlorozotocin, fotemustine, lomustine, nimustine, and ranimustine; antibiotics, such as the enediyne antibiotics (e.g., calicheamicin, especially calicheamicin gammall and calicheamicin omegall); dynemicin, including dynemicin A; bisphosphonates, such as clodronate; an esperamicin; as well as neocarzinostatin chromophore and related chromoprotein enediyne antibiotic chromophores, aclacinomysins, actinomycin, authramycin, azaserine, bleomycins, cactinomycin, carabacin, carminomycin, carzinophilin, chromomycinis, dactinomycin, daunorubicin, detorubicin, 6-diazo-5-oxo-L-norleucine, doxorubicin (including morpholino-doxorubicin, cyanomorpholino-doxorubicin, 2-pyrrolino-doxorubicin and deoxydoxorubicin), epirubicin, esorubicin, idarubicin, marcellomycin, mitomycins, such as mitomycin C, mycophenolic acid, nogalarinycin, olivomycins, peplomycin, potfiromycin, puromycin, quelamycin, rodorubicin, streptogrin, streptozocin, tubercidin, ubenimex, zinostatin, and zorubicin; anti-metabolites, such as methotrexate and 5-fluorouracil (5-FU); folic acid analogues, such as denopterin, pteropterin, and trimetrexate; purine analogs, such as fludarabine, 6-mercaptopurine, thiamiprine, and thioguanine; pyrimidine analogs, such as ancitabine, azacitidine, 6-azauridine, carmofur, cytarabine, dideoxyuridine, doxifluridine, enocitabine, and floxuridine; androgens, such as calusterone, dromostanolone propionate, epitostanol, mepitiostane, and testolactone; anti-adrenals, such as mitotane and trilostane; folic acid replenisher, such as frolinic acid; aceglatone; aldophosphamide glycoside; aminolevulinic acid; eniluracil; amsacrine; bestrabucil; bisantrene; edatraxate; defofamine; demecolcine; diaziquone; elformithine; elliptinium acetate; an epothilone; etoglucid; gallium nitrate; hydroxyurea; lentinan; lonidainine; maytansinoids, such as maytansine and ansamitocins; mitoguazone; mitoxantrone; mopidanmol; nitraerine; pentostatin; phenamet; pirarubicin; losoxantrone; podophyllinic acid; 2-ethylhydrazide; procarbazine; PSKpolysaccharide complex; razoxane; rhizoxin; sizofiran; spirogermanium; tenuazonic acid; triaziquone; 2,2',2"-trichlorotriethylamine; trichothecenes (especially T-2 toxin, verracurin A, roridin A and anguidine); urethan; vindesine; dacarbazine; mannomustine; mitobronitol; mitolactol; pipobroman; gacytosine; arabinoside ("Ara-C"); cyclophosphamide; taxoids, e.g., paclitaxel and docetaxel gemcitabine; 6-thioguanine; mercaptopurine; platinum coordination complexes, such as cisplatin, oxaliplatin, and carboplatin; vinblastine; platinum; etoposide (VP-16); ifosfamide; mitoxantrone; vincristine; vinorelbine; novantrone; teniposide; edatrexate; daunomycin; aminopterin; xeloda; ibandronate; irinotecan (e.g., CPT-11); topoisomerase inhibitor RFS 2000; difluoromethylornithine (DMFO); retinoids, such as retinoic acid; capecitabine; carboplatin, procarbazine, plicomycin, gemcitabine, navelbine, farnesyl-protein transferase inhibitors, transplatinum, and pharmaceutically acceptable salts, acids, or derivatives of any of the above.

B. Radiotherapy

[0204] Other factors that cause DNA damage and have been used extensively include what are commonly known as γ -rays, X-rays, and/or the directed delivery of radioisotopes to tumor cells. Other forms of DNA damaging factors are

also contemplated, such as microwaves, proton beam irradiation (U.S. Pat. Nos. 5,760,395 and 4,870,287), and UV-irradiation. It is most likely that all of these factors affect a broad range of damage on DNA, on the precursors of DNA, on the replication and repair of DNA, and on the assembly and maintenance of chromosomes. Dosage ranges for X-rays range from daily doses of 50 to 200 roentgens for prolonged periods of time (3 to 4 wk), to single doses of 2000 to 6000 roentgens. Dosage ranges for radioisotopes vary widely, and depend on the half-life of the isotope, the strength and type of radiation emitted, and the uptake by the neoplastic cells.

C. Immunotherapy

[0205] The skilled artisan will understand that additional immunotherapies (outside of the disclosed cell therapy) may be used in combination or in conjunction with methods of the embodiments. In the context of cancer treatment, immunotherapeutics, generally, rely on the use of immune effector cells and molecules to target and destroy cancer cells. Rituximab (RITUXAN®) is such an example. The immune effector may be, for example, an antibody specific for some marker on the surface of a tumor cell. The antibody alone may serve as an effector of therapy or it may recruit other cells to actually affect cell killing. The antibody also may be conjugated to a drug or toxin (chemotherapeutic, radionuclide, ricin A chain, cholera toxin, pertussis toxin, etc.) and serve as a targeting agent. Alternatively, the effector may be a lymphocyte carrying a surface molecule that interacts, either directly or indirectly, with a tumor cell target. Various effector cells include cytotoxic T cells and NK cells other than those having knockdown or knockout of TGF- β R2.

[0206] Antibody-drug conjugates have emerged as a breakthrough approach to the development of cancer therapeutics. Antibody-drug conjugates (ADCs) comprise monoclonal antibodies (MAbs) that are covalently linked to cell-killing drugs. This approach combines the high specificity of MAbs against their antigen targets with highly potent cytotoxic drugs, resulting in "armed" MAbs that deliver the payload (drug) to tumor cells with enriched levels of the antigen. Targeted delivery of the drug also minimizes its exposure in normal tissues, resulting in decreased toxicity and improved therapeutic index. The approval of two ADC drugs, ADCETRIS® (brentuximab vedotin) in 2011 and KADCYLA® (trastuzumab emtansine or T-DM1) in 2013 by FDA validated the approach. There are currently more than 30 ADC drug candidates in various stages of clinical trials for cancer treatment (Leal et al., 2014). As antibody engineering and linker-payload optimization are becoming more and more mature, the discovery and development of new ADCs are increasingly dependent on the identification and validation of new targets that are suitable to this approach and the generation of targeting MAbs. Two criteria for ADC targets are upregulated/high levels of expression in tumor cells and robust internalization.

[0207] In one aspect of immunotherapy, the tumor cell must bear some marker that is amenable to targeting, i.e., is not present on the majority of other cells. Many tumor markers exist and any of these may be suitable for targeting in the context of the present embodiments. Common tumor markers include CD20, carcinoembryonic antigen, tyrosinase (p97), gp68, TAG-72, HMFG, Sialyl Lewis Antigen, MucA, MucB, PLAP, laminin receptor, erb B, and p155. An

alternative aspect of immunotherapy is to combine anticancer effects with immune stimulatory effects. Immune stimulating molecules also exist including: cytokines, such as IL-2, IL-4, IL-12, GM-CSF, gamma-IFN, chemokines, such as MIP-1, MCP-1, IL-8, and growth factors, such as FLT3 ligand.

[0208] Examples of immunotherapies currently under investigation or in use are immune adjuvants, e.g., *Mycobacterium bovis*, *Plasmodium falciparum*, dinitrochlorobenzene, and aromatic compounds (U.S. Pat. Nos. 5,801,005 and 5,739,169; Hui and Hashimoto, 1998; Christodoulides et al., 1998); cytokine therapy, e.g., interferons of any kind, IL-1, GM-CSF, and TNF (Bukowski et al., 1998; Davidson et al., 1998; Hellstrand et al., 1998); gene therapy, e.g., TNF, IL-1, IL-2, and p53 (Qin et al., 1998; Austin-Ward and Villaseca, 1998; U.S. Pat. Nos. 5,830,880 and 5,846,945); and monoclonal antibodies, e.g., anti-CD20, anti-ganglioside GM2, and anti-p185 (Hollander, 2012; Hanibuchi et al., 1998; U.S. Pat. No. 5,824,311). It is contemplated that one or more anti-cancer therapies may be employed with the antibody therapies described herein.

[0209] In some embodiments, the immunotherapy may be an immune checkpoint inhibitor. Immune checkpoints either turn up a signal (e.g., co-stimulatory molecules) or turn down a signal. Inhibitory immune checkpoints that may be targeted by immune checkpoint blockade include adenosine A2A receptor (A2AR), B7-H3 (also known as CD276), B and T lymphocyte attenuator (BTLA), cytotoxic T-lymphocyte-associated protein 4 (CTLA-4, also known as CD152), indoleamine 2,3-dioxygenase (IDO), killer-cell immunoglobulin (KIR), lymphocyte activation gene-3 (LAG3), programmed death 1 (PD-1), T-cell immunoglobulin domain and mucin domain 3 (TIM-3) and V-domain Ig suppressor of T cell activation (VISTA). In particular, the immune checkpoint inhibitors target the PD-1 axis and/or CTLA-4.

D. Surgery

[0210] Approximately 60% of persons with cancer will undergo surgery of some type, which includes preventative, diagnostic or staging, curative, and palliative surgery. Curative surgery includes resection in which all or part of cancerous tissue is physically removed, excised, and/or destroyed and may be used in conjunction with other therapies, such as the treatment of the present embodiments, chemotherapy, radiotherapy, hormonal therapy, gene therapy, immunotherapy, and/or alternative therapies. Tumor resection refers to physical removal of at least part of a tumor. In addition to tumor resection, treatment by surgery includes laser surgery, cryosurgery, electrosurgery, and microscopically-controlled surgery (Mohs' surgery).

[0211] Upon excision of part or all of cancerous cells, tissue, or tumor, a cavity may be formed in the body. Treatment may be accomplished by perfusion, direct injection, or local application of the area with an additional anti-cancer therapy. Such treatment may be repeated, for example, every 1, 2, 3, 4, 5, 6, or 7 days, or every 1, 2, 3, 4, and 5 weeks or every 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 months. These treatments may be of varying dosages as well.

E. Other Agents

[0212] It is contemplated that other agents may be used in combination with certain aspects of the present embodi-

ments to improve the therapeutic efficacy of treatment. These additional agents include agents that affect the upregulation of cell surface receptors and GAP junctions, cytostatic and differentiation agents, inhibitors of cell adhesion, agents that increase the sensitivity of the hyperproliferative cells to apoptotic inducers, or other biological agents. Increases in intercellular signaling by elevating the number of GAP junctions would increase the anti-hyperproliferative effects on the neighboring hyperproliferative cell population. In other embodiments, cytostatic or differentiation agents can be used in combination with certain aspects of the present embodiments to improve the anti-hyperproliferative efficacy of the treatments. Inhibitors of cell adhesion are contemplated to improve the efficacy of the present embodiments. Examples of cell adhesion inhibitors are focal adhesion kinase (FAKs) inhibitors and Lovastatin. It is further contemplated that other agents that increase the sensitivity of a hyperproliferative cell to apoptosis, such as the antibody c225, could be used in combination with certain aspects of the present embodiments to improve the treatment efficacy.

VI. Vectors

[0213] The CD70-targeting CARs may be delivered to the recipient cell, including NK cell, by any suitable vector, including by a viral vector or by a non-viral vector. Examples of viral vectors include at least retroviral, lentiviral, adenoviral, or adeno-associated viral vectors. Examples of non-viral vectors include at least plasmids, transposons, lipids, nanoparticles, liposomes, combinations thereof, and so forth.

[0214] In cases wherein the NK cell is transduced with a vector encoding the CD70-targeting CAR and also requires transduction of another gene or genes into the cell, such as a suicide gene and/or cytokine and/or an optional therapeutic gene product, the CD70-targeting CAR, optional suicide gene, optional cytokine(s) may or may not be comprised on or with the same vector. In some cases, the CD70-targeting CAR, suicide gene, and/or cytokine(s) are expressed from the same vector molecule, such as the same viral vector molecule. In such cases, the expression of the CD70-targeting CAR, suicide gene, and/or cytokine(s) may or may not be regulated by the same regulatory element(s). When the CD70-targeting CAR, suicide gene, and/or cytokines are on the same vector, they may or may not be expressed as separate polypeptides. In cases wherein they are expressed as separate polypeptides, they may be separated on the vector by a 2A element or IRES element (or both kinds may be used on the same vector once or more than once), for example.

[0215] One of skill in the art would be well-equipped to construct a vector through standard recombinant techniques (see, for example, Sambrook et al., 2001 and Ausubel et al., 1996, both incorporated herein by reference) for the expression of the antigen receptors of the present disclosure.

A. Regulatory Elements

[0216] Expression cassettes included in vectors useful in the present disclosure in particular contain (in a 5'-to-3' direction) a eukaryotic transcriptional promoter operably linked to a protein-coding sequence, splice signals including intervening sequences, and a transcriptional termination/polyadenylation sequence. The promoters and enhancers that control the transcription of protein encoding genes in

eukaryotic cells may be comprised of multiple genetic elements. The cellular machinery is able to gather and integrate the regulatory information conveyed by each element, allowing different genes to evolve distinct, often complex patterns of transcriptional regulation. A promoter used in the context of the present disclosure includes constitutive, inducible, and tissue-specific promoters, for example. In cases wherein the vector is utilized for the generation of cancer therapy, a promoter may be effective under conditions of hypoxia.

B. Promoter/Enhancers

[0217] The expression constructs provided herein comprise a promoter to drive expression of the antigen receptor and other cistron gene products. A promoter generally comprises a sequence that functions to position the start site for RNA synthesis. The best known example of this is the TATA box, but in some promoters lacking a TATA box, such as, for example, the promoter for the mammalian terminal deoxynucleotidyl transferase gene and the promoter for the SV40 late genes, a discrete element overlying the start site itself helps to fix the place of initiation. Additional promoter elements regulate the frequency of transcriptional initiation. Typically, these are located in the region upstream of the start site, although a number of promoters have been shown to contain functional elements downstream of the start site as well. To bring a coding sequence “under the control of” a promoter, one positions the 5' end of the transcription initiation site of the transcriptional reading frame “downstream” of (i.e., 3' of) the chosen promoter. The “upstream” promoter stimulates transcription of the DNA and promotes expression of the encoded RNA.

[0218] The spacing between promoter elements frequently is flexible, so that promoter function is preserved when elements are inverted or moved relative to one another. In the tk promoter, for example, the spacing between promoter elements can be increased to 50 bp apart before activity begins to decline. Depending on the promoter, it appears that individual elements can function either cooperatively or independently to activate transcription. A promoter may or may not be used in conjunction with an “enhancer,” which refers to a cis-acting regulatory sequence involved in the transcriptional activation of a nucleic acid sequence.

[0219] A promoter may be one naturally associated with a nucleic acid sequence, as may be obtained by isolating the 5' non-coding sequences located upstream of the coding segment and/or exon. Such a promoter can be referred to as “endogenous.” Similarly, an enhancer may be one naturally associated with a nucleic acid sequence, located either downstream or upstream of that sequence. Alternatively, certain advantages will be gained by positioning the coding nucleic acid segment under the control of a recombinant or heterologous promoter, which refers to a promoter that is not normally associated with a nucleic acid sequence in its natural environment. A recombinant or heterologous enhancer refers also to an enhancer not normally associated with a nucleic acid sequence in its natural environment. Such promoters or enhancers may include promoters or enhancers of other genes, and promoters or enhancers isolated from any other virus, or prokaryotic or eukaryotic cell, and promoters or enhancers not “naturally occurring,” i.e., containing different elements of different transcriptional regulatory regions, and/or mutations that alter expression. For example, promoters that are most commonly used in

recombinant DNA construction include the β -lactamase (penicillinase), lactose and tryptophan (trp-) promoter systems. In addition to producing nucleic acid sequences of promoters and enhancers synthetically, sequences may be produced using recombinant cloning and/or nucleic acid amplification technology, including PCRTM, in connection with the compositions disclosed herein. Furthermore, it is contemplated that the control sequences that direct transcription and/or expression of sequences within non-nuclear organelles such as mitochondria, chloroplasts, and the like, can be employed as well.

[0220] Naturally, it will be important to employ a promoter and/or enhancer that effectively directs the expression of the DNA segment in the organelle, cell type, tissue, organ, or organism chosen for expression. Those of skill in the art of molecular biology generally know the use of promoters, enhancers, and cell type combinations for protein expression, (see, for example Sambrook et al. 1989, incorporated herein by reference). The promoters employed may be constitutive, tissue-specific, inducible, and/or useful under the appropriate conditions to direct high level expression of the introduced DNA segment, such as is advantageous in the large-scale production of recombinant proteins and/or peptides. The promoter may be heterologous or endogenous.

[0221] Additionally, any promoter/enhancer combination (as per, for example, the Eukaryotic Promoter Data Base EPDB, through world wide web at epd.isb-sib.ch/) could also be used to drive expression. Use of a T3, T7 or SP6 cytoplasmic expression system is another possible embodiment. Eukaryotic cells can support cytoplasmic transcription from certain bacterial promoters if the appropriate bacterial polymerase is provided, either as part of the delivery complex or as an additional genetic expression construct.

[0222] Non-limiting examples of promoters include early or late viral promoters, such as, SV40 early or late promoters, cytomegalovirus (CMV) immediate early promoters, Rous Sarcoma Virus (RSV) early promoters; eukaryotic cell promoters, such as, e. g., beta actin promoter, GAPDH promoter, metallothionein promoter; and concatenated response element promoters, such as cyclic AMP response element promoters (cre), serum response element promoter (sre), phorbol ester promoter (TPA) and response element promoters (tre) near a minimal TATA box. It is also possible to use human growth hormone promoter sequences (e.g., the human growth hormone minimal promoter described at GenBank®, accession no. X05244, nucleotide 283-341) or a mouse mammary tumor promoter (available from the ATCC, Cat. No. ATCC 45007). In certain embodiments, the promoter is CMV IE, dectin-1, dectin-2, human CD11c, F4/80, SM22, RSV, SV40, Ad MLP, beta-actin, MHC class I or MHC class II promoter, however any other promoter that is useful to drive expression of the therapeutic gene is applicable to the practice of the present disclosure.

[0223] In certain aspects, methods of the disclosure also concern enhancer sequences, i.e., nucleic acid sequences that increase a promoter's activity and that have the potential to act in cis, and regardless of their orientation, even over relatively long distances (up to several kilobases away from the target promoter). However, enhancer function is not necessarily restricted to such long distances as they may also function in close proximity to a given promoter.

C. Initiation Signals and Linked Expression

[0224] A specific initiation signal also may be used in the expression constructs provided in the present disclosure for efficient translation of coding sequences. These signals include the ATG initiation codon or adjacent sequences. Exogenous translational control signals, including the ATG initiation codon, may need to be provided. One of ordinary skill in the art would readily be capable of determining this and providing the necessary signals. It is well known that the initiation codon must be “in-frame” with the reading frame of the desired coding sequence to ensure translation of the entire insert. The exogenous translational control signals and initiation codons can be either natural or synthetic. The efficiency of expression may be enhanced by the inclusion of appropriate transcription enhancer elements.

[0225] In certain embodiments, the use of internal ribosome entry sites (IRES) elements are used to create multi-gene, or polycistronic messages. IRES elements are able to bypass the ribosome scanning model of 5' methylated Cap dependent translation and begin translation at internal sites. IRES elements from two members of the picornavirus family (polio and encephalomyocarditis) have been described, as well an IRES from a mammalian message. IRES elements can be linked to heterologous open reading frames. Multiple open reading frames can be transcribed together, each separated by an IRES, creating polycistronic messages. By virtue of the IRES element, each open reading frame is accessible to ribosomes for efficient translation. Multiple genes can be efficiently expressed using a single promoter/enhancer to transcribe a single message.

[0226] As detailed elsewhere herein, certain 2A sequence elements could be used to create linked- or co-expression of genes in the constructs provided in the present disclosure. For example, cleavage sequences could be used to co-express genes by linking open reading frames to form a single cistron. An exemplary cleavage sequence is the equine rhinitis A virus (E2A) or the F2A (Foot-and-mouth disease virus 2A) or a “2A-like” sequence (e.g., Thosea asigna virus 2A; T2A) or porcine teschovirus-1 (P2A). In specific embodiments, in a single vector the multiple 2A sequences are non-identical, although in alternative embodiments the same vector utilizes two or more of the same 2A sequences. Examples of 2A sequences are provided in US 2011/0065779 which is incorporated by reference herein in its entirety.

D. Origins of Replication

[0227] In order to propagate a vector in a host cell, it may contain one or more origins of replication sites (often termed “ori”), for example, a nucleic acid sequence corresponding to oriP of EBV as described above or a genetically engineered oriP with a similar or elevated function in programming, which is a specific nucleic acid sequence at which replication is initiated. Alternatively a replication origin of other extra-chromosomally replicating virus as described above or an autonomously replicating sequence (ARS) can be employed.

E. Selection and Screenable Markers

[0228] In some embodiments, cells such as NK cells comprising an anti-CD70 CAR-expressing construct of the present disclosure may be identified *in vitro* or *in vivo* by including a marker in the expression vector. Such markers

would confer an identifiable change to the cell permitting easy identification of cells containing the expression vector. Generally, a selection marker is one that confers a property that allows for selection. A positive selection marker is one in which the presence of the marker allows for its selection, while a negative selection marker is one in which its presence prevents its selection. An example of a positive selection marker is a drug resistance marker.

[0229] Usually the inclusion of a drug selection marker aids in the cloning and identification of transformants, for example, genes that confer resistance to neomycin, puromycin, hygromycin, DHFR, GPT, zeocin and histidinol are useful selection markers. In addition to markers conferring a phenotype that allows for the discrimination of transformants based on the implementation of conditions, other types of markers including screenable markers such as GFP, whose basis is colorimetric analysis, are also contemplated. Alternatively, screenable enzymes as negative selection markers such as herpes simplex virus thymidine kinase (tk) or chloramphenicol acetyltransferase (CAT) may be utilized. One of skill in the art would also know how to employ immunologic markers, possibly in conjunction with FACS analysis. The marker used is not believed to be important, so long as it is capable of being expressed simultaneously with the nucleic acid encoding a gene product. Further examples of selection and screenable markers are well known to one of skill in the art.

VII. Cells

[0230] The present disclosure encompasses immune cells of any kind that harbor a vector that encodes a CD70-targeting CAR and that also may encode at least one cytokine and at least one suicide gene. In some cases, different vectors encode the CAR vs. encodes the suicide gene and/or cytokine. Although conventional T cells, NK cells, gamma-delta T cells, NKT and invariant NK T cells, regulatory T cells, macrophages, B cells, tumor infiltrating lymphocytes, or a mixture thereof may be employed, in particular cases the cells are NK cells. The NK cells may be derived from cord blood, peripheral blood, induced pluripotent stem cells (iPSCs), hematopoietic stem cells (HSCs), or bone marrow. The NK cells may be derived from a cell line such as, but not limited to, NK-92 cells, for example. The NK cell may be a cord blood mononuclear cell, such as a CD56+ NK cell.

[0231] In some instances, the cell is not an immortalized cell line, but is instead a cell (e.g., a primary cell) obtained from an individual. For example, in some cases, the cell is an immune cell obtained from an individual. As an example, the cell is a T lymphocyte obtained from an individual. As another example, the cell is a cytotoxic cell obtained from an individual. As another example, the cell is a stem cell (e.g., peripheral blood stem cell) or progenitor cell obtained from an individual.

[0232] In particular embodiments, the cells of the disclosure may be specifically formulated and/or they may be cultured in a particular medium. The cells may be formulated in such a manner as to be suitable for delivery to a recipient without deleterious effects.

[0233] The medium in certain aspects can be prepared using a medium used for culturing animal cells as their basal medium, such as any of AIM V, X-VIVO-15, NeuroBasal, EGM2, TeSR, BME, BGJb, CMRL 1066, Glasgow MEM, Improved MEM Zinc Option, IMDM, Medium 199, Eagle

MEM, α MEM, DMEM, Ham, RPMI-1640, and Fischer's media, as well as any combinations thereof, but the medium may not be particularly limited thereto as far as it can be used for culturing animal cells. Particularly, the medium may be xeno-free or chemically defined.

[02334] The medium can be a serum-containing or serum-free medium, or xeno-free medium. From the aspect of preventing contamination with heterogeneous animal-derived components, serum can be derived from the same animal as that of the stem cell(s). The serum-free medium refers to medium with no unprocessed or unpurified serum and accordingly, can include medium with purified blood-derived components or animal tissue-derived components (such as growth factors).

[0235] The medium may contain or may not contain any alternatives to serum. The alternatives to serum can include materials which appropriately contain albumin (such as lipid-rich albumin, bovine albumin, albumin substitutes such as recombinant albumin or a humanized albumin, plant starch, dextrans and protein hydrolysates), transferrin (or other iron transporters), fatty acids, insulin, collagen precursors, trace elements, 2-mercaptoethanol, 3'-thioglycerol, or equivalents thereto. The alternatives to serum can be prepared by the method disclosed in International Publication No. 98/30679, for example (incorporated herein in its entirety). Alternatively, any commercially available materials can be used for more convenience. The commercially available materials include knockout Serum Replacement (KSR), Chemically-defined Lipid concentrated (Gibco), and Glutamax (Gibco).

[0236] In certain embodiments, the medium may comprise one, two, three, four, five, six, seven, eight, nine, ten, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more of the following: Vitamins such as biotin; DL Alpha Tocopherol Acetate; DL Alpha-Tocopherol; Vitamin A (acetate); proteins such as BSA (bovine serum albumin) or human albumin, fatty acid free Fraction V; Catalase; Human Recombinant Insulin; Human Transferrin; Superoxide Dismutase; Other Components such as Corticosterone; D-Galactose; Ethanolamine HCl; Glutathione (reduced); L-Carnitine HCl; Linoleic Acid; Linolenic Acid; Progesterone; Putrescine 2HCl; Sodium Selenite; and/or T3 (triiodo-L-thyronine). In specific embodiments, one or more of these may be explicitly excluded.

[0237] In some embodiments, the medium further comprises vitamins. In some embodiments, the medium comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or 13 of the following (and any range derivable therein): biotin, DL alpha tocopherol acetate, DL alpha-tocopherol, vitamin A, choline chloride, calcium pantothenate, pantothenic acid, folic acid nicotinamide, pyridoxine, riboflavin, thiamine, inositol, vitamin B12, or the medium includes combinations thereof or salts thereof. In some embodiments, the medium comprises or consists essentially of biotin, DL alpha tocopherol acetate, DL alpha-tocopherol, vitamin A, or combinations or salts thereof. In some embodiments, the medium further comprises proteins. In some embodiments, the proteins comprise albumin or bovine serum albumin, a fraction of BSA, catalase, insulin, transferrin, superoxide dismutase, or combinations thereof.

In some embodiments, the medium further comprises one or more of the following: corticosterone, D-Galactose, ethanolamine, glutathione, L-carnitine, linoleic acid, linolenic acid, progesterone, putrescine, sodium selenite, or triiodo-L-thyronine, or combinations thereof. In some embodiments, the medium comprises one or more of the following: a B-27[®] supplement, xeno-free B-27[®] supplement, GS21TM supplement, or combinations thereof. In some embodiments, the medium comprises or further comprises amino acids, monosaccharides, inorganic ions. In some embodiments, the amino acids comprise arginine, cystine, isoleucine, leucine, lysine, methionine, glutamine, phenylalanine, threonine, tryptophan, histidine, tyrosine, or valine, or combinations thereof. In some embodiments, the inorganic ions comprise sodium, potassium, calcium, magnesium, nitrogen, or phosphorus, or combinations or salts thereof. In some embodiments, the medium further comprises one or more of the following: molybdenum, vanadium, iron, zinc, selenium, copper, or manganese, or combinations thereof. In certain embodiments, the medium comprises or consists essentially of one or more vitamins discussed herein and/or one or more proteins discussed herein, and/or one or more of the following: corticosterone, D-Galactose, ethanolamine, glutathione, L-carnitine, linoleic acid, linolenic acid, progesterone, putrescine, sodium selenite, or triiodo-L-thyronine, a B-27[®] supplement, xeno-free B-27[®] supplement, GS21TM supplement, an amino acid (such as arginine, cystine, isoleucine, leucine, lysine, methionine, glutamine, phenylalanine, threonine, tryptophan, histidine, tyrosine, or valine), monosaccharide, inorganic ion (such as sodium, potassium, calcium, magnesium, nitrogen, and/or phosphorus) or salts thereof, and/or molybdenum, vanadium, iron, zinc, selenium, copper, or manganese. In specific embodiments, one or more of these may be explicitly excluded.

[0238] The medium can also contain one or more externally added fatty acids or lipids, amino acids (such as non-essential amino acids), vitamin(s), growth factors, cytokines, antioxidant substances, 2-mercaptoethanol, pyruvic acid, buffering agents, and/or inorganic salts. In specific embodiments, one or more of these may be explicitly excluded.

[0239] One or more of the medium components may be added at a concentration of at least, at most, or about 0.1, 0.5, 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 150, 180, 200, 250 ng/L, ng/ml, μ g/ml, mg/ml, or any range derivable therein.

[0240] In specific embodiments, the cells of the disclosure are specifically formulated. They may or may not be formulated as a cell suspension. In specific cases they are formulated in a single dose form. They may be formulated for systemic or local administration. In some cases the cells are formulated for storage prior to use, and the cell formulation may comprise one or more cryopreservation agents, such as DMSO (for example, in 5% DMSO). The cell formulation may comprise albumin, including human albumin, with a specific formulation comprising 2.5% human albumin. The cells may be formulated specifically for intravenous administration; for example, they are formulated for intravenous administration over less than one hour. In particular embodiments the cells are in a formulated cell suspension that is stable at room temperature for 1, 2, 3, or 4 hours or more from time of thawing.

[0241] In some cases, the immune cells have been expanded prior to use and/or prior to manufacture. In some

cases, the immune cells are NK cells that have been expanded in the presence of an effective amount of antigen-presenting cells, including universal antigen presenting cells (UAPCs), including in any suitable ratio. The cells may be cultured with the UAPCs at a ratio of 10:1 to 1:10; 9:1 to 1:9; 8:1 to 1:8; 7:1 to 1:7; 6:1 to 1:6; 5:1 to 1:5; 4:1 to 1:4; 3:1 to 1:3; 2:1 to 1:2; or 1:1, including at a ratio of 1:2, for example. In some cases, the NK cells are expanded in the presence of IL-2, such as at a concentration of 10-500, 10-400, 10-300, 10-200, 10-100, 10-50, 100-500, 100-400, 100-300, 100-200, 200-500, 200-400, 200-300, 300-500, 300-400, or 400-500 U/mL.

[0242] Following genetic modification with the vector(s), the NK cells may be immediately infused or may be stored. In certain aspects, following genetic modification, the cells may be propagated for days, weeks, or months *ex vivo* as a bulk population within about 1, 2, 3, 4, 5 days or more following gene transfer into cells. In a further aspect, the transfectants are cloned and a clone demonstrating presence of a single integrated or episomally maintained expression cassette or plasmid, and expression of the CD70-targeting CAR is expanded *ex vivo*. The clone selected for expansion demonstrates the capacity to specifically recognize and lyse CD70-expressing target cells. The recombinant immune cells may be expanded by stimulation with IL-2, or other cytokines that bind the common gamma-chain (e.g., IL-7, IL-12, IL-15, IL-21, and others). The recombinant immune cells may be expanded by stimulation with artificial antigen presenting cells. In a further aspect, the genetically modified cells may be cryopreserved.

[0243] Embodiments of the disclosure encompass cells that express one or more CD70-targeting CARs as encompassed herein. The NK cell comprises a recombinant nucleic acid that encodes one or more CD70-targeting CARs and one or more engineered nonsecretable, membrane bound TNF-alpha mutant polypeptides, in specific embodiments. In specific embodiments, in addition to expressing one or more CD70-targeting CARs, the cell also comprises a nucleic acid that encodes one or more therapeutic gene products.

[0244] The cells may be obtained from an individual directly or may be obtained from a depository or other storage facility. The cells as therapy may be autologous or allogeneic with respect to the individual to which the cells are provided as therapy.

[0245] The cells may be from an individual in need of therapy for a medical condition, and following their manipulation to express the CD70-targeting CAR, they may be provided back to the individual from which they were originally sourced. In some cases, the cells are stored for later use for the individual or another individual. In such cases, the cells may or may not be further modified prior to use.

[0246] The NK cells that harbor the CD70-targeting CAR that may be comprised in a population of cells, and that population may have a majority that are transduced with one or more CD70-targeting CARs. A cell population may comprise 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% of cells that are transduced with one or more CD70-targeting CARs.

[0247] The NK cells may be produced with the one or more CD70-targeting CARs for the intent of being modular

with respect to a specific purpose. For example, cells may be generated, including for commercial distribution, expressing a CD70-targeting CARs and a user may modify them to express one or more other genes of interest (including therapeutic genes) dependent upon their intended purpose (s). For instance, a party interested in treating CD70-positive cancer may modify them to express a second CAR for the same or a different antigen. Other modifications to the cells may include modifications that tailor the cells to the needs of the individual receiving the cells, including a need specific for a particular type of cancer with which the individual is afflicted.

VIII. Gene Editing of Cells

[0248] In particular embodiments, cells comprising an anti-CD70 CAR are gene edited to modify expression of one or more endogenous genes in the cell. In specific cases, cells are modified to have reduced levels of expression of one or more endogenous genes, including inhibition of expression of one or more endogenous genes (that may be referred to as knocked out). Such cells may or may not be expanded.

[0249] In particular cases, one or more endogenous genes of the cells are modified, such as disrupted in expression where the expression is reduced in part or in full. In specific cases, one or more genes are knocked down or knocked out using processes of the disclosure. In specific cases, multiple genes are knocked down or knocked out, and this may or may not occur in the same step in their production. The genes that are edited in the cells may be of any kind, but in specific embodiments the genes are genes whose gene products inhibit activity and/or proliferation of the cells, including antigen-specific, e.g., CD70-specific, CAR NK cells, such as those derived from cord blood, as one example. In specific cases the genes that are edited in the antigen-specific, e.g., CD70-specific, CAR cells allow the antigen-specific, e.g., CD70-specific, CAR cells to work more effectively in a tumor microenvironment. In specific cases, the genes are one or more of NKG2A, SIGLEC-7, LAG3, TIM3, CISH, FOXO1, TGFBR2, TIGIT, CD96, ADORA2, NR3C1, PD1, PDL-1, PDL-2, CD47, SIRPA, SHIP1, ADAM17, RPS6, 4EBP1, CD25, CD40, IL21R, ICAM1, CD95, CD80, CD86, IL10R, CD5, and CD7. In specific embodiments, the TGFBR2 gene is knocked out or knocked down in the antigen-specific, e.g., CD70-specific, CAR cells.

[0250] In some embodiments, the gene editing is carried out using one or more DNA-binding nucleic acids, such as alteration via an RNA-guided endonuclease (RGEN). For example, the alteration can be carried out using clustered regularly interspaced short palindromic repeats (CRISPR) and CRISPR-associated (Cas) proteins; in some embodiments, CpF1 is utilized instead of Cas9. In general, "CRISPR system" refers collectively to transcripts and other elements involved in the expression of or directing the activity of CRISPR-associated ("Cas") genes, including sequences encoding a Cas gene, a tracr (trans-activating CRISPR) sequence (e.g., tracrRNA or an active partial tracrRNA), a tracr-mate sequence (encompassing a "direct repeat" and a tracrRNA-processed partial direct repeat in the context of an endogenous CRISPR system), a guide sequence (also referred to as a "spacer" in the context of an endogenous CRISPR system), and/or other sequences and transcripts from a CRISPR locus.

[0251] The CRISPR/Cas nuclease or CRISPR/Cas nuclease system can include a non-coding RNA molecule (guide) RNA, which sequence-specifically binds to DNA, and a Cas protein (e.g., Cas9), with nuclease functionality (e.g., two nuclease domains). One or more elements of a CRISPR system can derive from a type I, type II, or type III CRISPR system, e.g., derived from a particular organism comprising an endogenous CRISPR system, such as *Streptococcus pyogenes*.

[0252] In some aspects, a Cas nuclease and gRNA (including a fusion of crRNA specific for the target sequence and fixed tracrRNA) are introduced into the cell. In general, target sites at the 5' end of the gRNA target the Cas nuclease to the target site, e.g., the gene, using complementary base pairing. The target site may be selected based on its location immediately 5' of a protospacer adjacent motif (PAM) sequence, such as typically NGG, or NAG. In this respect, the gRNA is targeted to the desired sequence by modifying the first 20, 19, 18, 17, 16, 15, 14, 14, 12, 11, or 10 nucleotides of the guide RNA to correspond to the target DNA sequence. In general, a CRISPR system is characterized by elements that promote the formation of a CRISPR complex at the site of a target sequence. Typically, "target sequence" generally refers to a sequence to which a guide sequence is designed to have complementarity, where hybridization between the target sequence and a guide sequence promotes the formation of a CRISPR complex. Full complementarity is not necessarily required, provided there is sufficient complementarity to cause hybridization and promote formation of a CRISPR complex.

[0253] The CRISPR system can induce double stranded breaks (DSBs) at the target site, followed by disruptions or alterations as discussed herein. In other embodiments, Cas9 variants, deemed "nickases," are used to nick a single strand at the target site. Paired nickases can be used, e.g., to improve specificity, each directed by a pair of different gRNAs targeting sequences such that upon introduction of the nicks simultaneously, a 5' overhang is introduced. In other embodiments, catalytically inactive Cas9 is fused to a heterologous effector domain such as a transcriptional repressor or activator, to affect gene expression.

[0254] The target sequence may comprise any polynucleotide, such as DNA or RNA polynucleotides. The target sequence may be located in the nucleus or cytoplasm of the cell, such as within an organelle of the cell. Generally, a sequence or template that may be used for recombination into the targeted locus comprising the target sequences is referred to as an "editing template" or "editing polynucleotide" or "editing sequence". In some aspects, an exogenous template polynucleotide may be referred to as an editing template. In some aspects, the recombination is homologous recombination.

[0255] Typically, in the context of an endogenous CRISPR system, formation of the CRISPR complex (comprising the guide sequence hybridized to the target sequence and complexed with one or more Cas proteins) results in cleavage of one or both strands in or near (e.g. within 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 50, or more base pairs from) the target sequence. The tracr sequence, which may comprise or consist of all or a portion of a wild-type tracr sequence (e.g. about or more than about 20, 26, 32, 45, 48, 54, 63, 67, 85, or more nucleotides of a wild-type tracr sequence), may also form part of the CRISPR complex, such as by hybridization along at least a portion of the tracr sequence to all or a portion of

a tracr mate sequence that is operably linked to the guide sequence. The tracr sequence has sufficient complementarity to a tracr mate sequence to hybridize and participate in formation of the CRISPR complex, such as at least 50%, 60%, 70%, 80%, 90%, 95% or 99% of sequence complementarity along the length of the tracr mate sequence when optimally aligned.

[0256] One or more vectors driving expression of one or more elements of the CRISPR system can be introduced into the cell such that expression of the elements of the CRISPR system direct formation of the CRISPR complex at one or more target sites. Components can also be delivered to cells as proteins and/or RNA. For example, a Cas enzyme, a guide sequence linked to a tracr-mate sequence, and a tracr sequence could each be operably linked to separate regulatory elements on separate vectors. Alternatively, two or more of the elements expressed from the same or different regulatory elements, may be combined in a single vector, with one or more additional vectors providing any components of the CRISPR system not included in the first vector. The vector may comprise one or more insertion sites, such as a restriction endonuclease recognition sequence (also referred to as a "cloning site"). In some embodiments, one or more insertion sites are located upstream and/or downstream of one or more sequence elements of one or more vectors. When multiple different guide sequences are used, a single expression construct may be used to target CRISPR activity to multiple different, corresponding target sequences within a cell.

[0257] A vector may comprise a regulatory element operably linked to an enzyme-coding sequence encoding the CRISPR enzyme, such as a Cas protein. Non-limiting examples of Cas proteins include Cas1, Cas1B, Cas2, Cas3, Cas4, Cas5, Cas6, Cas7, Cas8, Cas9 (also known as Csn1 and Csx12), Cas10, Csy1, Csy2, Csy3, Cse1, Cse2, Cse1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15, Csf1, Csf2, Csf3, Csf4, Cpf1 (Cas12a) homologs thereof, or modified versions thereof. These enzymes are known; for example, the amino acid sequence of *S. pyogenes* Cas9 protein may be found in the SwissProt database under accession number Q99ZW2.

[0258] The CRISPR enzyme can be Cas9 (e.g., from *S. pyogenes* or *S. pneumonia*). In some cases, Cpf1 (Cas12a) may be used as an endonuclease instead of Cas9. The CRISPR enzyme can direct cleavage of one or both strands at the location of a target sequence, such as within the target sequence and/or within the complement of the target sequence. The vector can encode a CRISPR enzyme that is mutated with respect to a corresponding wild-type enzyme such that the mutated CRISPR enzyme lacks the ability to cleave one or both strands of a target polynucleotide containing a target sequence. For example, an aspartate-to-alanine substitution (D10A) in the RuvC I catalytic domain of Cas9 from *S. pyogenes* converts Cas9 from a nuclease that cleaves both strands to a nickase (cleaves a single strand). In some embodiments, a Cas9 nickase may be used in combination with guide sequence(s), e.g., two guide sequences, which target respectively sense and antisense strands of the DNA target. This combination allows both strands to be nicked and used to induce NHEJ or HDR.

[0259] In some embodiments, an enzyme coding sequence encoding the CRISPR enzyme is codon optimized for

expression in particular cells, such as eukaryotic cells. The eukaryotic cells may be those of or derived from a particular organism, such as a mammal, including but not limited to human, mouse, rat, rabbit, dog, or non-human primate. In general, codon optimization refers to a process of modifying a nucleic acid sequence for enhanced expression in the host cells of interest by replacing at least one codon of the native sequence with codons that are more frequently or most frequently used in the genes of that host cell while maintaining the native amino acid sequence. Various species exhibit particular bias for certain codons of a particular amino acid. Codon bias (differences in codon usage between organisms) often correlates with the efficiency of translation of messenger RNA (mRNA), which is in turn believed to be dependent on, among other things, the properties of the codons being translated and the availability of particular transfer RNA (tRNA) molecules. The predominance of selected tRNAs in a cell is generally a reflection of the codons used most frequently in peptide synthesis. Accordingly, genes can be tailored for optimal gene expression in a given organism based on codon optimization.

[0260] In general, a guide sequence is any polynucleotide sequence having sufficient complementarity with a target polynucleotide sequence to hybridize with the target sequence and direct sequence-specific binding of the CRISPR complex to the target sequence. In some embodiments, the degree of complementarity between a guide sequence and its corresponding target sequence, when optimally aligned using a suitable alignment algorithm, is about or more than about 50%, 60%, 75%, 80%, 85%, 90%, 95%, 97%, 99%, or more.

[0261] Optimal alignment may be determined with the use of any suitable algorithm for aligning sequences, non-limiting example of which include the Smith-Waterman algorithm, the Needleman-Wunsch algorithm, algorithms based on the Burrows-Wheeler Transform (e.g. the Burrows Wheeler Aligner), Clustal W, Clustal X, BLAT, Novoalign (Novocraft Technologies, ELAND (Illumina, San Diego, Calif.), SOAP (available at soap.genomics.org.cn), and Maq (available at maq.sourceforge.net).

[0262] The CRISPR enzyme may be part of a fusion protein comprising one or more heterologous protein domains. A CRISPR enzyme fusion protein may comprise any additional protein sequence, and optionally a linker sequence between any two domains. Examples of protein domains that may be fused to a CRISPR enzyme include, without limitation, epitope tags, reporter gene sequences, and protein domains having one or more of the following activities: methylase activity, demethylase activity, transcription activation activity, transcription repression activity, transcription release factor activity, histone modification activity, RNA cleavage activity and nucleic acid binding activity. Non-limiting examples of epitope tags include histidine (His) tags, V5 tags, FLAG tags, influenza hemagglutinin (HA) tags, Myc tags, VSV-G tags, and thioredoxin (Trx) tags. Examples of reporter genes include, but are not limited to, glutathione-S-transferase (GST), horseradish peroxidase (HRP), chloramphenicol acetyltransferase (CAT) beta galactosidase, beta-glucuronidase, luciferase, green fluorescent protein (GFP), HcRed, DsRed, cyan fluorescent protein (CFP), yellow fluorescent protein (YFP), and auto-fluorescent proteins including blue fluorescent protein (BFP). A CRISPR enzyme may be fused to a gene sequence encoding a protein or a fragment of a protein that bind DNA

molecules or bind other cellular molecules, including but not limited to maltose binding protein (MBP), S-tag, Lex A DNA binding domain (DBD) fusions, GAL4A DNA binding domain fusions, and herpes simplex virus (HSV) BP16 protein fusions. Additional domains that may form part of a fusion protein comprising a CRISPR enzyme are described in US 20110059502, incorporated herein by reference.

IX. Methods of Treatment

[0263] In various embodiments CD70-targeting CAR constructs, nucleic acid sequences, vectors, host cells and so forth as contemplated herein and/or pharmaceutical compositions comprising the same are used for the prevention, treatment or amelioration of a cancerous disease, such as a tumorous disease. In particular embodiments, the pharmaceutical composition of the present disclosure may be particularly useful in preventing, ameliorating and/or treating cancer, including cancers that express CD70 and that may or may not be solid tumors, for example. The individual may utilize the treatment method of the disclosure as an initial treatment or after (or with) another treatment, such as following HSCT, for example. The immunotherapy methods may be tailored to the need of an individual with cancer based on the type and/or stage of cancer, and in at least some cases the immunotherapy may be modified during the course of treatment for the individual.

[0264] In some embodiments, the present disclosure provides methods for immunotherapy comprising administering an effective amount of the cells produced by methods of the present disclosure. In one embodiment, a medical disease or disorder is treated by transfer of cell populations produced by methods herein and that elicit an immune response. In certain embodiments of the present disclosure, cancer is treated by transfer of a cell population produced by methods of the disclosure and that elicits an immune response. Provided herein are methods for treating or delaying progression of cancer in an individual comprising administering to the individual an effective amount CD70-specific cell therapy. The present methods may be applied for the treatment of solid cancers or hematologic cancers.

[0265] Tumors for which the present treatment methods are useful include any malignant cell type, such as those found in a solid tumor or a hematological tumor. Exemplary solid tumors can include, but are not limited to, a tumor of an organ selected from the group consisting of acute myeloid leukemia, lymphoma, lung cancer, renal cancer, bladder cancer, melanoma, glioblastoma, breast cancer, head and neck cancer, mesothelioma, multiple myeloma, and pancreatic cancer.

[0266] In certain embodiments of the present disclosure, anti-CD70 CAR-expressing immune cells are delivered to an individual in need thereof, such as an individual that has cancer. The cells then enhance the individual's immune system to attack the cancer cells. In some cases, the individual is provided with one or more doses of the immune cells. In cases where the individual is provided with two or more doses of the immune cells, the duration between the administrations should be sufficient to allow time for propagation in the individual, and in specific embodiments the duration between doses is 1, 2, 3, 4, 5, 6, 7, or more days.

[0267] The NK cells for which the CD70-targeting CAR is utilized may be NK, T cells, or invariant NKT cells engineered for cell therapy for mammals, in particular embodiments. In such cases where the cells are NK cells, the NK

cell therapy may be of any kind and the NK cells may be of any kind. In specific embodiments, the cells are NK cells that have been engineered to express one or more CD70-targeting CARs and are provided to an individual in a therapeutically effective amount (in a range from 10^3 to 10^{10}) that ameliorates at least one symptom related to CD70-expressing cells in the individual. In specific embodiments, the cells are NK cells that are transduced with a CD70-targeting CAR. A therapeutically effective amount may be from 10^3 to 10^{10} , 10^3 to 10^9 , 10^3 to 10^8 , 10^3 to 10^7 , 10^3 to 10^6 , 10^3 to 10^5 , 10^3 to 10^4 , 10^4 to 10^{10} , 10^4 to 10^9 , 10^4 to 10^8 , 10^4 to 10^7 , 10^4 to 10^6 , 10^4 to 10^5 , 10^5 to 10^{10} , 10^5 to 10^9 , 10^5 to 10^8 , 10^5 to 10^7 , 10^5 to 10^6 , 10^6 to 10^{10} , 10^6 to 10^9 , 10^6 to 10^8 , 10^6 to 10^7 , 10^7 to 10^{10} , 10^7 to 10^9 , 10^7 to 10^8 , 10^8 to 10^{10} , 10^8 to 10^9 , or 10^9 to 10^{10} cells. Thus, in particular embodiments an individual having a CD70-positive cancer is provided once or multiple times a therapeutically effective amount of cells expressing one or more CD70-targeting CARs, including NK cells.

[0268] In particular embodiments, the present disclosure contemplates, in part, CD70 CAR-expressing cells, CD70-targeting CAR constructs, CD70-targeting CAR nucleic acid molecules and CD70-targeting CAR vectors that can administered either alone or in any combination using standard vectors and/or gene delivery systems, and in at least some aspects, together with a pharmaceutically acceptable carrier or excipient. In certain embodiments, subsequent to administration, the nucleic acid molecules or vectors may be stably integrated into the genome of the subject.

[0269] In specific embodiments, viral vectors may be used that are specific for certain cells or tissues and persist in NK cells, for example. Suitable pharmaceutical carriers and excipients are well known in the art. The compositions prepared according to the disclosure can be used for the prevention or treatment or delaying the above identified diseases.

[0270] Furthermore, the disclosure relates to a method for the prevention, treatment or amelioration of a tumorous disease comprising the step of administering to a subject in the need thereof an effective amount of cells that express a CD70-targeting CAR, a nucleic acid sequence, a vector, as contemplated herein and/or produced by a process as contemplated herein.

[0271] Possible indications for administration of the composition(s) of the exemplary CD70-targeting CAR cells are cancerous diseases, including tumorous diseases, including B cell malignancies, multiple myeloma, breast cancer, or lung cancer, for example. Exemplary indications for administration of the composition(s) of CD70-targeting CAR cells are cancerous diseases, including any malignancies that express CD70. The administration of the composition(s) of the disclosure is useful for all stages and types of cancer, including for minimal residual disease, early cancer, advanced cancer, and/or metastatic cancer and/or refractory cancer, for example.

[0272] Therapeutically effective amounts of the produced cells can be administered by a number of routes, including parenteral administration, for example, intravenous, intraperitoneal, intramuscular, intrasternal, intratumoral, intrathecal, intraventricular, through a reservoir, intraarticular injection, or infusion.

[0273] The therapeutically effective amount of the produced cells for use in adoptive cell therapy is that amount that achieves a desired effect in a subject being treated. For

instance, this can be the amount of immune cells necessary to inhibit advancement, or to cause regression of cancer.

[0274] The produced cell population can be administered in treatment regimens consistent with the disease, for example a single or a few doses over one to several days to ameliorate a disease state or periodic doses over an extended time to inhibit disease progression and prevent disease recurrence. The precise dose to be employed in the formulation will also depend on the route of administration, and the seriousness of the disease or disorder, and should be decided according to the judgment of the practitioner and each patient's circumstances. The therapeutically effective amount of cells will be dependent on the subject being treated, the severity and type of the affliction, and the manner of administration. In some embodiments, doses that could be used in the treatment of human subjects range from at least 1×10^3 , at least 1×10^4 , 3.8×10^4 , at least 3.8×10^5 , at least 3.8×10^6 , at least 3.8×10^7 , at least 3.8×10^8 , at least 3.8×10^9 , or at least 3.8×10^{10} T cells/ m^2 . In a certain embodiment, the dose used in the treatment of human subjects ranges from about 3.8×10^9 to about 3.8×10^{10} T cells/ m^2 . In additional embodiments, a therapeutically effective amount of T cells can vary from about 5×10^6 cells per kg body weight to about 7.5×10^8 cells per kg body weight, such as about 2×10^7 cells to about 5×10^8 cells per kg body weight, or about 5×10^7 cells to about 2×10^8 cells per kg body weight. The exact amount of T cells is readily determined by one of skill in the art based on the age, weight, sex, and physiological condition of the subject. Effective doses can be extrapolated from dose-response curves derived from in vitro or animal model test systems.

[0275] The disclosure further encompasses co-administration protocols with other compounds, e.g. bispecific antibody constructs, targeted toxins or other compounds, which act via immune cells. The clinical regimen for co-administration of the inventive compound(s) may encompass co-administration at the same time, before or after the administration of the other component. Particular combination therapies include chemotherapy, radiation, surgery, hormone therapy, or other types of immunotherapy.

[0276] Embodiments relate to a kit comprising a CD70-targeting CAR construct as defined herein, a nucleic acid sequence as defined herein, a vector as defined herein and/or a host as defined herein. It is also contemplated that the kit of this disclosure comprises a pharmaceutical composition as described herein above, either alone or in combination with further medicaments to be administered to an individual in need of medical treatment or intervention.

X. Kits of the Disclosure

[0277] Any of the compositions described herein may be comprised in a kit. In a non-limiting example, the kit comprises anti-CD70 CAR molecules, cells encompassing same, and/or reagents to generate same, and any of these may be comprised in suitable container means in a kit of the present disclosure. Kits may comprise immune cells, including NK cells, vectors, expression construct polynucleotides for insertion into a vector (whether viral or not), CD27 sequences of any kind and as encompassed herein, and so forth. Primers for amplification of any polynucleotide may be included. In some cases, the kit comprises cryopreserved cells, including NK cells. Any reagents for transfection or transduction of the cells may be included.

[0278] The compositions of the kits may be packaged either in aqueous media or in lyophilized form. The container means of the kits will generally include at least one vial, test tube, flask, bottle, syringe or other container means, into which one or more components may be placed, and preferably, suitably aliquoted. Where there are more than one component in the kit, the kit also may generally contain a second, third or other additional container into which the additional components may be separately placed. However, various combinations of components may be comprised in a vial. The kits of the present disclosure also will typically include a means for containing the anti-CD70 CAR molecules, cells encompassing same, and/or reagents to generate same in close confinement for commercial sale. Such containers may include injection or blow molded plastic containers into which the desired vials are retained.

[0279] When the components of the kit are provided in one and/or more liquid solutions, the liquid solution is an aqueous solution, with a sterile aqueous solution being particularly envisioned. The compositions may also be formulated into a syringeable composition. In which case, the container means may itself be a syringe, pipette, and/or other such like apparatus, from which the formulation may be applied to an infected area of the body, injected into an animal, and/or even applied to and/or mixed with the other components of the kit.

[0280] However, the components of the kit may be provided as dried powder(s). When reagents and/or components are provided as a dry powder, the powder can be reconstituted by the addition of a suitable solvent. It is envisioned that the solvent may also be provided in another container means.

[0281] Irrespective of the number and/or type of containers, the kits of the disclosure may also comprise, and/or be packaged with, an instrument for assisting with the injection/administration and/or placement of the ultimate composition within the body of an animal. Such an instrument may be a syringe, pipette, forceps, and/or any such medically approved delivery vehicle. In some embodiments, reagents or apparatuses or containers are included in the kit for ex vivo use.

EXAMPLES

[0282] The following examples are included to demonstrate preferred embodiments of the invention. It should be appreciated by those of skill in the art that the techniques disclosed in the examples that follow represent techniques discovered by the inventors to function well in the practice of the invention, and thus can be considered to constitute preferred modes for its practice. However, those of skill in the art should, in light of the present disclosure, appreciate that many changes can be made in the specific embodiments which are disclosed and still obtain a like or similar result without departing from the spirit and scope of the invention.

Example 1

CD27-Comprising Car to Target CD70-Positive Tumors

[0283] The inventors first confirmed CD70 expression on AML patient samples. A Tsne plot from mass cytometry data showed high expression of CD70 on primary AML samples (n=54) but not on healthy CB CD34+ cells (n=10) (FIG.

1A). As shown in FIG. 1, Tsne plots also showed various myeloid markers on lineage neg cells from CB (upper CB population from FIG. 1A) and AML samples (lower AML population from FIG. 1A). CD70 expression in AML samples was similar to other well-characterized myeloid markers.

[0284] FIG. 3 demonstrates that TrCD27 constructs showed higher transfection efficiency compared to flCD27 constructs. 293T cells were transfected with various CD27 constructs, as shown in figure, using GeneJuice as a transfection reagent. The transfection efficiency was determined by checking the surface expression of CD27 in 293T cells post virus collection using flow cytometry for CD27 constructs. Non transduced (NT) cells were used as control. We also included constructs containing only IL15 (i.e. IL15) and CD70scFv CAR construct containing single heavy and light chains derived from commercially available CD70 antibody sequence with IL15 (i.e. CD70-IL15). The surface expression of IgG was used to determine the transfection efficiency for the latter two constructs

[0285] FIG. 4 demonstrates that TrCD27 constructs showed higher transduction efficiency compared to flCD27 constructs. The retroviral supernatant collected from transfection experiment (FIG. 3) were used to transduce cord blood derived NK (CBNK) cells and RetroNectin was used to enhance the transduction efficiency. Forty-eight hours post transduction, the transduction efficiency was measured by checking the surface expression of CD27 on CBNK cells using flow cytometry. Supernatant from non-transduced (NT) cells were used as control. For IL15 and CD70 CAR-IL15 CAR constructs, the surface expression of IgG on CBNK cells was used to determine the transduction efficiency. Because flCD27 constructs under these conditions show poor transfection and transduction efficiency, focus on trCD27 constructs was emphasized.

[0286] In FIGS. 5A-5C, CBNK cells transduced with trCD27 constructs induced cancer cell apoptosis. Annexin V-LIVE/DEAD™ Fixable Aqua staining assay was performed to investigate if CBNK cells transduced with various trCD27 constructs could induce death of CD70-expressing cancer cells. Raji and Karpas cells were used as targets since both have relatively high surface expression of CD70 (FIG. 5A, FIG. 5B, respectively). NK cells transduced with TrCD27 constructs and cancer cells were co-cultured at a 1:1 ratio for 4 hrs and Annexin V-LIVE/DEAD™ Fixable Aqua staining was performed. Various trCD70 transduced CBNK cells (such as #3, #13, #15, #17, #19 #20, etc.) exerted superior killing of both Raji (black bar) and Karpas (grey bar) cells, when compared to NT, IL15 or CD70-IL15 CAR transduced CBNK cells (FIG. 5C).

[0287] In FIG. 6, CB-NK cells transduced with trCD27 constructs showed enhanced cytotoxic activity against cancer cells. The degranulation marker CD107a is a widely accepted marker for the cytotoxic activity of NK cells. TrCD27 CAR transduced CBNK cells and cancer cells (Raji and Karpas) were co-cultured at a 1:1 ratio in the presence of CD107a antibody for 6 hrs. After incubation, CD107a expression is assessed by flow cytometry. Most of the trCD70 transduced CB-NK cells showed enhanced cytotoxic activity against Raji (middle black bar) and Karpas (right gray bar) cells when compared to NT, IL15 or CD70-IL15 CAR transduced CB-NK cells. CB-NK cells in the absence of co-culture with cancer cells shows minimal basal expression level of CD107a (left green bar).

[0288] FIGS. 7A-7B demonstrate that CB-NK cells transduced with trCD27 constructs show enhanced cytotoxic activity against cancer cells. To analyze the real-time cytotoxic activity of trCD27 CAR-transduced CBNK cells against cancer cells, an IncuCyte cytotoxicity assay was performed. TrCD27 CAR-transduced CB-NK cells and Raji cells (FIG. 7A) or Karpas cells (FIG. 7B) were co-cultured at 1:1 ratio, and real-time cytotoxicity of NK cells against Raji and Karpas cells was measured every hour over a 12 hour period. All of the trCD70 CAR-transduced CB-NK cells used in this assay showed enhanced cytotoxic activity against Raji (left panel) and Karpas cells (right panel) when compared to NT, IL15 or CD70-IL15 CAR transduced CB-NK cells.

[0289] FIGS. 8A-8B demonstrate that CB-NK cells transduced with trCD27 CARs have effective anti-tumor activity against CD70-positive Raji cells in vivo. NSG mice were engrafted with firefly luciferase labelled Raji cells (Raji-FFluc) which have high CD70 expression. Mice were injected with 20,000 Raji cells/mouse and 5M CBNK cells/mouse. Bioluminescence imaging of the tumor showed that CB-NK cells transduced with trCD27 CARs were able to reduce tumor burden due to CD70 high Raji cells (FIG. 8A). Additionally, survival curve showed the significant survival benefit on NSG mice with Raji, when infused with single dose of CB-NK cells transduced with various trCD27 CARs when compared to non-transduced (NT) cells and cells transduced with IL15 or CD70-IL15 CARs (FIG. 8B). The trCD27 #23 CAR construct showed the most significant survival benefit. $p=0.002$ (Raji+NT vs Raji+CD27 #23); $p=0.0018$ (Raji+IL15 vs Raji+CD27 #23); $p=0.001$ (Raji+CD70-IL15 vs Raji+CD27 #23). Mantel-Cox test was used to measure p value.

[0290] FIGS. 9A-9B demonstrate that CB-NK cells transduced with trCD27 CARs have effective anti-tumor activity against CD70-positive AML cells in vivo. NSG mice were

engrafted with firefly luciferase labelled THP-1 cells (THP1-FFluc), which have high CD70 expression. Mice were injected with 100,000 THP1 cells/mouse and 5M CBNK cells/mouse. Bioluminescence imaging of the tumor showed that CB-NK cells transduced with trCD27 CARs were able to reduce tumor burden due to CD70 high THP-1 cells (FIG. 9A). Additionally, survival curves showed the significant survival benefit on NSG mice with THP-1, when infused with single dose of CB-NK cells transduced with various trCD27 CARs when compared to non-transduced (NT) cells and cells transduced with IL15 or CD70-IL15 CARs (FIG. 9B). The trCD27 #15 and trCD27 #23 CAR constructs showed the most significant survival benefit. $p=0.0027$ (THP-1+NT vs THP-1+CD27 #15 or #23); $p=0.001$ (THP-1+IL15 vs THP-1+CD27 #15 or #23); $p=0.001$ (THP-1+CD70-IL15 vs THP-1+CD27 #15 or #23). Mantel-Cox test was used to measure p value.

[0291] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the design as defined by the appended claims. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

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Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro
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Cys Ile Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His
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35 40 45
Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro
50 55 60
Cys Ile Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His
65 70 75 80
Cys Glu Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys
85 90 95
Thr Ile Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys
100 105 110
Arg Asp Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu
115 120 125
Thr Ala Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr His
130 135 140
Leu Pro Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His Met
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Trp Ala Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr Phe
 35 40 45

Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro
 50 55 60

Cys Ile Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His
 65 70 75 80

Cys Glu Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys
 85 90 95

Thr Ile Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys
 100 105 110

Arg Asp Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu
 115 120 125

Thr Ala Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr His
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Leu Pro Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His Met
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Gln Thr Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser Thr
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His Trp Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg Ile
 180 185 190

Leu Val Ile Phe Ser Gly Met Phe Leu Val Phe Thr Leu Ala Gly Ala
 195 200 205

Leu Phe Leu His Gln Arg Arg Lys Tyr Arg Ser Asn Lys Gly Glu Ser
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Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His Cys Glu Ser
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Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys Thr Ile Thr
65 70 75 80
Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys Arg Asp Lys
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Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu Thr Ala Arg
100 105 110
Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr His Leu Pro Tyr
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Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His Met Gln Thr Leu
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Ala Leu Phe Leu His
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 cttgttttca ccctggccgg ggccctgttc ctccatcaac gaaggaaata tagatcaaac 660
 aaaggagaaa gtctgtgga gctgcagag ccttgctggt acagctgccc caggaggagg 720
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 tga 783

<210> SEQ ID NO 10

<211> LENGTH: 113

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 10

Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly
 1 5 10 15
 Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr
 20 25 30
 Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys
 35 40 45
 Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys
 50 55 60
 Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg
 65 70 75 80
 Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala
 85 90 95

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Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
 100 105 110

Gly

<210> SEQ ID NO 11
 <211> LENGTH: 41
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 11

Arg Ser Lys Arg Ser Arg Leu Leu His Ser Asp Tyr Met Asn Met Thr
 1 5 10 15

Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln Pro Tyr Ala Pro
 20 25 30

Pro Arg Asp Phe Ala Ala Tyr Arg Ser
 35 40

<210> SEQ ID NO 12
 <211> LENGTH: 42
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 12

Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro Phe Met
 1 5 10 15

Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys Arg Phe
 20 25 30

Pro Glu Glu Glu Gly Gly Cys Glu Leu
 35 40

<210> SEQ ID NO 13
 <211> LENGTH: 780
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 13

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acaccggctc cgaagtctcg cccggagcgg cattattggg cacagggcaa gttgtgttgt	120
caaatgtgtg agccgggaac ctttctcgtg aaggattgcy atcagcatcg gaaggccgcg	180
cagtgcgacc catgtatacc aggggtctca ttttcccag atcaccatac gaggccgcac	240
tgtgagtctt gcaggcattg taattccggc ttgttggtcc gcaactgtac tattactgcy	300
aatgcagagt gtgctttag aaacggatgg cagtgcaggg acaaagaatg tacggagtgt	360
gatccactgc ctaaccccag tcttacagca agatcttcac aggcctcag cccgcacct	420
caaccaacac atcttcctta cgtgtcagaa atgttgagg cgcgaaaccg aggccatatg	480
cagaccctgg cggactttcg gcagctgcc gacgcacac ttagtacaca ctggccacca	540
caacgcagct tgtgtcttc cgatttcac cgcatactgg tcatcttttc tggaatgttc	600
cttgtgttca ccttggcagg agccctgttc cttcaccaga gacgcaagta caggtcaaac	660

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aagggtgaga gccccgttga acccgagag ccgtgtagat actcatgtcc tagagaagaa 720
gagggctcta ctatccctat tcaggaagat tatagaaaac ccgaaccgc gtgcagcccc 780

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<210> SEQ ID NO 14
<211> LENGTH: 22
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 14

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Met Leu Leu Leu Val Thr Ser Leu Leu Leu Cys Glu Leu Pro His Pro
1          5          10          15

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Ala Phe Leu Leu Ile Pro
20

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<210> SEQ ID NO 15
<211> LENGTH: 654
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

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<400> SEQUENCE: 15

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actccagccc ccaagagctg cccagagagg cactactggg ctccaggaaa gctgtgctgc 120
cagatgtgtg agccaggaac attcctcgtg aaggactgtg accagcatag aaaggctgct 180
cagtgtgatc cttgcatacc gggggctctc ttctctcctg accaccacac ccggccccac 240
tgtgagagct gtcggcactg taactctggt cttctcgttc gcaactgcac catcactgcc 300
aatgtctagt gtgcctgtcg caatggctgg cagtgcaggg acaaggagtg caccgagtgt 360
gatcctcttc caaaccttc gctgaccgct cggtcgtctc aggccttgag cccacaccct 420
cagccccccc acttacctta tgtcagttag atgctggagg ccaggacagc tgggcacatg 480
cagactctgg ctgacttcag gcagctgcct gcccggaact tctctaccca ctggccaccc 540
caaagatccc tgtgcagctc cgattttatt cgcttttggg tgctggtggt ggttggtgga 600
gtcctgggctt gctatagctt gctagtaaca gtggccttta ttattttctg ggtg 654

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<210> SEQ ID NO 16
<211> LENGTH: 2004
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

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<400> SEQUENCE: 16

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atgacaagag ttactaacag cccctctctc caagctcact tacaggctct ctacttagtc 60
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gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac 180
ctagaacctc gctggaaaag accttacaca gtctctgtga ccacccccac cgccctcaaa 240
gtagacggca tcgcagcttg gatacacgcc gccacgtga aggctgccga ccccggggggt 300
ggaccatcct ctagactgcc atgctcgagg atggcacggc cacatccctg gtggctgtgc 360
gttctgggga ccctgggtggg gctctcagct actccagccc ccaagagctg cccagagagg 420

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cactactggg ctcagggaaa gctgtgctgc cagatgtgtg agccaggaac attcctcgtg	480
aaggactgtg accagcatag aaaggctgct cagtgtgato cttgcatacc gggggtctcc	540
ttctctcctg accaccacac cgggccccac tgtgagagct gteggcactg taactctggt	600
cttctcgttc gcaactgcac catcactgcc aatgctgagt gtgcctgtcg caatggctgg	660
cagtgcaggg acaaggagtg caccgagtgt gatcctcttc caaacccctc gctgaccgct	720
cggtcgtctc aggcctgag cccacaccct cagccccc accctaccta tgtcagtgag	780
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gcccggactc tctctaccca ctggccaccc caaagatccc tgtgcagctc cgattttatt	900
cgcttttggg tgctgggtgt ggttggtgga gtctggctt gctatagctt gctagtaaca	960
gtggccttta ttattttctg ggtgaaacgg ggcagaaaga aactcctgta tatattcaaa	1020
caaccattta tgagaccagt acaactact caagaggag atggctgtag ctgccgattt	1080
ccagaagaag aagaaggagg atgtgaactg aaacgcgtga agttcagcag gagcgagac	1140
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gaggagtacg atgttttggg caaaagacgt ggcggggacc ctgagatggg gggaaagccg	1260
agaagggaaga accctcagga aggcctgtac aatgaactgc agaaagataa gatggcgag	1320
gcctacagtg agattgggat gaaaggcgag cgccggaggg gcaaggggca cgatggcctt	1380
taccagggtc tcagtacagc caccaaggac acctacgacg cccttcacat gcaggccctg	1440
ccccctcgcg gaccgcagtg tactaattat gctctcttga aattggctgg agatgttgag	1500
agcaatcccg ggcccatcg ccttagcaag cccacactgc ggagcatcag catccagtgc	1560
tacctgtgcc tgctgtgtaa cagccacttc ctgaccgagg ccggcatcca cgtgttcac	1620
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gacctgaaga agatcgagga cctgatccag agcatgcaca tcgacgccac cctgtacacc	1740
gagagcgacg tgcacccag ctgcaagggt accgccatga agtgctttct gctggaactg	1800
cagggtgatca gcctggaaag cggcgacgcc agcatccacg acaccgtgga gaacctgatc	1860
atcctggcca acaacagcct gagcagcaac ggcaacgtga ccgagagcgg ctgcaaagag	1920
tgcgaggaa tggaagagaa gaacatcaaa gagtttctgc agagcttcgt gcacatcgtg	1980
cagatgttca tcaacaccag ctga	2004

<210> SEQ ID NO 17

<211> LENGTH: 2013

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 17

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cagcacgaag tctggagacc tctggcgga gcctaccaag aacaactgga ccgaccggtg	120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaaagg accttacaca gtctgtgtga ccacccccac cgccctcaaa	240
gtagacggca tcgcagcttg gatacacgcc gccacgtga aggctgccga ccccgggggt	300
ggaccatcct ctgactgcc atgctcgagg atgcttctcc tggtgacaag ccttctgctc	360

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tgtgagttac cacaccagc attcctcctg atcccagcta caccggtcc gaagtccctgc	420
ccggagcggc attattgggc acagggcaag ttgtgttgtc aaatgtgtga gccgggaacc	480
tttctcgtga aggattgcca tcagcatcgg aaggccgcgc agtgcgaccc atgtatacca	540
ggggtctcat tttcccaga tcaccatacg aggcgcact gtgagtcttg caggcattgt	600
aattccggct tgttggtccg caactgtact attactgcga atgcagagtg tgcttgtaga	660
aacggatggc agtgcaggga caaagaatgt acggagtgtg atccactgcc taaccccagt	720
cttacagcaa gatcttcaca ggccctcagc ccgcatactc aaccaacaca tcttccttac	780
gtgtcagaaa tgttgagggc gcgaaccgca ggccatatgc agaccctggc ggactttcgg	840
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gatttcaccc gcttttgggt gctggtggtg gttggtggag tctggcttg ctatagcttg	960
ctagtaacag tggcctttat tattttctgg gtgaaacggg gcagaaagaa actcctgtat	1020
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tgccgatttc cagaagaaga agaaggagga tgtgaactga aacgcgtgaa gttcagcagg	1140
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gatggccttt accagggtct cagtacagcc accaaggaca cctacgcgc ccttcacatg	1440
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gatgttgaga gcaatcccg ggccatgcgc attagcaagc cccacctgcg gagcatcagc	1560
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gtgttcaccc tgggtgctt cagcgccgga ctgcccaga ccgaggccaa ctgggtgaac	1680
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ctgtacaccg agagcgacgt gcaccccagc tgcaaggtag cggccatgaa gtgctttctg	1800
ctggaactgc aggtgatcag cctggaaagc ggcgacgcca gcatccaga caccgtggag	1860
aacctgatca tctcggccaa caacagcctg agcagcaacg gcaacgtgac cgagagcggc	1920
tgcaaagagt gcgaggaact ggaagagaag aacatcaaag agtttctgca gagcttcgtg	1980
cacatcgtgc agatgttcat caacaccagc tga	2013

<210> SEQ ID NO 18

<211> LENGTH: 1950

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 18

atgacaagag ttactaacag cccctctctc caagctcact tacaggtctt ctacttagtc	60
cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga ccgaccggtg	120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaaagg accttacaca gtctctgtga ccacccccac cggcctcaaa	240
gtagacggca tcgcagcttg gatacacgcc gccacgtga aggtgcgca ccccgggggt	300

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ggaccatcct ctagactgcc atgctcgagg atggcacggc cacatccctg gtggctgtgc	360
gttctgggga ccttgggtgg gctctcagct actccagccc ccaagagctg cccagagagg	420
cactactggg ctcaggga aa gctgtgctgc cagatgtgtg agccaggaac attcctcgtg	480
aaggactgtg accagcatag aaaggctgct cagtgtgatc cttgcatacc gggggctctc	540
ttctctcctg accaccacac ccggcccccac tgtgagagct gtcggcactg taactctggt	600
cttctcgttc gcaactgcac catcactgcc aatgctgagt gtgcctgtcg caatggctgg	660
cagtgcaggg acaaggagtg caccgagtgt gatcctcttc caaaccttc gctgaccgct	720
cggctgtctc aggcctgag cccacaccct cagccacccc acttacctta tgtcagttag	780
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cgcttttggg tgctgggtgt ggttgggtga gtcctggctt gctatagctt gctagtaaca	960
gtggccttta ttattttctg ggtgctttgc gcacgccac gccgcagccc cgccaagaa	1020
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gcagacgccc ccgctacca gcagggccag aaccagctct ataacgagct caatctagga	1140
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aagccgagaa ggaagaacct tcaggaaggc ctgtacaatg aactgcagaa agataagatg	1260
gcgaggcct acagttagat tgggatgaaa ggcgagcgcc ggaggggcaa ggggcacgat	1320
ggcctttacc aggttctcag tacagccacc aaggacacct acgacgacct tcacatgcag	1380
gccttgcccc ctccgggacc gcagtgtact aattatgctc tcttgaaatt ggctggagat	1440
gttgagagca atcccgggc catgcgcat agcaagcccc acctgaggag catcagcatc	1500
cagtgtctacc tgtgctgct gctgaacagc cacttctga ccgaggccgg catccacgtg	1560
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tacaccgaga gcgacgtgca cccagctgc aagggtgacc ccatgaagtg ctttctgctg	1740
gaactgcagg tgatcagcct ggaaagcggc gacgccagca tccacgacac cgtggagAAC	1800
ctgatcatcc tggccaacaa cagcctgagc agcaacggca acgtgaccga gagcggtgc	1860
aaagagtgcg aggaactgga agagaagaac atcaaagagt ttctgcagag ctctgtgcac	1920
atcgtgcaga tgttcacaa caccagctga	1950

<210> SEQ ID NO 19

<211> LENGTH: 1959

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 19

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cagcacaag tctggagacc tctggcggca gcctaccaag aacaactgga ccgaccggtg	120
gtacctcacc cttaccgagt cgccgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaagg accttacaca gtctgtgtga ccacccccc cgccctcaaa	240
gtagacggca tcgacgcttg gatacacgcc gccacgtga aggtgcccga ccccggggt	300

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ggaccatcct ctagactgcc atgctcgagg atgcttctcc tggtgacaag ccttctgctc	360
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cgggagcggc attattgggc acagggcaag ttgtgttgtc aaatgtgtga gccgggaacc	480
tttctcgtga aggattgcga tcagcatcgg aaggccgcgc agtgcgaccc atgtatacca	540
ggggtctcat tttcccaga tcaccatacg aggccgact gtgagtcttg caggcattgt	600
aattccggct tgttggtccg caactgtact attactgcga atgcagagtg tgctttaga	660
aacggatggc agtgcaggga caaagaatgt acggagtgtg atccactgcc taaccccagt	720
cttacagcaa gatcttcaca ggccctcagc ccgcatectc aaccaacaca tcttccttac	780
gtgtcagaaa tgttgaggc gcgaaccgca ggccatagc agaccctggc ggactttcgg	840
cagctgccag cacgcacact tagtacacac tggccaccac aacgcagctt gtgctcttcc	900
gatttcaccc gcttttgggt gctggtggtg gttggtggag tcctggcttg ctatagcttg	960
ctagtaacag tggcctttat tattttcttg gtgctttgag caegcccacg ccgcagcccc	1020
gcccagaag atggcaaagt ctacatcaac atgccaggca ggggcaaacg cgtgaagttc	1080
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gtggagaacc tgatcatcct ggccaacaac agcctgagca gcaacggcaa cgtgaccgag	1860
agcggctgca aagagtgcga ggaactggaa gagaagaaca tcaaagagtt tctgcagagc	1920
ttcgtgcaca tcgtgcagat gttcatcaac accagctga	1959

<210> SEQ ID NO 20

<211> LENGTH: 2034

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 20

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cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga ccgaccggtg	120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaaagg accttacaca gtctctgtga ccacccccac cgcctcctaaa	240
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ggaccatcct ctagactgcc atgctcgagg atggcacggc cacatccctg gtggctgtgc	360
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cactactggg ctcaggga aa gctgtgctgc cagatgtgtg agccaggaac attcctcgtg	480
aaggactgtg accagcatag aaaggtctgt cagtgtgatc cttgcatacc gggggctctc	540
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cttctcgttc gcaactgcac catcactgcc aatgctgagt gtgcctgtcg caatggctgg	660
cagtgcaggg acaaggagtg caccgagtgat gatcctcttc caaacccctc gctgaccgct	720
cggctgtctc agggccctgag cccacaccct cagcccccac acttacctta tgtcagttag	780
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gcccggactc tctctaccca ctggccaccc caaagatccc tgtgcagctc cgattttatt	900
cgcttttggg tgctgggtgt ggttggtgga gtccctggctt gctatagctt gctagtaaca	960
gtggccttta ttattttctg ggtgtacttc ctgggcccgc tggctccctg ggggcgaggg	1020
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ctccagggtc agaggctgga tgtctacagc gacctcaaca cacagaggcc gtattacaaa	1140
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cgccggaggg gcaaggggca cgatggcctt taccagggtc tcagtacagc caccaaggac	1440
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gctctcttga aattggcttg agatgttgag agcaatcccg ggcccatgcg cattagcaag	1560
ccccacctgc ggagcatcag catccagtgc tacctgtgcc tgctgctgaa cagccacttc	1620
ctgaccgagg ccggcatcca cgtgttcac cttggctgct tcagcgccgg actgcccaag	1680
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agcatgcaca tcgacgccac cctgtacacc gagagcgacg tgcaccccag ctgcaagggtg	1800
accgcatga agtgctttct gctggaactg cagggtgatc gcctggaaa cggcgacgcc	1860
agcatccacg acaccgtgga gaacctgatc atcctggcca acaacagcct gagcagcaac	1920
ggcaacgtga ccgagagcgg ctgcaagag tgcgaggaac tggaagagaa gaacatcaaa	1980
gagtttctgc agagcttcgt gcacatcgtg cagatgttca tcaacaccag ctga	2034

<210> SEQ ID NO 21

<211> LENGTH: 2043

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 21

atgacaagag ttactaacag cccctctctc caagctcact tacaggctct ctacttagtc	60
cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga ccgaccggtg	120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaaagg accttacaca gtccctgtga ccacccccac cggccctcaaa	240

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gtagacggca tgcagcttg gatacacgcc gccacgtga aggetgccga ccccggggt	300
ggaccatcct ctgactgcc atgctcgagg atgcttctcc tggtgacaag ccttctgctc	360
tgtgagttac cacaccagc attcctctg atcccagcta caccggctcc gaagtctgc	420
ccggagcggc attattgggc acagggaag ttgtgtgtc aaatgtgtga gccgggaacc	480
tttctcgtga aggattgcga tcagcatcgg aaggccgcgc agtgcgaccc atgtatacca	540
ggggtctcat tttcccaga tcaccatacg aggcgcact gtgagtctt caggcattgt	600
aattccggct tgttggtccg caactgtact attactgcga atgcagagtg tgctttaga	660
aacggatggc agtgacggga caaagaatgt acggagtgtg atccactgcc taaccccagt	720
cttacagcaa gatcttcaca ggccctcagc ccgcatcctc aaccaacaca tcttccttac	780
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gatttcatcc gcttttgggt gctggtggtg gttggtggag tcttggett ctatagcttg	960
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ggcgacgcca gcatccagc caccgtggag aacctgatca tcttgccaa caacagcctg	1920
agcagcaacg gcaacgtgac cgagagcggc tgcaaaagat gcgaggaact ggaagagaag	1980
aacatcaaa agtttctgca gagcttcgtg cacatcgtgc agatgttcat caacaccagc	2040
tga	2043

<210> SEQ ID NO 22

<211> LENGTH: 2031

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 22

atgacaagag ttactaacag cccctctctc caagctcact tacaggctct ctacttagtc	60
cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga ccgaccggtg	120

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gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaagg accttacaca gtccctgctga cccccccac cgccctcaaa	240
gtagacggca tcgcagcttg gatacacgcc gccacgtga aggctgccga ccccggggt	300
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aaggactgtg accagcatag aaaggctgct cagtgtgatc cttgcatacc gggggtctcc	540
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cttctcgttc gcaactgcac catcactgcc aatgctgagt gtgcctgtcg caatggctgg	660
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gccatgaagt gctttctgct ggaactgcag gtgatcagcc tggaaagcgg cgacgccagc	1860
atccacgaca ccgtggagaa cctgatcacc ctggccaaca acagcctgag cagcaacggc	1920
aacgtgaccg agagcggctg caaagagtgc gaggaactgg aagagaagaa catcaaagag	1980
ttctgcaga gcttcgtgca catcgtgcag atgttcacca acaccagctg a	2031

<210> SEQ ID NO 23

<211> LENGTH: 2040

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 23

atgacaagag ttactaacag cccctctctc caagctcact tacaggctct ctacttagtc	60
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cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga cgcaccggtg	120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaaagg accttacaca gtctgtctga ccacccccac cgcctcaaaa	240
gtagacggca tcgcagcttg gatacacgcc gccacagtga aggctgccga ccccgggggt	300
ggaccatcct ctagactgcc atgctcgagg atgcttctcc tggtgacaag ccttctgtctc	360
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aattccggct tgttggtccg caactgtact attactgcga atgcagagtg tgctttaga	660
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aaggtgaccg ccatgaagtg ctttctgctg gaactgcagg tgatcagcct ggaaagcggc	1860
gacgccagca tccacgacac cgtggagaac ctgatcatcc tggccaacaa cagcctgagc	1920
agcaacggca acgtgaccga gagcggctgc aaagagtgcg aggaactgga agagaagaac	1980
atcaaagagt ttctgcagag cttcgtgcac atcgtgcaga tgttcacaa caccagctga	2040

<210> SEQ ID NO 24

<211> LENGTH: 1986

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 24

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gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaaagg accttacaca gtctgtgtga cccccccac cgccctcaaa	240
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cactactggg ctcagggaag gctgtgtgtc cagatgtgtg agccaggaa attcctcgtg	480
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cttctcggtc gcaactgcac catcactgcc aatgctgagt gtgcctgtcg caatggctgg	660
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ctgagcagca acggcaacgt gaccgagagc ggctgcaaag agtgcgagga actggaagag	1920
aagaacatca aagagtttct gcagagcttc gtgcacatcg tgcagatgtt catcaacacc	1980
agctga	1986

<210> SEQ ID NO 25

<211> LENGTH: 1995

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

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<400> SEQUENCE: 25

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gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
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ggaccatctc ctgactgcc atgctcgagg atgcttctcc tggtgacaag ccttctgttc	360
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ccggagcggc attattgggc acagggcaag ttgtgtgtgc aaatgtgtga gccgggaacc	480
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aacaacagcc tgagcagcaa cggcaacgtg accgagagcg gctgcaaaga gtgcgaggaa	1920
ctggaagaga agaacatcaa agagtttctg cagagcttcg tgcacatcgt gcagatgttc	1980
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<210> SEQ ID NO 26

<211> LENGTH: 1860

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 26

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gacgccagca tccacgacac cgtggagaac ctgatcatcc tggccaacaa cagcctgagc   1740
agcaacggca acgtgaccga gagcggtgc aaagagtgcg aggaactgga agagaagaac   1800
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<210> SEQ ID NO 27

<211> LENGTH: 1866

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 27

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gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
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ctgagcgaca acggcaacgt gaccgagagc ggctgcaaag agtgcgagga actggaagag	1800
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<210> SEQ ID NO 28

<211> LENGTH: 1980

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 28

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cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga cgcaccggtg	120
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agcaacggca acgtgaccga gagcggtgc aaagagtgcg aggaactgga agagaagaac	1920
atcaaagagt ttctgcagag ctcgtgcac atcgtgcaga tgttcatcaa caccagctga	1980

<210> SEQ ID NO 29

<211> LENGTH: 1989

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 29

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ctagaacctc gctggaaagg accttacaca gtctgtgtga ccacccccac cgcctcaaaa	240
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agcctgagca gcaacggcaa cgtgaccgag agcggctgca aagagtgcga ggaactggaa	1920
gagaagaaca tcaagagtt tctgcagagc ttcgtgcaca tcgtgcagat gttcatcaac	1980
accagctga	1989

<210> SEQ ID NO 30

<211> LENGTH: 1998

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 30

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ctagaacctc gctggaaagg accttacaca gtctgtctga ccacccccac cgccctcaaa	240
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gacgtgcacc ccagctgcaa ggtgaccgcc atgaagtgtt ttctgtctga actgcaggtg	1800
atcagcctgg aaagcgcgca cgccagcatc caccacaccg tggagaacct gatcatcctg	1860
gccacaaca gcctgagcag caacggcaac gtgaccgaga gcggctgcaa agagtgcgag	1920
gaactggaag agaagaacat caaagagttt ctgcagagct tcgtgcacat cgtgcagatg	1980
ttcatcaaca ccagctga	1998

<210> SEQ ID NO 31

<211> LENGTH: 2007

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

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<400> SEQUENCE: 31

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gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac      180
ctagaacctc gctggaagg accttacaca gtctgtgtga ccacccccac cgccctcaaa      240
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tttctcgtga aggattgcga tcagcatcgg aaggccgcgc agtgcgaccc atgtatacca      540
ggggtctcat tttcccaga tcaccatacg aggcgcact gtgagtcttg caggcattgt      600
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atcatcctgg ccaacaacag cctgagcagc aacggcaacg tgaccgagag cggtgcaaa      1920
gagtgcgagg aactggaaga gaagaacatc aaagagtctt tgcagagctt cgtgcacatc      1980
gtgcagatgt tcataacac cagctga      2007

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<210> SEQ ID NO 32

<211> LENGTH: 1932

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 32

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ctagaacctc gctggaaagg accttacaca gtctgtgtga ccacccccac cgccctcaaa     240
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aacagcctga gcagcaacgg caacgtgacc gagagcggct gcaaagagtg cgaggaaactg    1860
gaagagaaga acatcaaaga gtttctgcag agcttcgtgc acatcgtgca gatgttcac    1920
aacaccagct ga                                         1932
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<210> SEQ ID NO 33

<211> LENGTH: 1941

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 33

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aattccggct tgttggtccg caactgtact attactgcga atgcagagtg tgctttaga      660
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aggaagaacc ctcaggaagg cctgtacaat gaactgcaga aagataagat gccggaggcc   1260
tacagtgaga ttgggatgaa aggcgagcgc cggaggggga aggggcacga tggcctttac   1320
caggggtctc gtacagccac caaggacacc tacgacgccc ttcacatgca ggccctgccc   1380
cctcgcggac cgcagtgtac taattatgct ctcttgaat tggctggaga tgttgagagc   1440
aatccccggc ccatgcgcat tagcaagccc cacctcgcca gcatcagcat ccagtgtctac   1500
ctgtgectgc tgctgaacag caacttctg accgaggccg gcatecagct gttcactctg   1560
ggctgcttca gcgcgggact gcccaagacc gaggccaaact gggtgaaagt gatcagcgac   1620
ctgaagaaga tcgaggacct gatccagagc atgcacatcg acgccaccct gtacaccgag   1680
agcgacgtgc accccagctg caaggtgacc gccatgaagt gctttctgct ggaactgcag   1740
gtgatcagcc tggaaagcgg cgacgccagc atccacgaca ccgtggagaa cctgatcatc   1800
ctggccaaca acagcctgag cagcaacggc aacgtgaccg agagcggctg caaagagtgc   1860
gaggaactgg aagagaagaa catcaaagag tttctgcaga gcttcgtgca catcgtgcag   1920
atgttcatca acaccagctg a                                     1941

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<210> SEQ ID NO 34

<211> LENGTH: 2016

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 34

atgacaagag ttactaacag cccctctctc caagctcact tacaggctct ctacttagtc	60
cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga ccgaccggtg	120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac	180
ctagaacctc gctggaaagg accttacaca gtctgtgtga ccacccccac cgccctcaaa	240
gtagacggca tcgcagcttg gatacacgcc gcccacgtga aggctgccga ccccgggggt	300
ggaccatcct ctagactgcc atgctcgagg atggcacggc cacatccctg gtggctgtgc	360
gttctgggga ccttgggtgg gctctcagct actccagccc ccaagagctg cccagagagg	420
cactactggg ctcagggaaa gctgtgtgtc cagatgtgtg agccaggaa attcctcgtg	480
aaggactgtg accagcatag aaaggctgct cagtgtgatc cttgcatacc gggggtctcc	540
ttctctctg accaccacac cgggccccac tgtgagagct gtcggcactg taactctggt	600
cttctcgttc gcaactgcac catcactgcc aatgctgagt gtgcctgtcg caatggctgg	660
cagtgcaggg acaaggagtg caccgagtgt gatcctcttc caaacccctc gctgaccgct	720
cggctcgtctc aggcctgag cccacaccct cagcccaccc acttacctta tgtcagttag	780
atgctggagg ccaggacagc tgggcacatg cagactctgg ctgacttcag gcagctgcct	840
gcccggactc tctctaccca ctggccaccc caaagatccc tgtgcagctc cgattttatt	900
cgcacccctg tgatctcttc tggaatgttc cttgttttca ccttggccgg ggccctgttc	960
ctccattact tcctgggccc gctggtcctt cgggggagag gggctgcgga ggcagcgacc	1020
cggaaacagc gtatcactga gaccgagtcg ccttatcagg agctccaggg tcagaggctc	1080
gatgtctaca gcgacctcaa cacacagagg ccgtattaca aaaaacgcgt gaagttcagc	1140
aggagcgcag acgccccgcg gtaccagcag ggccagaacc agctctataa cgagctcaat	1200
ctaggacgaa gagaggagta cgatgttttg gacaaaagac gtggccggga ccttgagatg	1260
gggggaaagc cgagaaggaa gaacctcag gaaggcctgt acaatgaact gcagaaagat	1320
aagatggcgg aggcctacag tgagattggg atgaaaggcg agcgccggag gggcaagggg	1380
cacgatggcc tttaccaggg tctcagtaca gccaccaagg acacctacga cgcccttcac	1440
atgcaggccc tgccccctcg cggaccgcag tgtactaatt atgctctctt gaaattggct	1500
ggagatgttg agagcaatcc cgggcccatt cgcattagca agccccacct gcggagcatt	1560
agcatccagt gctacctgtg cctgctgtcg aacagccact tcctgaccga ggccggcatc	1620
cacgtgttca tcctgggctg cttcagcgcc ggactgcccc agaccgaggc caactgggtg	1680
aacgtgatca gcgacctgaa gaagatcgag gacctgatcc agagcatgca catcgacgcc	1740
accctgtaca ccgagagcga cgtgcacccc agctgcaagg tgaccgcat gaagtgtttt	1800
ctgctggaac tgcagggtgat cagcctggaa agcggcgacg ccagcatcca cgacaccgtg	1860
gagaacctga tcctcctggc caacaacagc ctgagcagca acggcaacgt gaccgagagc	1920
ggctgcaaag agtgcgagga actggaagag aagaacatca aagagtttct gcagagcttc	1980
gtgcacatcg tgcagatgtt catcaacacc agctga	2016

<210> SEQ ID NO 35

<211> LENGTH: 2025

<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 35

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atgacaagag ttactaacag cccctctctc caagctcact tacaggctct ctacttagtc      60
cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga ccgaccgggtg    120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac    180
ctagaacctc gctggaaagg accttacaca gtctgtgtga cccccccac cgccctcaaa    240
gtagacggga tcgcagcttg gatacacgcc gcccacgtga aggctgccga ccccgggggt    300
ggaccatcct ctagactgcc atgctcgagg atgcttctcc tggtgacaag ccttctgtctc    360
tgtgagttac cacaccagc attcctctg atccagcta caccggctcc gaagtctgtc    420
cgggagcggc attattgggc acagggaag ttgtgtgtgc aaatgtgtga gccgggaacc    480
tttctctgta aggattgaga tcagcatcgg aaggccgcgc agtgcgaccc atgtatacca    540
ggggtctcat tttcccaga tcaccatacg aggcgcact gtgagtcttg caggcattgt    600
aattccggct tggtggtcgc caactgtact attactgca atgcagagtg tgctttaga    660
aacggatggc agtgacggga caaagaatgt acggagtgtg atccactgcc taaccccagt    720
cttacagcaa gatcttcaca ggccctcagc ccgcatactc aaccaacaca tcttccttac    780
gtgtcagaaa tggtggaggc gcgaaccgca ggccatagc agaccctggc ggactttcgg    840
cagctgccag cagcacact tagtacacac tggccaccac aacgcagctt gtgctcttcc    900
gatttcatcc gcatactggt catcttttct ggaatgttcc ttgtgttcac cctggcagga    960
gccctgttcc ttcactactt cctgggccgg ctggtcctcc gggggcgagg ggctgcggag   1020
gcagcgaccc ggaacacagc tatcactgag accgagtcgc cttatcagga gctccagggt   1080
cagaggtcgg atgtctacag cgacctcaac acacagaggc cgtattacaa aaaacgcgtg   1140
aagttcagca ggagcgcaga cgcgcccgcg taccagcagg gccagaacca gctctataac   1200
gagctcaatc taggacgaag agaggagtac gatgttttgg acaaaagacg tggccgggac   1260
cctgagatgg ggggaaagcc gagaaggaa aaccctcagg aaggcctgta caatgaactg   1320
cagaaagata agatggcgga ggccctacag gagattggga tgaaaggcga gcgccggagg   1380
ggcaaggggc acgatggcct ttaccagggt ctcagtagag ccaccaagga cacctacgac   1440
gcccttcaca tgcaggccct gcccctcgc ggaccgcagt gtactaatta tgctctcttg   1500
aaattggctg gagatgttga gagcaatccc gggcccatgc gcattagcaa gcccacctg   1560
cggagcatca gcattccagt ctacctgtgc ctgctgtgta acagccactt cctgaccgag   1620
gccggcatcc acgtgttcat cctgggctgc ttcagcgccg gactgccccaa gaccgaggcc   1680
aactgggtga acgtgatcag cgacctgaag aagatcgagg acctgatcca gagcatgcac   1740
atcgacgcca cctgtacac cgagagcgac gtgcacccca gctgcaagggt gaccgccatg   1800
aagtgttttc tgctggaact gcagggtgac agcctggaaa gcggcgacgc cagcatccac   1860
gacaccgtgg agaacctgat catcctggcc aacaacagcc tgagcagcaa cggaacgtg   1920
accgagagcg gctgcaaaga gtgcgaggaa ctggaagaga agaacatcaa agagtttctg   1980
cagagcttcg tgcacatcgt gcagatgttc atcaacacca gctga                        2025

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<210> SEQ ID NO 36

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<211> LENGTH: 2013
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 36

atgacaagag ttactaacag cccctctctc caagctcact tacaggctct ctacttagtc      60
cagcacgaag tctggagacc tctggcggca gcctaccaag aacaactgga ccgaccggtg      120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac      180
ctagaacctc gctggaagg accttacaca gtccctgtga ccacccccac cgccctcaaa      240
gtagacggca tcgcagcttg gatacacgcc gccacgtga aggctgcga ccccgggggt      300
ggaccatcct ctagactgcc atgctcgagg atggcacggc cacatccctg gtggctgtgc      360
gttctgggga ccctgggtgg gctctcagct actccagccc ccaagagctg ccagagagg      420
cactactggg ctcagggaag gctgtgctgc cagatgtgtg agccaggaa attcctctgt      480
aaggactgtg accagcatag aaaggctgct cagtgtgatc cttgcatacc gggggtctcc      540
ttctctcctg accaccacac ccggcccccac tgtgagagct gtcggcactg taactctggt      600
cttctcgttc gcaactgcac catcactgcc aatgctgagt gtgcctgtcg caatggctgg      660
cagtgcaggg acaaggagtg caccgagtgt gatcctcttc caaaccttc gctgaccgct      720
cggtcgtctc aggcacctgag cccacaccct cagccacccc acttacctta tgtcagtgag      780
atgctggagg ccaggacagc tgggcacatg cagactctgg ctgacttcag gcagctgcct      840
gcccggaact tctctaccca ctggccaccc caaagatccc tgtgcagctc cgattttatt      900
cgcacacctg tgatctctc tggaatgttc cttgttttca ccctggccgg ggcctgttc      960
ctccatagcg cgaacgaacg ctgcaaaagc aaagtgggtc cgtgccgcca gaaacagtgg      1020
cgcaccagct ttgatagcaa aaaactggat ctgaactata accattttga aagcatggaa      1080
tggagccatc gcagccgccc cggccgcatt tggggcatga aacgcgtgaa gttcagcagg      1140
agcgcagacg ccccccgcga ccagcagggc cagaaccagc tctataacga gctcaatcta      1200
ggacgaagag aggagtacga tgttttgac aaaagacgtg gccgggaccc tgagatgggg      1260
ggaaagccga gaaggaagaa ccctcaggaa ggcctgtaca atgaactgca gaaagataag      1320
atggcggagg cctacagtga gattgggatg aaaggcagc gccggagggg caaggggcac      1380
gatggccttt accagggtct cagtacagcc accaaggaca cctacgagc ccttcacatg      1440
caggccctgc cccctcgcgg accgcagtgt actaattatg ctctcttgaa attggctgga      1500
gatgttgaga gcaatcccg gcccctgcgc attagcaagc cccacctgcg gagcatcagc      1560
atccagtgtc acctgtgcct gctgtgaac agccaactcc tgaccgaggc cgcatccac      1620
gtgttcaccc tgggtgtctt cagcgccgga ctgcccaaga ccgaggccaa ctgggtgaac      1680
gtgatcagcg acctgaagaa gatcgaggac ctgatccaga gcatgcacat cgacgccacc      1740
ctgtacacgg agagcgacgt gcaccccagc tgcaaggatga ccgccatgaa gtgctttctg      1800
ctggaactgc aggtgatcag cctggaaagc ggcgacgcca gcatccacga caccgtggag      1860
aacctgatca tcctggccaa caacagcctg agcagcaacg gcaacgtgac cgagagcggc      1920
tgcaaagagt gcgaggaact ggaagagaag aacatcaaag agtttctgca gagcttcgtg      1980
cacatcgtgc agatgttcat caacaccagc tga                                     2013

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<210> SEQ ID NO 37
<211> LENGTH: 2022
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 37
atgacaagag ttactaacag cccctctctc caagctcact tacaggctct ctacttagtc      60
cagcacgaag tctggagacc tctggcgga gcctaccaag aacaactgga ccgaccggtg      120
gtacctcacc cttaccgagt cggcgacaca gtgtgggtcc gccgacacca gactaagaac      180
ctagaacctc gctggaaagg accttacaca gtctgtctga ccacccccac cgcctcctc      240
gtagacggca tcgcagcttg gatacacgcc gccacgtga aggctgccga ccccgggggt      300
ggaccatcct ctagactgcc atgctcgagg atgcttctcc tggtgacaag ccttctgctc      360
tgtgagttac cacaccagc attcctctc atcccagcta caccggtcc gaagtctgc      420
ccggagcggc attattgggc acagggaag ttgtgtgtc aaatgtgtga gccgggaacc      480
tttctcgtga aggattgcga tcagcatcgg aaggccgcgc agtgcgaccc atgtatacca      540
ggggtctcat tttcccaga tcaccatacg aggcgcact gtgagtctt caggcattgt      600
aattccggct tgttggtccg caactgtact attactgcga atgcagagtg tgctttaga      660
aacggatggc agtgacggga caaagaatgt acggagtgtg atccactgcc taaccccagt      720
cttacagcaa gatcttcaca ggccctcagc ccgcacctc aaccaacaca tcttcttac      780
gtgtcagaaa tgttgagggc gcgaaccgca ggccatatgc agaccctggc ggactttcgg      840
cagctgccag cagcacact tagtacacac tggccaccac aacgcagctt gtgctcttc      900
gatttcatcc gcatactggt catcttttct ggaatgttcc ttgtgttcac cctggcagga      960
gcctgttcc ttcacagcgc gaacgaacgc tgcaaaagca aagtgggtgc gtgccgccag      1020
aaacagtggc gcaccagctt tgatagcaaa aaactggatc tgaactataa ccattttgaa      1080
agcatggaat ggagccatcg cagccgccgc ggcgcattt ggggcagaa acgcgtgaag      1140
ttcagcagga gcgcagacgc ccccgctac cagcagggcc agaaccagct ctataacgag      1200
ctcaatctag gacgaagaga ggagtacgat gttttggaca aaagacgtgg ccgggaccct      1260
gagatggggg gaaagccgag aaggaagaac cctcaggaag gcctgtacaa tgaactgcag      1320
aaagataaga tggcggaggc ctacagtga attgggatga aaggcgagcg ccggaggggc      1380
aaggggcacg atggccttta ccagggtctc agtacagcca ccaaggacac ctacgacgcc      1440
cttcacatgc aggcctgcc cctcgcgga ccgcagtgtg ctaattatgc tctcttgaaa      1500
ttggctggag atgttgagag caatcccgga cccatgcgca ttagcaagcc ccacctgcgg      1560
agcatcagca tccagtgcta cctgtgctc ctgtgaaca gccacttct gaccgaggcc      1620
ggcatccacg tgttcactc gggctgcttc agcgcggac tgcccaagac cgaggccaac      1680
tgggtgaacg tgatcagcga cctgaagaag atcgaggacc tgatccagag catgcacatc      1740
gacgcccccc tgtacaccga gacgcagctg caccacagct gcaagggtgac cgccatgaag      1800
tgctttctgc tggaaactgca ggtgatcagc ctggaaagcg gcgacgccag catccacgac      1860
accgtggaga acctgatcat cctggccaac aacagcctga gcagcaacgg caacgtgacc      1920
gagagcggct gcaaaagtg cgaggaactg gaagagaaga acatcaaaga gtttctgcag      1980

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agcttcgtgc acatcgtgca gatgttcac aacaccagct ga 2022

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<210> SEQ ID NO 38
<211> LENGTH: 218
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 38

Met Ala Arg Pro His Pro Trp Trp Leu Cys Val Leu Gly Thr Leu Val
1          5          10          15

Gly Leu Ser Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His Tyr
20          25          30

Trp Ala Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr Phe
35          40          45

Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro
50          55          60

Cys Ile Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His
65          70          75          80

Cys Glu Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys
85          90          95

Thr Ile Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys
100         105         110

Arg Asp Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu
115         120         125

Thr Ala Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr His
130         135         140

Leu Pro Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His Met
145         150         155         160

Gln Thr Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser Thr
165         170         175

His Trp Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg Phe
180         185         190

Trp Val Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser Leu Leu
195         200         205

Val Thr Val Ala Phe Ile Ile Phe Trp Val
210         215

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<210> SEQ ID NO 39
<211> LENGTH: 667
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 39

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
1          5          10          15

Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr
20          25          30

Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
35          40          45

Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
50          55          60

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Trp 65	Lys	Gly	Pro	Tyr	Thr 70	Val	Leu	Leu	Thr	Thr 75	Pro	Thr	Ala	Leu	Lys 80
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Ala
Arg	Pro	His	Pro	Trp	Trp	Leu	Cys	Val	Leu	Gly	Thr	Leu	Val	Gly	Leu
Ser	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His	Tyr	Trp	Ala
Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr	Phe	Leu	Val
Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp	Pro	Cys	Ile
Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	His	Cys	Glu
Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	Cys	Thr	Ile
Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	Cys	Arg	Asp
Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	Leu	Thr	Ala
Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	His	Leu	Pro
Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	Met	Gln	Thr
Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	Thr	His	Trp
Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	Phe	Trp	Val
Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr
Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu
Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu
Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly	Gly	Cys
Glu	Leu	Lys	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr
Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg
Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met
Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu
Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys
Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu
Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu

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465		470		475		480									
Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala
				485					490					495	
Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His
			500					505					510		
Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser
		515					520					525			
His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His	Val	Phe	Ile	Leu	Gly	Cys	Phe
	530					535					540				
Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn	Val	Ile	Ser
545					550					555					560
Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala
			565						570					575	
Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala
			580					585					590		
Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly
		595					600					605			
Asp	Ala	Ser	Ile	His	Asp	Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn
	610					615					620				
Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu
625					630					635					640
Cys	Glu	Glu	Leu	Glu	Glu	Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe
			645					650					655		
Val	His	Ile	Val	Gln	Met	Phe	Ile	Asn	Thr	Ser					
			660					665							

<210> SEQ ID NO 40
 <211> LENGTH: 670
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 40

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala
1				5					10					15	
Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr
		20						25					30		
Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly
		35				40						45			
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg
	50					55					60				
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys
65					70					75				80	
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala
			85					90					95		
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Leu
		100						105				110			
Leu	Leu	Val	Thr	Ser	Leu	Leu	Leu	Cys	Glu	Leu	Pro	His	Pro	Ala	Phe
		115					120					125			
Leu	Leu	Ile	Pro	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His
	130					135					140				
Tyr	Trp	Ala	Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr

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145					150						155					160
Phe	Leu	Val	Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp	
				165					170						175	
Pro	Cys	Ile	Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	
			180					185					190			
His	Cys	Glu	Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	
		195					200					205				
Cys	Thr	Ile	Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	
	210				215						220					
Cys	Arg	Asp	Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	
225					230					235					240	
Leu	Thr	Ala	Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	
			245						250					255		
His	Leu	Pro	Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	
			260					265					270			
Met	Gln	Thr	Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	
	275						280					285				
Thr	His	Trp	Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	
	290					295					300					
Phe	Trp	Val	Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	
305				310						315					320	
Leu	Val	Thr	Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Lys	Arg	Gly	Arg	Lys	
			325						330					335		
Lys	Leu	Leu	Tyr	Ile	Phe	Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	
			340					345					350			
Thr	Gln	Glu	Glu	Asp	Gly	Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	
	355					360						365				
Gly	Gly	Cys	Glu	Leu	Lys	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	
	370					375					380					
Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	
385					390					395					400	
Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	
			405						410					415		
Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	
			420					425					430			
Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	
	435						440					445				
Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	
	450					455					460					
Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	
465					470					475					480	
Gln	Ala	Leu	Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	
			485					490						495		
Lys	Leu	Ala	Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	
			500					505					510			
Lys	Pro	His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	
		515					520					525				
Leu	Asn	Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His	Val	Phe	Ile	Leu	
	530					535					540					
Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn	
545					550					555					560	

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Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile Gln Ser Met His
565 570 575

Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His Pro Ser Cys Lys
580 585 590

Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln Val Ile Ser Leu
595 600 605

Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu Asn Leu Ile Ile
610 615 620

Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val Thr Glu Ser Gly
625 630 635 640

Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile Lys Glu Phe Leu
645 650 655

Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn Thr Ser
660 665 670

<210> SEQ ID NO 41
 <211> LENGTH: 649
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 41

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
1 5 10 15

Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Tyr
20 25 30

Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
35 40 45

Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
50 55 60

Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
65 70 75 80

Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
85 90 95

Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Ala
100 105 110

Arg Pro His Pro Trp Trp Leu Cys Val Leu Gly Thr Leu Val Gly Leu
115 120 125

Ser Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His Tyr Trp Ala
130 135 140

Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr Phe Leu Val
145 150 155 160

Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro Cys Ile
165 170 175

Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His Cys Glu
180 185 190

Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys Thr Ile
195 200 205

Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys Arg Asp
210 215 220

Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu Thr Ala
225 230 235 240

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Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	His	Leu	Pro	245	250	255
Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	Met	Gln	Thr	260	265	270
Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	Thr	His	Trp	275	280	285
Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	Phe	Trp	Val	290	295	300
Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr	305	310	315
Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Leu	Cys	Ala	Arg	Pro	Arg	Arg	Ser	325	330	335
Pro	Ala	Gln	Glu	Asp	Gly	Lys	Val	Tyr	Ile	Asn	Met	Pro	Gly	Arg	Gly	340	345	350
Lys	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	355	360	365
Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	370	375	380
Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	385	390	395
Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	405	410	415
Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	420	425	430
Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	435	440	445
Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	450	455	460
Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp	465	470	475
Val	Glu	Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	485	490	495
Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe	500	505	510
Leu	Thr	Glu	Ala	Gly	Ile	His	Val	Phe	Ile	Leu	Gly	Cys	Phe	Ser	Ala	515	520	525
Gly	Leu	Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	530	535	540
Lys	Lys	Ile	Glu	Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	545	550	555
Tyr	Thr	Glu	Ser	Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	565	570	575
Cys	Phe	Leu	Leu	Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	580	585	590
Ser	Ile	His	Asp	Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	595	600	605
Leu	Ser	Ser	Asn	Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	610	615	620
Glu	Leu	Glu	Glu	Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	625	630	635

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Ile Val Gln Met Phe Ile Asn Thr Ser
645

<210> SEQ ID NO 42
<211> LENGTH: 652
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 42

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
1 5 10 15
Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr
20 25 30
Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
35 40 45
Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
50 55 60
Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
65 70 75 80
Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
85 90 95
Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Leu
100 105 110
Leu Leu Val Thr Ser Leu Leu Leu Cys Glu Leu Pro His Pro Ala Phe
115 120 125
Leu Leu Ile Pro Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His
130 135 140
Tyr Trp Ala Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr
145 150 155 160
Phe Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp
165 170 175
Pro Cys Ile Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro
180 185 190
His Cys Glu Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn
195 200 205
Cys Thr Ile Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln
210 215 220
Cys Arg Asp Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser
225 230 235 240
Leu Thr Ala Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr
245 250 255
His Leu Pro Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His
260 265 270
Met Gln Thr Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser
275 280 285
Thr His Trp Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg
290 295 300
Phe Trp Val Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser Leu
305 310 315 320
Leu Val Thr Val Ala Phe Ile Ile Phe Trp Val Leu Cys Ala Arg Pro
325 330 335

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Arg	Arg	Ser	Pro	Ala	Gln	Glu	Asp	Gly	Lys	Val	Tyr	Ile	Asn	Met	Pro
			340					345					350		
Gly	Arg	Gly	Lys	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala
		355					360					365			
Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg
	370					375					380				
Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu
385					390					395					400
Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn
				405					410					415	
Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met
			420					425					430		
Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly
		435					440					445			
Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala
	450					455					460				
Leu	Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu
465					470					475					480
Ala	Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro
			485						490					495	
His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn
			500					505					510		
Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His	Val	Phe	Ile	Leu	Gly	Cys
		515					520					525			
Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn	Val	Ile
	530					535					540				
Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp
545				550						555					560
Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr
			565					570						575	
Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser
			580					585					590		
Gly	Asp	Ala	Ser	Ile	His	Asp	Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala
		595					600					605			
Asn	Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys
	610					615					620				
Glu	Cys	Glu	Glu	Leu	Glu	Glu	Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser
625				630						635					640
Phe	Val	His	Ile	Val	Gln	Met	Phe	Ile	Asn	Thr	Ser				
			645						650						

<210> SEQ ID NO 43

<211> LENGTH: 677

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 43

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala
1			5						10					15	

Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr
		20					25					30			

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Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly
	35						40					45			
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg
	50					55					60				
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys
65					70					75					80
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala
			85						90					95	
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Ala
			100					105					110		
Arg	Pro	His	Pro	Trp	Trp	Leu	Cys	Val	Leu	Gly	Thr	Leu	Val	Gly	Leu
		115					120					125			
Ser	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His	Tyr	Trp	Ala
	130					135					140				
Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr	Phe	Leu	Val
145					150					155					160
Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp	Pro	Cys	Ile
				165					170					175	
Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	His	Cys	Glu
			180					185					190		
Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	Cys	Thr	Ile
		195					200					205			
Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	Cys	Arg	Asp
	210					215					220				
Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	Leu	Thr	Ala
225					230					235					240
Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	His	Leu	Pro
			245						250					255	
Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	Met	Gln	Thr
			260					265					270		
Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	Thr	His	Trp
		275					280					285			
Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	Phe	Trp	Val
	290					295					300				
Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr
305					310					315					320
Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Tyr	Phe	Leu	Gly	Arg	Leu	Val	Pro
			325						330					335	
Arg	Gly	Arg	Gly	Ala	Ala	Glu	Ala	Ala	Thr	Arg	Lys	Gln	Arg	Ile	Thr
			340					345					350		
Glu	Thr	Glu	Ser	Pro	Tyr	Gln	Glu	Leu	Gln	Gly	Gln	Arg	Ser	Asp	Val
		355					360					365			
Tyr	Ser	Asp	Leu	Asn	Thr	Gln	Arg	Pro	Tyr	Tyr	Lys	Lys	Arg	Val	Lys
	370					375					380				
Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln
385					390					395					400
Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu
			405					410						415	
Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg
			420					425					430		
Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met

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435					440					445					
Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly
450						455					460				
Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp
465					470					475					480
Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg	Gly	Pro	Gln
				485						490					495
Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp	Val	Glu	Ser	Asn
			500						505					510	
Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	Ser	Ile	Ser	Ile
		515					520					525			
Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe	Leu	Thr	Glu	Ala
	530				535						540				
Gly	Ile	His	Val	Phe	Ile	Leu	Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro	Lys
545					550					555					560
Thr	Glu	Ala	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu
				565						570					575
Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser
			580					585						590	
Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu
		595					600					605			
Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp
	610					615					620				
Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn
625				630						635					640
Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu
				645					650					655	
Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln	Met
		660						665					670		
Phe	Ile	Asn	Thr	Ser											
		675													

<210> SEQ ID NO 44

<211> LENGTH: 680

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 44

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala
1				5					10					15	
Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr
		20						25					30		
Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly
		35					40					45			
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg
	50					55					60				
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys
65				70						75				80	
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala
			85						90					95	
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Leu

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100						105						110					
Leu	Leu	Val	Thr	Ser	Leu	Leu	Leu	Cys	Glu	Leu	Pro	His	Pro	Ala	Phe		
	115						120					125					
Leu	Leu	Ile	Pro	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His		
	130						135				140						
Tyr	Trp	Ala	Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr		
145					150					155					160		
Phe	Leu	Val	Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp		
			165						170					175			
Pro	Cys	Ile	Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro		
			180					185					190				
His	Cys	Glu	Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn		
		195					200					205					
Cys	Thr	Ile	Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln		
	210					215					220						
Cys	Arg	Asp	Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser		
225					230					235					240		
Leu	Thr	Ala	Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr		
			245						250					255			
His	Leu	Pro	Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His		
		260						265					270				
Met	Gln	Thr	Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser		
	275						280					285					
Thr	His	Trp	Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg		
	290					295					300						
Phe	Trp	Val	Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu		
305					310					315					320		
Leu	Val	Thr	Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Tyr	Phe	Leu	Gly	Arg		
			325						330					335			
Leu	Val	Pro	Arg	Gly	Arg	Gly	Ala	Ala	Glu	Ala	Ala	Thr	Arg	Lys	Gln		
		340					345						350				
Arg	Ile	Thr	Glu	Thr	Glu	Ser	Pro	Tyr	Gln	Glu	Leu	Gln	Gly	Gln	Arg		
	355					360						365					
Ser	Asp	Val	Tyr	Ser	Asp	Leu	Asn	Thr	Gln	Arg	Pro	Tyr	Tyr	Lys	Lys		
	370					375					380						
Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly		
385					390					395					400		
Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr		
		405							410					415			
Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys		
		420						425					430				
Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys		
	435					440						445					
Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg		
	450					455					460						
Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala		
465					470					475					480		
Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg		
			485						490					495			
Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp	Val		
		500						505					510				

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Glu Ser Asn Pro Gly Pro Met Arg Ile Ser Lys Pro His Leu Arg Ser
 515 520 525
 Ile Ser Ile Gln Cys Tyr Leu Cys Leu Leu Leu Asn Ser His Phe Leu
 530 535 540
 Thr Glu Ala Gly Ile His Val Phe Ile Leu Gly Cys Phe Ser Ala Gly
 545 550 555 560
 Leu Pro Lys Thr Glu Ala Asn Trp Val Asn Val Ile Ser Asp Leu Lys
 565 570 575
 Lys Ile Glu Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr
 580 585 590
 Thr Glu Ser Asp Val His Pro Ser Cys Lys Val Thr Ala Met Lys Cys
 595 600 605
 Phe Leu Leu Glu Leu Gln Val Ile Ser Leu Glu Ser Gly Asp Ala Ser
 610 615 620
 Ile His Asp Thr Val Glu Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu
 625 630 635 640
 Ser Ser Asn Gly Asn Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu
 645 650 655
 Leu Glu Glu Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile
 660 665 670
 Val Gln Met Phe Ile Asn Thr Ser
 675 680

<210> SEQ ID NO 45
 <211> LENGTH: 676
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 45

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
 1 5 10 15
 Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Tyr
 20 25 30
 Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
 35 40 45
 Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
 50 55 60
 Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
 65 70 75 80
 Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
 85 90 95
 Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Ala
 100 105 110
 Arg Pro His Pro Trp Trp Leu Cys Val Leu Gly Thr Leu Val Gly Leu
 115 120 125
 Ser Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His Tyr Trp Ala
 130 135 140
 Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr Phe Leu Val
 145 150 155 160
 Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro Cys Ile
 165 170 175

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Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	His	Cys	Glu	180	185	190
Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	Cys	Thr	Ile	195	200	205
Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	Cys	Arg	Asp	210	215	220
Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	Leu	Thr	Ala	225	230	235
Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	His	Leu	Pro	245	250	255
Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	Met	Gln	Thr	260	265	270
Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	Thr	His	Trp	275	280	285
Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	Phe	Trp	Val	290	295	300
Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr	305	310	315
Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Ser	Ala	Asn	Glu	Arg	Cys	Lys	Ser	325	330	335
Lys	Val	Val	Pro	Cys	Arg	Gln	Lys	Gln	Trp	Arg	Thr	Ser	Phe	Asp	Ser	340	345	350
Lys	Lys	Leu	Asp	Leu	Asn	Tyr	Asn	His	Phe	Glu	Ser	Met	Glu	Trp	Ser	355	360	365
His	Arg	Ser	Arg	Arg	Gly	Arg	Ile	Trp	Gly	Met	Lys	Arg	Val	Lys	Phe	370	375	380
Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	385	390	395
Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	405	410	415
Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	420	425	430
Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	435	440	445
Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	450	455	460
Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	465	470	475
Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg	Gly	Pro	Gln	Cys	485	490	495
Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp	Val	Glu	Ser	Asn	Pro	500	505	510
Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	515	520	525
Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	530	535	540
Ile	His	Val	Phe	Ile	Leu	Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	545	550	555
Glu	Ala	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	565	570	575

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Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp
			580					585					590		
Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu
		595					600					605			
Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp	Thr
	610					615					620				
Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn	Gly
	625				630					635					640
Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu	Lys
			645						650					655	
Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln	Met	Phe
			660					665					670		
Ile	Asn	Thr	Ser												
			675												

<210> SEQ ID NO 46
 <211> LENGTH: 679
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 46

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala
1				5					10					15	
Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr
		20						25					30		
Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly
		35					40					45			
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg
	50				55						60				
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys
	65				70					75					80
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala
			85						90					95	
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Leu
		100						105					110		
Leu	Leu	Val	Thr	Ser	Leu	Leu	Leu	Cys	Glu	Leu	Pro	His	Pro	Ala	Phe
		115				120					125				
Leu	Leu	Ile	Pro	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His
		130				135					140				
Tyr	Trp	Ala	Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr
	145				150					155					160
Phe	Leu	Val	Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp
			165					170						175	
Pro	Cys	Ile	Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro
			180					185					190		
His	Cys	Glu	Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn
		195						200					205		
Cys	Thr	Ile	Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln
	210					215					220				
Cys	Arg	Asp	Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser
	225				230					235					240

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Leu	Thr	Ala	Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	245	250	255
His	Leu	Pro	Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	260	265	270
Met	Gln	Thr	Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	275	280	285
Thr	His	Trp	Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	290	295	300
Phe	Trp	Val	Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	305	310	315
Leu	Val	Thr	Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Ser	Ala	Asn	Glu	Arg	325	330	335
Cys	Lys	Ser	Lys	Val	Val	Pro	Cys	Arg	Gln	Lys	Gln	Trp	Arg	Thr	Ser	340	345	350
Phe	Asp	Ser	Lys	Lys	Leu	Asp	Leu	Asn	Tyr	Asn	His	Phe	Glu	Ser	Met	355	360	365
Glu	Trp	Ser	His	Arg	Ser	Arg	Arg	Gly	Arg	Ile	Trp	Gly	Met	Lys	Arg	370	375	380
Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	385	390	395
Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	405	410	415
Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	420	425	430
Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	435	440	445
Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	450	455	460
Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	465	470	475
Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg	Gly	485	490	495
Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp	Val	Glu	500	505	510
Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	Ser	Ile	515	520	525
Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe	Leu	Thr	530	535	540
Glu	Ala	Gly	Ile	His	Val	Phe	Ile	Leu	Gly	Cys	Phe	Ser	Ala	Gly	Leu	545	550	555
Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	565	570	575
Ile	Glu	Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	580	585	590
Glu	Ser	Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	595	600	605
Leu	Leu	Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	610	615	620
His	Asp	Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	625	630	635
Ser	Asn	Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu			

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645          650          655
Glu Glu Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val
660          665          670

Gln Met Phe Ile Asn Thr Ser
675

<210> SEQ ID NO 47
<211> LENGTH: 661
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 47

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
1          5          10          15

Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr
20          25          30

Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
35          40          45

Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
50          55          60

Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
65          70          75          80

Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
85          90          95

Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Ala
100         105         110

Arg Pro His Pro Trp Trp Leu Cys Val Leu Gly Thr Leu Val Gly Leu
115         120         125

Ser Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His Tyr Trp Ala
130         135         140

Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr Phe Leu Val
145         150         155         160

Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro Cys Ile
165         170         175

Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His Cys Glu
180         185         190

Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys Thr Ile
195         200         205

Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys Arg Asp
210         215         220

Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu Thr Ala
225         230         235         240

Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr His Leu Pro
245         250         255

Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His Met Gln Thr
260         265         270

Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser Thr His Trp
275         280         285

Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg Ile Leu Val
290         295         300

Ile Phe Ser Gly Met Phe Leu Val Phe Thr Leu Ala Gly Ala Leu Phe

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305	310	315	320
Leu His Lys Arg Gly Arg Lys Lys Leu Leu Tyr Ile Phe Lys Gln Pro	325	330	335
Phe Met Arg Pro Val Gln Thr Thr Gln Glu Glu Asp Gly Cys Ser Cys	340	345	350
Arg Phe Pro Glu Glu Glu Glu Gly Gly Cys Glu Leu Lys Arg Val Lys	355	360	365
Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln	370	375	380
Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu	385	390	395
Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg	405	410	415
Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met	420	425	430
Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly	435	440	445
Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp	450	455	460
Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg Gly Pro Gln	465	470	475
Cys Thr Asn Tyr Ala Leu Leu Lys Leu Ala Gly Asp Val Glu Ser Asn	485	490	495
Pro Gly Pro Met Arg Ile Ser Lys Pro His Leu Arg Ser Ile Ser Ile	500	505	510
Gln Cys Tyr Leu Cys Leu Leu Leu Asn Ser His Phe Leu Thr Glu Ala	515	520	525
Gly Ile His Val Phe Ile Leu Gly Cys Phe Ser Ala Gly Leu Pro Lys	530	535	540
Thr Glu Ala Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu	545	550	555
Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser	565	570	575
Asp Val His Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu	580	585	590
Glu Leu Gln Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp	595	600	605
Thr Val Glu Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn	610	615	620
Gly Asn Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu	625	630	635
Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met	645	650	655
Phe Ile Asn Thr Ser	660		

<210> SEQ ID NO 48

<211> LENGTH: 664

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 48

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala	1	5	10	15
Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr	20	25	30	
Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly	35	40	45	
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg	50	55	60	
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys	65	70	75	80
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala	85	90	95	
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Leu	100	105	110	
Leu	Leu	Val	Thr	Ser	Leu	Leu	Leu	Cys	Glu	Leu	Pro	His	Pro	Ala	Phe	115	120	125	
Leu	Leu	Ile	Pro	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His	130	135	140	
Tyr	Trp	Ala	Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr	145	150	155	160
Phe	Leu	Val	Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp	165	170	175	
Pro	Cys	Ile	Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	180	185	190	
His	Cys	Glu	Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	195	200	205	
Cys	Thr	Ile	Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	210	215	220	
Cys	Arg	Asp	Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	225	230	235	240
Leu	Thr	Ala	Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	245	250	255	
His	Leu	Pro	Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	260	265	270	
Met	Gln	Thr	Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	275	280	285	
Thr	His	Trp	Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	290	295	300	
Ile	Leu	Val	Ile	Phe	Ser	Gly	Met	Phe	Leu	Val	Phe	Thr	Leu	Ala	Gly	305	310	315	320
Ala	Leu	Phe	Leu	His	Lys	Arg	Gly	Arg	Lys	Lys	Leu	Leu	Tyr	Ile	Phe	325	330	335	
Lys	Gln	Pro	Phe	Met	Arg	Pro	Val	Gln	Thr	Thr	Gln	Glu	Glu	Asp	Gly	340	345	350	
Cys	Ser	Cys	Arg	Phe	Pro	Glu	Glu	Glu	Glu	Gly	Gly	Cys	Glu	Leu	Lys	355	360	365	
Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	370	375	380	
Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	385	390	395	400

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Asp	Val	Leu	Asp	Lys 405	Arg	Arg	Gly	Arg	Asp 410	Pro	Glu	Met	Gly	Gly	Lys 415
Pro	Arg	Arg	Lys 420	Asn	Pro	Gln	Glu	Gly 425	Leu	Tyr	Asn	Glu	Leu	Gln	Lys
Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys 445	Gly	Glu	Arg
Arg	Arg	Gly	Lys	Gly	His	Asp 455	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala
Thr	Lys	Asp	Thr	Tyr	Asp 470	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg 480
Gly	Pro	Gln	Cys	Thr 485	Asn	Tyr	Ala	Leu	Leu 490	Lys	Leu	Ala	Gly	Asp 495	Val
Glu	Ser	Asn	Pro 500	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	Ser
Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys 520	Leu	Leu	Leu	Asn	Ser	His	Phe	Leu
Thr	Glu	Ala	Gly	Ile	His	Val 535	Phe	Ile	Leu	Gly	Cys 540	Phe	Ser	Ala	Gly
Leu	Pro	Lys	Thr	Glu	Ala	Asn 550	Trp	Val	Asn	Val 555	Ile	Ser	Asp	Leu	Lys 560
Lys	Ile	Glu	Asp	Leu 565	Ile	Gln	Ser	Met	His 570	Ile	Asp	Ala	Thr	Leu	Tyr 575
Thr	Glu	Ser	Asp 580	Val	His	Pro	Ser	Cys 585	Lys	Val	Thr	Ala	Met 590	Lys	Cys
Phe	Leu	Leu	Glu	Leu	Gln	Val	Ile 600	Ser	Leu	Glu	Ser	Gly 605	Asp	Ala	Ser
Ile	His	Asp	Thr	Val	Glu	Asn 615	Leu	Ile	Ile	Leu	Ala 620	Asn	Asn	Ser	Leu
Ser	Ser	Asn	Gly	Asn 630	Val	Thr	Glu	Ser	Gly 635	Cys	Lys	Glu	Cys	Glu	Glu 640
Leu	Glu	Glu	Lys	Asn 645	Ile	Lys	Glu	Phe	Leu 650	Gln	Ser	Phe	Val	His 655	Ile
Val	Gln	Met	Phe 660	Ile	Asn	Thr	Ser								

```
<210> SEQ ID NO 49
<211> LENGTH: 619
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide
```

<400> SEQUENCE: 49

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala
1				5					10					15	
Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr
			20					25					30		
Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly
		35					40					45			
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg
	50					55					60				
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys
65					70					75				80	

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Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala		85	90	95
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Ala		100	105	110
Arg	Pro	His	Pro	Trp	Trp	Leu	Cys	Val	Leu	Gly	Thr	Leu	Val	Gly	Leu		115	120	125
Ser	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His	Tyr	Trp	Ala		130	135	140
Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr	Phe	Leu	Val		145	150	155
Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp	Pro	Cys	Ile		165	170	175
Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	His	Cys	Glu		180	185	190
Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	Cys	Thr	Ile		195	200	205
Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	Cys	Arg	Asp		210	215	220
Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	Leu	Thr	Ala		225	230	235
Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	His	Leu	Pro		245	250	255
Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	Met	Gln	Thr		260	265	270
Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	Thr	His	Trp		275	280	285
Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	Ile	Leu	Val		290	295	300
Ile	Phe	Ser	Gly	Met	Phe	Leu	Val	Phe	Thr	Leu	Ala	Gly	Ala	Leu	Phe		305	310	315
Leu	His	Lys	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr		325	330	335
Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg		340	345	350
Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met		355	360	365
Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu		370	375	380
Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys		385	390	395
Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu		405	410	415
Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu		420	425	430
Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala		435	440	445
Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His		450	455	460
Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser		465	470	475

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His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His	Val	Phe	Ile	Leu	Gly	Cys	Phe
				485					490					495	
Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn	Val	Ile	Ser
			500					505					510		
Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala
		515					520					525			
Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala
	530					535					540				
Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly
545				550						555					560
Asp	Ala	Ser	Ile	His	Asp	Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn
			565						570					575	
Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu
		580					585						590		
Cys	Glu	Glu	Leu	Glu	Glu	Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe
	595					600						605			
Val	His	Ile	Val	Gln	Met	Phe	Ile	Asn	Thr	Ser					
	610					615									

<210> SEQ ID NO 50
 <211> LENGTH: 622
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 50

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala
1				5					10					15	
Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr
		20						25					30		
Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly
		35				40						45			
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg
	50				55						60				
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys
65				70					75					80	
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala
			85						90					95	
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Leu
		100						105					110		
Leu	Leu	Val	Thr	Ser	Leu	Leu	Leu	Cys	Glu	Leu	Pro	His	Pro	Ala	Phe
		115				120					125				
Leu	Leu	Ile	Pro	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His
	130					135					140				
Tyr	Trp	Ala	Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr
145					150					155					160
Phe	Leu	Val	Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp
			165						170					175	
Pro	Cys	Ile	Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro
		180						185					190		
His	Cys	Glu	Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn
	195						200						205		

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Cys Thr Ile Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln	210	215	220	
Cys Arg Asp Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser	225	230	235	240
Leu Thr Ala Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr	245	250	255	
His Leu Pro Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His	260	265	270	
Met Gln Thr Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser	275	280	285	
Thr His Trp Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg	290	295	300	
Ile Leu Val Ile Phe Ser Gly Met Phe Leu Val Phe Thr Leu Ala Gly	305	310	315	320
Ala Leu Phe Leu His Lys Arg Val Lys Phe Ser Arg Ser Ala Asp Ala	325	330	335	
Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu	340	345	350	
Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp	355	360	365	
Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu	370	375	380	
Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile	385	390	395	400
Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr	405	410	415	
Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met	420	425	430	
Gln Ala Leu Pro Pro Arg Gly Pro Gln Cys Thr Asn Tyr Ala Leu Leu	435	440	445	
Lys Leu Ala Gly Asp Val Glu Ser Asn Pro Gly Pro Met Arg Ile Ser	450	455	460	
Lys Pro His Leu Arg Ser Ile Ser Ile Gln Cys Tyr Leu Cys Leu Leu	465	470	475	480
Leu Asn Ser His Phe Leu Thr Glu Ala Gly Ile His Val Phe Ile Leu	485	490	495	
Gly Cys Phe Ser Ala Gly Leu Pro Lys Thr Glu Ala Asn Trp Val Asn	500	505	510	
Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile Gln Ser Met His	515	520	525	
Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His Pro Ser Cys Lys	530	535	540	
Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln Val Ile Ser Leu	545	550	555	560
Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu Asn Leu Ile Ile	565	570	575	
Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val Thr Glu Ser Gly	580	585	590	
Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile Lys Glu Phe Leu	595	600	605	
Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn Thr Ser				

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610	615	620
<210> SEQ ID NO 51		
<211> LENGTH: 659		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Synthetic Polypeptide		
<400> SEQUENCE: 51		
Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala		
1 5 10 15		
Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr		
20 25 30		
Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly		
35 40 45		
Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg		
50 55 60		
Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys		
65 70 75 80		
Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala		
85 90 95		
Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Ala		
100 105 110		
Arg Pro His Pro Trp Trp Leu Cys Val Leu Gly Thr Leu Val Gly Leu		
115 120 125		
Ser Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His Tyr Trp Ala		
130 135 140		
Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr Phe Leu Val		
145 150 155 160		
Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro Cys Ile		
165 170 175		
Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His Cys Glu		
180 185 190		
Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys Thr Ile		
195 200 205		
Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys Arg Asp		
210 215 220		
Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu Thr Ala		
225 230 235 240		
Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr His Leu Pro		
245 250 255		
Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His Met Gln Thr		
260 265 270		
Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser Thr His Trp		
275 280 285		
Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg Ile Leu Val		
290 295 300		
Ile Phe Ser Gly Met Phe Leu Val Phe Thr Leu Ala Gly Ala Leu Phe		
305 310 315 320		
Leu His Arg Ser Lys Arg Ser Arg Leu Leu His Ser Asp Tyr Met Asn		
325 330 335		
Met Thr Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln Pro Tyr		

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340					345					350					
Ala	Pro	Pro	Arg	Asp	Phe	Ala	Ala	Tyr	Arg	Ser	Arg	Val	Lys	Phe	Ser
	355						360					365			
Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr
	370					375					380				
Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys
	385					390					395				400
Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn
			405						410					415	
Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu
			420					425					430		
Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly
	435					440						445			
His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr
	450					455					460				
Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr
	465					470					475				480
Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly
			485						490					495	
Pro	Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys
			500					505					510		
Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	Ile
	515					520						525			
His	Val	Phe	Ile	Leu	Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu
	530					535					540				
Ala	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu
	545					550					555				560
Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val
			565					570						575	
His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu
			580					585					590		
Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp	Thr	Val
	595					600						605			
Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn
	610					615					620				
Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu	Lys	Asn
	625					630					635				640
Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln	Met	Phe	Ile
			645					650					655		

Asn Thr Ser

<210> SEQ ID NO 52

<211> LENGTH: 662

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 52

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala
1				5					10					15	

Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr
		20						25					30		

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Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly
	35						40					45			
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg
	50					55					60				
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys
	65				70					75					80
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala
			85						90					95	
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Leu
		100						105					110		
Leu	Leu	Val	Thr	Ser	Leu	Leu	Leu	Cys	Glu	Leu	Pro	His	Pro	Ala	Phe
		115					120					125			
Leu	Leu	Ile	Pro	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His
	130					135					140				
Tyr	Trp	Ala	Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr
	145				150					155					160
Phe	Leu	Val	Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp
			165						170					175	
Pro	Cys	Ile	Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro
		180						185					190		
His	Cys	Glu	Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn
		195					200					205			
Cys	Thr	Ile	Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln
	210				215						220				
Cys	Arg	Asp	Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser
	225				230					235					240
Leu	Thr	Ala	Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr
			245						250					255	
His	Leu	Pro	Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His
		260						265					270		
Met	Gln	Thr	Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser
		275					280					285			
Thr	His	Trp	Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg
	290				295						300				
Ile	Leu	Val	Ile	Phe	Ser	Gly	Met	Phe	Leu	Val	Phe	Thr	Leu	Ala	Gly
	305				310					315					320
Ala	Leu	Phe	Leu	His	Arg	Ser	Lys	Arg	Ser	Arg	Leu	Leu	His	Ser	Asp
			325					330						335	
Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly	Pro	Thr	Arg	Lys	His	Tyr
		340						345					350		
Gln	Pro	Tyr	Ala	Pro	Pro	Arg	Asp	Phe	Ala	Ala	Tyr	Arg	Ser	Arg	Val
		355					360					365			
Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn
	370					375					380				
Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val
	385				390					395					400
Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg
			405					410						415	
Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys
		420						425						430	

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Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg			
		435					440					445						
Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys			
	450					455					460							
Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg	Gly	Pro			
465					470					475					480			
Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp	Val	Glu	Ser			
			485						490					495				
Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	Ser	Ile	Ser			
			500					505					510					
Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe	Leu	Thr	Glu			
		515				520						525						
Ala	Gly	Ile	His	Val	Phe	Ile	Leu	Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro			
	530					535					540							
Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile			
545				550						555					560			
Glu	Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu			
			565					570						575				
Ser	Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu			
			580					585					590					
Leu	Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His			
		595					600					605						
Asp	Thr	Val	Glu	Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser			
	610				615					620								
Asn	Gly	Asn	Val	Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu			
625				630						635					640			
Glu	Lys	Asn	Ile	Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln			
		645						650					655					
Met	Phe	Ile	Asn	Thr	Ser													
			660															

<210> SEQ ID NO 53

<211> LENGTH: 665

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 53

Met	Thr	Arg	Val	Thr	Asn	Ser	Pro	Ser	Leu	Gln	Ala	His	Leu	Gln	Ala			
1				5					10					15				
Leu	Tyr	Leu	Val	Gln	His	Glu	Val	Trp	Arg	Pro	Leu	Ala	Ala	Ala	Tyr			
		20						25					30					
Gln	Glu	Gln	Leu	Asp	Arg	Pro	Val	Val	Pro	His	Pro	Tyr	Arg	Val	Gly			
		35				40						45						
Asp	Thr	Val	Trp	Val	Arg	Arg	His	Gln	Thr	Lys	Asn	Leu	Glu	Pro	Arg			
	50				55						60							
Trp	Lys	Gly	Pro	Tyr	Thr	Val	Leu	Leu	Thr	Thr	Pro	Thr	Ala	Leu	Lys			
65				70					75					80				
Val	Asp	Gly	Ile	Ala	Ala	Trp	Ile	His	Ala	Ala	His	Val	Lys	Ala	Ala			
			85				90						95					
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Ala			
		100					105						110					

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Arg	Pro	His	Pro	Trp	Trp	Leu	Cys	Val	Leu	Gly	Thr	Leu	Val	Gly	Leu
		115					120					125			
Ser	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His	Tyr	Trp	Ala
	130					135					140				
Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr	Phe	Leu	Val
145					150					155					160
Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp	Pro	Cys	Ile
				165					170					175	
Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	His	Cys	Glu
			180					185					190		
Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	Cys	Thr	Ile
		195					200					205			
Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	Cys	Arg	Asp
	210					215					220				
Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	Leu	Thr	Ala
225					230					235					240
Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	His	Leu	Pro
				245					250					255	
Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	Met	Gln	Thr
			260					265					270		
Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	Thr	His	Trp
		275					280					285			
Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	Phe	Trp	Val
	290					295					300				
Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr
305					310					315					320
Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	Ser	Arg	Leu	Leu
				325					330					335	
His	Ser	Asp	Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly	Pro	Thr	Arg
			340					345					350		
Lys	His	Tyr	Gln	Pro	Tyr	Ala	Pro	Pro	Arg	Asp	Phe	Ala	Ala	Tyr	Arg
		355					360					365			
Ser	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln
	370					375					380				
Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu
385					390					395					400
Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly
			405					410						415	
Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln
			420					425					430		
Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu
		435					440					445			
Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr
	450					455					460				
Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro
465					470					475					480
Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp
				485				490						495	
Val	Glu	Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg
			500					505					510		
Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe

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515	520	525
Leu Thr Glu Ala Gly Ile His Val Phe Ile Leu Gly Cys Phe Ser Ala		
530	535	540
Gly Leu Pro Lys Thr Glu Ala Asn Trp Val Asn Val Ile Ser Asp Leu		
545	550	555 560
Lys Lys Ile Glu Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu		
565	570	575
Tyr Thr Glu Ser Asp Val His Pro Ser Cys Lys Val Thr Ala Met Lys		
580	585	590
Cys Phe Leu Leu Glu Leu Gln Val Ile Ser Leu Glu Ser Gly Asp Ala		
595	600	605
Ser Ile His Asp Thr Val Glu Asn Leu Ile Ile Leu Ala Asn Asn Ser		
610	615	620
Leu Ser Ser Asn Gly Asn Val Thr Glu Ser Gly Cys Lys Glu Cys Glu		
625	630	635 640
Glu Leu Glu Glu Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His		
645	650	655
Ile Val Gln Met Phe Ile Asn Thr Ser		
660	665	

<210> SEQ ID NO 54

<211> LENGTH: 668

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 54

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala		
1	5	10 15
Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr		
20	25	30
Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly		
35	40	45
Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg		
50	55	60
Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys		
65	70	75 80
Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala		
85	90	95
Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Leu		
100	105	110
Leu Leu Val Thr Ser Leu Leu Leu Cys Glu Leu Pro His Pro Ala Phe		
115	120	125
Leu Leu Ile Pro Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His		
130	135	140
Tyr Trp Ala Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr		
145	150	155 160
Phe Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp		
165	170	175
Pro Cys Ile Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro		
180	185	190
His Cys Glu Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn		

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195					200					205					
Cys	Thr	Ile	Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln
210						215					220				
Cys	Arg	Asp	Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser
225					230					235					240
Leu	Thr	Ala	Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr
				245					250					255	
His	Leu	Pro	Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His
			260					265					270		
Met	Gln	Thr	Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser
		275					280					285			
Thr	His	Trp	Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg
	290					295					300				
Phe	Trp	Val	Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu
305					310					315					320
Leu	Val	Thr	Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	Ser
				325					330					335	
Arg	Leu	Leu	His	Ser	Asp	Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly
			340					345					350		
Pro	Thr	Arg	Lys	His	Tyr	Gln	Pro	Tyr	Ala	Pro	Pro	Arg	Asp	Phe	Ala
		355					360					365			
Ala	Tyr	Arg	Ser	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala
	370					375					380				
Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg
385					390					395					400
Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu
				405					410					415	
Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn
			420					425					430		
Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met
		435				440						445			
Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly
	450					455					460				
Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala
465					470					475					480
Leu	Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu
				485					490					495	
Ala	Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser	Lys	Pro
			500					505					510		
His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu	Leu	Asn
	515					520						525			
Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His	Val	Phe	Ile	Leu	Gly	Cys
	530					535					540				
Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn	Val	Ile
545					550					555					560
Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile	Gln	Ser	Met	His	Ile	Asp
			565					570						575	
Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His	Pro	Ser	Cys	Lys	Val	Thr
		580						585					590		
Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu	Gln	Val	Ile	Ser	Leu	Glu	Ser
	595					600						605			

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Gly Asp Ala Ser Ile His Asp Thr Val Glu Asn Leu Ile Ile Leu Ala
610 615 620

Asn Asn Ser Leu Ser Ser Asn Gly Asn Val Thr Glu Ser Gly Cys Lys
625 630 635 640

Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile Lys Glu Phe Leu Gln Ser
645 650 655

Phe Val His Ile Val Gln Met Phe Ile Asn Thr Ser
660 665

<210> SEQ ID NO 55

<211> LENGTH: 643

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 55

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
1 5 10 15

Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Tyr
20 25 30

Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
35 40 45

Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
50 55 60

Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
65 70 75 80

Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
85 90 95

Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Ala
100 105 110

Arg Pro His Pro Trp Trp Leu Cys Val Leu Gly Thr Leu Val Gly Leu
115 120 125

Ser Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His Tyr Trp Ala
130 135 140

Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr Phe Leu Val
145 150 155 160

Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro Cys Ile
165 170 175

Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His Cys Glu
180 185 190

Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys Thr Ile
195 200 205

Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys Arg Asp
210 215 220

Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu Thr Ala
225 230 235 240

Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr His Leu Pro
245 250 255

Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His Met Gln Thr
260 265 270

Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser Thr His Trp
275 280 285

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Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg Ile Leu Val
 290                295                300

Ile Phe Ser Gly Met Phe Leu Val Phe Thr Leu Ala Gly Ala Leu Phe
305                310                315                320

Leu His Leu Cys Ala Arg Pro Arg Arg Ser Pro Ala Gln Glu Asp Gly
                325                330                335

Lys Val Tyr Ile Asn Met Pro Gly Arg Gly Lys Arg Val Lys Phe Ser
                340                345                350

Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr
                355                360                365

Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys
                370                375                380

Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn
385                390                395                400

Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu
                405                410                415

Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly
                420                425                430

His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr
                435                440                445

Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg Gly Pro Gln Cys Thr
                450                455                460

Asn Tyr Ala Leu Leu Lys Leu Ala Gly Asp Val Glu Ser Asn Pro Gly
465                470                475                480

Pro Met Arg Ile Ser Lys Pro His Leu Arg Ser Ile Ser Ile Gln Cys
                485                490                495

Tyr Leu Cys Leu Leu Leu Asn Ser His Phe Leu Thr Glu Ala Gly Ile
                500                505                510

His Val Phe Ile Leu Gly Cys Phe Ser Ala Gly Leu Pro Lys Thr Glu
                515                520                525

Ala Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu
                530                535                540

Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val
545                550                555                560

His Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu
                565                570                575

Gln Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val
                580                585                590

Glu Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn
                595                600                605

Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn
                610                615                620

Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile
625                630                635                640

Asn Thr Ser

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<210> SEQ ID NO 56

<211> LENGTH: 646

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 56

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Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
1      5      10      15
Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr
20     25     30
Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
35     40     45
Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
50     55     60
Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
65     70     75     80
Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
85     90     95
Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Leu
100    105    110
Leu Leu Val Thr Ser Leu Leu Leu Cys Glu Leu Pro His Pro Ala Phe
115    120    125
Leu Leu Ile Pro Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His
130    135    140
Tyr Trp Ala Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr
145    150    155    160
Phe Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp
165    170    175
Pro Cys Ile Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro
180    185    190
His Cys Glu Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn
195    200    205
Cys Thr Ile Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln
210    215    220
Cys Arg Asp Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser
225    230    235    240
Leu Thr Ala Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr
245    250    255
His Leu Pro Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His
260    265    270
Met Gln Thr Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser
275    280    285
Thr His Trp Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg
290    295    300
Ile Leu Val Ile Phe Ser Gly Met Phe Leu Val Phe Thr Leu Ala Gly
305    310    315    320
Ala Leu Phe Leu His Leu Cys Ala Arg Pro Arg Arg Ser Pro Ala Gln
325    330    335
Glu Asp Gly Lys Val Tyr Ile Asn Met Pro Gly Arg Gly Lys Arg Val
340    345    350
Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn
355    360    365
Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val
370    375    380
Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro Arg

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385	390	395	400
Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys	405	410	415
Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg	420	425	430
Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys	435	440	445
Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg Gly Pro	450	455	460
Gln Cys Thr Asn Tyr Ala Leu Leu Lys Leu Ala Gly Asp Val Glu Ser	465	470	475
Asn Pro Gly Pro Met Arg Ile Ser Lys Pro His Leu Arg Ser Ile Ser	485	490	495
Ile Gln Cys Tyr Leu Cys Leu Leu Leu Asn Ser His Phe Leu Thr Glu	500	505	510
Ala Gly Ile His Val Phe Ile Leu Gly Cys Phe Ser Ala Gly Leu Pro	515	520	525
Lys Thr Glu Ala Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile	530	535	540
Glu Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu	545	550	555
Ser Asp Val His Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu	565	570	575
Leu Glu Leu Gln Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His	580	585	590
Asp Thr Val Glu Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser	595	600	605
Asn Gly Asn Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu	610	615	620
Glu Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln	625	630	635
Met Phe Ile Asn Thr Ser	645		

<210> SEQ ID NO 57

<211> LENGTH: 671

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 57

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala	1	5	10	15
Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr	20	25	30	
Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly	35	40	45	
Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg	50	55	60	
Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys	65	70	75	80
Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala				

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85								90				95			
Asp	Pro	Gly	Gly	Gly	Pro	Ser	Ser	Arg	Leu	Pro	Cys	Ser	Arg	Met	Ala
			100				105						110		
Arg	Pro	His	Pro	Trp	Trp	Leu	Cys	Val	Leu	Gly	Thr	Leu	Val	Gly	Leu
			115				120						125		
Ser	Ala	Thr	Pro	Ala	Pro	Lys	Ser	Cys	Pro	Glu	Arg	His	Tyr	Trp	Ala
			130				135						140		
Gln	Gly	Lys	Leu	Cys	Cys	Gln	Met	Cys	Glu	Pro	Gly	Thr	Phe	Leu	Val
			145				150						155		
Lys	Asp	Cys	Asp	Gln	His	Arg	Lys	Ala	Ala	Gln	Cys	Asp	Pro	Cys	Ile
			165				170						175		
Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	His	Cys	Glu
			180				185						190		
Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	Cys	Thr	Ile
			195				200						205		
Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	Cys	Arg	Asp
			210				215						220		
Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	Leu	Thr	Ala
			225				230						235		
Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	His	Leu	Pro
			245				250						255		
Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	Met	Gln	Thr
			260				265						270		
Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	Thr	His	Trp
			275				280						285		
Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	Ile	Leu	Val
			290				295						300		
Ile	Phe	Ser	Gly	Met	Phe	Leu	Val	Phe	Thr	Leu	Ala	Gly	Ala	Leu	Phe
			305				310						315		
Leu	His	Tyr	Phe	Leu	Gly	Arg	Leu	Val	Pro	Arg	Gly	Arg	Gly	Ala	Ala
			325				330						335		
Glu	Ala	Ala	Thr	Arg	Lys	Gln	Arg	Ile	Thr	Glu	Thr	Glu	Ser	Pro	Tyr
			340				345						350		
Gln	Glu	Leu	Gln	Gly	Gln	Arg	Ser	Asp	Val	Tyr	Ser	Asp	Leu	Asn	Thr
			355				360						365		
Gln	Arg	Pro	Tyr	Tyr	Lys	Lys	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp
			370				375						380		
Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn
			385				390						395		
Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg
			405				410						415		
Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly
			420				425						430		
Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu
			435				440						445		
Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu
			450				455						460		
Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His
			465				470						475		
Met	Gln	Ala	Leu	Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu
			485				490						495		

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Leu Lys Leu Ala Gly Asp Val Glu Ser Asn Pro Gly Pro Met Arg Ile
    500                      505                      510

Ser Lys Pro His Leu Arg Ser Ile Ser Ile Gln Cys Tyr Leu Cys Leu
    515                      520                      525

Leu Leu Asn Ser His Phe Leu Thr Glu Ala Gly Ile His Val Phe Ile
    530                      535                      540

Leu Gly Cys Phe Ser Ala Gly Leu Pro Lys Thr Glu Ala Asn Trp Val
    545                      550                      555                      560

Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile Gln Ser Met
    565                      570                      575

His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His Pro Ser Cys
    580                      585                      590

Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln Val Ile Ser
    595                      600                      605

Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu Asn Leu Ile
    610                      615                      620

Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val Thr Glu Ser
    625                      630                      635                      640

Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile Lys Glu Phe
    645                      650                      655

Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn Thr Ser
    660                      665                      670

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<210> SEQ ID NO 58
<211> LENGTH: 674
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 58

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Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
 1      5      10      15

Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Tyr
 20     25     30

Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
 35     40     45

Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
 50     55     60

Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
 65     70     75     80

Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
 85     90     95

Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Leu
100    105    110

Leu Leu Val Thr Ser Leu Leu Leu Cys Glu Leu Pro His Pro Ala Phe
115    120    125

Leu Leu Ile Pro Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His
130    135    140

Tyr Trp Ala Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr
145    150    155    160

Phe Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp
165    170    175

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Pro	Cys	Ile	Pro	Gly	Val	Ser	Phe	Ser	Pro	Asp	His	His	Thr	Arg	Pro	180	185	190
His	Cys	Glu	Ser	Cys	Arg	His	Cys	Asn	Ser	Gly	Leu	Leu	Val	Arg	Asn	195	200	205
Cys	Thr	Ile	Thr	Ala	Asn	Ala	Glu	Cys	Ala	Cys	Arg	Asn	Gly	Trp	Gln	210	215	220
Cys	Arg	Asp	Lys	Glu	Cys	Thr	Glu	Cys	Asp	Pro	Leu	Pro	Asn	Pro	Ser	225	230	235
Leu	Thr	Ala	Arg	Ser	Ser	Gln	Ala	Leu	Ser	Pro	His	Pro	Gln	Pro	Thr	245	250	255
His	Leu	Pro	Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	260	265	270
Met	Gln	Thr	Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	275	280	285
Thr	His	Trp	Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	290	295	300
Ile	Leu	Val	Ile	Phe	Ser	Gly	Met	Phe	Leu	Val	Phe	Thr	Leu	Ala	Gly	305	310	315
Ala	Leu	Phe	Leu	His	Tyr	Phe	Leu	Gly	Arg	Leu	Val	Pro	Arg	Gly	Arg	325	330	335
Gly	Ala	Ala	Glu	Ala	Ala	Thr	Arg	Lys	Gln	Arg	Ile	Thr	Glu	Thr	Glu	340	345	350
Ser	Pro	Tyr	Gln	Glu	Leu	Gln	Gly	Gln	Arg	Ser	Asp	Val	Tyr	Ser	Asp	355	360	365
Leu	Asn	Thr	Gln	Arg	Pro	Tyr	Tyr	Lys	Lys	Arg	Val	Lys	Phe	Ser	Arg	370	375	380
Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	385	390	395
Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	405	410	415
Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	420	425	430
Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	435	440	445
Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	450	455	460
Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	465	470	475
Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr	Asn	485	490	495
Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly	Pro	500	505	510
Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr	515	520	525
Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His	530	535	540
Val	Phe	Ile	Leu	Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala	545	550	555
Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile	565	570	575

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Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
580 585 590

Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
595 600 605

Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu
610 615 620

Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
625 630 635 640

Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile
645 650 655

Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
660 665 670

Thr Ser

<210> SEQ ID NO 59
 <211> LENGTH: 670
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 59

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
1 5 10 15

Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr
20 25 30

Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
35 40 45

Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
50 55 60

Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
65 70 75 80

Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
85 90 95

Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Ala
100 105 110

Arg Pro His Pro Trp Trp Leu Cys Val Leu Gly Thr Leu Val Gly Leu
115 120 125

Ser Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His Tyr Trp Ala
130 135 140

Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr Phe Leu Val
145 150 155 160

Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp Pro Cys Ile
165 170 175

Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro His Cys Glu
180 185 190

Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn Cys Thr Ile
195 200 205

Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln Cys Arg Asp
210 215 220

Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser Leu Thr Ala
225 230 235 240

Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr His Leu Pro

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245							250					255			
Tyr	Val	Ser	Glu	Met	Leu	Glu	Ala	Arg	Thr	Ala	Gly	His	Met	Gln	Thr
			260				265						270		
Leu	Ala	Asp	Phe	Arg	Gln	Leu	Pro	Ala	Arg	Thr	Leu	Ser	Thr	His	Trp
			275				280						285		
Pro	Pro	Gln	Arg	Ser	Leu	Cys	Ser	Ser	Asp	Phe	Ile	Arg	Ile	Leu	Val
			290				295						300		
Ile	Phe	Ser	Gly	Met	Phe	Leu	Val	Phe	Thr	Leu	Ala	Gly	Ala	Leu	Phe
			305							315			320		
Leu	His	Ser	Ala	Asn	Glu	Arg	Cys	Lys	Ser	Lys	Val	Val	Pro	Cys	Arg
				325						330			335		
Gln	Lys	Gln	Trp	Arg	Thr	Ser	Phe	Asp	Ser	Lys	Lys	Leu	Asp	Leu	Asn
			340						345			350			
Tyr	Asn	His	Phe	Glu	Ser	Met	Glu	Trp	Ser	His	Arg	Ser	Arg	Arg	Gly
			355			360						365			
Arg	Ile	Trp	Gly	Met	Lys	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala
			370			375						380			
Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu
						390						395			
Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp
			405						410			415			
Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu
			420						425			430			
Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile
			435			440						445			
Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr
			450			455						460			
Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met
			465						475			480			
Gln	Ala	Leu	Pro	Pro	Arg	Gly	Pro	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu
			485						490			495			
Lys	Leu	Ala	Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly	Pro	Met	Arg	Ile	Ser
			500						505			510			
Lys	Pro	His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	Leu
			515			520						525			
Leu	Asn	Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His	Val	Phe	Ile	Leu
			530			535						540			
Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	Asn
			545			550			555			560			
Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile	Gln	Ser	Met	His
			565						570			575			
Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His	Pro	Ser	Cys	Lys
			580						585			590			
Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu	Gln	Val	Ile	Ser	Leu
			595			600						605			
Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp	Thr	Val	Glu	Asn	Leu	Ile	Ile
			610			615						620			
Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn	Val	Thr	Glu	Ser	Gly
			625			630			635			640			
Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu	Lys	Asn	Ile	Lys	Glu	Phe	Leu
			645						650			655			

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Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn Thr Ser
660 665 670

<210> SEQ ID NO 60
<211> LENGTH: 673
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 60

Met Thr Arg Val Thr Asn Ser Pro Ser Leu Gln Ala His Leu Gln Ala
1 5 10 15

Leu Tyr Leu Val Gln His Glu Val Trp Arg Pro Leu Ala Ala Ala Tyr
20 25 30

Gln Glu Gln Leu Asp Arg Pro Val Val Pro His Pro Tyr Arg Val Gly
35 40 45

Asp Thr Val Trp Val Arg Arg His Gln Thr Lys Asn Leu Glu Pro Arg
50 55 60

Trp Lys Gly Pro Tyr Thr Val Leu Leu Thr Thr Pro Thr Ala Leu Lys
65 70 75 80

Val Asp Gly Ile Ala Ala Trp Ile His Ala Ala His Val Lys Ala Ala
85 90 95

Asp Pro Gly Gly Gly Pro Ser Ser Arg Leu Pro Cys Ser Arg Met Leu
100 105 110

Leu Leu Val Thr Ser Leu Leu Leu Cys Glu Leu Pro His Pro Ala Phe
115 120 125

Leu Leu Ile Pro Ala Thr Pro Ala Pro Lys Ser Cys Pro Glu Arg His
130 135 140

Tyr Trp Ala Gln Gly Lys Leu Cys Cys Gln Met Cys Glu Pro Gly Thr
145 150 155 160

Phe Leu Val Lys Asp Cys Asp Gln His Arg Lys Ala Ala Gln Cys Asp
165 170 175

Pro Cys Ile Pro Gly Val Ser Phe Ser Pro Asp His His Thr Arg Pro
180 185 190

His Cys Glu Ser Cys Arg His Cys Asn Ser Gly Leu Leu Val Arg Asn
195 200 205

Cys Thr Ile Thr Ala Asn Ala Glu Cys Ala Cys Arg Asn Gly Trp Gln
210 215 220

Cys Arg Asp Lys Glu Cys Thr Glu Cys Asp Pro Leu Pro Asn Pro Ser
225 230 235 240

Leu Thr Ala Arg Ser Ser Gln Ala Leu Ser Pro His Pro Gln Pro Thr
245 250 255

His Leu Pro Tyr Val Ser Glu Met Leu Glu Ala Arg Thr Ala Gly His
260 265 270

Met Gln Thr Leu Ala Asp Phe Arg Gln Leu Pro Ala Arg Thr Leu Ser
275 280 285

Thr His Trp Pro Pro Gln Arg Ser Leu Cys Ser Ser Asp Phe Ile Arg
290 295 300

Ile Leu Val Ile Phe Ser Gly Met Phe Leu Val Phe Thr Leu Ala Gly
305 310 315 320

Ala Leu Phe Leu His Ser Ala Asn Glu Arg Cys Lys Ser Lys Val Val
325 330 335

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<210> SEQ ID NO 61
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 61
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atggcacggc cacatccctg gtggtgtggtc gttctgggga ccctgggtggg gctctca 57

<210> SEQ ID NO 62
 <211> LENGTH: 66
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 62

atgcttctcc tgggtacaag ccttctgtc tgtgagttac cacaccagc attcctctg 60

atccca 66

<210> SEQ ID NO 63
 <211> LENGTH: 516
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 63

gctactccag cccccaagag ctgcccagag aggcactact gggctcaggg aaagctgtgc 60

tgccagatgt gtgagccagg aacattctc gtgaaggact gtgaccagca tagaaaggct 120

gctcagtggtg atccttgcat accgggggtc tccttctctc ctgaccacca caccggccc 180

cactgtgaga gctgtcgga ctgtaactct ggtcttctc ttcgcaactg caccatcact 240

gccaatgctg agtgtgcctg tcgcaatggc tggcagtgca gggacaagga gtgcaccgag 300

tgtgatctc ttccaaacc ttcgtgacc gctcggctgt ctcaggccct gagccacac 360

cctcagccca cccacttacc ttatgtcagt gagatgctgg aggccaggac agctgggac 420

atgcagactc tgggtgactt caggcagctg cctgcccga ctctctctac ccactggcca 480

ccccaaagat cctgtgcag ctccgatttt attcgc 516

<210> SEQ ID NO 64
 <211> LENGTH: 516
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 64

gtacacccg ctccgaagtc ctgcccggag cggcattatt gggcacaggg caagttgtgt 60

tgtcaaatgt gtgagccggg aacctttctc gtgaaggatt gcgatcagca tcggaaggcc 120

gcgagtgctg acccatgtat accaggggtc tcattttccc cagatcacca tacgaggccg 180

cactgtgagt cttgcaggca ttgtaattcc ggcttggtgg tccgcaactg tactattact 240

gcgaatgcag agtgtgcttg tagaaacgga tggcagtgca gggacaaaga atgtacggag 300

tgtgatccac tgcctaacc cagtcttaca gcaagatctt cacaggccct cagcccgcat 360

cctcaaccaa cacatcttcc ttacgtgtca gaaatgttg aggcgcgaac cgcaggccat 420

atgcagacc tggcgactt tcggcagctg ccagcacgca cacttagtac aactggcca 480

ccacaacgca gcttggtgtc ttccgatttc atccgc 516

<210> SEQ ID NO 65
 <211> LENGTH: 81
 <212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 65

ttttgggtgc tgggtggtggt tgggtggagtc ctggcttgct atagcttgct agtaacagtg      60
gcctttatta tttctggtg      81

<210> SEQ ID NO 66
<211> LENGTH: 63
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 66

atccttgtag tcttctctgg aatgttctt gttttcacc tggtcggggc cctgttctc      60
cat      63

<210> SEQ ID NO 67
<211> LENGTH: 340
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 67

cgcgtgaagt tcagcaggag cgcagacgcc cccgcgtacc agcaggggcca gaaccagctc      60
tataacgagc tcaatctagg acgaagagag gactacgatg ttttgacaa aagacgtggc      120
cgggaccctg agatgggggg aaagccgaga aggaagaacc ctacggaagg cctgtacaat      180
gaactgcaga aagataagat ggcggaggcc tacagtgaga ttgggatgaa aggcgagcgc      240
cggaggggca aggggcacga tggcctttac cagggtctca gtacagccac caaggacacc      300
tacgacgccc ttcacatgca ggccctgccc cctcgcgagc      340

<210> SEQ ID NO 68
<211> LENGTH: 126
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 68

aaacggggca gaaagaaact cctgtatata ttcaacaac catttatgag accagtacaa      60
actactcaag aggaagatgg ctgtagctgc cgatttcag aagaagaaga aggaggatgt      120
gaactg      126

<210> SEQ ID NO 69
<211> LENGTH: 72
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 69

ctttgcgcac gccacgcgc cagccccgcc caagaagatg gcaaagtcta catcaacatg      60
ccaggcaggg gc      72

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<210> SEQ ID NO 70
<211> LENGTH: 153
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 70

agcgcgaacg aacgctgcaa aagcaaagtg gtgccgtgcc gccagaaaca gtggcgcacc 60
agctttgata gcaaaaaact ggatctgaac tataaccatt ttgaaagcat ggaatggagc 120
catcgcagcc gccgcggcgc catttggggc atg 153

<210> SEQ ID NO 71
<211> LENGTH: 156
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polynucleotide

<400> SEQUENCE: 71

tacttctctgg gccggctggt ccctcggggg cgaggggctg cggaggcagc gacccggaaa 60
cagcgtatca ctgagaccga gtcgccttat caggagctcc agggtcagag gtcggatgtc 120
tacagcgacc tcaacacaca gaggccgtat taaaa 156

<210> SEQ ID NO 72
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 72

Leu Cys Ala Arg Pro Arg Arg Ser Pro Ala Gln Glu Asp Gly Lys Val
1 5 10 15
Tyr Ile Asn Met Pro Gly Arg Gly
20

<210> SEQ ID NO 73
<211> LENGTH: 51
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 73

Ser Ala Asn Glu Arg Cys Lys Ser Lys Val Val Pro Cys Arg Gln Lys
1 5 10 15
Gln Trp Arg Thr Ser Phe Asp Ser Lys Lys Leu Asp Leu Asn Tyr Asn
20 25 30
His Phe Glu Ser Met Glu Trp Ser His Arg Ser Arg Arg Gly Arg Ile
35 40 45
Trp Gly Met
50

<210> SEQ ID NO 74
<211> LENGTH: 52
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 74

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Tyr Phe Leu Gly Arg Leu Val Pro Arg Gly Arg Gly Ala Ala Glu Ala
1           5           10          15
Ala Thr Arg Lys Gln Arg Ile Thr Glu Thr Glu Ser Pro Tyr Gln Glu
          20          25          30
Leu Gln Gly Gln Arg Ser Asp Val Tyr Ser Asp Leu Asn Thr Gln Arg
          35          40          45
Pro Tyr Tyr Lys
          50

```

1-55. (canceled)

56. A polynucleotide that encodes an anti-CD70 chimeric antigen receptor (CAR) comprising a signal peptide, an anti-CD70 antigen binding domain, transmembrane domain, and at least one intracellular domain, wherein the anti-CD70 antigen binding domain does not comprise an antibody.

57. The polynucleotide of claim **56**, wherein the anti-CD70 antigen binding domain is comprised in an extracellular domain of CD27 and comprises, consists of, or consists essentially of SEQ ID NO:2, SEQ ID NO:4, or SEQ ID NO:5.

58. The polynucleotide of claim **56**, wherein the transmembrane domain is from CD28, the alpha chain of the T-cell receptor, beta chain of the T-cell receptor, zeta chain of the T-cell receptor, CD3 zeta, CD3 epsilon, CD3 gamma, CD3 delta, CD45, CD4, CD5, CD8, CD9, CD 16, CD22, CD33, CD37, CD64, CD80, CD86, CD 134, CD137, CD154, ICOS/CD278, GITR/CD357, NKG2D, DAP10, or DAP12.

59. The polynucleotide of claim **56**, wherein the at least one intracellular domain comprises CD3zeta; an ITAM-containing signaling domain; and/or a costimulatory domain selected from the group consisting of CD27, CD28, 4-1BB, DAP12, NKG2D, OX-40 (CD134), DAP10, CD40L, 2B4, DNAM, CS1, CD48, NKp30, NKp44, NKp46, NKp80, or a combination thereof.

60. The polynucleotide of claim **56**, wherein the signal peptide is from CD27 or granulocyte-macrophage colony-stimulating factor receptor (GMSCF-R).

61. The polynucleotide of claim **56**, wherein the signal peptide comprises, consists of, or consists essentially of SEQ ID NO:6; the antigen binding domain comprises, consists of, or consists essentially of SEQ ID NO:2, SEQ ID NO:4, or SEQ ID NO:5; the transmembrane domain comprises, consists of, or consists essentially of SEQ ID NO:3 or SEQ ID NO:7; and the intracellular domain comprises one or more of SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:11, or SEQ ID NO:12.

62. The polynucleotide of claim **56**, wherein the CAR comprises one or more of the following:

- (a) CD27 signal peptide (SP), CD27 extracellular domain (EC), CD27 transmembrane domain (TMD), DAP12 intracellular domain (ICD), and CD3zeta;
- (b) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, DAP12 ICD;
- (c) CD27 SP, CD27 EC, CD28 TMD, DAP12 ICD, and CD3zeta;

- (d) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, DAP12 ICD, and CD3zeta;
- (e) CD27 SP, CD27 EC, CD27 TMD, Natural killer group 2 member D (NKG2D) ICD, and CD3zeta;
- (f) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, NKG2D ICD, and CD3zeta;
- (g) CD27 SP, CD27 EC, CD28 TMD, Natural killer group 2 member D (NKG2D) ICD, and CD3zeta;
- (h) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, NKG2D ICD, and CD3zeta;
- (i) CD27 SP, CD27 EC, CD27 TMD, 4-1BB ICD, and CD3zeta;
- (j) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, 4-1BB ICD, and CD3zeta;
- (k) CD27 SP, CD27 EC, CD28 TMD, 4-1BB ICD, and CD3zeta;
- (l) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, 4-1BB ICD, and CD3zeta;
- (m) CD27 SP, CD27 EC, CD27 TMD, DAP10 ICD, and CD3zeta;
- (n) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, DAP10 ICD, and CD3zeta;
- (o) CD27 SP, CD27 EC, CD28 TMD, DAP10 ICD, and CD3zeta;
- (p) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, DAP10 ICD, and CD3zeta;
- (q) CD27 SP, CD27 full length (FL), CD27 TMD, CD27 ICD, and CD3zeta;
- (r) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, and CD3zeta;
- (s) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, CD28 ICD, and CD3zeta;
- (t) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, CD28 ICD, and CD3zeta;
- (u) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, 4-1BB ICD, and CD3zeta;
- (v) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, 4-1BB ICD, and CD3zeta;
- (w) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, DAP10 ICD, and CD3zeta;
- (x) GMSCF-R SP, CD27 FL, CD27 TMD, CD27 ICD, DAP10 ICD, and CD3zeta;
- (y) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, DAP12 ICD, and CD3zeta;

- (z) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, DAP12 ICD, and CD3zeta;
 - (aa) CD27 SP, CD27 FL, CD27 TMD, CD27 ICD, NKG2D ICD, and CD3zeta;
 - (bb) GMSCF-R SP, codon optimized CD27 FL, codon optimized CD27 TMD, codon optimized CD27 ICD, NKG2D ICD, and CD3zeta;
 - (cc) CD27 SP, CD27 EC, CD27 TMD, and CD3zeta;
 - (dd) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, and CD3zeta;
 - (ee) CD27 SP, CD27 EC, CD28 TMD, and CD3zeta;
 - (ff) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD28 TMD, and CD3zeta;
 - (gg) CD27 SP, CD27 EC, CD27 TMD, CD28 ICD, and CD3zeta;
 - (hh) GMSCF-R SP, codon optimized CD27 EC, codon optimized CD27 TMD, CD28 ICD, and CD3zeta;
 - (ii) CD27 SP, CD27 EC, CD28 TMD, CD28 ICD, and CD3zeta; or
 - (jj) GMSCF-R SP, codon optimized CD27 EC, CD28 TMD, and CD3zeta.
- 63.** The polynucleotide of claim **56**, wherein the polynucleotide comprises SEQ ID NO:30.
- 64.** An immune cell comprising the polynucleotide of claim **56**.
- 65.** The immune cell of claim **64**, wherein the immune cell is a Natural Killer (NK) cell, T cell, gamma delta T cells, invariant NKT (iNKT) cell, B cell, macrophage, MSCs, or dendritic cell.
- 66.** The immune cell of claim **65**, wherein the NK cell is derived from cord blood, peripheral blood, induced pluripotent stem cells, bone marrow, from a cell line, or a mixture thereof.

- 67.** The immune cell of claim **65**, wherein the NK cell is a CD56+ NK cell.
- 68.** The immune cell of claim **65**, wherein the NK cells express one or more exogenously provided cytokines comprising IL-15, IL-2, IL-12, IL-18, IL-21, IL-7, or a combination thereof.
- 69.** A method of killing CD70-positive cells in an individual, the method comprising the step of administering to the individual a therapeutically effective amount of cells harboring the polynucleotide of claim **56**.
- 70.** The method of claim **69**, wherein the CD70-positive cells are T regulatory cells or CD70 expressing cancer cells.
- 71.** The method of claim **69**, wherein the cells are allogeneic or autologous with respect to the individual.
- 72.** The method of claim **69**, wherein the cells are administered to the individual once or more than once.
- 73.** The method of claim **69**, further comprising the step of providing to the individual an effective amount of an additional therapy comprising one or more antibodies, surgery, gene therapy, immunotherapy, or hormone therapy.
- 74.** The method of claim **69**, wherein the cells are administered to the individual by injection, intravenously, intraarterially, intraperitoneally, intratracheally, intratumorally, intramuscularly, endoscopically, intralesionally, intracranially, percutaneously, subcutaneously, regionally, by perfusion, in a tumor microenvironment, or a combination thereof.
- 75.** The method of claim **69**, further comprising the step of identifying CD70-positive cells in the individual.

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